Machine learning for gravitational-wave detector data: a literature review catalogue for the gravyflow model zoo

Michael Norman

Status: **Work in progress** 19 October 2026

Abstract. We catalogue 313 machine-learning models for gravitational-wave detector data, 138 from the first catalogue (seeded by a thesis literature review) and 175 added by completeness sweeps, and extract 3160 reported metrics from the papers, 676 of them read from figures. For each model we place its headline results on a small set of comparison axes that the literature supports: efficiency at a false-alarm rate normalised per month and in hertz, sensitive distance or volume at a false-alarm rate, Gravity Spy accuracy and macro-F1 with the dataset release, waveform overlap for denoisers, calibration and bias for parameter estimation, and sensitivity depth for continuous waves. Only 26 models are directly comparable on a shared contract and 79 more conditionally; 208 report metrics that cannot be placed on a shared axis. We list 381 suspected errors or internal inconsistencies across 228 papers, record which models release code (106) and trained weights (29), and set out the programme for reproducing these models in gravyflow. What the papers state about speed is recorded with the hardware it was measured on: 172 models give a latency (22), an inference cost (126) or a training cost (120), in 492 records each checked against the paper; there is no efficiency ratio across hardware, and no comparable MLGWSC-1 entry states a latency. Every model carries a placeholder for our reproduction, which is pending.

Contents

1 Scope and method 2
2 A short history and taxonomy 2
3 Compact-binary detection 3
4 Early warning 17
5 Burst and unmodelled transients 19
6 Anomaly and multi-signal detection 24
7 Continuous and long-duration waves 26
8 Glitch classification and data quality 28
9 Denoising and signal extraction 33
10 Parameter estimation 37
11 Space-based detectors 42
12 Other: alerts, generators, lensing, backgrounds 43
13 Latency and compute cost 45
14 Code and reproducibility 59
15 Reproduction programme in gravyflow 61
16 Open questions 67
References 69
A Diagram notation 84

1 Scope and method

Sources. The starting point was the literature review of the owner's thesis, which contributed 46 of the original 138 entries; arXiv and web searches completed the first catalogue. The completeness sweeps then proposed 186 further papers. Four were dropped at the sweep (a review, a population-level classifier, a likely duplicate and an ID fix); of the remaining 182, 175 are kept and 7 are dropped here because they do not train on or apply to gravitational-wave detector data (Table 2). Duplicates in the first catalogue had been merged before.

Metric extraction. Each paper was read by an extraction agent from its PDF, with figures rendered as page images. For every reported metric the agent recorded the value, units, the conditions under which it was measured (noise, population, SNR range, test-set size), its location in the paper (table, figure or section) and whether it was stated in the text or read from a figure; read-from-figure values carry the reading uncertainty noted in the catalogue. The agents also recorded code repositories (checked on the hosting site for licence, last commit and trained weights) and every internal inconsistency they found, with its evidence.

Grading. The synthesiser then placed the headline metrics of each paper on the shared axes of Table 1, converted false-alarm rates to per month (30-day months, the MLGWSC-1 convention) and to hertz where the paper states a rate or a fraction over disjoint segments of stated length, and graded each model comparable, conditionally comparable or not comparable, saying what differs. Suspected errors highlighted in this review are a selection of the extracted inconsistencies, stated neutrally; they are reading notes, not verdicts, and each should be checked against the paper before it is relied on. All 381 extracted notes are kept in the machine-readable catalogue.

Archive. PDFs were downloaded from arXiv only; 276 of the 313 models have a paper in the private archive. The remaining 37 have no arXiv version: their dossiers rest on the publisher's or INSPIRE's abstract page, and they are marked as such. The archive is private and is not redistributed.

Numbers. Every number in the tables is the paper's, located by table, figure or section; "read-from-figure" marks values read from a plot. No value was invented or interpolated by the synthesiser beyond the unit conversions stated.

Table 1. Comparison axes and grades used throughout.

Axis	Meaning
eff@FAR	Detection efficiency (TPR) at a FAR that is a rate per unit time; far_hz and far_per_month given.
FAR@eff	FAR at fixed efficiency.
reach@FAR	Sensitive distance or volume (or hrss50 for bursts) at a stated FAR; the search-grade comparison.
eff@FAP	TPR at a per-window or per-sample false-alarm probability: not a rate; comparable only with the same window length, noise and SNR distribution.
AUC	Area under ROC; quoted only where noise and SNR distribution are stated, otherwise noted as such.
GS	Glitch classification on a Gravity Spy release: accuracy and macro-F1 with release, class count and split.
overlap	Denoising/extraction: overlap or match with the injected waveform.
range	Noise regression: BNS inspiral-range or VT gain on real data.
PE	Parameter estimation: calibration (P-P / coverage), bias or JSD against a sampler, sampling efficiency.
depth@pfa	Continuous waves: sensitivity depth at a false-alarm probability per frequency band.
events	Real catalogue events recovered (supplementary; never a substitute for efficiency at FAR).
other	A headline metric outside the shared axes.
comparable	Same contract as a published reference (MLGWSC-1, LVK injection sets, Gravity Spy split with re-trained baselines, CW depth at pfa), so numbers can be set side by side.
conditional	On a shared axis but with a stated difference (noise, population, detector count, dataset release, relative-only gain).
not comparable	No shared axis: per-window fractions, private toy sets, qualitative or unread.

Table 2. Added models dropped because they are not trained on or applied to gravitational-wave detector data.

Key	arXiv	Reason (dossier)
badaracco2020	2005.09289	Not applied to GW strain (uses seismic correlations; Virgo is not NN-limited so C_NN is modelled, not measured); one hour of data; universality of c(N)=a(1-1/bN) is unexplained; no real cancellation test.
baltus2022light	none	Thesis body not read. The ML content is a compilation of Baltus et al. 2021 (PRD 103, 102003; baltus2021), the CBMI 2021 paper (baltus2021b), Baltus et al. 2022 (PRD 106, 042002; baltus2022), plus a sub-solar-mass matched-filter search and a sky-localisation chapter whose papers are not in the catalogue.
boudart2023	none	Thesis body not read. Its ML chapters correspond to catalogue entries boudart2022 (ALBUS, arXiv 2201.08727) and boudart2022glitch (arXiv 2210.04588).
khan2021	2112.07669	Trained and tested on noiseless NRHybSur3dq8 surrogate waveforms; no detector noise or detector data is used anywhere (PyCBC comparison uses a flat PSD only to set the likelihood). It is a waveform-manifold regression study.
mitra2023	2310.15649	No detector noise, no SNR/distance dependence, no detection stage; axisymmetric sims lack realistic prompt convection; one progenitor; classification of EOS labels not regression. Not an ML-on-detector-data model: the network is trained and tested on noise-free simulated waveforms.
puecher2024	2408.10678	Trained on NR simulation parameters, not on detector data; classifier outputs depend on NR physics and EoS coverage (authors state this is the dominant uncertainty).
whittaker2022	2201.06461	Not applied to detector data, noise, detection or parameter inference; the proof of principle is on a synthetic generator derived from the same 123 fits.

2 A short history and taxonomy

Detection on whitened windows (2017-2019). Two groups showed in 2017-2018 that a small convolutional network on one-second windows of whitened strain can approach matched-filter sensitivity for binary black holes at fixed per-window false-alarm probabilities [1], [2], and that a slightly deeper version of the same approach runs on real O1 noise [3]. A critique followed: per-window fractions are not search false-alarm rates, and networks must be judged on continuous data [4].

Search-grade evaluation (2020-2022). Schafer et al. reported sensitive distance against false-alarm rate on months of continuous simulated data and studied training strategies that keep the network usable at low false-alarm rates [5], [6], [7]. MLGWSC-1 made this contract a benchmark with four datasets, the last on real O3a noise, and found that the best entries came close to PyCBC on simulated noise but fell further behind on real noise [8].

Production searches (2023-2026). Networks then ran on real detector data with measured backgrounds and injection studies, several over full observing runs: Aframe and its BNS extension [9], [10], [11], AresGW on MLGWSC-1 and O3 [12], [13], Sage on the MLGWSC-1 O3a month [14], SNR-series networks [15], [16] and anomaly pipelines [17]. Machine learning inside the cWB burst pipeline followed a parallel route, for generic bursts and for binary black holes [18], [19].

Glitches, denoising and inference. Gravity Spy set the image format and labels for glitch classification [20], [21], later extended to O3 [22], [23]. Denoising began with recurrent autoencoders [24] and noise regression from witness channels [25]. Inference moved from point regressors to normalising-flow posteriors validated against samplers on real events [26], [27], [28].

Taxonomy. We organise the catalogue by task (compact-binary detection, early warning, bursts, anomalies, continuous waves, glitches, denoising, parameter estimation, space-based detectors, other) and, within each task, by family. A second axis cuts across tasks: the input representation (whitened strain, time-frequency images, matched-filter SNR series, pipeline triggers or alert products, auxiliary witness channels). The representation determines most of the adaptation work in gravyflow (Section Section 15).

Table 3. Models per section, with grades.

Section	Models	Added	Comparable	Conditional	Not comparable
Compact-binary detection	92	40	12	24	56
Early warning	9	3	0	3	6
Burst and unmodelled transients	33	12	5	10	18
Anomaly and multi-signal detection	11	7	2	1	8
Continuous and long-duration waves	18	14	5	5	8
Glitch classification and data quality	56	32	0	14	42
Denoising and signal extraction	32	19	0	13	19
Parameter estimation	33	28	2	9	22
Space-based detectors	14	12	0	0	14
Other: alerts, generators, lensing, backgrounds	15	8	0	0	15

3 Compact-binary detection

Compact-binary coalescence (CBC) detection is the largest category. The early papers asked whether a network could match matched filtering on short whitened windows of simulated noise; the later ones ask whether a network-based search, run over months of real data with a measured background, reaches the sensitive distance of the production pipelines. Only the second kind of result can be compared across papers.

3.1 Families and lineage

3.1.1 Deep Filtering and 1D classifiers on whitened strain

The first wave fed one-second windows of whitened strain to small 1D networks and compared them with matched filtering at fixed per-window false-alarm probabilities [1], [2] real O1 noise and two-detector coincidence followed [3], [29]. Later work varied the waveform space (binary neutron stars, [30], [31] precession, [32] eccentricity and higher modes, [33]), the architecture or training [34], [35], [36], and the output (Bayesian networks, [37] ensembles, [38] four classes, [39]).

> **Our results (to be added)**
> gravyflow reproduction: *pending* for 37 models: [29], [2], [1], [3], [33], [40], [41], [42], [43], [44], [45], [35], [30], [46], [47], [48], [31], [37], [49], [34], [50], [51], [52], [38], [53], [39], [32], [54], [55], [56], [57], [58], [36], [59], [60], [61], [62]. Results will be placed on the shared axes of this section beside the published values.

3.1.2 Per-timestep (WaveNet) ensembles

Gebhard et al. argued that a detector must run on continuous data and be judged by its false positives per unit time [4]. The WaveNet-style ensembles of Huerta, Wei and collaborators label each time step and add coincidence across models and detectors [63], [64], [65], later with higher modes and graph layers [66], [67].

> **Our results (to be added)**
> gravyflow reproduction: *pending* for 7 models: [4], [68], [64], [63], [65], [66], [67]. Results will be placed on the shared axes of this section beside the published values.

Type	Num Filters / Neurons / Heads	Filter / Pool / Head Size	Stride	Dilation	Activation Function
Conv Stage 1	16	16, 4	1, 4	1	ReLU
Conv Stage 2	32	8, 4	1, 4	4	ReLU
Conv Stage 3	64	8, 4	1, 4	4	ReLU
Flatten	-	-	-	-	-
Dense Head	64, 2	-	-	-	ReLU, SoftMax

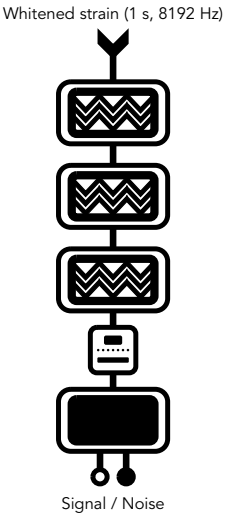


Figure 1: George & Huerta 2018 Deep Filtering (shallow net) [1], collapsed blocks. Architecture from: Paper Fig. 5 (layer list of the 3-convolution net, about 2 MB) and the text of Sec. Neural networks; the 4-convolution deeper net is Fig. 6. Diagram confidence: **high**; 5 details the paper leaves unstated are listed in the diagram spec, and are not drawn. Notation: Appendix A.

Type	Num Filters / Neurons / Heads	Filter / Pool / Head Size	Activation Function	Dropout
Conv Stage 1	8	64, 32, 8	ELU	-
Conv Stage 2	16	32, 16, 6	ELU	-
Conv Stage 3	32	16, 16, 4	ELU	-
Dense Head	64, 64, 2	-	ELU, SoftMax	0.5, 0.5

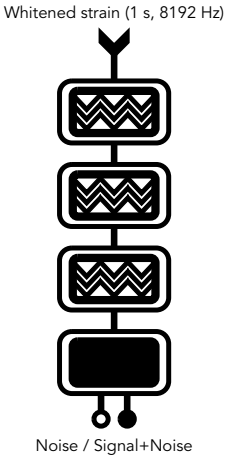


Figure 2: Gabbard et al. 2018 CNN [2], collapsed blocks. Architecture from: Paper Table I (neurons, filter size, max-pool size, dropout and activation per layer). The caption's pooling positions (layers 1, 5, 8) contradict the table's own columns; this diagram follows the table (pooling after layers 2, 4 and 6). Diagram confidence: **high**; 4 details the paper leaves unstated are listed in the diagram spec, and are not drawn. Notation: Appendix A.

	Type	Num Filters / Neurons / Heads	Filter / Pool / Head Size	Dilation	Activation Function
	Convolutional	512	1	-	-
×12 Dilated Conv Blocks	Dilated Conv Block	512	2	1-2048	ReLU
	Convolutional	1	1	-	Sigmoid

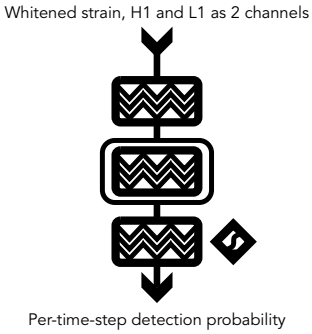


Figure 3: Gebhard et al. 2019 fully convolutional detector [4], collapsed blocks. Architecture from: Paper Sec. IV A (Model architecture) and Fig. 3 (schematic of the dilated stack): input 1x1 convolution, 12 dilated convolution blocks, output 1x1 convolution, sigmoid. Diagram confidence: **high**; 2 details the paper leaves unstated are listed in the diagram spec, and are not drawn. Notation: Appendix A.

3.1.3 Search-grade evaluation and the MLGWSC-1 benchmark

Schafer et al. measured sensitive distance against FAR on continuous simulated data [5], [6], [7], and the first Machine Learning Gravitational-Wave Search Mock Data Challenge (MLGWSC-1) fixed that contract for four datasets, the last on real O3a noise [8]. Entries and successors on the same datasets include AresGW [12] (later run over O3 itself [13]), the updated TPI FSU Jena search [69], Sage [14], a matched-filter hybrid [70] and the transformer pipelines GW-Whisper and CASTOR [71], [72].

Type	Num Filters / Neurons / Heads	Filter / Pool / Head Size	Dilation	Activation Function
Convolutional (L1)	-	-	-	ReLU
Gated Dilated Residual Block (L1)	-	3	-	Tanh x Sigmoid
Addition (L1)	-	-	-	ReLU
Convolutional (L1)	-	1	-	ReLU
Convolutional (L1)	-	1	-	-
Convolutional (H1)	-	-	-	ReLU
Gated Dilated Residual Block (H1)	-	3	-	Tanh x Sigmoid
Addition (H1)	-	-	-	ReLU
Convolutional (H1)	-	1	-	ReLU
Convolutional (H1)	-	1	-	-
Concatenate				
Convolutional	-	3	-	ReLU
Convolutional	-	3	-	Sigmoid

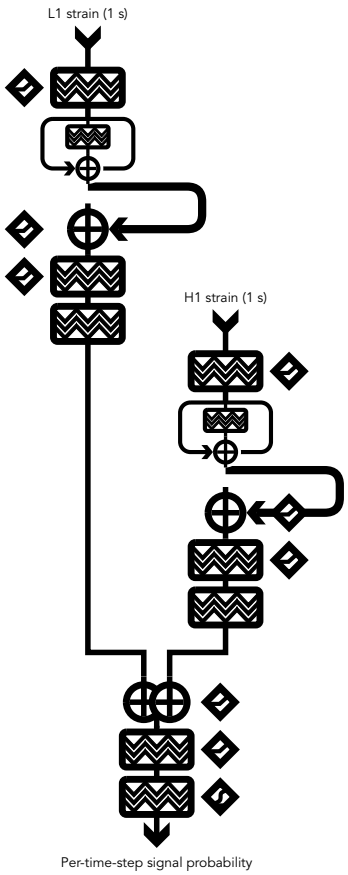
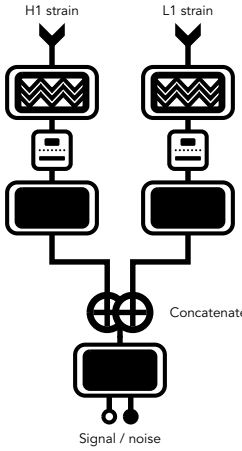


Figure 4: WaveNet detector, Model I (Wei et al. 2021) [63], collapsed blocks. Architecture from: Fig. 1 (two WaveNet lanes for L and H, concatenation, two final convolutions, sigmoid) and Section II text (kernel 3 non-causal dilated convolutions; Model II changes). Diagram confidence: **medium**; 8 details the paper leaves unstated are listed in the diagram spec, and are not drawn. Notation: Appendix A.

Type	Num Filters / Neurons / Heads	Filter / Pool / Head Size	Stride	Activation Function	Dropout
Convolutional ×6 + pooling	8-32	64-16, pool 4-2	1, pool = size	ELU	-
Flatten	-	-	-	-	-
Dense ×2	64, 64	-	-	ELU	yes
Concatenate					
Dense Head	128, 64, 2	-	-	ELU, SoftMax	-



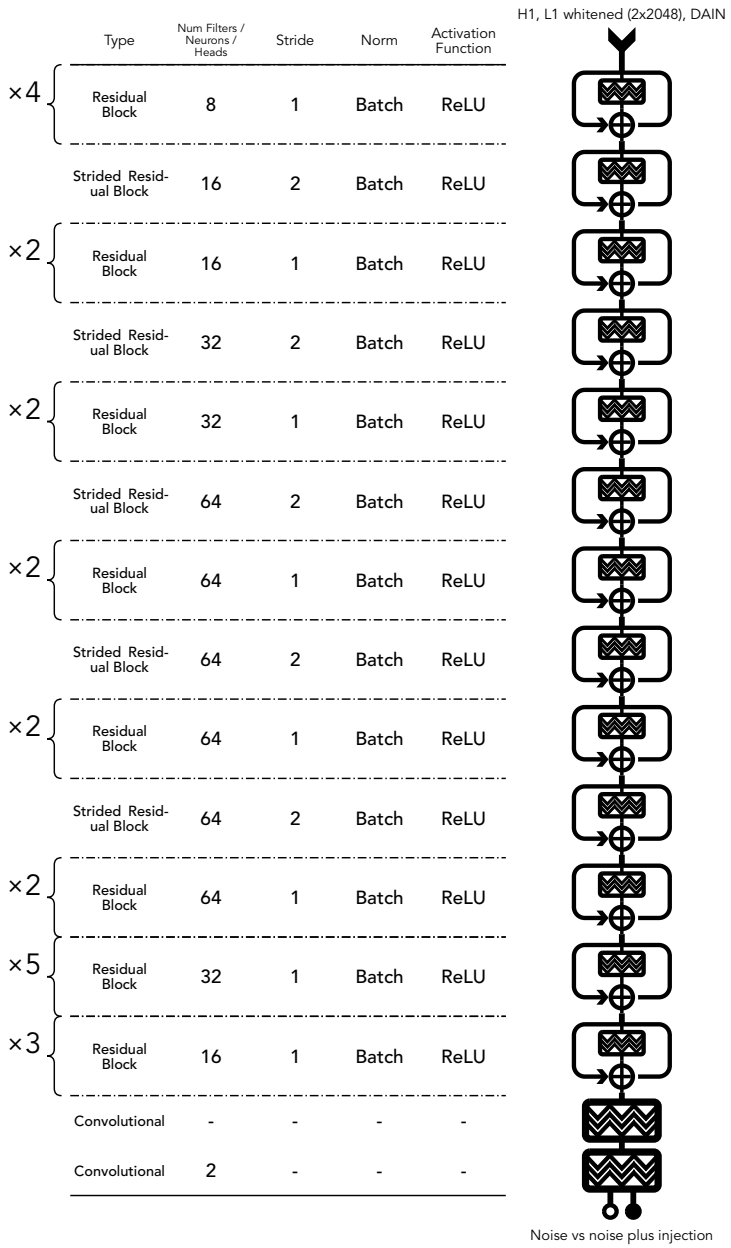
Type	Num Filters / Neurons / Heads	Filter / Pool / Head Size	Stride	Activation Function	Dropout
Convolutional ×6 + pooling	8-32	64-16, pool 4-2	1, pool = size	ELU	-
Flatten	-	-	-	-	-
Dense ×2	64, 64	-	-	ELU	yes
Concatenate					
Dense Head	128, 64, 2	-	-	ELU, SoftMax	-

Figure 5: Schafer & Nitz 2021 two-detector coincident network [7], collapsed blocks. Architecture from: Paper Fig. 3 (two-detector architecture, sub-networks A, B and C) with the single-detector sub-network from Table I (final dense layer removed). Diagram confidence: **high**; 7 details the paper leaves unstated are listed in the diagram spec, and are not drawn. Notation: Appendix A.

> **Our results (to be added)**
> gravyflow reproduction: *pending* for 10 models: [5], [6], [7], [8], [12], [13], [69], [14], [70], [72]. Results will be placed on the shared axes of this section beside the published values.

3.1.4 Production-style searches on observing-run data

Aframe brought a 1D ResNet to real-time and offline O3 searches with one-year and hundred-year time-slide backgrounds and injections drawn from the GWTC-3 sensitivity population [9], [10], and was extended to binary neutron stars [11]. Other searches with measured backgrounds, mostly on O3 data (O2 for [73], O1 for [74], [75], O1–O3 for [76], simulated noise for [77]), include Q-transform ResNets [73], [78], [79], SNR-series networks [15], [16], [77], and lightweight or alternative-physics detectors [74], [75], [76], [80], [81].



	Type	Num Filters / Neurons / Heads	Stride	Norm	Activation Function
×4	Residual Block	8	1	Batch	ReLU
	Strided Residual Block	16	2	Batch	ReLU
×2	Residual Block	16	1	Batch	ReLU
	Strided Residual Block	32	2	Batch	ReLU
×2	Residual Block	32	1	Batch	ReLU
	Strided Residual Block	64	2	Batch	ReLU
×2	Residual Block	64	1	Batch	ReLU
	Strided Residual Block	64	2	Batch	ReLU
×2	Residual Block	64	1	Batch	ReLU
	Strided Residual Block	64	2	Batch	ReLU
×2	Residual Block	64	1	Batch	ReLU
×5	Residual Block	32	1	Batch	ReLU
×3	Residual Block	16	1	Batch	ReLU
	Convolutional	-	-	-	-
	Convolutional	2	-	-	-

Figure 6: AresGW 54-layer 1D ResNet (Nousi et al.) [12], collapsed blocks. Architecture from: Table I (blocks, filters, strided blocks, input shapes), Fig. 3 and Sec. II B (two convolutions per block, batch norm and ReLU after each convolution, stride-2 shortcut layer in blocks 5, 8, 11, 14, 17, two final convolutions to 2 channels). Diagram confidence: **medium**; 7 details the paper leaves unstated are listed in the diagram spec, and are not drawn. Notation: Appendix A.

> **Our results (to be added)**
> gravyflow reproduction: *pending* for 15 models: [73], [82], [78], [77], [81], [74], [9], [79], [15], [10], [16], [76], [80], [75], [11]. Results will be placed on the shared axes of this section beside the published values.

3.1.5 Matched-filter front ends

A template bank inside or in front of the network: fixed [83] or trainable [84] template layers, matched filtering after denoising [85], SNR-map classifiers [86], and overlap statistics against reconstructed signals [87].

> **Our results (to be added)**
> gravyflow reproduction: *pending* for 6 models: [83], [84], [85], [87], [88], [86]. Results will be placed on the shared axes of this section beside the published values.

3.1.6 Time-frequency images

Spectrogram or Q-transform images fed to image backbones (ResNets, a U-Net, a COCO-pretrained YOLO detector) [89], [90], [91], [92], including Einstein Telescope studies [93], [94].

Type	Num Filters / Neurons / Heads	Filter / Pool / Head Size	Stride	Activation Function	Dropout
Conv Stage 1	16	33, 17, pool 4	32, 16; 1, pool 4	ELU	-
Conv Stage 2	16-32	17, 16, 9, 8; pool 3	1, pool 3	ELU	-
Conv Stage 3	32-64	9, 8, 9, 8; pool 2	1, pool 2	ELU	-
Flatten	-	-	-	-	-
Dense Head	128, 128, 2	-	-	ELU	yes (rate n/s)

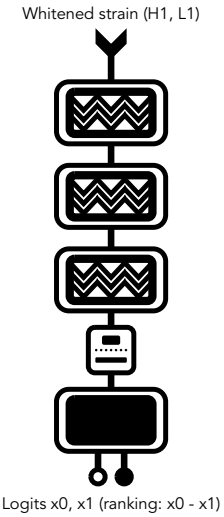


Figure 7: Zelenka et al. 2024 coherent two-detector 1D CNN (TPI FSU Jena MLGWSC-1 entry) [69], collapsed blocks. Architecture from: Paper Table II (base network: layer, kernel size, output shape, activation); output statistic from Sec. III E. Diagram confidence: **high**; 5 details the paper leaves unstated are listed in the diagram spec, and are not drawn. Notation: Appendix A.

Type	Num Filters / Neurons / Heads	Filter / Pool / Head Size	Stride	Norm	Activation Function
Multiscale Frontend (H1)	32-128	4-256	-	-	-
Multiscale Frontend (L1)	32-128	4-256	-	-	-
Stack detectors (2D feature map)					
CBAM-ResNet50 Backend	64-512	1x1, 3x3	-	-	ReLU
Embedding	512	-	-	-	-
3 separate FC heads	512 x 3	-	-	-	Sigmoid

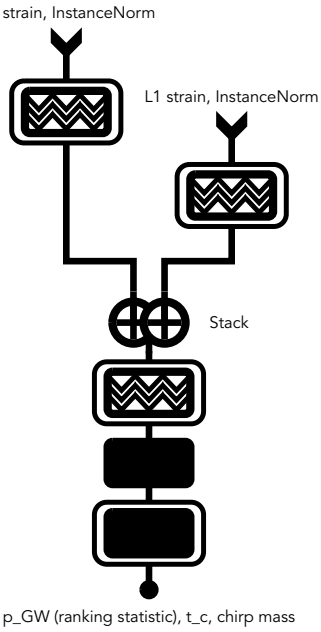


Figure 8: Sage (Nagarajan and Messenger 2025) [14], collapsed blocks. Architecture from: Sec. II G text, Fig. 8(a) (multi-scale residual block) and Fig. 8(b) (frontend stack); backend and heads from the Backend Classifier and Point-estimate Outputs paragraphs. Diagram confidence: **medium**; 8 details the paper leaves unstated are listed in the diagram spec, and are not drawn. Notation: Appendix A.

Our results (to be added)
gravyflow reproduction: *pending* for 11 models: [89], [95], [96], [97], [93], [90], [94], [98], [92], [91], [99]. Results will be placed on the shared axes of this section beside the published values.

3.1.7 Re-ranking inside existing pipelines

Classifiers, density estimators or learned signal-consistency tests that re-rank the triggers of GstLAL, PyCBC or GRB-triggered searches, or combine several pipelines' outputs, rather than read strain [100], [101], [102], [103], [104], [105].

Our results (to be added)
gravyflow reproduction: *pending* for 6 models: [100], [101], [102], [103], [105], [104]. Results will be placed on the shared axes of this section beside the published values.

3.2 Comparison on the shared axes

Rows are the models graded comparable or conditionally comparable (36 of 92); each axis entry is one row. FAR is converted to per month (30 d) and to Hz where the paper states a rate.

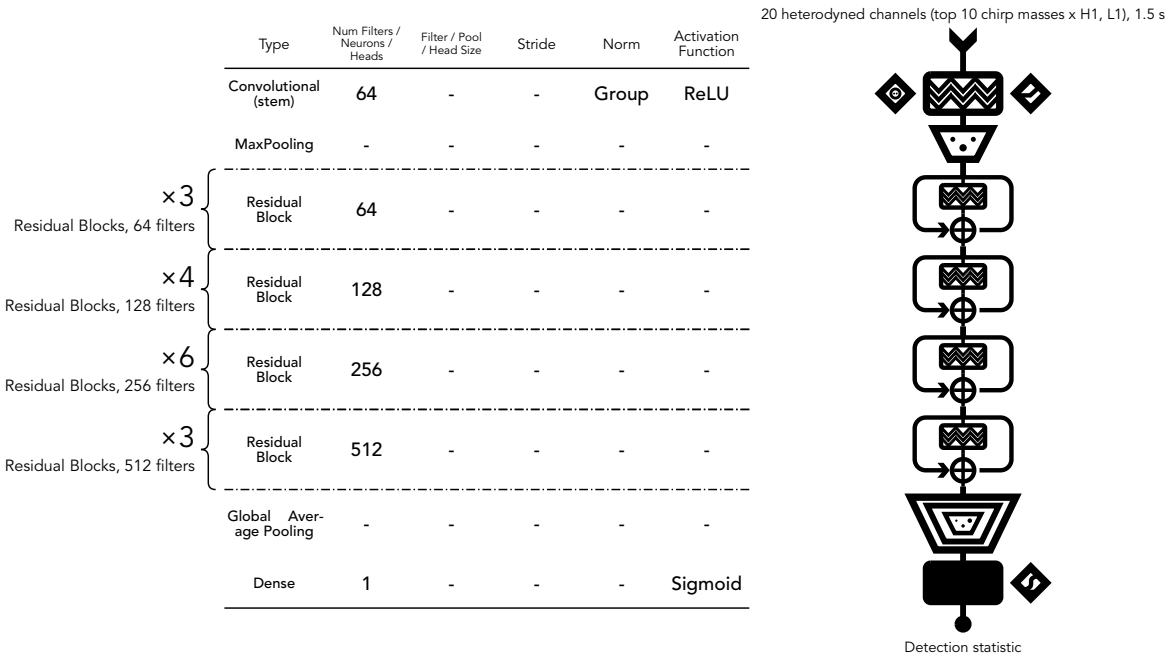


Figure 9: Aframe-BNS (Gupta et al. 2026) [11], collapsed blocks. Architecture from: Paper Sec. II A and II C (text): 'modified ResNet-34' with 1D convolutions, group norm and a single sigmoid output, identical to Aframe BBH except for 2k=20 input channels; Fig. 4a shows the heterodyne pre-processing only. Stage layout [3,4,6,3] and filters 64-512 are the standard ResNet-34 definition, not stated in this paper. Diagram confidence: **medium**; 6 details the paper leaves unstated are listed in the diagram spec, and are not drawn. Notation: Appendix A.

Model	Axis	Published value (location; how obtained)	FAR	Grade: what differs	gravyflow
`gabbard2018` [2]	eff@FAP	CNN vs matched filter TPR at FAP 0.1, rho_opt 4: ~0.44 vs ~0.31; at FAP 1e-3, rho_opt 6: ~0.49 (both) (Fig 3; read-from-figure)	-	conditional Per-SNR balanced test sets of 1 s windows in Gaussian aLIGO-design noise, non-spinning BBH; FAP is per window, so no rate. The paired matched-filter curve on identical data is the paper's comparison.	*pending*
`bresten2019` [40]	AUC	mixed-SNR (2.1-18) test: raw CNN 0.50, TDA 0.76, raw+TDA 0.79 (Fig 4; stated)	-	conditional White (uncoloured) Gaussian noise with a stated SNR range; AUC comparable only to other white-noise toy studies.	*pending*
`gebhard2019` [4]	eff@FAR	~89 % of injections (SNR 5-20) at one false positive per 19.5 min (Sec VII A; stated)	2.22e+03 /month; 0.000855 Hz	conditional Real O1 noise, but false positives are counted over 16,384 separate 8 s test examples, not continuous data, and the paper warns this rate cannot be compared directly to a false-alarm rate; ~2.2e3 per month, over a uniform-SNR injection set rather than a population.	*pending*
	events	all GWTC-1 events except GW170817 (Sec VII C; stated)	-		*pending*
`jadhav2020` [102]	reach@FAR	VT ratio MLStat/base 1.32 / 1.27 / 1.30 / 1.29 at IFAR 0.1 / 1 / 10 / 100 yr (chirp mass 5-50 Msun) (Fig 3 bottom; read-from-figure)	0.0821 /month; 3.17e-08 Hz	conditional PyCBC re-ranking with a glitch classifier on O1/O2 chunks; VT ratios relative to the base statistic.	*pending*
	events	GW151012 IFAR 5.84 to 218.1 yr; GW170729 0.73 to 55.8 yr (Sec IV; stated)	-		*pending*
`schafer2020` [5]	reach@FAR	sensitive distance ~130 Mpc at 10 per month (BNS); PyCBC Live reconstructed ~245 Mpc at 0.5 per month (Abstract, Sec V, Fig 4; stated)	10 /month; 3.86e-06 Hz	comparable Continuous ~101 d of Gaussian aLIGO-design H1+L1 data with BNS injections, sensitive distance vs FAR: the template for the later MLGWSC-1 metric. Gaussian noise only.	*pending*
`menendezvazquez2021` [73]	eff@FAR	L1-H1 SNR for 50 / 80 / 99 % efficiency: 11 / 12 / 19 (low mass), 5 / 6 / 7 (high mass) at ~25 false positives per day (Table III; stated)	750 /month; 0.000289 Hz	conditional Threshold set for ~25 false positives per day on a test set; the real O2 scan gives 24-178 per day (34-47 for L1-H1), so the rate is nominal.	*pending*
	events	8 of 11 O1-O2 catalogue events (Table IV; stated)	-		*pending*
`schafer2021a` [6]	reach@FAR	sensitive distance ~2150 Mpc vs matched filter ~2340 Mpc at 1 per month (single detector, BBH 10-50 Msun) (Sec III A, Fig 5; stated (ML); read-from-figure (matched filter))	1 /month; 3.86e-07 Hz	conditional Single detector, Gaussian design noise, with a matched-filter baseline on the same data; sensitive distance at a true FAR. Distances differ from MLGWSC-1 by detector count and population.	*pending*
	eff@FAP	efficiency 0.90-0.94 at SNR 9, FAP 1e-4 (Sec III B; stated)	-		*pending*
`schafer2021b` [7]	reach@FAR	two-detector coincidence retains 92.4 % of the time-coincident matched-filter distance down to ~4e-4 per month; ~2200 Mpc at 4e-4/month (Sec II H, Fig 2; stated (ratio); read-from-figure (distance))	0.0004 /month; 1.54e-10 Hz	conditional Gaussian design noise, H1+L1, ~200 yr time-shift background; ML and matched filter on the same data.	*pending*
`wei2021ensemble` [63]	eff@FAR	no efficiency; 3 false positives in 200 h of August 2017 H1+L1 data (~1 per 2.7 d) (Abstract, Sec IV; stated)	10.8 /month; 4.17e-06 Hz	conditional A true FAR on real data but with no injection campaign, so no efficiency to pair it with.	*pending*

Model	Axis	Published value (location; how obtained)	FAR	Grade: what differs	gravyflow
	events	GW170104, 0729, 0809, 0814, 0817, 0818, 0823, GW190412 (0608 and 190521 after fine-tuning) (Sec III B, Figs 4, 7; stated)	-		*pending*
`xia2021` [34]	AUC	ConvNet5 0.91 vs basic 0.86 (SNR 7-7.5); 0.98 vs 0.96 (SNR 7-15) (Table IV; stated)	-	conditional Gaussian noise, stated SNR ranges, 500-sample test sets (AUC uncertainty ~0.01-0.02).	*pending*
`andrews2022` [82]	eff@FAP	TAP ~0.86 at 3 sigma (FAP 1.3e-3) and ~0.81 at 4 sigma, H1+L1, real O1 noise (Fig 3 left; read-from-figure)	-	conditional Real O1 noise with a BBH injection population, but FAP is a fraction of background windows without time slides.	*pending*
`mcisaac2022` [103]	reach@FAR	VT gain of the tuned chi^2 at FAR 1/yr: ~4 % (100-300 Msun) to ~11 % (300-600 Msun); band labels +2 to +17 % (Fig 5, Sec V; stated)	0.0821 /month; 3.17e-08 Hz	conditional Relative VT gain inside a PyCBC IMBH search on 8 held-out days.	*pending*
`mlgwsc2022` [8]	reach@FAR	DS4 (real O3a): PyCBC 1544, Virgo-AUTh 1092, MFCNN 650, CNN-Coinc 0, TPI 0 Mpc at 1/month (Table IV; stated)	1 /month; 3.86e-07 Hz	comparable The reference contract: one month of H1+L1 test data per dataset, IMRPhenomXPHM injections, FAR per month from injection-free background, chirp-distance-weighted sensitive distance.	*pending*
	reach@FAR	DS1 (Gaussian): PyCBC 2492, TPI 2363, AUTh 2116 Mpc at 1/month (Table IV; stated)	1 /month; 3.86e-07 Hz		*pending*
`andrescarcasona2023` [78]	eff@FAR	network SNR for 80 / 90 / 95 % efficiency, H1-L1-V1: 17.3 / 21.1 / 25.8 (Table IV; stated)	-	conditional Real O3 search with time-slide FAR down to 1/152.6 yr, but the threshold behind Table IV is not clearly stated.	*pending*
	other	O3 merger-rate upper limits 560 to 3e6 Gpc^-3 yr^-1 (asymmetric, sub-solar) (Sec IV; stated)	-		*pending*
`beveridge2023` [77]	eff@FAR	TPR ~0.18 / ~0.48 at network SNR 8 / 10 at 1 per 2 months (1.9e-7 Hz) (Fig 8; read-from-figure)	0.5 /month; 1.93e-07 Hz	conditional Simulated coloured Gaussian noise with simulated glitches; a true FAR from 6e6 s of background, but the alert threshold lies in the extrapolated tail.	*pending*
	eff@FAR	34 % of 10,000 GWTC-3-like injections at FAR < 1 per 2 months (Sec III B 1; stated)	0.5 /month; 1.93e-07 Hz		*pending*
`jadhav2023` [81]	reach@FAR	sensitive distance at 1/month on 8.8 d of O3: 140 / 996 / 2173 Mpc for chirp mass 1-10 / 10-40 / 40-80 Msun (Sec VI B, Fig 16; stated)	1 /month; 3.86e-07 Hz	conditional Real O3 data with injections and a sensitive distance at a true FAR, but mass bins and the distance definition differ from MLGWSC-1.	*pending*
	AUC	CBC vs PyCBC background triggers 0.976 (0.989 after GAN training) (Fig 14b; stated)	-		*pending*
	events	GW190519 and GW190521_074359 at FAR < 1/20,350 yr (Sec VI; stated)	-		*pending*
`nousi2022` [12]	reach@FAR	MLGWSC-1 DS4: ~1575 Mpc at 1/month (PyCBC ~1545, cWB ~1230 on the same figure) (Fig 10; read-from-figure)	1 /month; 3.86e-07 Hz	comparable MLGWSC-1 DS4 with the challenge evaluation: directly comparable with the challenge entries.	*pending*
`tian2023` [66]	eff@FAR	TPR ~0.95 at SNR 8.4 and ~0.55 at SNR 6.2 for 2-3 false positives per year (coloured Gaussian HLV, O4/O5 PSDs) (Fig 2, Table 1; read-from-figure)	0.246 /month; 9.51e-08 Hz	conditional A year of synthetic coloured Gaussian HLV noise: a FAR, but for Gaussian noise only.	*pending*
`tian2024gnn` [67]	AUC	ROC AUC 0.982 / 0.973 / 0.965 (2 / 3 / 4 models) at network SNR 8.4 (Table 1; stated)	-	conditional Decade of synthetic coloured Gaussian noise; the real-data month is a tuned ensemble with manual vetting.	*pending*
	eff@FAR	TPR 0.45 / 0.33 / 0.27 at SNR 6.2 for 7 / 3 / 2 false positives per decade (synthetic HLV) (Fig 1 insets; read-from-figure)	0.0246 /month; 9.51e-09 Hz		*pending*
`trovato2023` [74]	eff@FAR	efficiency at FAR 1e-5 Hz: TCN/IT 0.49 / 0.75 / 0.91 at SNR 9.5 / 11.5 / 14.5; CNN 0.43 / 0.68 / 0.85 (Fig 5; read-from-figure)	25.9 /month; 1e-05 Hz	conditional Single-detector L1 O1 test segments; FAR is the test-set false-alarm count per second of segments, not time slides; model selected on the test set.	*pending*
	eff@FAR	TCN ~0.76 at ~1.7e-6 Hz, the lowest FAR its curve reaches (averaged over SNR 8-20) (Fig 4; read-from-figure)	4.41 /month; 1.7e-06 Hz		*pending*
`badger2024` [87]	eff@FAR	overlap statistic TPR ~0.06 vs matched filter ~0.22 at 12/yr (astrophysical population, simulated A+ Gaussian, single detector) (Fig 4; read-from-figure)	0.986 /month; 3.8e-07 Hz	conditional Single-detector simulated A+ noise with FAR from one month of noise; a paired matched-filter baseline on the same injections.	*pending*
`koloniari2024` [13]	events	34 of 43 published O3 events in range at p_astro > 0.5; 28 at FAR < 1/yr; 8 new candidates (Sec VI A, Sec VII, Tables I, III, VII; stated)	-	conditional Real O3 search with a 10 yr time-shifted background but no injection-based sensitivity: events recovered is the only efficiency.	*pending*
`marx2024` [9]	reach@FAR	sensitive volume at 1/yr: Aframe ~10.4 vs PyCBC-BBH ~9.5, GstLAL ~9.8 Gpc^3 (35+35 Msun); 20+10 Msun: ~0.7 vs ~1.4 (Fig 4; read-from-figure)	0.0821 /month; 3.17e-08 Hz	comparable O3 data with a year of time slides and its own injections drawn from the GWTC-3 sensitivity prior, against GWTC-3 pipeline volumes from the public data release: comparable with the LVK searches on sensitive volume at FAR.	*pending*
	events	9 of 9 GWTC-3 candidates in the test period (Table IV; stated)	-		*pending*
`menendez2024` [79]	eff@FAR	network SNR for 50 / 80 / 99 % efficiency at 1/week: 10 / 12 / 20 (high mass), 20 / 24 / 39 (low mass) (Table 2; stated)	4.29 /month; 1.65e-06 Hz	conditional Real O3 HLV data with time-slide FAR down to 1/152.6 yr; efficiency in network SNR rather than distance.	*pending*
	events	26 of 50 three-detector O3 events at 1/week (22 of 31 in range) (Sec 5; stated)	-		*pending*
`wang2023tf` [91]	reach@FAR	U-Net sensitive distance ~230 Mpc at 1/month, ~1000 at 100/month (own O3a time-shifted background) (Fig 4; read-from-figure)	1 /month; 3.86e-07 Hz	conditional Sensitive distance in the MLGWSC-1 style but on 32.5 d of its own background from 10 shifts of 3.25	*pending*

Model	Axis	Published value (location; how obtained)	FAR	Grade: what differs	gravyflow
				d, and MLGWSC-1 curves are overplotted from a different data set.	
	events	all O1/O2 BBH and 59 of 73 O3 BBH (union over detectors) (Sec III C; stated)	-		*pending*
`zelenka2024` [69]	reach@FAR	MLGWSC-1 DS4: updated TPI FSU Jena 1316 Mpc at 1/month (original submission 0) (Sec IV C 2, Fig 4; stated)	1 /month; 3.86e-07 Hz	comparable MLGWSC-1 DS4 with the challenge evaluation, but the update is selected as the best of six runs on the authors' own dataset-4 realisation (a different seed from the challenge evaluation set, drawn from the same O3a real-noise file).	*pending*
`beveridge2025o3` [15]	reach@FAR	sensitive hyper-volume per mass pair vs GWTC-3 pipelines on 83,815 O3 injections (p_astro >= 0.5, FAR < 2/day): comparable above chirp mass ~25 Msun, lower below (Table I, Fig 5; stated)	60 /month; 2.31e-05 Hz	comparable Real O3 data with the GWTC-3 injection campaign and the LVK detection criterion: search-grade.	*pending*
	events	2,916 of 10,472 week-4/6/8 injections (PyCBC-BBH 3,751) at FAR < 1 per 2 months per the Table III caption; the Sec III A text gives FAR < 2 per month (Table III; stated)	0.5 /month; 1.93e-07 Hz		*pending*
`marx2025aframeO3` [10]	reach@FAR	sensitive volume at 1/yr, 35+35 Msun: Aframe ~10.8 vs GstLAL ~9.8, PyCBC-BBH ~9.5 Gpc^3; 35+20: ~4.8 vs ~5.5 / ~5.4 (Fig 2; read-from-figure)	0.0821 /month; 3.17e-08 Hz	comparable Offline O3 search over 202.4 d against GWTC-3 pipeline curves for the same four mass distributions.	*pending*
`mcleod2025` [16]	reach@FAR	BNS sensitive distance ~172 Mpc at 1 per 2 months (PyCBC ~175, MBTA ~172, GstLAL ~163) on the GWTC-3 O3 week-4 injections (Fig 7; read-from-figure)	0.5 /month; 1.93e-07 Hz	comparable Real O3 noise and the official GWTC-3 BNS injection set: comparable with the LVK pipelines on that set, though their FARs are taken without the p_astro cut and come from searches that also cover NSBH and BBH.	*pending*
	eff@FAR	34 % of 2,800 BNS injections at FAR < 1 per 2 months (Sec V A; stated)	0.5 /month; 1.93e-07 Hz		*pending*
`nagarajan2025` [14]	reach@FAR	MLGWSC-1 DS4: Sage-Broad 1638 Mpc at 1/month vs PyCBC 1539, AresGW 1574, Aframe 1441 (Table V; stated)	1 /month; 3.86e-07 Hz	comparable MLGWSC-1 DS4 with the challenge evaluation; best-of-three seeds reported.	*pending*
`tiki2025` [80]	reach@FAR	median sensitive volume 5.4 Gpc^3 at median FAR 20.5/yr; 4.6 Gpc^3 at a stricter threshold where three models have 0 and one has 1 false alarm in a year of background (O3b, 10,000 BBH) (Abstract, Sec 3; stated)	1.68 /month; 6.5e-07 Hz	conditional Real O3 noise with shifted-noise FAR and a fixed BBH population; population and FAR definition differ from MLGWSC-1 and Aframe.	*pending*
	events	8-9 of 13 O3b GWTC-3 events, 0-1 false triggers in 22.7 d (Sec 3; stated)	-		*pending*
`wang2025exotic` [75]	reach@FAR	sensitive distance ~640 Mpc at 10/yr (GR injections), real O1 noise, 4-day test (Fig 6; read-from-figure; the text says ~1000 Mpc)	0.821 /month; 3.17e-07 Hz	conditional Volume-averaged distance in the MLGWSC-1 style, but on 4 days of O1 data with a 0.1 s-step time-shift background and the paper's own BBH prior (m1 5-100 Msun, z < 2).	*pending*
`yamamoto2025` [70]	reach@FAR	MLGWSC-1 DS1: 2428 Mpc at 1/month (PyCBC 2492), 2743 at 100/month (PyCBC 2687) (Table III; stated)	1 /month; 3.86e-07 Hz	comparable MLGWSC-1 DS1 with the challenge script: directly comparable with the DS1 entries.	*pending*
`chatterjee2026` [72]	reach@FAR	MLGWSC-1 at 1/month: CASTOR 1482 Mpc (DS3), 1350 Mpc (DS4) (Sec V, Fig 4; stated)	1 /month; 3.86e-07 Hz	comparable MLGWSC-1 DS3/DS4 with the public evaluation code.	*pending*
	events	14 of 31 confident O3b BBH at FAR <= 1/month (Table II; stated)	-		*pending*
`gupta2026` [11]	reach@FAR	BNS sensitive volume at 1/yr: Aframe-BNS 0.0065 vs MBTA 0.0078, PyCBC-Hyperbank 0.0077, GstLAL 0.0059 Gpc^3 (1.4+1.4 Msun) (Fig 4; read-from-figure)	0.0821 /month; 3.17e-08 Hz	comparable Real O3 noise with 10 yr time slides, compared with GWTC-3 matched-filter numbers; Aframe FAR carries a factor-2 trials penalty.	*pending*
	events	O3 MDC: 671 of 1790 injections at FAR < 1/month (MBTA 648, GstLAL 528) (Table II; stated)	1 /month; 3.86e-07 Hz		*pending*
`insley2026flows` [104]	eff@FAR	+35 injections (+0.78 %) recovered at FAR < 1/yr over the original histogram method (HLV +3.85 %), one O3 week; the gain combines the modified sampling and the flow, which alone changes two-detector recoveries by <~0.05 % (Sec V A, Fig 4; stated)	0.0821 /month; 3.17e-08 Hz	conditional Normalising flows replace a histogram inside the PyCBC ranking statistic; the gain is relative to PyCBC on one week of O3 data.	*pending*

MLGWSC-1 contract. Sensitive distance (Mpc) at FAR 100, 10 and 1 per month on the four MLGWSC-1 datasets, from the challenge paper and from later papers that used its evaluation code. Dataset 4 is real O3a noise; datasets 1-3 are simulated. Values marked read-from-figure are read from plotted curves.

DS	Source	Search	D@100	D@10	D@1	Location; how	gravyflow
1	`mlgwsc2022` [8]	A MFCNN	1587	852	779	MLGWSC-1 Table IV; stated	*pending*
1	`mlgwsc2022` [8]	B PyCBC	2687	2551	2492	MLGWSC-1 Table IV; stated	*pending*
1	`mlgwsc2022` [8]	C CNN-Coinc	1372	1045	1002	MLGWSC-1 Table IV; stated	*pending*
1	`mlgwsc2022` [8]	D TPI FSU Jena	2635	2472	2363	MLGWSC-1 Table IV; stated	*pending*
1	`mlgwsc2022` [8]	E Virgo-AUTh	2512	2318	2116	MLGWSC-1 Table IV; stated	*pending*
1	`yamamoto2025` [70]	MF + CNN hybrid	2743	2582	2428	Table III; stated	*pending*
2	`mlgwsc2022` [8]	A MFCNN	1532	582	449	MLGWSC-1 Table IV; stated	*pending*
2	`mlgwsc2022` [8]	B PyCBC	1720	1600	1544	MLGWSC-1 Table IV; stated	*pending*
2	`mlgwsc2022` [8]	C CNN-Coinc	554	444	374	MLGWSC-1 Table IV; stated	*pending*
2	`mlgwsc2022` [8]	D TPI FSU Jena	1712	1544	1455	MLGWSC-1 Table IV; stated	*pending*
2	`mlgwsc2022` [8]	E Virgo-AUTh	1609	1410	1242	MLGWSC-1 Table IV; stated	*pending*
3	`mlgwsc2022` [8]	A MFCNN	885	473	356	MLGWSC-1 Table IV; stated	*pending*
3	`mlgwsc2022` [8]	B PyCBC	1734	1630	1577	MLGWSC-1 Table IV; stated	*pending*
3	`mlgwsc2022` [8]	C CNN-Coinc	620	468	402	MLGWSC-1 Table IV; stated	*pending*
3	`mlgwsc2022` [8]	D TPI FSU Jena	1728	1578	1488	MLGWSC-1 Table IV; stated	*pending*

DS	Source	Search	D@100	D@10	D@1	Location; how	gravyflow
3	mlgwsc2022 [8]	E Virgo-AUTh	1647	1495	1241	MLGWSC-1 Table IV; stated	*pending*
3	mlgwsc2022 [8]	F cWB	1462	1360	1252	MLGWSC-1 Table IV; stated	*pending*
3	chatterjee2026 [72]	CASTOR	-	-	1482	Sec V; stated	*pending*
3	chatterjee2024 [71]	GW-Whisper	1180	1030	890	Fig 2a; read-from-figure	*pending*
4	mlgwsc2022 [8]	A MFCNN	1269	999	650	MLGWSC-1 Table IV; stated	*pending*
4	mlgwsc2022 [8]	B PyCBC	1722	1610	1544	MLGWSC-1 Table IV; stated	*pending*
4	mlgwsc2022 [8]	C CNN-Coinc	961	621	0	MLGWSC-1 Table IV; stated	*pending*
4	mlgwsc2022 [8]	D TPI FSU Jena	258	0	0	MLGWSC-1 Table IV; stated	*pending*
4	mlgwsc2022 [8]	E Virgo-AUTh	1609	1400	1092	MLGWSC-1 Table IV; stated	*pending*
4	mlgwsc2022 [8]	F cWB	1407	1332	1229	MLGWSC-1 Table IV; stated	*pending*
4	nagarajan2025 [14]	Aframe (private comm.)	1708	1570	1441	Table V; stated	*pending*
4	nousi2022 [12]	AresGW	1815	1630	1575	Fig 10; read-from-figure	*pending*
4	chatterjee2026 [72]	CASTOR	-	-	1350	Sec V; stated	*pending*
4	chatterjee2024 [71]	GW-Whisper	1185	1025	905	Fig 2b; read-from-figure	*pending*
4	nagarajan2025 [14]	Sage-Broad (best of 3)	1859	1722	1638	Table V; stated	*pending*
4	nagarajan2025 [14]	Sage-Limited	1852	1711	1553	Table V; stated	*pending*
4	zelenka2024 [69]	TPI FSU Jena (updated)	-	-	1316	Sec IV C 2; stated	*pending*
4	nagarajan2025 [14]	TPI-FSU Jena 2024	1584	1424	1316	Table V; stated	*pending*

3.3 Latency and compute cost

51 of the 92 models in this section state a latency, inference cost or training cost; 24 papers were read and state none, and 17 were not read for runtime. Headline: the model's latency if it states one, else its inference cost; normalised values in brackets (Section 13).

Model	Headline: latency if stated, else inference cost	Hardware	Grade; scope; mode	Location	Training cost (hardware; grade; location)
george2018plb [29]	Inference cost: 85 milliseconds per second of data (each CNN) (0.085 compute-s per s)	1x CPU (model unstated)	stated; network; unstated	Sec. 3 (Results), final paragraph	-
george2018 [1]	Inference cost: 6.7 milliseconds per input of 1 second duration (simple classifier and predictor, 2 MB each) (0.0067 compute-s per s)	1x Intel Core i7-6500U CPU (2 cores, per Fig. 14 caption)	stated; network; batch	Sec. V (Discussion); Fig. 14	about 10 hours in total; unstated (Tesla P100 GPUs thanked in acknowledgments; Wolfram Language/ MXNet) (stated; Sec. III (architecture and training), 'we trained it for about a total of 10 hours')
george2018b [3]	Inference cost: 85 milliseconds per second of data (each CNN) (0.085 compute-s per s)	1x CPU core (model unstated)	stated; network; unstated	Sec. III (Results), final paragraph	-
rebei2018 [33]	training cost only	-	-	-	thirty minutes (training stage, TensorFlow + Horovod distributed training); NVIDIA Tesla V100 GPUs on Blue Waters and the NCSA Innovative Systems Lab (count unstated) (stated; Sec. VI (list of algorithm improvements))
fan2019 [41]	training cost only	-	-	-	nearly nine hours; 1x NVIDIA GTX 1080 Ti (11 GB) (stated; Sec. III (results, after Table I); Sec. IV conclusion ('approximately nine hours'))
gebhard2019 [4]	training cost only	-	-	-	approximately 30 hours for 64 epochs; 5x NVIDIA Tesla V100 (32 GB each) (stated; Sec. V B (training), text after the optimiser description)
kim2019 [101]	Inference cost: tens of milliseconds to score ~45,000 samples (6 features each)	unstated	stated; network; batch	Abstract; Sec. II; Sec. V	a few hours for ~165,000 samples x 6 features; unstated (stated; Sec. V discussion)
lin2019 [42]	Inference cost: approximately 0.027 seconds per sample (4 s single-detector sample at 16384 Hz) (0.0135 compute-s per s)	1 core of an Intel Xeon E7-4830 2.13 GHz CPU (one-core computation, Sec. 5)	stated; unstated; unstated	Sec. 4 'Efficiency' paragraph; Sec. 5	-
deighan2020 [35]	training cost only	-	-	-	under 24 hours per network (N_TS and N_epochs set as large as possible such that training a single network completes in under 24 hours); NVIDIA Tesla V100 (count per network unstated) (stated; Sec. 3.3)
li2020 [46]	Inference cost: 18.765 milliseconds per sample (0.03 s detection window)	12x Intel Xeon E7-4830 CPUs, 2.31 GHz	stated; unstated; unstated	Sec. 4 (end, after Sec. 4.2)	approximately 430 minutes (experiment SE 5, Fig. 8, training set1); unstated (stated; Sec. 4 (end of discussion of improvements))
schafer2020 [5]	Latency: 10.2 s between signal arrival and alert (conservative estimate at FAR 10 per month, true positives) (10.2 s)	unstated	stated; unstated; streaming	Abstract; Sec. V B; Sec. VI	roughly 5 hours per training epoch plus validation; 1x NVIDIA V100 (32 GB), 17 GB memory at mini-batch 32 (stated; Sec. IV B)
wang2020 [83]	training cost only	-	-	-	within 6 hours; 4x NVIDIA GeForce GTX 1080Ti (11GB) (stated; Sec. III B (Training strategy))
dodia2021specgrav [95]	training cost only	-	-	-	about 19 minutes; 1x GPU with 2 GB memory (model unstated) (stated; Abstract; Results section)
gunny2021 [68]	Latency: 1 s (1 s)	unstated (V100 16GB GPUs per Methods B.4, count varies with Fig. 9 configuration)	stated; end to end; streaming	Sec. Discussion/ conclusion; Methods B.4 and Fig. 9	<20 minutes on a single GPU; 1x GPU (model unstated) (stated; Discussion (retraining paragraph))

Model	Headline: latency if stated, else inference cost	Hardware	Grade; scope; mode	Location	Training cost (hardware; grade; location)
huerta2021 [64]	Inference cost: 7 min for the whole of August 2017 (H1 and L1) (0.01004 compute-s per s)	64x NVIDIA V100 (16 nodes x 4, HAL cluster); ensemble of 4 networks, 16 GPUs per network	stated; end to end; batch	Sec. Results, Fig. 2a, Fig. 6	-
krastev2021 [31]	Inference cost: ~3 milliseconds to process 10 s of GW data (with both the detection and regression CNNs)	GPUs (model and count unstated)	stated; network; unstated	Sec. 3.4	-
lin2021 [37]	Latency: about 20 seconds (20 s)	moderate GPU-equipped personal computer (AMD RX 480 GPU, Intel i5 6500 CPU as stated for training; inference machine not separately specified)	stated; end to end; unstated	Abstract; Sec. III B; Sec. V	about 50 hours (150 epochs, batch size 256); 1x AMD RX 480 GPU with Intel i5 6500 CPU (stated; Sec. II D (hardware paragraph))
schafer2021a [6]	Inference cost: about 1 hour to evaluate an entire month of data (0.00139 compute-s per s)	1x NVIDIA RTX 2070 SUPER	stated; network; batch	Sec. II B	-
schafer2021b [7]	Inference cost: < 12 hours to generate O(1000) months of time-shifted background (sub-network C on stored sub-network A/B outputs)	1x NVIDIA RTX 2070 Super	stated; network; batch	Sec. III D (Background Estimation)	-
verma2021 [49]	training cost only	-	-	-	O(1) hour per rho_opt case; 1x GPU (model unstated) (stated; Sec. V)
wei2021ensemble [63]	Inference cost: within 14 hours for 200 hours of data (0.28 compute-s per s)	4x NVIDIA V100 (one HAL node)	stated; end to end; batch	Sec. IV (Results, summary); Fig. 5 caption	-
alhassan2022 [93]	Inference cost: 4 minutes and 57 seconds per hour of data min:s per hour of ET mock data (0.0825 compute-s per s)	Core i7 MacBook Pro (16 GB, 2.6 GHz), CPU, not parallelised	stated; end to end; unstated	Sec. 4 (continuous-data inference) and Sec. 5 (conclusions)	-
aveiro2022 [90]	training cost only	-	-	-	close to 14 hours hours (total training time, including data loading and caching); 1x NVIDIA V100 PCIe 16GB, 32 vCPU cores, 64 GB RAM (stated; Sec. IV (Model training))
chaturvedi2022 [65]	Inference cost: within 50 seconds for one month (August 2017) of H1+L1 advanced LIGO data (0.00299 compute-s per s)	160x NVIDIA A100 (20 ThetaGPU nodes x 8 GPUs)	stated; unstated; batch	Abstract; Sec. 1; Fig. 7 caption; Sec. 3.4	2 hours per model; 192x NVIDIA V100 (32 Summit nodes) (stated; Abstract; Sec. 2 (AI ensemble construction); Sec. 3.4)
ma2022 [38]	Inference cost: less than 2 milliseconds ('mini-seconds' in the paper; typical delay time of the ensemble model)	1x NVIDIA 1080Ti GPU	stated; network; unstated	Sec. III B (Ensemble model), before subsection 3 'Sub-ensemble model construction'	-
mcisaac2022 [103]	training cost only	-	-	-	~24 hours (main training step, training set analysed 25 times); 8 CPU cores (CPU model unstated) (stated; Sec. IV C (Training))
mlgwsc2022 [8]	Inference cost: 43331 s per month of two-detector data (mean of foreground 42842 and background 43820) (0.1337 compute-s per s)	8x NVIDIA RTX 2070 Super (all 8 GPUs used), 2x Intel Xeon Silver 4215	stated; end to end; batch	Tab. IV (A: MFCNN, dataset 1); Sec. V C	-
alhassan2023pymerger [94]	Inference cost: 1.9 minutes per hour of data (TSDCD, three sub-detectors) min per hour of data (0.0317 compute-s per s)	Core i7 MacBook Pro (16 GB, 2.6 GHz) CPU	stated; end to end; unstated	Sec. 5; Sec. 6 (PyMerger description); conclusions	-
andrescarcasona2023 [78]	Inference cost: of the order of 2,000 CPU-hours for the entire O3 scan CPU-hours (0.5376 compute-s per s)	Intel Xeon E5-2680 v4 @ 2.40 GHz (core count unstated)	stated; end to end; batch	Sec. IV (Results)	-
beveridge2023 [77]	training cost only	-	-	-	7 hours hours; 1x NVIDIA A100 GPU (stated; Sec. II (model training))
jadhav2023 [81]	Inference cost: ~16 hours run-time for multi-GPU inference on images of 8.8 days of H1 and L1 data (0.303 compute-s per s)	4x NVIDIA Tesla P100	stated; network; batch	Sec. VI B	~11 hours (VAE, 600 epochs); 4x NVIDIA A100 Tensor Core (stated; Sec. III (VAE training))
nousi2022 [12]	Inference cost: less than 2 hours to evaluate one month of test data	1x NVIDIA A6000 GPU (same as training)	stated; end to end; batch	Sec. I (Introduction)	31 hours (including validation), 14 epochs on 12 days of training data; 1x NVIDIA A6000 GPU with tensor cores (stated; Sec. I (Introduction))
tian2023 [66]	Inference cost: 165 seconds (0.00535 compute-s per s)	1024x NVIDIA A100 (Polaris)	stated; network; batch	Sec. 3 Results, 'AI performance'	22 hours; 96x NVIDIA V100 (distributed) (stated; Abstract; Methods 'Training methodology')
tian2024gnn [67]	Inference cost: 3.19 hours (0.00466 compute-s per s)	128x NVIDIA A100 (Polaris)	stated; network; batch	Methods 'AI inference at scale'	1.7 hours; 256x NVIDIA A100 (Polaris) (stated; Abstract; Intro; Methods 'Distributed training at scale')
trovato2023 [74]	Inference cost: about 4 hours per model for the full 4,216,489 s data set (0.00342 compute-s per s)	NVIDIA Tesla V100S (count unstated)	stated; end to end; batch	Sec. 5.2	220 seconds per epoch (CNN); 1x NVIDIA Tesla K40d (stated; Sec. 3.4)
badger2024 [87]	Inference cost: 24.6 s to reconstruct all 35 O3b confident events s	AMD EPYC 7502 CPU	stated; network; batch	Sec. III / Sec. IV	at most 454 s to learn a dictionary (range 5 s to 454 s) s; Intel E5-2698 v4 CPU (footnote 1) (stated; Sec. II (dictionary creation; footnote 1) and Sec. IV)
marx2024 [9]	Latency: 3.114 (+0.006 / -0.001) seconds (3.114 s)	1x NVIDIA A30 GPU (24 GB)	stated; end to end; streaming	Tab. III; Sec. VIII	approximately 43 hours; 1x NVIDIA 16 GB Tesla V100 (stated; Sec. VIII)
menendez2024 [79]	Inference cost: of the order of 2000 CPU hours per search (entire O3 analysis) (0.5376 compute-s per s)	Intel Xeon E5-2680 v4 2.4 GHz CPU (core count unstated)	stated; end to end; batch	Sec. 5 (Results)	typically two hours per CNN (up to 12 epochs); GPU (model and count unstated; Keras/TensorFlow) (stated;

Model	Headline: latency if stated, else inference cost	Hardware	Grade; scope; mode	Location	Training cost (hardware; grade; location)
					Sec. 3 (Neural network definition and training))
wang2023tf [91]	Inference cost: 0.1868 seconds for 173 GW signals (approximately 1 ms each)	unstated	stated; network; batch	Sec. II (identification time); footnote 7	within 6 hours; 4x NVIDIA GeForce RTX A6000 (48GB) (stated; Sec. II (Methodology, U-Net training))
zelenka2024 [69]	Inference cost: ~40 times ratio of estimated PyCBC runtime to this work's runtime (datasets 2-4)	8x Intel Xeon Silver 4215 cores + 8x NVIDIA RTX 2070 (challenge hardware, for the ML runs)	stated; end to end; batch	Sec. IV B	-
mcleod2025 [16]	Latency: ~0.5 seconds (inference step) (0.5 s)	1x NVIDIA A100 GPU	stated; network; streaming	Sec. V E	typically 3 hours; 1x NVIDIA A100 GPU (stated; Sec. III (training and loss description))
nagarajan2025 [14]	Inference cost: ≈ 16 hours to generate triggers on 1 month of testing data (0.02222 compute-s per s)	1x NVIDIA Quadro RTX 8000 GPU plus 16 CPU cores	stated; end to end; batch	Sec. III (preamble, before III A); Sec. V 'Evaluation speed'	≈ 2 hours per epoch, excluding validation; 1x NVIDIA Quadro RTX 8000 GPU, 16 CPU cores (stated; Sec. III (preamble, before III A))
tiki2025 [80]	Inference cost: under 5.0 hours for 22.7 days of noise livetime (plus 13 events) (0.00918 compute-s per s)	1x NVIDIA A100 (Delta)	stated; end to end; batch	Sec. 2.3 'Benchmark on real O3b data'; footnote 1	under nine hours (75 epochs); 1x NVIDIA H100 (DeltaAI) (stated; Sec. 2.3 'Benchmark on real O3b data')
wang2025exotic [75]	training cost only	-	-	-	within six hours (200 epochs); 2x NVIDIA GeForce RTX A6000 (48GB) (stated; Sec. II (Method, training))
yamamoto2025 [70]	Inference cost: 2802 s runtime, foreground, MLGWSC-1 dataset 1	Intel Xeon Platinum 8468 (2.1 GHz, 48 core) + 1x NVIDIA H100 (GPU count implied single, stated as 'a GPU NVIDIA H100')	stated; end to end; batch	Tab. III	-
chatterjee2026 [72]	Inference cost: ≈ 0.9 hrs for one month of continuous MLGWSC-1 data (0.00125 compute-s per s)	1x NVIDIA A100 80 GB	stated; unstated; batch	Tab. I; Sec. IV A	≈ 1.5 hrs per run; 1x NVIDIA A100 80 GB (stated; Tab. I; Sec. III B 2)
gupta2026 [11]	Latency: ~4.7 s median end to end, quoted for Aframe's online deployment (data arrival to GraceDB), not measured for this BNS model; network latency ≈172 µs per sample (batch of 128 in 22 ms) (4.7 s)	1x NVIDIA A30 (24 GB) for online inference	stated; end to end; streaming	Sec. V B	approximately 96 hours; 1x NVIDIA H200 (140 GB) (stated; Sec. V A)
insley2026flows [104]	training cost only	-	-	-	0.25 CPU hours (2-detector flow, 500,000 samples); CPU (count and model unstated) (stated; Tab. III)
mobilia2026 [86]	Inference cost: ~3 ms per image (TT-SNR map)	off-the-shelf CPUs, no particular optimisation (model unstated)	stated; network; unstated	Conclusions (Sec. IX)	~8 hours; 2x Intel Xeon E5-2650 v4 @ 2.20 GHz (24 cores total), CPU only (stated; Sec. V (Simulation 1), training paragraph)
rai2026 [99]	Latency: < 45 ms total pipeline (classifier + FastSHAP explainer)	standard laptop, Intel Core i7 class CPU, no GPU	stated; network; unstated	Sec. 4.2; Tab. 3; Tab. A.4	-
weerasinghe2026 [62]	Inference cost: 3.04 ms per event (100 ms post-merger window, ensemble of 5 CNNs)	unstated (GPU with host-device transfers implied)	stated; unstated; unstated	Abstract (3.0 ms); Tab. 1; Sec. 5.3.1	4 hours per ensemble member (about 6 min per epoch, average 37 epochs); 1x NVIDIA RTX 4070 Laptop GPU (stated; Sec. 3.4 (training))

3.4 What cannot be compared, and why

Most CBC papers report a true-positive rate at a per-window false-alarm probability, an accuracy on a balanced set, or an AUC over a mixture of SNRs. None of these converts to a false-alarm rate without the window stride and the clustering rule, and none fixes the injected population, so a 98 % accuracy in one paper and a 0.9 TPR in another say nothing about each other. Real-event recoveries are listed for completeness but are not efficiencies: windows are often placed on the known event, and the denominator is the catalogue, not an injection set. Comparable rows are those on the MLGWSC-1 contract or on the LVK injection campaigns (sensitive distance or volume at a stated FAR); conditional rows have a true FAR but a different noise model, population or detector count.

The 56 models below have no shared axis; their headline metric is listed with the reason.

Model	Headline metric (location; how)	Why not comparable	gravyflow
kim2015grb [100]	efficiency at per-trigger FAP ~1e-4: ANN 0.56-0.74 vs detection statistic 0.50-0.64 (Fig 7; read-from-figure)	GRB-triggered coherent CBC search on S5/VSR1, per-trigger FAP; the MLR is built from the same samples it is scored on.	*pending*
george2018plb [29]	sensitivity ~45 / ~77 / ~93 % at SNR 5 / 6 / 7 vs matched filter ~52 / ~87 / ~97 % at <1 % false alarms per window (Fig 1 left; read-from-figure)	Proceedings version of george2018b; per-window FAP.	*pending*
george2018 [1]	100 % TPR for SNR >= 10 (shallow) / >= 9 (deep) at ~0.5 % false alarms per 1 s window (Results, Figs 7, 8; stated)	One operating point per network on Gaussian ZDHP noise; the "false alarm rate" is a per-window fraction.	*pending*
george2018b [3]	100 % TPR for SNR > 10 at <1 % false alarms per window (single detector) (Sec III, Fig 4; stated); GW150914 recovered with masses inside published errors (Sec III; stated)	Real O1 noise but one operating point; the two-detector 0.01 % squares the single-detector fraction.	*pending*
rebei2018 [33]	100 % sensitivity for SNR >= 10 on eccentric higher-mode NR waveforms (Fig 8; stated)	Sensitivity on a few NR waveforms; the one-false-alarm-per-three-months rate on real LIGO noise (Sec VI A) is not tied to the Fig 8 curves.	*pending*
fan2019 [41]	TPR 98.7 % at network SNR 8, 100 % at >= 9, 0.4 % false alarms per 2 s window (HLV) (Sec III A, Fig 4; stated)	Fixed-SNR balanced test sets in coloured Gaussian noise; one operating point.	*pending*
kim2019 [101]	RF 0.666 / 0.689 vs GstLAL 0.633 / 0.658 (H1 / L1 triggers) (Fig 1; stated)	Trigger-level re-ranking inside GstLAL with GstLAL software injections labelled mock data of GW150914 but spanning recovered chirp masses ~1-40 Msun (Figs 2-3) against time-slide triggers; the FAR in Hz is a GstLAL-style conversion of a trigger FPR.	*pending*

Model	Headline metric (location; how)	Why not comparable	gravyflow
lin2019 [42]	TPR >97.2 % at per-sample FPR >= 1e-3 (BNS, rho 8-12) (Sec 4, Fig 3; stated)	Per-sample FPR with 1/7000 resolution in Gaussian noise; rho mixture undefined.	*pending*
luo2019 [43]	none extracted	Not read (journal page unresolvable); no metric.	*pending*
schmitt2019 [44]	none extracted	Not read (ACM paywall); no metric.	*pending*
chauhan2020 [45]	100 % accuracy at SNR 4 on a 400-sample test set (Sec 3; stated)	Tiny balanced set; training accuracy 1.00 after one epoch; no false-alarm context.	*pending*
deighan2020 [35]	accuracy 0.71 (seed) to 0.79 (GA-optimised), white noise (Sec 5.3; stated)	Architecture-search study on a white-noise toy problem.	*pending*
krastev2020 [30]	BNS TPR at FAP 1e-2: 0.33 / 0.63 / 0.88 at rho 8 / 10 / 12 (Fig 4; read-from-figure)	Per-sample FAP from ~1,667 noise samples (resolution ~6e-4) in Gaussian noise; the FAP 1e-3 curves rest on 1-2 samples.	*pending*
li2020 [46]	wavelet-packet CNN (SE5) 0.947 at FPR 1e-3, rho_opt 2 (Table 3; stated)	Not a compact-binary search: damped sinusoids in white noise, 0.03 s windows.	*pending*
santos2020 [47]	ring-down time from score decay, e.g. GW150914 3.87 ms (LIGO ~4 ms) (Table 1; stated)	No FAR, TPR or distance; the score-decay "ring-down" is not a strain measurement.	*pending*
wang2020 [83]	TPR at per-sample FAP 1e-3: 0.47 / 0.72 / 0.90 at rho_opt 4 / 5 / 6 (Fig 3; read-from-figure); GW150914, GW151012, GW151226 and eight O2 events flagged (Figs 4-5; stated)	Real O1 two-detector noise but per-sample FAP on 1610 noise samples, so FAP 1e-3 rests on 1-2 samples; the O1 scan triggers carry no FAR.	*pending*
alvares2021 [89]	three-detector classifier ROC AUC 0.87 at 2000 Mpc (Fig 5; stated); 34 of 37 GWTC-2 BBH above threshold 0.5 (Table III; stated)	Balanced validation at one injection distance; noise from one 500 s segment; no FAR.	*pending*
dodia2021residual [48]	detection ratio ~100 % at SNR 11 (BBH), 20 (BNS), no FAR stated (Figs 6-7; stated)	Detection ratio without a false-alarm level.	*pending*
dodia2021specgrav [95]	9 of 11 GWTC-1 events correctly identified (Result, Table 1; stated)	Three-class classifier on small balanced spectrogram sets in real H1 noise; real events classified, not searched; no FAR.	*pending*
fan2021_2d [96]	none extracted	Not read (ICPR paywall); no metric.	*pending*
g2net2022_acm [97]	none extracted	Not read; G2Net-style balanced AUC at best.	*pending*
gunny2021 [68]	throughput 6.4e-5 s per second of data on 4 V100 (DeepClean); end-to-end ~1 s latency claimed (Figs 3, 9; read-from-figure)	Inference-as-a-service systems paper; no detection performance for BBHnet.	*pending*
huerta2021 [64]	ensemble TPR 0.37-0.81 at SNR 10.6 across FPR (unit unstated) ~3e-6 to ~4e-4 (Fig 4 inset; read-from-figure); all four August 2017 BBHs found, zero false positives in the month (Abstract, Fig 2b; stated)	FPR with an undefined unit; the paper does not reconcile the ROC's FPR of 1e-6 to 1e-3 with zero false positives over the month.	*pending*
krastev2021 [31]	BNS TPR at FAP 1e-2: 0.29 / 0.63 / 0.89 at rho 8 / 10 / 12 (real O2 L1 noise) (Fig 3; read-from-figure); 11 of 11 GWTC-1 events classified with p > 0.995 (windows placed on the known event times) (Fig 4; stated)	Per-10 s-sample FAP; the paper's FAR conversion is order-of-magnitude; real events are classified, not searched.	*pending*
lin2021 [37]	window TPR >90 % for rho_opt > 8 counting "awareness"; false awareness 19.6 % of noise windows (Sec III A, Fig 4; stated); 7 of 7 O2 L1 BBHs (Sec III B; stated)	Three-state output in which awareness counts as detection; event-level false rate 47.7 %.	*pending*
verma2021 [49]	level-1 accuracy ~99 % (99.43 % from the matrix), FAP 0.33 % per sample (Sec V, Fig 5; stated)	Balanced synthetic single-detector set, SNR 10-20 with the test SNR unstated.	*pending*
yan2021 [84]	FNR ~3.7e-5 (matched filter) vs ~3.3e-5 (MNet-Shallow) at per-segment FPR 3e-4, SNR 9 (Fig 13; read-from-figure)	Fixed-SNR single-detector 0.6 s segments of real O2 L1 noise, 40-50 Msun only; FPR per overlapping segment.	*pending*
zhan2021 [50]	accuracy 96.4 % at SNR 10 (called sensitivity; recall 98.4 %, FPR 5.6 %) (Fig 2; stated)	Glitch-response probe on Gaussian design noise.	*pending*
alhassan2022 [93]	ResNet-101 accuracy 0.62 (SNR 4-5) to 0.985 (SNR > 8), single ET detector (Table 3; stated)	Einstein Telescope single-detector STFT classification in Gaussian noise coloured to an ET PSD (model not named), SNR-binned balanced sets; the mock-run false positives are per 5 s window.	*pending*
aveiro2022 [90]	YOLOv5 precision 0.922, recall 0.823, mAP@0.5 0.945 (BNS CQT images, SNR 10-20) (Abstract, Table II; stated); GW170817 H1 at confidence 0.91; L1 only after ad hoc capping (Sec V C; stated)	Object-detection metrics on synthetic H1 Gaussian noise; no FAR.	*pending*
chaturvedi2022 [65]	79 time-slide false positives in 5 yr (~1.3 per month); all four August 2017 BBHs found (Sec 3.1, 3.3; stated)	Same ensemble as huerta2021 with TensorRT; time-slide counts are not independent trials and no efficiency is paired with them.	*pending*
jiang2022 [51]	none extracted	Not read; no metric.	*pending*
jiang2022real [52]	41 of 41 O1-O3a BBH; FAR < 1 per 2 months in O2 replay (abstract) (abstract; stated)	Abstract-level only.	*pending*
ma2022 [38]	FAP at TAP 0.8: 0.0042 (single) vs 0.00185 (harmonic ensemble), Hanford, per 1 s window (Sec III.3, Fig 5; stated); 9 of 10 O1/O2 BBH (GW170818 missed) (Sec IV B; stated)	Per-window FAP on a bespoke 20 %-signal test set in O1 noise; the "1 per 11.24 days" FAR is extrapolated by multiplying H and L probabilities.	*pending*
menendez2022sub [53]	none extracted	Not read; no metric.	*pending*
qiu2022 [39]	TAP saturates at FAP 1e-3 for SNR >= 8 (BBH), >= 10 (NSBH), >= 13 (BNS), real L1 noise (Sec 3.1, Fig 4; stated)	Per-segment FAP on real L1 O2/O3b noise free of known events and hardware injections, single detector, four-class softmax.	*pending*
verma2022 [32]	aligned vs precessing (second stage): 96.2 % (synthetic), 90.5 % (H1 PSD), 92.3 % (L1 PSD) (Figs 5-7; stated); three-class noise / aligned / precessing 93.7 / 93.5 / 93.6 % (Fig 8; stated)	Per-detector models on Gaussian noise coloured by O3 PSDs; classification at an unstated threshold.	*pending*
yan2022 [54]	none extracted	Not read; no metric.	*pending*
zhang2022 [55]	detection ratio ~89.6 % at FAP 4.88e-4 (abstract) (abstract; stated)	Abstract only (APS paywall); the abstract converts the FAP to about one false alarm per 18.2 h.	*pending*
alhassan2023pymerger [94]	three-sub-detector ResNet accuracy 0.785 / 0.84 / 0.995 at SNR 4-5 / 5-6 / 6-7 (Gaussian, ET PSD) (Sec 4, Fig 5; stated)	Einstein Telescope triangle in Gaussian noise; balanced SNR bins; no FAR curve.	*pending*
garg2024 [56]	none extracted	Qualitative only; no numbers.	*pending*
kumar2023 [57]	none extracted	Not read; no metric.	*pending*
ma2024tempfree [85]	TAP ~0.8 at per-sample FAP 1e-4 in most mass sub-spaces (denoiser + discriminator) (Fig 9; read-from-figure); denoised overlap > 0.9 for >= 97.5 % of samples (heaviest class), ~60 % (lightest) (Sec IV A 1, Fig 4; stated)	Per-sample FAP on 10,000 noise samples per sub-space test set; the real-data background test uses only the second half of August 2017 because the first half trained the discriminator.	*pending*
sasaoka2023_cbc [98]	none extracted	Not read; no metric.	*pending*
sasaoka2023attr [92]	ensemble TAP at network SNR 8, FAP 1e-3 per 4 s: ~0.94 (BBH), ~0.60 (NSBH), ~0.12 (BNS), real O3 HLV noise (Fig 3; read-from-figure)	Per-4 s-segment FAP (~one false positive per hour) with SNR computed on truncated signals.	*pending*
sharma2023 [58]	none extracted	Not read; no metric.	*pending*
meijer2024 [36]	AUC 0.949 (aligned) / 0.940 (precessing) / 0.953 (curriculum), signal vs blip at matched SNR (Sec V A; stated)	Balanced simulated IMBH-vs-blip set with SNR distributions forced to coincide; no FAR.	*pending*
sharma2024 [59]	none extracted	Not read; no metric.	*pending*
ussipov2024 [60]	source-type accuracy up to 0.96 (abstract) (abstract; stated)	Abstract-level only.	*pending*

Model	Headline metric (location; how)	Why not comparable	gravyflow
ashton2025 [105]	pipeline-combination AUC: max-IFAR 0.925, LR 0.944, MLP 0.967 (O3 replay MDC) (Fig 1; stated)	Learns on pipeline candidate lists, not strain; AUC on an injection-rich MDC split.	*pending*
ma2025et [88]	none extracted	Not read; no numbers in the abstract.	*pending*
silver2025 [76]	57 of 75 GWTC-3 BBH at a threshold with ~63 time-shifted false positives per year (derived) (Sec III C-D; stated)	Real-data event recovery with thresholds tuned on the same GWTC-3 search; no injection-based sensitivity at a FAR.	*pending*
gupta2026q [61]	validation AUC 0.9209 (G2Net subset) (abstract; stated)	Abstract-level; G2Net synthetic data with an unstated SNR distribution.	*pending*
mobilia2026 [86]	Sim4 TAP at per-window FAP 1e-4: CNN ~0.30 vs reweighted SNR ~0.13 vs SNR ~6e-3 (Sec VIII, Fig 13; stated)	Simulated Gaussian noise with synthetic glitches; FAP 1e-4 rests on about one of ~12,000 noise samples.	*pending*
rai2026 [99]	mean per-event AUC 0.915 +- 0.086 on 66 GWTC events (per 0.25 s window) (Sec 3.2; stated)	Per-window classification of catalogue events; FPR 0.0061 per 0.25 s window is ~0.024 Hz as a rate.	*pending*
weerasinghe2026 [62]	ROC AUC 0.999999; 99.998 % efficiency at 1 % FPR per 100 ms segment (Sec 5.1; stated)	Synthetic post-merger damped sinusoids; saturated metrics on 1000-fold augmentations of a small waveform set.	*pending*

3.5 Suspected errors in the literature

134 suspected errors or internal inconsistencies in 74 papers, selected from the extraction notes (258 notes for this section in the catalogue). Each is a reading of the paper to be checked, not a verdict.

- [100] Sec 2.1 names S6/VSR2-3 data while the analysed GRBs are S5/VSR1.
- [100] Sigma conversions are two-sided (FAP 1e-4 called 3.89 sigma).
- [29] "1 per 100 s" for 1 % of overlapping 1 s windows stepped 0.2 s is a factor-5 slip.
- [29] Text says 100 % above SNR 10; Fig 1 reads ~99.5 % at 10.
- [2] CNN above the matched filter at FAP 0.1 for rho 3-5 (Fig 3), as the paper itself reports, shows its bank-maximised SNR benchmark is not the optimal statistic the paper presents it as.
- [2] Waveform count conflict: 4e5 series with 25 noise draws each implies 8000 signals, the text says 1000.
- [1] Very-high-SNR error bound given as <3 % (Fig 10 caption) and <5 % (text).
- [3] "1 % = one per 100 s of noise" is inconsistent with 1 s windows stepped by 0.2 s (one per ~20 s).
- [40] Raw CNN AUC 0.50 (Fig 6) vs 0.99 (Fig 7) on the same signal and SNR range points to an unconverged baseline.
- [40] Text says the combined features are "better in every case"; Fig 7 gives raw and combined both AUC 0.99.
- [41] Fig 4 calls a per-sample fraction a "false alarm rate".
- [41] Dataset size given as 3,000/2,000 and as 100,000 training sets.
- [4] Results section: 89 % at one per 19.5 min; conclusion: 86 % at one per 40 min for the same setup.
- [101] The abstract's "about 10 %" efficiency gain is an absolute TPR difference (60-70 % relative).
- [101] The 18 % sensitivity gain at ~1e-11 Hz is not reproducible from the stated ranges (21 % in radius).
- [42] Matched-filter TPR 0.78 at rho 8 and FPR 1e-3 while the CNN reaches ~0.985 is implausible for an adequate bank in Gaussian noise.
- [42] "More than 960 times faster" is a factor-10 slip: 261 s / 0.027 s = 9,667.
- [42] Sub-populations fall below the headline 97.2 % (Figs 4-5).
- [45] A 100 % sensitivity at SNR 4 with perfect training accuracy after one epoch points to leakage or an easy set.
- [45] Lowest SNR stated as 1.5 and as 4.
- [35] "11 % increase" is relative; parameters reduced by 78 % (abstract, Sec 5.3), 79 % (Sec 5.4, Fig 8) or about 80 % (Fig 6).
- [102] Abstract says all GWTC-1 CBCs are recovered; GW170818 is not (10 of 11).
- [102] Most MLStat IFARs in Table I are inferred by assuming a constant background reduction, not measured.
- [30] "Saturation at SNR >= 18" is approximate: BNS at FAP 1e-3 reads ~0.99 at SNR 16 (Fig 4).
- [46] CNN sensitivity 0.947 at FPR 1e-3 and rho 2 (Table 3) exceeds the ~0.14 Neyman-Pearson bound for a known signal in Gaussian noise at that SNR; the matched filter's ~0.05 (Fig 7) is over the whole test set, not at rho 2.
- [47] The positive class is a fit to the events later scored (leakage); KS = 1 is on training patterns.
- [5] The PyCBC Live reference is reconstructed from another paper by an SNR-threshold assumption, not run on the same data.
- [83] GW170817 (total mass ~2.7 Msun) is reported detected by a network trained on 10-150 Msun.
- [83] The same quantity is called matched-filter SNR (Fig 2) and optimal SNR (Fig 3), with different values (0.64 vs 0.60 at SNR 4, FAP 0.1).
- [89] Table II labels the 0.72 / 0.82 accuracies as AUC.
- [89] Abstract "as much as 70 %" improvement vs 0.72 to 0.82 in the main comparison.
- [95] A first model with 100 % training and validation accuracy on simulated noise missed most real events; its validation set did not test real-noise performance.
- [68] The one-second detection latency is claimed without any detection result.
- [64] Text says FPR rises as the threshold rises (direction reversed).
- [64] FPR range stated as 1e-6 to 1e-3; the figure spans ~3e-6 to ~4e-4.
- [64] FPR 1e-6 (unit not stated), if per 0.5 s step, implies ~5 false peaks per month against the zero reported.
- [31] FAP 1e-3 is equated with O(10^3) false alarms per month; at the 0.3 s stride it is ~8.6e3.
- [31] BBH saturation SNR given as 10 and as ~12 in the same section.
- [37] Abstract >95 % vs text >90 %.
- [37] Event TPR at rho 2 (~43 %) is below the false event rate (47.7 %).
- [73] O2 trigger rates (34-47 per day) exceed the tuned 25 per day.
- [73] Events with secondary masses below the stated training range are detected (GW170104, GW170809).
- [6] Headline retention 91.5 % (v2) vs 97.5 % (v1 and README); both are radius ratios (0.915^3 = 0.77 in volume).
- [7] Percentages are minimum radius ratios presented as sensitivity fractions.
- [49] Fig 6 caption describes a per-patch ROC while the axes are noise-vs-signal.
- [49] Patch-2 accuracy 97.3 % stated vs 97.4 % from the printed matrix.
- [63] One training noise segment (GPS 1186725888, 14 August 2017) falls inside the August 5-16 2017 benchmark period.
- [63] Summaries claiming "a single GPU" conflict with the 4-V100 node used.
- [34] Introduction "12 % higher accuracy" is relative (0.84 vs 0.75 is 9 points absolute).
- [84] The claimed outperformance rests on ~37 vs ~32 missed injections out of ~1e6, within Poisson scatter.
- [84] Matched-filter templates are selected on the test set (best of 30 draws).
- [50] Fig 2 calls the accuracy "sensitivity".
- [93] ResNet per-class recalls (Table 3) imply accuracy ~0.77 at SNR > 8, not the 0.985 stated.
- [93] Conclusion quotes an 85 % mean accuracy; Table 3 averages 0.796.
- [93] Mock-run detection fractions (0.29, 0.46) disagree with Table 3 recalls (0.59, 0.57).

- [82] Text says 90 % to 70 % between 3 and 4 sigma; Fig 3 reads ~0.86 and ~0.81.
- [82] The matched-filter benchmark TAP equals FAP (chance level), impossible for a correct matched filter on injected signals.
- [90] Conclusion says GW170817 detected "in both the H1 and the L1 detectors"; L1 required a post-hoc cap on spectrogram values.
- [65] "No misclassifications" on August 2017 against 1.3 false positives per month on time slides.
- [38] The coincident FAR assumes independent, exactly simultaneous H and L false triggers.
- [103] Abstract "~11 % above 300 Msun" while Fig 5 shows +17 % in the highest band and non-monotonic bands.
- [103] Fig 5 colourbar floor (~1.03; lowest tick 1.04) is above its lowest band label (+2 %).
- [8] Text says CNN-Coinc retains >= 50 % of the leading ML distance down to 10/month on DS4; Table IV gives 621 vs 1400 Mpc (44 %).
- [39] FAP 1e-3 per 0.2 s segment is ~1.3e4 false alarms per month, not ~1e3 as stated.
- [39] One misclassified GWTC-3 event counted as BBH (GW200210_092254) has SNR 8.0 although the Table 1 caption says all misclassified GWTC-3 BBH have SNR < 7 (the body text sets this event aside).
- [32] Confusion-matrix rows sum to 109.2 % (Fig 5) and 101.7 % (Fig 7).
- [32] 498 recovered injections but 214 + 206 = 420 classified.
- [55] The first catalogue gave 4.88e-4 as a FAR in Hz; the abstract calls it a probability (~1.5e-5 Hz as a rate).
- [94] The "FPR" on continuous mock data is the missed-injection fraction.
- [94] Text counts (594 of 722 undetected at SNR 4-5) disagree with Fig 9 (0.526).
- [78] Table IV H1-V1 row is non-monotonic (26.8 > 21.1) and repeats the H1-L1-V1 90 % and 95 % values.
- [78] The 95 % CL zero-event FP limit is 3/N = 4.0e-4, not 2e-4.
- [78] Inclination and polarisation labels are swapped in Table I.
- [77] The 1-per-2-months threshold (~1.9e-7 Hz) sits just above the 1.67e-7 Hz background resolution, in the range where FARs come from a linear tail fit.
- [77] The max-SNR baseline is ~0 below FAR 0.05 Hz, which says more about the simulated glitch rate than about matched filtering.
- [81] The 40-80 Msun bin exceeds the 5-60 Msun training range.
- [81] The FAR bound comes from 0.9 s time slides of one 8.8-day chunk, which are strongly correlated.
- [85] "Half a month" of data is 12.15 days.
- [85] FAP 1e-4 equals one false positive in 10,000 samples.
- [85] Same PRD article number (109 043009) as the MSNRnet-1 citation in the sweep: one citation is wrong.
- [12] The first catalogue attributed DS1-DS4 distances (2116/1242/1241/1092) that are the Virgo-AUTh MLGWSC-1 values, not this paper's.
- [92] SNR is computed on the 4 s truncation, so NSBH/BNS SNRs are not comparable with full-signal SNRs elsewhere.
- [66] The introduction and the Figures-of-merit paragraph name ROC and PR AUC as figures of merit but no AUC is printed.
- [67] Abstract "zero false positives" in February 2020 follows removal of 7 of 13 triggers as noise after data-quality and spectrogram follow-up.
- [74] "Two false alarms per day" is equated with 1 in 1.7e5 trials (2 per day is 1 in ~4.3e4 s).
- [74] Expected maximum false alarms 29/43/55 do not follow from 0/1/2 test false alarms (Poisson 95 % gives ~22/35/46).
- [87] Sec IV's TPR >= 0.9 at 12/yr for astrophysical M1 ~ 40 Msun, q ~ 1 systems at SNR >= 12.5 conflicts with the Fig 7 astrophysical iso-lines (0.32 near SNR 10-11, 0.98 near SNR 15-17).
- [87] All FARs below 12/yr are extrapolated from one month of data.
- [13] The printed FAR fit (Eq 6) gives 13.1/yr at Rs 16, not the 0.34/yr in the text.
- [13] Table I GW200301_211019 FAR is 10x the Eq 6 value (80380 vs 8380/yr).
- [13] Five of eight Virgo-LIGO events said to exceed Rs 3.5; Table V has four.
- [9] Text gives TPR 0.053 at FPR ~1e-6 and SNR 6.23; Fig 5 reads ~0.43.
- [9] Table IV FAR floor printed as < 0.97/yr while the text says one per year.
- [36] Same classifier and glitch set give TN 9762 vs 9763 on two test sets.
- [79] Table A1 has 25 events below the 1/week FAR against the 26 in the text.
- [79] Table A1 floors at 0.01/yr although the time slides reach 1/152.6 yr.
- [91] Conclusion says all FAPs were better than 0.1 %; Sec III A gives 0.26 % for the Hanford O3 model.
- [91] Real-event counts use the union of two detectors without adjusting the false-alarm probability.
- [69] Run and epoch selection by sensitive distance used the authors' own dataset-4 realisation (seed 4261537, not the challenge seed, but drawn from the same ~3-month O3a noise file), which may bias the 1316 Mpc upward.
- [69] "Virgo-AUTh at 87 % of this value" matches neither Virgo-AUTh variant in Fig 4 (83 % or 119 %).
- [69] Fig 4 caption and legend disagree on which lines are dashed.
- [105] AUC differences are reported while the 90 % permutation bands overlap.
- [15] GWTC-3 candidate counts (90 / 79 / 63) do not reconcile across sections.
- [15] Missed-event accounting (54 + 23 > 75) does not close.
- [10] Text calls 35+35 sensitivity "similar"; Fig 2 shows Aframe above the leading matched-filter pipelines (GstLAL, PyCBC-BBH) there only above ~0.3/yr and ~30-50 % below them at 20+10.
- [10] Comparison volumes are taken from the LVK GWTC-3 data release, not recomputed on Aframe's injections and background; the paper does not discuss the differences.
- [16] Fig 7 caption names week 3 (the background week) as the injection week.
- [16] GW170817 FAR 6.9e-13 Hz is a Gaussian-tail extrapolation from a separate one-week O2 background (the one-week O3 background reaches only 1.8e-8 Hz).
- [14] Table V cWB row (1364 / 1441 / 1461 Mpc at 100 / 10 / 1 per month) rises as the FAR falls and does not match MLGWSC-1 Table IV for DS4 (1407 / 1332 / 1229).
- [14] Table V re-quotes MFCNN (1269 / 985 / 588) and PyCBC (1722 / 1607 / 1539) differently from MLGWSC-1 Table IV (1269 / 999 / 650; 1722 / 1610 / 1544).
- [14] Headline gains use the best of three seeds; the seed mean is lower (Fig 11).
- [76] Thresholds were set from performance in the real search, so 57/75 is in-sample.
- [76] "False positives" are defined in Sec III B as events the pipeline does not detect.
- [76] Extrapolated FARs (~1/350,000 yr) go far beyond the ~100 yr background.
- [80] The same configuration is reported with ~1.5 % different volumes in Tables 3 and 4.
- [80] Table 3 evaluates each model at its lowest zero-FP threshold (0.995 for the reference), not at the 0.9999 fixed for the final O3b model, so its volumes are not at the production operating point.
- [75] Text says ~1000 Mpc at 10/yr; Fig 6 reads ~650 Mpc there (1000 Mpc near 400/yr).
- [75] The Fig 7 FAR axis is labelled in years rather than per year (Fig 6 uses per year).
- [70] "Comparable to PyCBC" at 1/month hides that the method is 2.5 % below PyCBC there but above it at 10-100/month.
- [72] Abstract says about five months of O3b; the live time is 95.3 d (~3.1 months).
- [72] Per-month and per-year FARs in Table II are not related by a constant factor.
- [72] The discussion's "roughly a factor of two" over GW-Whisper is 1.66 and 1.49 in distance.

Type	Num Filters / Neurons / Heads	Filter / Pool / Head Size	Stride	Activation Function
Convolutional + MaxPooling ×4	32-256	16-8	(pool 4)	ReLU
Flatten	-	-	-	-
Dense Head	128, 2	-	-	ReLU, SoftMax

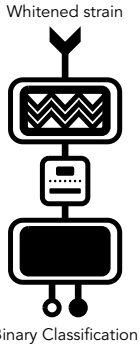


Figure 10: Baltus et al. 2021 early-inspiral CNN (CBMI proceedings) [107], collapsed blocks. Architecture from: Fig. 3 of the proceedings paper (best-performing architecture); all MaxPool layers have kernel size and stride 4. Diagram confidence: **high**; 4 details the paper leaves unstated are listed in the diagram spec, and are not drawn. Notation: Appendix A.

- [11] 10,000 training vs 20,000 validation waveforms is reversed from the usual split (Sec II D 2).
- [104] Four-detector FAR range "0.5 to 3.03e-4" has no unit and is in reverse order.
- [104] "Three orders of magnitude" storage reduction holds for three detectors, not two.
- [86] TAP values at FAP 1e-4 are quoted from ~1 noise sample.
- [99] Table 3 sets a per-window FPR (0.0061 per 0.25 s window) in the same row as Aframe's FAR < 1/yr, with only a general caption caveat about differing evaluation methods.
- [99] ECE reduction given as 15 % (Sec 3.3) and 95.6 % (Table 3).
- [62] Headline efficiency and the FAR calibration use different thresholds (1 % vs 0.3 % FPR).
- [62] AUC is said to be computed to FPR 1e-3 yet reported as a full AUC.
- [62] Uncertainty grows with SNR (slope +0.25), so the SNR labels are unusable.

4 Early warning

Early-warning models try to flag a compact binary inspiral (mostly binary neutron star or neutron star-black hole) before merger, from a partial inspiral whose signal-to-noise ratio (SNR) grows with time.

4.1 Families and lineage

4.1.1 Partial-inspiral classifiers

Baltus et al. trained one CNN per binary neutron star mass class on long windows truncated before merger [106], [107], then a single three-detector network for all binary neutron star masses [108] later work changed the network or the deployment target [109], [110], [111] and added auxiliary-channel noise subtraction [112].

> **Our results (to be added)**
> gravyflow reproduction: *pending* for 7 models: [106], [107], [112], [108], [110], [109], [111]. Results will be placed on the shared axes of this section beside the published values.

4.1.2 Forecasting curves

Forecasting networks report the output probability against time to merger for single loud injections [113], [114].

> **Our results (to be added)**
> gravyflow reproduction: *pending* for 2 models: [114], [113]. Results will be placed on the shared axes of this section beside the published values.

4.2 Comparison on the shared axes

Rows are the models graded comparable or conditionally comparable (3 of 9); each axis entry is one row. FAR is converted to per month (30 d) and to Hz where the paper states a rate.

Model	Axis	Published value (location; how obtained)	FAR	Grade: what differs	gravyflow
`yu2021early` [112]	eff@FAR	BNS TAR 0.38 at d_eff 40 Mpc and 0.09 at 60 Mpc at 1 false alarm per 7.1 h (0.01 per 256 s segment) (Fig 9; read-from-figure)	101 /month; 3.91e-05 Hz	conditional Simulated non-stationary O3-like noise with witness-channel subtraction; the per-segment fraction converts exactly to a rate because segments are disjoint.	*pending*
	eff@FAR	NSBH TAR 0.43 at d_eff 160 Mpc at the same FAR (Fig 10; read-from-figure)	101 /month; 3.91e-05 Hz		*pending*
`baltus2022` [108]	eff@FAP	50 % TAP at partial-inspiral SNR ~17 (Gaussian O3 and O4), ~20 (real O3) at FAP 1 % per 300 s frame (Sec III A, Fig 6; stated)	-	conditional Efficiency is at a per-frame FAP while the rate on real data is measured separately; the two are not paired.	*pending*
	other	real O3b trigger rate 277.5 / day (single), 1.71 / day with 10 consecutive triggers (Sec III A; stated)	-		*pending*

Model	Axis	Published value (location; how obtained)	FAR	Grade: what differs	gravyflow
alfaidi2024 [110]	eff@FAR	LSTM TAP ~0.44 vs exact-template matched filter ~0.79 at FAR ~1e-3 Hz (SNR 2-20, single detector, Gaussian) (Fig 4; read-from-figure)	2.59e+03 /month; 0.001 Hz	conditional A FAR in Hz of a statistic maximised over 4 s windows, with an exact-template matched filter as upper bound; binary black holes in Gaussian noise only.	*pending*

4.3 Latency and compute cost

8 of the 9 models in this section state a latency, inference cost or training cost; 1 papers were read and state none, and 0 were not read for runtime. Headline: the model's latency if it states one, else its inference cost; normalised values in brackets (Section 13).

Model	Headline: latency if stated, else inference cost	Hardware	Grade; scope; mode	Location	Training cost (hardware; grade; location)
baltus2021 [106]	Inference cost: ~0.005 s to analyse one 50 s frame (0.0001 compute-s per s)	1x Nvidia GeForce RTX 2070 SUPER	stated; network; unstated	Sec. IV A (Performance of the CNNs); footnote 4	-
baltus2021b [107]	Inference cost: 0.005 s to analyze a 50 second frame (0.0001 compute-s per s)	1x Nvidia GeForce RTX 2070 SUPER	stated; network; unstated	Sec. IV (Results and discussion)	-
wei2021 [114]	Inference cost: 8 ms per prediction (INT8 quantized ResNet-50, batch of 16 images) (0.008 compute-s per s)	Intel Xeon E5-2620 CPU (AVX2), count unstated	stated; network; batch	Sec. 4	-
yu2021early [112]	Inference cost: ~30 ms to classify one 256 s time series (CNN-class) (0.000117 compute-s per s)	1x NVIDIA Tesla P100-PCIE-16GB	stated; network; unstated	Fig. 12, Sec. VI (Discussion)	-
baltus2022 [108]	Latency: around 2 s to load 300 s of data for 3 detectors with GWpy on the LIGO servers, the limiting step; each analysis step also needs about 1.14 s of pre-processing (PSD 0.13 s, whitening 0.01 s, peak removal 1 s) and 0.01 s (GTX 750 GPU) or 0.5 s (CPU) of prediction (≈3.2 s per step with the GPU, ≈3.6 s with the CPU, derived)	LIGO servers (unstated)	stated; end to end; streaming	Sec. III A	-
alfaidi2024 [110]	training cost only	-	-	-	33 hours (100 epochs, batch 1000, 1 million training time series); 1x NVIDIA GeForce RTX 3090 (24 GB) (stated; Sec. 4.2 (Training the LSTM Model))
martins2024 [109]	Inference cost: 27.53 ± 3.69 ms per sample (9.18e-05 compute-s per s)	1x AMD EPYC 7551P 32-core (all 32 cores)	stated; network; unstated	Tab. 3, Sec. 5.2	~10^15 to 10^16 (read off a log axis) cumulative training FLOPs (FindCNN and GWaveNet, all five curriculum steps); 1x NVIDIA Tesla V100 (estimated; Fig. 4, Sec. 5.1)
papigkiotis2026 [111]	Inference cost: ~0.1 s per 40 s observational time window (TSFEL feature extraction) (0.025 compute-s per s)	CPU (model and count unstated)	stated; unstated; streaming	Sec. V C 1	-

4.4 What cannot be compared, and why

Early-warning results are quoted against partial-inspiral SNR at a per-frame false-alarm probability of about 1 %, a rate of order one per hour for frames of a minute. Only papers that convert to a rate over disjoint segments (or measure one) are graded conditional; lead times for single loud injections are not comparable.

The 6 models below have no shared axis; their headline metric is listed with the reason.

Model	Headline metric (location; how)	Why not comparable	gravyflow
baltus2021 [106]	heavy-BNS TAP ~0.99 at 125 Mpc, ~0.9 at 175 Mpc at FAP 1 % per frame (single detector, Gaussian) (Fig 5; read-from-figure)	Per-frame FAP of 1 % (frames 30-80 s) in Gaussian design noise; the realistic-population test detects 5 of ~90.	*pending*
baltus2021b [107]	light-BNS TAP ~1 to 15 Mpc, 0.2 at 40 Mpc at FAP 1 % per frame (Sec IV, Fig 4; stated)	Conference version of baltus2021 (arXiv 2105.13664, which the first catalogue missed).	*pending*
wei2021 [114]	none extracted	Forecasting curves for single injections at SNR 30; no FAR, ROC or efficiency.	*pending*
wei2021forecast [113]	BNS flagged up to 30 s before merger at SNR 30-40; GW170817 10 s before merger (Fig 2-3; stated)	Lead time for single high-SNR injections; no FAR or population efficiency.	*pending*
martins2024 [109]	Test accuracy at FAP 1 % per 300 s sample: GWaveNet 76.2 % vs FindCNN 66.8 % (2D: 79.4 vs 69.5) (Table 2; stated)	Same data recipe as Baltus et al.; per-sample FAP on a balanced set.	*pending*
papigkiotis2026 [111]	test accuracy ~90 % (ET-like), >97 % (CE-like) at low FAP (abstract) (abstract; stated)	Per-sample FAP in Gaussian ET/CE noise (Tables VI-VII); the per-day FAR curves of the ranking statistic (Fig 18) are not paired with a population efficiency.	*pending*

4.5 Suspected errors in the literature

11 suspected errors or internal inconsistencies in 7 papers, selected from the extraction notes (28 notes for this section in the catalogue). Each is a reading of the paper to be checked, not a verdict.

- [106] Network numbering differs between the text and the Fig 5 legend.
- [106] "Comparable to matched filtering" on the realistic population while Fig 7 shows 5 found and ~85 missed.
- [114] "Decreases the top-1 error rate by 10 %" with no baseline error stated.
- [113] Abstract "up to 30 s" holds only at injected SNR 30-40; the one real event is flagged 10 s ahead.
- [112] Stationary-noise BNS range quoted as ~80 Mpc at TAR 0.4; Fig 9 reads TAR ~0.31 there.
- [112] Abstract lead times (100 s, 10 s) vs body (~87 s, 12 s).
- [110] Efficiency at 1/day and 1/6 h is quoted although Fig 4 starts near 6e-5 Hz (1 per 4.6 h).
- [110] Test-point arithmetic (256,000 vs 2,560,000) does not close.

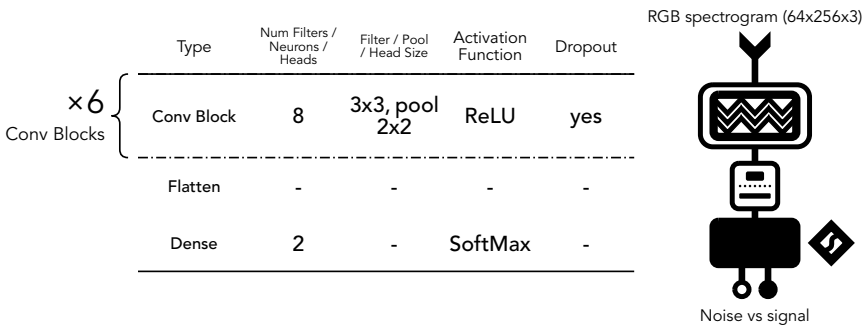


Figure 11: Astone et al. 2018 CCSN CNN [115], collapsed blocks. Architecture from: Paper Sec. on the Convolutional Neural Network and Fig. 4 (block diagram): six repeats of zero padding, 3x3 convolution with 8 filters and ReLU, max pooling and dropout, then Flatten and a 2-neuron softmax layer. Input from Figs. 2 and 3 (RGB image from cWB spectrograms of H1, L1 and V1). Diagram confidence: **high**; 4 details the paper leaves unstated are listed in the diagram spec, and are not drawn. Notation: Appendix A.

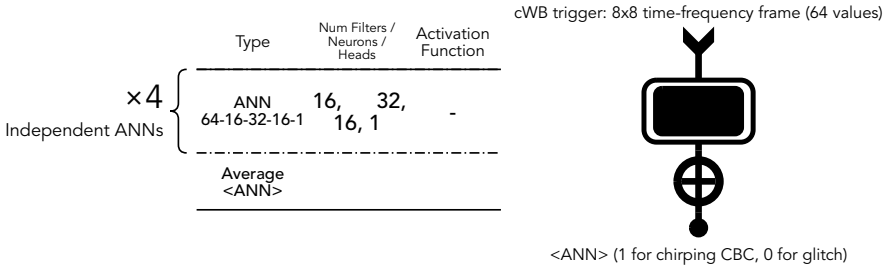


Figure 12: cWB ANN classifier (Vinciguerra et al. 2017) [129], collapsed blocks. Architecture from: Section 2 (8x8 frame input, multilayer perceptrons, average of 4 ANNs, Fig. 2) and Table B2 (architecture IN 64, H 16/32/16, OUT 1). Diagram confidence: **high**; 3 details the paper leaves unstated are listed in the diagram spec, and are not drawn. Notation: Appendix A.

- [109] Text says FindCNN is ~2 ms faster on GPU; Table 3 gives ~12 ms.
- [109] Table 1 prints PISNR "342" for D2, apparently a typo for ~34.2.
- [111] FPR is defined as a false-discovery rate.

5 Burst and unmodelled transients

Burst searches look for short transients without a template: core-collapse supernovae (CCSN), generic sine-Gaussian and white-noise-burst morphologies, cosmic-string cusps and long magnetar-like transients. Two traditions coexist: classifiers on simulated data, and machine-learning stages added to coherent WaveBurst (cWB) on real observing-run data.

5.1 Families and lineage

5.1.1 Core-collapse supernova classifiers

From the first CNN on cWB images of phenomenological CCSN waveforms [115] through real-noise versions [116] and multi-class models on numerical waveforms [117], [118], [119], [120], to equation-of-state and frequency-slope classifiers [121], [122] and contrastive models [123], [124].

> **Our results (to be added)**
> gravyflow reproduction: *pending* for 14 models: [115], [117], [118], [125], [126], [116], [119], [120], [122], [127], [128], [121], [123], [124]. Results will be placed on the shared axes of this section beside the published values.

5.1.2 Machine learning inside cWB

Neural networks, linear discriminants and support-vector machines, boosted trees, Gaussian mixtures and genetic programming applied to cWB triggers (summary statistics or time-frequency patterns) [18], [19], [129], [130], [131], [132], [133], on real observing-run data except for the recoloured S6D-VSR3 noise of [129] and a CNN on wavelet maps that re-ranks PyCBC IMBH and higher-mode triggers on O3a data [134]. Several of these report results in search-grade units (FAR, iFAR or sensitive volume).

> **Our results (to be added)**
> gravyflow reproduction: *pending* for 8 models: [129], [130], [131], [132], [19], [133], [18], [134]. Results will be placed on the shared axes of this section beside the published values.

Type	Num Filters / Neurons / Heads	Filter / Pool / Head Size	Stride	Norm	Activation Function
Residual Block 1	64	42, 36	1	-	-
Residual Block 2	128	42, 36	1	-	-
Residual Block 3	128	42, 36	1	-	-
Global Average Pooling	-	-	-	-	-
Dense Head	256, 256	-	-	Batch	-
Dense	2	-	-	-	SoftMax
Convolutional ×3	56-214	32-16	4-2	-	-
Global Average Pooling	-	-	-	-	-
Convolutional ×2	64	8, 5	1	-	-
Global Average Pooling	-	-	-	-	-
Concatenate					
Dense	256	-	-	-	-
Dense	2	-	-	-	SoftMax

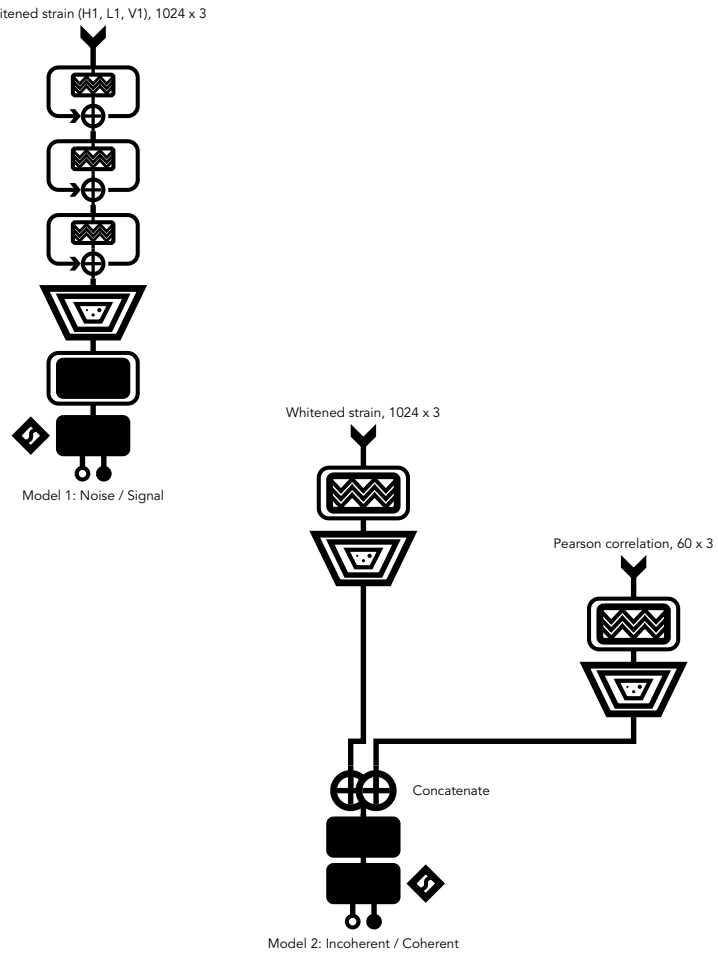


Figure 13: MLy (Skliris, Norman, Sutton 2024): coincidence and coherence models [135], collapsed blocks. Architecture from: Paper Fig. 1 (Model 1, coincidence) and Fig. 2 (Model 2, coherence); filter counts, kernels and strides from the figures, activation, normalisation and pooling from Sec. III C and Sec. III C 2. Diagram confidence: **high**; 5 details the paper leaves unstated are listed in the diagram spec, and are not drawn. Notation: Appendix A.

5.1.3 Unmodelled neural searches

Coherence and autoencoder networks on real data with time-slide backgrounds: MLy [135], GWAK [136], the Wavelet Detection Filter with a graph network [137], and, outside that setting, short-window anomaly detectors on CodaBench challenge data [138], [139] or on synthetic Gaussian noise [140].

> **Our results (to be added)**
> gravyflow reproduction: *pending* for 7 models: [141], [140], [136], [135], [139], [137], [138]. Results will be placed on the shared axes of this section beside the published values.

5.1.4 Long-duration and exotic transients

Minute-long transients, post-glitch CW-like transients and cosmic strings [142], [143], [144], [145].

> **Our results (to be added)**
> gravyflow reproduction: *pending* for 4 models: [142], [145], [144], [143]. Results will be placed on the shared axes of this section beside the published values.

5.2 Comparison on the shared axes

Rows are the models graded comparable or conditionally comparable (15 of 33); each axis entry is one row. FAR is converted to per month (30 d) and to Hz where the paper states a rate.

Model	Axis	Published value (location; how obtained)	FAR	Grade: what differs	gravyflow
`astone2018` [115]	eff@FAR	CNN 5.9 / 60.7 / 91.8 % vs cWB 1.7 / 40.8 / 86.9 % at network SNR 8 / 12 / 20 (Table II; stated)	181 /month; 7e-05 Hz	conditional Phenomenological CCSN in Gaussian noise; efficiency is over all injections but capped by a cWB pre-selection; per-SNR, not per distance.	*pending*
`chan2020` [117]	eff@FAP	TAP 86 % (H+L+VK) / 68 % (HLVK) at FAP 0.1 %, magnetorotational, 50 kpc (Sec IV, Fig 6; stated)	-	conditional Three-class CCSN in Gaussian design noise; FAP is a per-sample misclassification fraction, one CNN per distance.	*pending*

Model	Axis	Published value (location; how obtained)	FAR	Grade: what differs	gravyflow
	eff@FAP	TAP 76 % / 55 % at FAP 10 %, neutrino-driven, 10 kpc (Sec IV, Conclusion; stated)	-		*pending*
`lopez2021b` [116]	AUC	0.896 (blind set, phenomenological) / 0.79 (3D simulations), real O2 noise (Fig 12, Sec VI C; stated)	-	conditional CCSN in real O2 noise; "FAR_CNN" is FP/(FP+TP) on a balanced set; efficiency averaged over the injected SNR/distance mix.	*pending*
	eff@FAP	TPR ~0.55 at FPR 2 %; ~0.84 at FPR 20 % (blind set) (Fig 12; read-from-figure)	-		*pending*
`mishra2021` [132]	eff@FAR	BBH: ~0.90 (O1) / ~0.88 (O2) ML vs ~0.62 / ~0.65 standard cWB at 1/yr; IMBH ~0.81 / ~0.81 vs ~0.65 / ~0.65 (Fig 2 top; read-from-figure)	0.0821 /month; 3.17e-08 Hz	conditional Non-neural (XGBoost) cWB post-processing, O1/O2 time slides; efficiency is over recovered, not injected, events.	*pending*
	events	all 7 cWB O1/O2 BBH events recovered with lower FAR (Table II; stated)	-		*pending*
`alvarez2022cwbml` [19]	reach@FAR	sensitive hyper-volume O3a 0.822 vs 0.676 Gpc^3 yr (ML vs standard cWB); GstLAL 1.22, PyCBC-BBH 1.2, MBTA 0.885 on the same population (Table IV; stated)	0.164 /month; 6.34e-08 Hz	comparable Non-neural (XGBoost) cWB post-processing on full O3 HL; hyper-volume under the GWTC-2 Power Law + Peak population is the LVK search metric, so it is comparable with the matched-filter pipelines quoted beside it. The Fig 3 efficiencies use recovered (not injected) events as the denominator.	*pending*
	eff@FAR	BBH set: ~0.95 vs ~0.77 (ML vs standard) at 1/yr; ~0.73 vs ~0.57 at 0.01/yr (Fig 3 top; read-from-figure)	0.0821 /month; 3.17e-08 Hz		*pending*
	events	5 further GWTC-3 BBHs at FAR < 1/yr that standard cWB vetoed (Table II; stated)	-		*pending*
`lopez2022gmm` [133]	eff@FAR	Gaussian pulse: gain >20 points over cWB (GA0d1 ~0.70 vs ~0.43); WNB ~0.89-0.905 vs ~0.78-0.79 (Fig 1, Sec III D; read-from-figure)	0.000821 /month; 3.17e-10 Hz	conditional cWB post-processing (non-neural GMM) on 104.9 d of O3a HL; efficiency over an hrss grid, not a sensitive volume.	*pending*
	events	GW190521 iFAR >200 yr (cWB+GMM) vs 65 yr (standard) (Table IV; stated)	-		*pending*
`moreno2022` [140]	eff@FAP	LSTM-AE TPR 46.1 % (BBH) / 23.7 % (BNS) at single-detector FPR 0.1; 43.3 % (BBH) at dual FPR 0.01 (Tables 1-2; stated)	-	conditional Synthetic (ggwd-style) data, SNR uniform 5-20, FPR per 2.5 s window; AUC conditionally comparable because noise and SNR distribution are stated.	*pending*
	AUC	LSTM-AE 0.73 (BBH) / 0.59 (BNS) single; 0.79 / 0.63 dual; supervised CNN 0.92 / 0.77 single, 0.96 / 0.82 dual (Figs 4, 6; stated)	-		*pending*
`szczepanczyk2022` [18]	reach@FAR	hrss50 ratio XGB/standard cWB: Gaussian pulses 0.20-0.33, SG 0.84-0.95, WNB 0.91-0.95 (HL); volume gain 1.2-27x (Table I, Sec IV B, Fig 5; stated)	0.000821 /month; 3.17e-10 Hz	comparable O3 all-sky burst search by a cWB team (non-neural XGBoost on cWB), compared with the LVK O3 search; real O3 background, LVK morphologies: the reference point for burst hrss50.	*pending*
	events	16 known CBCs at iFAR >= 1 yr (standard cWB 14), O3 HL (Sec IV A; stated)	-		*pending*
`raikman2024` [136]	reach@FAR	hrss50 0.55-1.83e-22 Hz^-1/2 (SG), 0.70-0.89e-22 (WNB) (Table II; stated)	0.000821 /month; 3.17e-10 Hz	comparable GWAK on full O3 with ~10,000 yr of time slides; LVK burst morphologies, so hrss50 is comparable with cWB/MLy at the same iFAR.	*pending*
	eff@FAR	SNR at 50 % efficiency: SG 512-1024 Hz ~8.5, BBH ~11.5, CCSN ~21 (Fig 1; read-from-figure)	0.0821 /month; 3.17e-08 Hz		*pending*
`skliris2024` [135]	reach@FAR	hrss50 MLy vs cWB: SGE70Q3 1.9 vs 1.5; SGE235Q100 1.4 vs 0.9; WNB100 1.7 vs 1.2 (1e-22 Hz^-1/2) (Table I; stated)	0.0821 /month; 3.17e-08 Hz	comparable O2 real background (~1e3 time shifts), O2 LVK burst injection set; compared with cWB hrss50 from the LIGO-Virgo O2 search for the same waveform types (private communication), not on the same data: MLy uses only 1-25 Aug 2017 HLV data.	*pending*
	eff@FAR	>=50 % at network SNR 19, >=90 % at 27 (WNB/CCSN/BBH) (Fig 6; stated)	0.0821 /month; 3.17e-08 Hz		*pending*
`fernandes2025` [134]	reach@FAR	sensitive VT ratio ML/base, PyCBC-IMBH: 1.07 (100-200 Msun) to 1.45 (400-600 Msun) at IFAR 100 yr (Fig 5; read-from-figure)	0.000821 /month; 3.17e-10 Hz	comparable Classifier re-ranking of PyCBC IMBH and higher-mode searches on O3a chunks 1-4; VT ratio at IFAR is a search-level metric, but relative to its own baseline only.	*pending*
	reach@FAR	PyCBC-HM: 0.87 to 1.29 at IFAR 100 yr (Fig 6; read-from-figure)	0.000821 /month; 3.17e-10 Hz		*pending*
`pracchia2025` [128]	reach@FAR	distance at 50 % efficiency: 0.64 kpc (s18), 1.83 kpc (m39), 0.15 kpc (mesa20_pert) (Table I; stated)	0.0164 /month; 6.34e-09 Hz	conditional Single detector, Gaussian O4-design noise; threshold one background event per 5 simulated years (Gaussian only).	*pending*
`cuoco2026` [137]	eff@FAR	GNN vs deterministic WDF: all classes 0.70 vs 0.27 at 1/day (simulated); 0.28 vs 0.08 (real O4b); 0 at 1/month on real O4b (Tables II-III; stated)	30 /month; 1.16e-05 Hz	conditional Own simulated and 2.16 d real O4b sets, mixed CBC/CCSN classes; recovered fraction, not sensitive distance.	*pending*
`ratner2026` [138]	eff@FAR	recall 0.88 (BBH) / 0.86 (SGLF) at <=1 false alarm per year, no trigger (Table I; stated)	0.0821 /month; 3.17e-08 Hz	conditional CodaBench anomaly-challenge 50 ms windows of cleaned real noise; FAR 1/yr extrapolated from 50k backgrounds; PR-AUC depends on the anomaly rate.	*pending*
	AUC	PR-AUC 0.91 (BBH) / 0.89 (SGLF) (Table I; stated)	-		*pending*
`sun2026` [123]	AUC	CS-CAE 0.976 / supervised CNN 0.984 / CAE 0.916 (in-distribution); 0.995 / 0.993 / 0.979 (numerical CCSN) (Figs 2-3; stated)	-	conditional Simulated ET Gaussian noise with phenomenological CCSN; AUC conditionally comparable within ET CCSN studies; the distance has no stated FAR.	*pending*
	reach@FAR	~50 % efficiency at ~120 kpc (ET), threshold not stated (Sec III B, Fig 4; stated)	-		*pending*

5.3 Latency and compute cost

12 of the 33 models in this section state a latency, inference cost or training cost; 20 papers were read and state none, and 1 were not read for runtime. Headline: the model's latency if it states one, else its inference cost; normalised values in brackets (Section 13).

Model	Headline: latency if stated, else inference cost	Hardware	Grade; scope; mode	Location	Training cost (hardware; grade; location)
`nigam2020` [125]	training cost only	-	-	-	73700 s; CPU i7 8750h (stated; Tab. III (Deep Filtering))
`lopez2021b` [116]	Inference cost: 3 ms per 2 s long image (0.0015 compute-s per s)	1x Nvidia Quadro P5000 (named in the same sentence as training)	stated; network; batch	Sec. VII (Discussion)	1h 18 min train, validate and test, Mini Inception-ResNet, 5 epochs per SNR (Gaussian noise data set); 1x Nvidia Quadro P5000 (stated; Sec. VI A)
`iess2023` [119]	training cost only	-	-	-	~15 s per epoch (1D CNN, L1 dataset); 1x Tesla K40 GPU (stated; Sec. 5 (before Sec. 5.1), Fig. 3)
`meijer2023cosmicstrings` [145]	Inference cost: 10 milliseconds per example (three data streams of 8 seconds) (0.00125 compute-s per s)	Intel E5-2670 CPU and 1x NVIDIA Tesla V100 16GB GPU (the machine the CNNs were run on, Sec. IV B)	stated; network; unstated	Sec. V A (Numerical Results); Sec. VI (Conclusions)	-
`modafferi2023` [144]	Inference cost: c_CNN ≈ 4 × 10^-4 s per sample, averaged over a batch of 10^4 via Model.predict_on_batch	1x Tesla V100-PCIE-32GB	stated; network; batch	Sec. III E	about 1.8 hours, synthetic data only; 1x Tesla V100-PCIE-32GB (stated; Sec. III E (Timings))
`attadio2024` [143]	Inference cost: 20 minutes (denoise O(5000) maps)	1x NVIDIA L40S 48 GB	stated; network; batch	Sec. IV F (Computational load); App. C	3 hours (train the denoiser with O(2000) maps); 1x NVIDIA L40S 48 GB (stated; Sec. IV F (Computational load); App. C)
`morales2024` [122]	Inference cost: about 5 s for testing (test set of the phenomenological-waveform analysis)	1x NVIDIA T4 (Google Colaboratory)	stated; network; batch	Sec. 3.1.3	344 s ≈ 5.73 min total computing time, final training of the optimized ResNet50; 1x NVIDIA T4 (Google Colaboratory) (stated; Sec. 3.1.3)
`nunes2024` [127]	training cost only	-	-	-	around 90 s per epoch; unstated (stated; Sec. IV D (1.00 s window dataset))
`raikman2024` [136]	Inference cost: 4,500 GPU hours (5.13e-05 compute-s per s)	GPU (model and count unstated)	stated; unstated; batch	Sec. III C (Computational limitations)	-
`skliris2024` [135]	Latency: order of 1 second s (1 s)	unstated	stated; end to end; streaming	Abstract / Sec. I	~1 hour per model; 1x Tesla V100-SXM2-16GB (stated; Sec. V B (Inference and Training Times))
`cuoco2026` [137]	Latency: about three seconds (conditioning reads ahead of what it emits) (3 s)	1 core of a 3.0 GHz AMD EPYC 7313 (conditioning and search, about 10 ms per s of data); post-candidate steps on that core plus 1x NVIDIA RTX A5000	stated; end to end; streaming	Sec. VII (Conclusion, penultimate paragraph)	-
`ratner2026` [138]	Inference cost: 5 hours for three billion windows	1x T4 GPU	stated; network; batch	Sec. VI (Impact of trigger)	under 3 hours (0.2%, 100 event case) to over 12 hours (0.02% case) hours; 1x T4 (printed as 'T4 TPU') (stated; App. D (Training and Model selection))

5.4 What cannot be compared, and why

Burst papers use different waveform families (phenomenological or numerical CCSN, sine-Gaussians, white-noise bursts), quote efficiency against network SNR, distance or hrss, and often define "FAR" as a false-discovery fraction on a balanced set. Results are comparable only within a morphology at a stated FAR: hrss50 on the LVK morphologies (cWB, MLy, GWAK) or volume gains of cWB post-processing on the same observing run.

The 18 models below have no shared axis; their headline metric is listed with the reason.

Model	Headline metric (location; how)	Why not comparable	gravyflow
`vinciguerra2017` [129]	cWB baseline efficiency by distance shell for wide-mass CBC: 72.3 % (45-65 Mpc) to 10.2 % (340-505 Mpc) (Table 3; stated)	Signal-morphology classification inside cWB on recoloured S6D-VSR3 noise; the reported efficiencies are the baseline pipeline, not the classifier.	*pending*
`cavaglia2020` [130]	specificity 0.962 +- 0.011, FNR 0.036 +- 0.008 (Dimmelmeier, 3.16 kpc) (Sec V, Fig 2; stated)	Genetic-programming classifier of cWB single-detector triggers (non-neural); balanced trigger test sets, no rate.	*pending*
`iess2020` [118]	signal-vs-glitch accuracy 95.4 % (1-D) / 96.3 % (2-D), Virgo O3 PSD; 7-class 87.9 % (Fig 7, Fig 8, Sec 6.2; stated)	Accuracy is conditional on a WDF trigger chosen with knowledge of the injection time; Gaussian noise and two parametrised glitch families; no FAR.	*pending*
`nigam2020` [125]	8-class accuracy 100 % (Deep Filtering, CNN, BiLSTM); RF 97.75 % (Table II; stated)	Eight synthetic transient classes in white noise with no noise class; accuracy only.	*pending*
`antelis2021` [131]	background FAR at cWB rho=5 cut from 0.0297 to 1.78e-4 Hz (L1) and 0.0253 to 1.95e-4 Hz (H1); L1H1 9.76e-6 Hz to the 1/3.3 yr floor (Sec III.B, Figs 5-6; stated); CCSN he3.5 detection efficiency vs distance essentially unchanged after ML (Fig 6; stated)	Non-neural (SVM) cWB follow-up for CCSN on 3-day O3a windows; the FARs are trigger rates at one cWB threshold, and the efficiency is unchanged, so no efficiency-at-FAR gain is shown.	*pending*
`lopez2021a` [126]	70 % efficiency at "FAR" < 5 % for SNR > 15 (abstract level) (abstract; stated)	Conference version of lopez2021b; "FAR" there is a false-discovery fraction, not a rate. Full text not read.	*pending*
`marianer2021` [141]	20-class accuracy 97.3 % on regenerated O1+O2 Gravity Spy-style spectrograms (1,197 test) (Secs 2.2-2.3; stated); hrss50 for flagging: 10e-22 (WNB 100 Hz) to 40e-22 Hz^-1/2 (sweeps); no FAR attached (Table 4; stated)	Outlier search; hrss50 is for "flagged as outlier" with no time-normalised false-alarm rate.	*pending*
`boudart2022` [142]	none extracted	No quantitative metric: qualitative recovery limit at visibility 12-16; detection statistic deferred.	*pending*

Model	Headline metric (location; how)	Why not comparable	gravyflow
`iess2023` [119]	merged per-trigger TPR 99.3 % (L1), 99.1 % (H1), 79.8 % (V1), 97.8 % (3-detector) (Table 2, Sec 5; stated)	Trigger classification at a fixed 1 kpc on real O2 noise; noise class is whatever WDF triggers; no FAR.	*pending*
`meijer2023cosmicstrings` [145]	accuracy 0.788, TPR 0.755 at FPR 0.18 (cusp vs blip, ET) (Table I; stated)	Balanced cusp-versus-glitch classification in simulated ET noise; no noise class or FAR.	*pending*
`modafferi2023` [144]	p_det > 90 % at p_FA 1e-7 (synthetic); two-stage CNN filter ~80x faster with ~1-3 % p_det change (Sec V, Fig 3; stated)	Long-duration (post-glitch Vela) transient CW search on F-statistic atoms; p_FA per template, no link to transient FAR.	*pending*
`sasaoka2023_ccsn` [120]	10-class accuracy 0.984 (1 kpc), 0.603 (5 kpc), 0.332 (10 kpc), real O3 noise (Sec III A, Fig 5; 1 and 10 kpc stated, 5 kpc derived as the mean of the Fig 5b diagonal)	Read paper is arXiv 2310.09551 (PRD), not the catalogued PoS; per-distance multi-class accuracy; ~4 % of pure-noise samples are flagged as signal, a false alarm every ~25 s (Sec III C), with no efficiency at a usable FAR.	*pending*
`attadio2024` [143]	efficiency 0.90 at FAP 0.02 (masked-loss denoiser) (Eq 23, Fig 10; stated)	Long (1200 s) magnetar transients, single detector, simulated O4 noise; FAP is FP over all samples per map.	*pending*
`morales2024` [122]	3-class HFF-slope accuracy 0.987 / 0.933 / 0.853 at 1 / 5 / 10 kpc (L1, real O3b) (Table 6; stated)	Classification of one waveform per class at fixed galactic distances; no FAR or noise class.	*pending*
`nunes2024` [127]	time-series classifier accuracy 98 %, FPR 0.10 % (one noise sample flagged), 4.5 % of signals missed, all at SNR <= 10 (network SNR > 5) (abstract, Sec IV C, Fig 7; stated)	Rotational CCSN (Richers et al. catalogue) at 5-20 kpc in real LIGO-Virgo O3a noise (the spectrogram model uses O2); FPR is a per-sample fraction on a balanced validation set, not a rate.	*pending*
`yan2025deepgrav` [139]	challenge TNR 0.9708 (development) / 0.9832 (final), first of 272 teams (abstract, Fig 3; stated)	NSF HDR A3D3 anomaly-challenge score: TNR at an undisclosed threshold and TPR; trained on both signal classes, so not anomaly detection in the unsupervised sense.	*pending*
`casallaslagos2026` [121]	5-class EOS accuracy 98.58 % (1 kpc), 52.43 % (5 kpc), real O3b noise (abstract; stated; not reproduced by Table 6 or by Tables 7-8, which give 0.90-0.95 at 1 kpc and 0.80-0.83 at 5 kpc)	Classifies already-triggered cWB maps of five fixed waveforms by equation of state; no false alarms enter, no FAR.	*pending*
`wang2026ccsn_scl` [124]	rotational TPR ~1.0 at FPR 1e-4 (10 and 200 kpc); ~0.80 at 1200 kpc (Sec IV-B, Fig 5; stated)	Synthetic Gaussian HLVK spectrograms; FPR is per sample over the other classes; no real noise or glitches.	*pending*

5.5 Suspected errors in the literature

41 suspected errors or internal inconsistencies in 27 papers, selected from the extraction notes (96 notes for this section in the catalogue). Each is a reading of the paper to be checked, not a verdict.

- [115] The FAR of 7e-5 Hz in Table II is not derived from the CNN false-alarm fraction of 3-4 % (Eq 9 is a dimensionless fraction).
- [130] Table II shows efficiency falling after cleaning (about 10 % at 2 sigma, 30 % at 3 sigma) while the abstract claims improved reach; the two use different operating points.
- [117] Abstract attributes the 52 % / 83 % TAPs to waveforms; the body attributes them to the two networks.
- [117] Training-set size given as 1.8e5 per distance and as about 4e5 per class.
- [118] Text says total accuracies are consistently above 90 %, but Table 1 has 83.2 % (ET, s25, 1-D).
- [125] Logistic-regression accuracy 89.67 % disagrees with its own confusion matrix (89.17 %).
- [125] Three models at 100 % down to SNR 5 suggests a degenerate benchmark or leakage.
- [131] Abstract quotes FAR reduction by 10-100x; Table IV ranges 11-1019x, with L1H1 saturated at the background floor.
- [131] L1H1 noise rejection given as 99.4 % (text) and 99.0 % (Table II).
- [116] Efficiency floor of ~58-66 % at SNR 1-6 (Fig 14) is incompatible with a ~5-10 % false-positive fraction.
- [132] The 26 % / 16 % gains are absolute efficiency differences, not relative gains.
- [132] Efficiency denominator excludes injections cWB never recovers.
- [19] Efficiency denominator is the number of cWB-recovered injections, not all injected, so curves reach 1.0.
- [19] The "about 20 %" gain is +18 % relative for BBH but about +27 % relative for IMBH at 1/yr (Fig 3).
- [133] Background test fraction given as 30 % (Sec III B) and 20 % (Sec IV).
- [140] Text equates FPR 1e-6 with one false alarm per three months, which holds only for 8 s windows, not the 2.5 s windows used.
- [140] Dual-detector BBH-CNN TPR (25.6 % at 1e-2) is below single-detector (66 % at 1e-1) although dual AUC is higher.
- [18] Table I Gaussian pulse tau=1.0 ms HL volume (1.5 vs 0.11, ratio 13.6) contradicts the 2.3-4.5 range in the text; likely a typo.
- [18] Fig 1 caption says 10 HLV events at iFAR >= 1 yr; the text says 8.
- [119] Abstract "~80 % for Virgo" is the merged model on the triple-coincident subset; single V1 models are 68-75 %.
- [145] Matched filtering against the exact injected template is reported worse than random (Fig 16-17), which is implausible for the optimal filter.
- [120] PoS abstract gives 96.9 % at 1 kpc against 98.4 % in the arXiv/PRD version.
- [143] FAP is defined as FP/(all samples), not FP/(FP+TN), which understates the false-positive rate.
- [122] Binned-regression accuracy rises from 1 to 5 kpc (0.55 to 0.85, Table 8), implausible for an amplitude-limited task.
- [136] hrss50 takes only three distinct values across eight SG morphologies, suggesting a coarse amplitude grid.
- [136] Abstract claims three CBCs detected; none passes the paper's 1/100 yr threshold.
- [135] Text says MLy is 10-50 % worse in hrss than cWB; Table I gives up to +56 %.
- [135] The 1/yr threshold is extrapolated from time lags of days to weeks of data.
- [134] Text gives up to ~70 % VT gain at IFAR 100 yr; Fig 5 reads ~45 % there (70 % matches the 1000 yr marker).
- [134] PyCBC-HM VT ratio at 100 yr lies below both 10 yr and 1000 yr in every mass bin, unexplained.
- [128] Table I column headings (10/50/90 % exclusion distance) are defined only in the Fig 6 caption; values fall as the percentage rises.
- [139] TNR without the matching TPR or threshold is not a detection performance.
- [139] Augmentation counts do not close (n in {3,5,10} x 200,000 vs the stated 300,000).
- [121] Abstract macro AUC 0.97 and 0.98 at 1 kpc: 0.97 is the TW2 value (Table 8), TW1 gives 0.80 macro / 0.89 micro (Sec 7, Table 7), and no macro AUC of 0.98 appears.
- [121] Abstract 52.43 % at 5 kpc against 0.80-0.83 in Tables 7-8.
- [121] Table 1 GPS times give each window 14 days, not one week, and make TW1 overlap TW2, contradicting the calendar dates in the same table.
- [137] "The ordering is the same, with the GNN above Rmor" on real O4 (Sec V C) fails for binary neutron stars in Table III: GNN 0.17 against Rmor 0.21 at 100/day (0.23 against 0.25 at 1000/day, where the caption says the rankings agree).
- [137] Fig 7 AUCs below 0.5 (BNS 0.06) are coincidence-saturated, not discrimination AUCs.
- [138] Abstract quotes recall 0.91 (BBH) and 0.85 (SGLF); Table I gives 0.88 and 0.86 (0.91 is the PR-AUC).
- [123] The 120 kpc reach is quoted at an "adopted operating threshold" with no FPR or FAR.
- [124] Fig 7 plots FPR down to 1e-5 with 50,000 noise samples (resolution 2e-5).

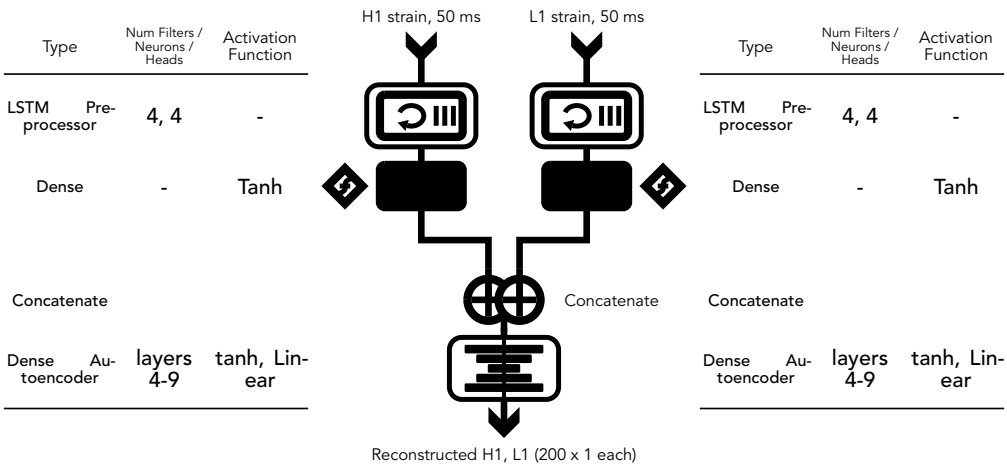


Figure 14: GWAK LSTM autoencoder (BBH signal autoencoder) [146], collapsed blocks. Architecture from: Sec. 3.1, Fig. 2 (LSTM autoencoder: per-detector two LSTM(4) layers and tanh linear, concatenation, tanh linear layers 4-8 with a latent in the middle, linear layer 9); the dense autoencoders for background and glitches are Fig. 3 and are not drawn. Diagram confidence: **medium**; 4 details the paper leaves unstated are listed in the diagram spec, and are not drawn. Notation: Appendix A.

6 Anomaly and multi-signal detection

Anomaly and multi-signal detectors are trained on noise (or on noise plus a few known classes) and flag what does not fit. Their false-alarm behaviour is set by the definition of normal data, which differs from paper to paper.

6.1 Families and lineage

6.1.1 Autoencoder anomaly detectors

Recurrent or convolutional autoencoders trained on noise (GWAK also on signal and glitch classes) [146], [147], [148], [149], [150], [151], scaled up to a multi-level pipeline over O3-O4 with a measured background [17] and to Einstein Telescope mock data [152].

> **Our results (to be added)**
> gravyflow reproduction: *pending* for 8 models: [151], [146], [147], [150], [148], [149], [152], [17]. Results will be placed on the shared axes of this section beside the published values.

6.1.2 Foundation-model features

Pretrained audio or vision transformers used as feature extractors [71], [153].

> **Our results (to be added)**
> gravyflow reproduction: *pending* for 2 models: [71], [153]. Results will be placed on the shared axes of this section beside the published values.

6.1.3 Other approaches

Entries that do not belong to the families above.

> **Our results (to be added)**
> gravyflow reproduction: *pending* for 1 models: [154]. Results will be placed on the shared axes of this section beside the published values.

6.2 Comparison on the shared axes

Rows are the models graded comparable or conditionally comparable (3 of 11); each axis entry is one row. FAR is converted to per month (30 d) and to Hz where the paper states a rate.

Model	Axis	Published value (location; how obtained)	FAR	Grade: what differs	gravyflow
`raikman2023` [146]	eff@FAR	SNR at 50 % efficiency: BBH ~15, SG 512-1024 Hz ~9.5, WNB ~22, CCSN ~27 (Fig 13; read-from-figure)	0.0821 /month; 3.17e-08 Hz	conditional About 71 min of O3a H1+L1 noise; the FAR uses 1 yr of time slides whose source data and reuse are not stated.	*pending*
`chatterjee2024` [71]	reach@FAR	sensitive distance 890 Mpc (DS3) / 905 Mpc (DS4) (Fig 2a-b; read-from-figure)	1 /month; 3.86e-07 Hz	comparable MLGWSC-1 DS3/DS4 following the challenge evaluation protocol: directly comparable with the challenge entries.	*pending*
	events	18 of 31 O3b GWTC-3 BBHs recovered (Sec 1.1, Fig 3; stated)	-		*pending*

Model	Axis	Published value (location; how obtained)	FAR	Grade: what differs	gravyflow
inguglia2026 [17]	reach@FAR	VT relative to LVK pipelines at FAR < 1/yr: O3a 20-100 Msun 0.20-0.54 of matched filter, 0.48-0.56 of cWB; 100-160 Msun 0.54-1.28 and 0.63-1.05 (Sec 5; stated)	0.0821 /month; 3.17e-08 Hz	comparable Real O3-O4 HL data with time-slide FAR, injection-calibrated, and VT against reweighted LVK sensitivity injections: search-grade, although training overlaps part of the O3a test period.	*pending*
	eff@FAR	injection recovery at FAR < 1/yr, O3a: 0.07 / 0.39 / 0.72 / 0.92 at network SNR 10 / 15 / 25 / 40 (Table 1; stated)	0.0821 /month; 3.17e-08 Hz		*pending*
	events	47 detections at FAR < 1/yr in O3a-O4b (44 shared with cWB), none new (Sec 4, Table 3; stated)	-		*pending*

6.3 Latency and compute cost

2 of the 11 models in this section state a latency, inference cost or training cost; 7 papers were read and state none, and 2 were not read for runtime. Headline: the model's latency if it states one, else its inference cost; normalised values in brackets (Section 13).

Model	Headline: latency if stated, else inference cost	Hardware	Grade; scope; mode	Location	Training cost (hardware; grade; location)
cain2025 [148]	Inference cost: 164.4 ± 1.0 ms per 32-second segment (total processing) (0.00514 compute-s per s)	1x AMD Ryzen 9 9950X CPU, 64 GB RAM, no GPU	stated; end to end; unstated	Sec. 4.4 (Computational Efficiency)	-
inguglia2025 [152]	training cost only	-	-	-	less than 1 hour (supervised or weakly supervised training/validation, 60 epochs, 120,000 spectrograms); 1x Nvidia RTX 4000 Ada 20 GB (one of 4 in a machine with 2x16-core CPU at 4.1 GHz) (stated; Sec. 3 (end of the model/training paragraphs, before Sec. 3.1))

6.4 What cannot be compared, and why

The anomaly literature has no shared definition of normal data, so AUCs and recalls depend on which anomalies were chosen. Only the O3-O4 pipeline with time slides and sensitive volume is search-grade.

The 8 models below have no shared axis; their headline metric is listed with the reason.

Model	Headline metric (location; how)	Why not comparable	gravyflow
morawski2021 [151]	TPR at 5 % per-segment FPR: 96 % (simulated aLIGO), 41 % (simulated aVirgo); 52 % L1, 50 % H1, 27 % V1 in real O2 noise (Tables 1-2; stated)	1 s segments with injected BBHs (matched-filter SNR 4-40) against noise; an MSE threshold at a per-segment FPR of 5 %, not a FAR.	*pending*
fayad2024 [147]	0.89 (LSTM-VAE) vs 0.76 (AE), 10 events vs 10 noise samples (Fig 6; stated)	n = 20 test samples, so AUC carries roughly +-0.1; no SNR distribution, no FAR.	*pending*
imran2024 [150]	none extracted	Abstract-level: autoencoder on Gravity Spy spectrograms; no numbers.	*pending*
cain2025 [148]	ROC-AUC 0.994, AP 0.967 (102 O4 events vs 399 noise segments, H1) (Fig 5, Sec 4.4; stated); synthetic BBH SNR 8-25, aLIGO PSD: AUC 0.806 (Table 1; stated)	Catalogue events versus random science segments with an F1-maximising threshold chosen on the test set; the authors show the O1-O4 skill partly tracks run-specific whitening (AUC 0.78 to 0.44 after uniform re-whitening).	*pending*
guo2025 [149]	per-class AUC vs SNR, e.g. IMRPhenomD 0.561 / 0.855 / 0.993 / 1.000 at SNR 5 / 10 / 15 / 20 (mock Gaussian noise) (Table I; stated); real 4-OGC events detected: 0.04 / 0.24 / 0.80 / 1.0 at single-detector SNR 5 / 10 / 15 / 20 at FAR ~1e-3 Hz (Fig 17; read-from-figure)	Single-detector anomaly flagger at ~3.6 alarms per hour; the FAR is several orders above search thresholds.	*pending*
inguglia2025 [152]	IMBH-forming BBH: 23 % (unsupervised) / 100 % (weakly supervised) at nominal 4.5/yr (Sec 4.2, Fig 8; stated)	ET mock data, single channel, glitches not considered; FAR is a nominal Gaussian-tail statement, not a background count.	*pending*
ma2025overlap [154]	none extracted	Not read beyond the abstract: separation of overlapping signals for PE; no numbers.	*pending*
cirfeta2026 [153]	recall at 1 % exceedance: harmonic comb >0.98, scattered light 0.58, blips 0.00 at SNR ~380-430 (Sec V A, Table III; stated)	Glitch-morphology anomaly ranking on O4a with injected SNRs of 200-430; recall at an empirical calibration-pool exceedance, not a FAR.	*pending*

6.5 Suspected errors in the literature

15 suspected errors or internal inconsistencies in 7 papers, selected from the extraction notes (35 notes for this section in the catalogue). Each is a reading of the paper to be checked, not a verdict.

- [146] Supernova sensitivity quoted three ways (SNR 22 at 1/hour; 20-25 at 1-2/month; 50 % at ~27 at 1/yr).
- [71] Catalogue summary "94-98 % per class" does not match Fig 4b (0.90-0.99).
- [71] O3b FARs down to <1e-20 per month come from exponential extrapolation of a one-month background.
- [148] Threshold described as optimised on validation data for recall at precision >90 % (Sec 3) and as F1-maximising (Sec 4.4); all 102 signals are in the test set, so it was apparently set on test data.
- [148] Event counts disagree: 126 O4 (102 usable) + 90 O1-O3 events in Sec 4.2 (216, or 192 usable) vs 207 O1-O4 events in Sec 6.1.
- [148] Claimed chirp band to 800 Hz exceeds the Nyquist frequency of the 1024 Hz data.
- [149] Table III caption gives 3.75 per hour for 1e-3 per second (3.6).
- [149] Conclusions report 1482 anomalies in 88.74 h; the body reports 270 in 546 min.
- [152] Abstract 4.5/yr vs conclusions 4.6/yr; the 5-sigma threshold used is ~6.3 sigma on the fitted Gaussian.
- [152] Weak-supervision injections overlap the IMBH test population.
- [153] Blip failure SNR given as ~380 (text) and ~430 (Table III).
- [153] Version-1 catalogue counts are retained but superseded by the author.
- [17] Table 3 has 46 rows against the 47 detections claimed.
- [17] Gate accounting in Sec 6 (53 above threshold, 48 pass, "only two" rejected) does not close.
- [17] Training uses 32 of the 143 O3a blocks that are also searched.

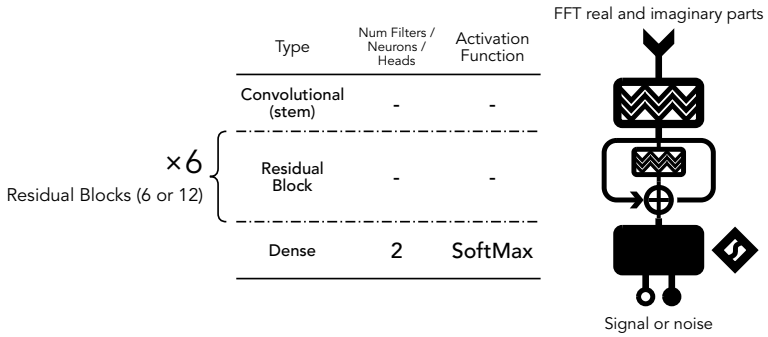


Figure 15: Dreissigacker et al. 2019 CW ResNet [155], collapsed blocks. Architecture from: Sec. III A text and Fig. 1 (stem block, n residual blocks, end block); residual block 'built according to' He et al. (Ref. 48). No layer table is given. Diagram confidence: **medium**; 5 details the paper leaves unstated are listed in the diagram spec, and are not drawn. Notation: Appendix A.

7 Continuous and long-duration waves

Continuous-wave (CW) models search for long, nearly monochromatic signals from spinning neutron stars, or classify the candidates of a semi-coherent search. The field has a shared figure of merit, the sensitivity depth at a fixed false-alarm probability per frequency band, which makes comparison more direct than in the transient categories.

7.1 Families and lineage

7.1.1 Deep-learning CW searches against WEAVE

Dreissigacker et al. set the benchmark of sensitivity depth at 1 % false alarm per 50 mHz against the WEAVE coherent search [155], [156] large-kernel CNNs and vision transformers later reached or approached matched-filter sensitivity in targeted searches (detection probability 87-90 % at the matched-filter depth) and in one-day directed and all-sky searches (78-89 %) [157], [158], while ten-day all-sky CNNs and U-Nets remain well short of it (33-69 %) [159], [160].

> **Our results (to be added)**
> gravyflow reproduction: *pending* for 9 models: [155], [156], [161], [162], [157], [159], [160], [158], [163]. Results will be placed on the shared axes of this section beside the published values.

7.1.2 Candidate follow-up and clustering

Networks that act on the output of a semi-coherent search: Viterbi maps [164], Einstein@Home candidate clusters [165], [166], TD-Fstat distributions [167], FrequencyHough candidates [168] and excess-power selection [169] a related network classifies time-resampled, double-Fourier-transformed strain data directly rather than search candidates [170].

> **Our results (to be added)**
> gravyflow reproduction: *pending* for 7 models: [167], [164], [165], [166], [169], [170], [168]. Results will be placed on the shared axes of this section beside the published values.

7.1.3 Long and intermittent signals

Post-merger remnant transients and intermittent backgrounds [171], [172].

> **Our results (to be added)**
> gravyflow reproduction: *pending* for 2 models: [171], [172]. Results will be placed on the shared axes of this section beside the published values.

7.2 Comparison on the shared axes

Rows are the models graded comparable or conditionally comparable (10 of 18); each axis entry is one row. FAR is converted to per month (30 d) and to Hz where the paper states a rate.

Model	Axis	Published value (location; how obtained)	FAR	Grade: what differs	gravyflow
`dreissigacker2019` [155]	depth@pfa	D90 10.8 vs WEAVE 11.4 (20 Hz) and 7.7 vs 9.7 (1000 Hz), T = 1e5 s, pfa 1 % per 50 mHz (Tables IV, VI; stated)	-	comparable Single detector, Gaussian noise; depth at a per-band false-alarm probability against a matched-filter (WEAVE) depth on the same setup: the CW benchmark convention.	*pending*
`bayley2020` [164]	depth@pfa	depth ~10 Hz^-1/2 at 95 % efficiency, 1 % FAP per sub-band (S6 MDC) (Abstract, Fig 10b; stated)	-	conditional D95 (not D90) on S6 mock data with a SOAP + CNN pipeline.	*pending*
`dreissigacker2020` [156]	depth@pfa	all-sky DNN ~7 % (20 Hz) to ~51 % (1000 Hz) less sensitive in h0 than WEAVE (DNN D90 14.9 to 9.0 vs WEAVE 16.0 to 13.6, T = 1e5 s) (Tables IV-V, abstract; stated)	-	comparable Two detectors, Gaussian and realistic noise; D90 at pfa 1 % per 50 mHz against WEAVE on the same setup.	*pending*

Model	Axis	Published value (location; how obtained)	FAR	Grade: what differs	gravyflow
`yamamoto2021cw` [169]	depth@pfa	D95 of at least ~80 Hz^-1/2 at overall FAP 1 % (headline), 20-120 across the parameter scan (Sec VI, Fig 6; stated)	-	conditional Single-detector Gaussian data; the 95 % check skips the coherent follow-up.	*pending*
`yamamoto2022cw` [170]	depth@pfa	D95 43.9 Hz^-1/2 at FAP ~0.66 % (Gaussian); 3.62 with a unit-amplitude line at FAP ~10 % (Eqs 4.2, 4.4; stated)	-	conditional Single 100 Hz bin of simulated data; the line and no-line depths are at different FAPs.	*pending*
`joshi2023` [157]	depth@pfa	p_det 87.4-89.0 % at the matched-filter D90 (86.2 / 81.8 Hz^-1/2), pfa 1 %, targeted 10-day search (Table IV; stated)	-	comparable White Gaussian H1+L1, single template; p_det at the matched-filter depth is the direct comparison.	*pending*
`joshi2024large` [159]	depth@pfa	all-sky p_det at WEAVE depth 67.0 / 52.8 / 47.6 / 41.5 / 38.8 % (20 / 100 / 200 / 500 / 1000 Hz); D90 29.3 to 18.3 Hz^-1/2 (Tables V-VI; stated)	-	comparable Simulated white Gaussian H1+L1, 10 days; WEAVE depths rescaled by sqrt(T).	*pending*
`cheung2025unet` [160]	depth@pfa	D90 29.97 (20 Hz) to 17.10 (1000 Hz) Hz^-1/2 at 1 % per 50 mHz; WEAVE 40.28 to 34.50 on a different setup (Table II; stated)	-	conditional Simulated Gaussian H1+L1 data (T = 921,600 s); the WEAVE reference is joshi2024large's depth rescaled to this span, not a search on this data.	*pending*
`joshi2025vit` [158]	depth@pfa	p_det at the F-statistic depth: targeted 87.3-89.7 %, directed 84.8-88.7 %, all-sky 88.3 % (20 Hz) to 78.2 % (1000 Hz) (Table IV; stated)	-	comparable Same benchmark family as joshi2023/2024 (white Gaussian H1+L1), against F-statistic depths; targeted over 10 days, directed and all-sky over 1 day (joshi2024large: 10 days).	*pending*
`tenorio2025g2net_cw` [163]	AUC	G2Net CW Kaggle private AUC: 1st 0.860, 2nd 0.854, SOAP 0.604 (Fig 5; stated)	-	conditional Competition data with SNR ~ Gamma(5, 9) in mixed Gaussian and real O3 noise; AUC comparable across entries of this competition only.	*pending*

7.3 Latency and compute cost

14 of the 18 models in this section state a latency, inference cost or training cost; 2 papers were read and state none, and 2 were not read for runtime. Headline: the model's latency if it states one, else its inference cost; normalised values in brackets (Section 13).

Model	Headline: latency if stated, else inference cost	Hardware	Grade; scope; mode	Location	Training cost (hardware; grade; location)
`dreissigacker2019` [155]	Inference cost: 58.8 s (full all-sky DNN search, T = 1e5 s) (0.000588 compute-s per s)	unstated	stated; network; batch	Tab. VII	~1 days per network (T = 1e5 s case); mostly Nvidia GTX 750 GPUs (ATLAS cluster) (stated; Sec. IV E (Timing); Sec. IV A text on Fig. 3)
`miller2019` [171]	Inference cost: about 2 minutes (CNNs analysing full data set and generating triggers) min per 1 week of data, 100-1900 Hz (0.000198 compute-s per s)	unstated	stated; unstated; batch	Sec. VI (Search for a remnant of GW170817)	~30 minutes min; unstated (stated; Sec. V (Search design, shifting-training discussion))
`morawski2020cw` [167]	training cost only	-	-	-	1 hour (1D CNN), 150 epochs hours; 1x NVIDIA Tesla K40 (same machine for both CNNs) (stated; Sec. 3.2)
`bayley2020` [164]	Latency: ~13 hours to efficiency curve and candidates after end of run, parallelised over 200 CPUs with Viterbi-map network trained and tested (4.68e+04 s)	200 CPUs (+1 GPU)	stated; end to end; batch	Sec. VI	~240 hours, generating training data (single CPU); 1 CPU (stated; Tab. II)
`beheshtipour2020b` [166]	training cost only	-	-	-	about 40 hours of training; a high-end 32 GB GPU (model unstated; Atlas cluster GPUs) (stated; Sec. III C and Appendix)
`dreissigacker2020` [156]	Inference cost: roughly two times faster than the matched-filtering search (training, executing and following up, T = 1e5 s) ratio of total computing cost	unstated	stated; end to end; batch	Sec. VI (Discussion)	10 days per network; mixed NVIDIA GPUs (RTX2060, 2070, 2080(Ti), GV100, GTX1660(Ti)) on the ATLAS cluster (stated; Sec. III B (DNN training and validation))
`yamamoto2021cw` [169]	Inference cost: 6.4e6 x (FAP_EP/1e-2) s total NN time (Eq. 5.4) (0.3815 compute-s per s)	1x GeForce GTX 1080Ti	stated; network; batch	Sec. V A, Eq. 5.4	-
`yamamoto2022cw` [170]	Inference cost: 1.02e8 s total CNN time (Eq. 5.6; abstract: O(1e8) s) (6.08 compute-s per s)	1x GeForce GTX 1080Ti	stated; network; batch	Sec. V, Eq. 5.6	-
`joshi2023` [157]	training cost only	-	-	-	4-32 hours, to reach matched-filtering sensitivity (range across targeted-search cases, ten-day, two detectors); NVIDIA A100-SXM4 40 GB (count unstated) (stated; Abstract; Sec. IV D and Fig. 5)
`yamamoto2023` [172]	Inference cost: 7.66 x 10^-2 s for 500 test data (residual network; shallower CNN 2.54 x 10^-2 s, deeper CNN 7.96 x 10^-2 s; the non-Gaussian-statistic grid search baseline takes 1.13 x 10^4 s on CPU)	GPU Quadro GV100 (16.6 TFLOPS single precision)	stated; network; batch	Tab. IV; Sec. IV C	-
`joshi2024large` [159]	Inference cost: ~27 s to evaluate ~50 000 input windows across 20-1000 Hz (batch size 256) for a ten-day, two-detector (H1, L1) search (3.13e-05 compute-s per s)	unstated for inference (training on NVIDIA A100-SXM4 40 GB)	stated; network; batch	Sec. IV C	~25 days per DNN; NVIDIA A100-SXM4 40 GB (count unstated) (stated; Sec. III D (hardware in Sec. III B))
`cheung2025unet` [160]	training cost only	-	-	-	~5 days; 1x NVIDIA Tesla V100 16 GB (stated; Sec. III C; Introduction)
`joshi2025vit` [158]	training cost only	-	-	-	less than 1 day (targeted-search ViTs, ten-day timespan); NVIDIA A100-SXM4 40 GB (count unstated) (stated; Sec. III C; Conclusions)
`tenorio2025g2net_cw` [163]	Inference cost: 5 days (JunKoda), cK ~ 50 s per band days on a GPU for the full dataset	1x GPU (model unstated)	stated; unstated; unstated	Sec. 5 (Comparison to CW searches), after Eq. 8	-

7.4 What cannot be compared, and why

CW results are comparable when they share the convention of sensitivity depth at a false-alarm probability per frequency band and the same data span (or are rescaled to a common span, as [159] and [163] do with the square root of the span and [160] with its fourth root). Candidate-clustering efficiencies and toy-model detectability thresholds are not on that axis.

The 8 models below have no shared axis; their headline metric is listed with the reason.

Model	Headline metric (location; how)	Why not comparable	gravyflow
`miller2019` [171]	50 % efficiency at h0 ~3.6e-23 (150-300 Hz) at per-map FAP ~1 % (CNN) vs ~0.01 % (FrequencyHough) (Figs 3, 6; read-from-figure)	Long-transient maps in real O2 L1 noise; the CNN and FrequencyHough are compared at FAPs 100x apart.	*pending*
`morawski2020cw` [167]	CW-vs-rest AUC 0.99 (1D and 2D CNN); TPR 96.3 % (1D) / 96.8 % (2D) at 1 % FPR (Fig 7, Table 2; stated)	Classifier of TD-Fstat outputs in white Gaussian noise with injected SNR 8-20.	*pending*
`beheshtipour2020` [165]	cluster-identification efficiency 97 % (1137 of 1171) on Einstein@Home result images (Sec III; stated)	Post-processing of search candidates; efficiency for loud, visible signals.	*pending*
`beheshtipour2020b` [166]	cascade efficiency 92 % above normalised amplitude 1/60 vs deterministic clustering 83 % (Sec IV-V; stated)	Einstein@Home O2 candidate clustering; comparable only with the deterministic clustering it replaces.	*pending*
`dominguez2023` [161]	none extracted	Qualitative only; the abstract reports degradation on real O3 data.	*pending*
`duraisamy2023` [162]	none extracted	Not read; no abstract.	*pending*
`yamamoto2023` [172]	minimum detectable SNR rho_90 ~0.7 to ~3.0 (residual network; non-Gaussian statistic 0.7 to 2.8) vs duty cycle at FAP 5 % (Fig 4; read-from-figure)	Toy intermittent-background model in white noise; no detector data conventions.	*pending*
`dicesare2026fh` [168]	p_det 0.969 (L1) / 0.998 (H1) at p_fa 0.012 / 0.002 for injections at h0 7.76e-26, below the O3 follow-up sensitivity (Table 5; stated)	Follow-up classifier of FrequencyHough candidates; p_det and p_fa are counted per Hough map in real O3 data.	*pending*

7.5 Suspected errors in the literature

18 suspected errors or internal inconsistencies in 13 papers, selected from the extraction notes (40 notes for this section in the catalogue). Each is a reading of the paper to be checked, not a verdict.

- [155] "Nearly rivaling matched filtering" is a fixed-depth comparison where matched filtering is 90 % by construction.
- [171] 500 maps of 2000 s are said to span ~11.5 days and 2000 maps ~23 days (should be ~46).
- [171] 64,000 noise maps are built from 2,000 unique ones.
- [167] Stated dataset total (292,500 at 2,500 per class per configuration, i.e. 39 configurations) does not divide evenly over the 6 training bands.
- [164] 95 % sensitivities given as SNR 85 / 115 (Sec VI A) and 90 / 120 (Sec VII).
- [165] 1345 returned - 1137 correct = 208 wrong clusters, but the text reports 174 false alarms.
- [166] Conclusion: deterministic clustering drops below 50 % at 1/80 sqrt(Hz); Fig 9's lowest bin reads ~0.57.
- [169] Headline D95 $\gtrsim$ 80 is a bound over (FAP_EP, r_NN) settings within a computing budget; no single operating point is named (Fig 6 spans ~20-120).
- [170] Abstract FAP ~0.5 % vs 0.66 % from the 20,000-event test.
- [170] The 8.2 % depth ratio compares FAP 10 % with FAP 0.66 %.
- [170] Efficiency "one or two orders of magnitude" vs SOAP's three orders in Table VII.
- [157] Matched-filter parity claimed at 88.3 % mean against 90 % by construction (~1.7 points lower).
- [159] Table VI compares 10-day depths with unrescaled 1e6 s depths of the earlier work.
- [159] p_fa calibration changes between validation (0.76 % per 38 mHz) and test (0.48 %).
- [159] Abstract claims high sensitivity at all frequencies for the single DNN, which reaches 30-40 % p_det at the matched-filter depth.
- [160] "Comparable sensitivity at 20 and 200 Hz" is relative to the joshi2024large DNN, though the preceding sentence introduces the matched-filter depths, at which p_det is 69 % and 46 % (Table II).
- [158] Targeted reference time and D90 (86.1 / 81.6) differ slightly from joshi2023 (86.2 / 81.8) although described as the same benchmark.
- [163] Legend AUC order (KDL 0.789 7th, MC Digital 0.791 8th) disagrees with the ranking.

8 Glitch classification and data quality

Deep-learning glitch classification grew up around Gravity Spy, whose citizen-science labels and image format became the de facto benchmark. Later work changed the backbone, the input view, the class set and the data release, which is why accuracies from different papers rarely share a test set.

8.1 Families and lineage

8.1.1 Gravity Spy supervised classifiers

Gravity Spy introduced multi-duration Omega-scan images and a CNN trained on citizen-science labels [20], [21], [173] transfer learning [174], [175], [176], new releases and classes [22], [23], [177], and new architectures [178], [179], [180], a wavelet-scattering input with a random forest [181] and GAN data augmentation [182] followed.

> **Our results (to be added)**
> gravyflow reproduction: *pending* for 22 models: [21], [174], [20], [173], [175], [176], [22], [183], [182], [184], [178], [23], [185], [181], [186], [180], [179], [177], [187], [188], [189], [190]. Results will be placed on the shared axes of this section beside the published values.

8.1.2 Unsupervised and similarity methods

Embeddings and clustering for unseen classes [191], [192], [193], [194], [195], [196], [197], [198], [199], [200].

Type	Num Filters / Neurons / Heads	Filter / Pool / Head Size	Stride	Activation Function
Conv Stage 1 (2D)	128	5x5, pool 2x2	1, 2	ReLU
Conv Stage 2 (2D)	128	5x5, pool 2x2	1, 2	ReLU
Flatten	-	-	-	-
Dense Head	256, 20	-	-	ReLU, SoftMax

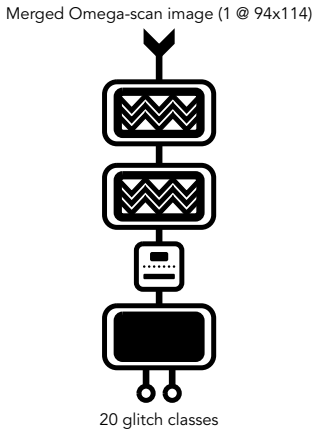


Figure 16: Gravity Spy 2017 image CNN [20], collapsed blocks. Architecture from: Paper Fig. 6 (feature-map sizes, 5x5 kernels, 2x2 pools, 256 hidden units, 20 outputs) and Sec. 3.3.1 text (ReLU, softmax output). Diagram confidence: **high**; 5 details the paper leaves unstated are listed in the diagram spec, and are not drawn. Notation: Appendix A.

×2 Conv Stages

Type	Num Filters / Neurons / Heads	Filter / Pool / Head Size	Stride	Activation Function
Conv Stage	128	5x5, 2x2	-	ReLU
Dense Head	256, 20	-	-	SoftMax

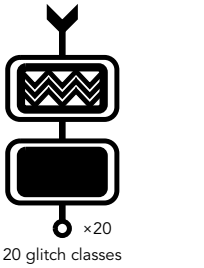


Figure 17: Bahaadini et al. 2017 merged-view CNN [21], collapsed blocks. Architecture from: Table 1 (right column, merged view model) and Sec. 3.2; Fig. 3 of the paper. The best model (96.89%). Diagram confidence: **high**; 3 details the paper leaves unstated are listed in the diagram spec, and are not drawn. Notation: Appendix A.

Our results (to be added)
gravyflow reproduction: *pending* for 11 models: [191], [192], [200], [201], [193], [194], [199], [195], [198], [196], [197]. Results will be placed on the shared axes of this section beside the published values.

8.1.3 Signal-versus-glitch vetoes

Classifiers that separate astrophysical signals from glitches for a search [202], [203], [204], [205], [206], [207], [208], [209], [210].

Our results (to be added)
gravyflow reproduction: *pending* for 10 models: [204], [202], [209], [203], [208], [207], [206], [210], [211], [205]. Results will be placed on the shared axes of this section beside the published values.

8.1.4 Auxiliary-channel data quality

Models that read witness channels rather than strain [212], [213], [214], [215].

Our results (to be added)
gravyflow reproduction: *pending* for 4 models: [215], [213], [212], [214]. Results will be placed on the shared axes of this section beside the published values.

8.1.5 Early and simulated-glitch classifiers

Classifiers of simulated morphologies in Gaussian noise [216], [217], [218], the first test of the 2015 PCAT, PC-LIB and WDF-ML methods on Advanced LIGO engineering-run data [219], a CNN glitch detector on cross-correlated O3a time-frequency maps [220], tabular metadata models [221], [222] and glitch generators [223], [224].

Our results (to be added)
gravyflow reproduction: *pending* for 9 models: [216], [219], [218], [217], [220], [223], [221], [224], [222]. Results will be placed on the shared axes of this section beside the published values.

8.2 Comparison on the shared axes

Rows are the models graded comparable or conditionally comparable (14 of 56); each axis entry is one row. FAR is converted to per month (30 d) and to Hz where the paper states a rate.

Model	Axis	Published value (location; how obtained)	FAR	Grade: what differs	gravyflow
bahaadini2017 [21]	GS	merged-view CNN accuracy 96.89 % (20 classes, Gravity Spy, 7,730 glitches, 75/12.5/12.5 split) (Tables 2-3; stated)	-	conditional Gravity Spy 20-class set (observing run not stated) with its own split; accuracy only (no macro-F1).	*pending*
george2017tl [174]	GS	top-1 98.84 % (ResNet50, Inception-V3), 22 classes, Gravity Spy O1, random 80/20 split (Table 1; stated)	-	conditional arXiv version of PRD 97 101501(R); george2018glitch is the shorter NIPS 2017 workshop version with the same headline numbers.	*pending*
zevin2017 [20]	GS	average test accuracy 97.1 % (20 classes, Gravity Spy O1, 7,718 glitches); per-class recall mean ~0.90 (Sec 4.1 stated; recall mean computed from Fig 7)	-	conditional The Gravity Spy reference; accuracy is sample-weighted.	*pending*
george2018glitch [175]	GS	top-1 accuracy 98.84 % (Inception-V3, ResNet50), 22 classes, Gravity Spy O1 ~8,500 images, random 80/20 split (Table 1; stated)	-	conditional Gravity Spy O1 22-class set with a random per-class 80/20 train/test split (the best models were trained on different splits); a validation set is named only in the Table 1 early-stopping note and is not described apart from the test set.	*pending*
soni2021 [22]	GS	validation accuracy 98.8 % (expert) / 97.5 % (volunteer), 23 classes, Gravity Spy O3 (9,631 expert / 9,496 volunteer examples) (Sec 2.2-2.3; stated)	-	conditional Validation, not held-out test, accuracies on the O3 23-class set.	*pending*
yan2022gan [182]	GS	weighted F1 0.9885 (Inception-V3, GAN N=5000, no pre-training) vs 0.9880 (pre-trained baseline), Gravity Spy 1.0, 21 classes (Table 4; stated)	-	conditional Gravity Spy v1.0 own split, 5-run means.	*pending*
bini2023 [209]	reach@FAR	cWB XGBoost+AE vs XGBoost: sensitivity-volume gain at IFAR 50 yr +8 to +44 % (SG), +33 % (cosmic strings) (Sec 5.2, Fig 4; stated)	0.00164 /month; 6.34e-10 Hz	conditional Glitch-rejection feature inside cWB on 40 d of O3b with ~380 yr of time slides; relative to the cWB XGBoost baseline.	*pending*
fernandes2023 [178]	GS	macro-F1 97.18 %, accuracy 98.68 % (ResNet18 baseline), 22 classes, Gravity Spy O1+O2 held-out test (Table I; stated)	-	conditional One of the few Gravity Spy papers to report macro-F1; uses the public Gravity Spy v1.0 stratified 70/15/15 split.	*pending*
glanzer2023 [23]	GS	O3 model train 99.9 % / validation 98.8 % (23 classes, 9,631 samples); O1-O2 model train 98.2 % (Sec 2.3; stated)	-	conditional Data-release paper quoting the published Gravity Spy accuracies; no test accuracy.	*pending*
licciardi2024 [181]	GS	random-forest accuracy 91.03 % (WST + Q-transform) vs 87.41 % (Q-transform), Gravity Spy 1.1.0, 21 classes (Table II; stated)	-	conditional Gravity Spy 1.1.0 after the authors' own cleaning; accuracy only.	*pending*
srivastava2024 [180]	GS	accuracy 92.26 %, F1 92.13 % (24 classes: Gravity Spy O1/O2 + 2 O3a classes) (Sec 3; stated)	-	conditional Custom 24-class set mixing Gravity Spy images with authors' Q-transforms; F1 averaging unstated.	*pending*
wu2024 [179]	GS	accuracy 0.941 / F1 0.931 vs retrained Gravity Spy 0.906 / 0.904 on the same split (23 classes, O3, confidence > 0.9) (Tables 2, 4; stated)	-	conditional Re-trains the previous Gravity Spy model on the same split, so the internal comparison is fair; split differs from other papers.	*pending*
zevin2024 [177]	GS	re-quoted: 97.1 % (2017, 20 classes), 98.2 % (2018, 22 classes); O3 model train 99.9 / val 99.8 % (Sec 3.2, Table 1; stated)	-	conditional Review quoting earlier results.	*pending*
vandyke2025 [205]	AUC	CoBiTS 0.953 vs ResNet18 0.900 vs newSNR 0.945 (all scenarios, H1 O3, 4,000 test segments) (Fig 4; stated)	-	conditional Single-detector signal-vs-glitch discrimination with a matched-filter statistic on the same segments.	*pending*
	eff@FAP	TPR at FPP 1e-3: 0.82 vs 0.70 vs 0.66 (Fig 4; read-from-figure)	-		*pending*

8.3 Latency and compute cost

20 of the 56 models in this section state a latency, inference cost or training cost; 28 papers were read and state none, and 8 were not read for runtime. Headline: the model's latency if it states one, else its inference cost; normalised values in brackets (Section 13).

Model	Headline: latency if stated, else inference cost	Hardware	Grade; scope; mode	Location	Training cost (hardware; grade; location)
george2017tl [174]	Inference cost: a few milliseconds ms per input spectrogram	GPU (model unstated)	stated; network; unstated	Sec. 4 (Discussion)	less than 20 minutes (fewer than 10 epochs to over 98% accuracy) minutes; unstated (stated; Sec. 3 (results after fine-tuning))
cavaglia2018 [215]	training cost only	-	-	-	in minutes minutes (RF and GP codes, training and running together); LIGO computing clusters (stated; Sec. 1 (Introduction) and Conclusions)
george2018glitch [175]	training cost only	-	-	-	less than 20 minutes (fewer than 10 epochs to over 98% accuracy) minutes; unstated (stated; Sec. 3 (Results))
llorensmonteagudo2018 [218]	Inference cost: about 36 hours hours to reconstruct 3000 glitches, each in one 16384-sample patch, by all three learned dictionaries (43.2 compute-s per s)	1x AMD Phenom II x4 processor, 12 GB RAM	stated; network; batch	Sec. 6 (Conclusions)	-
razzano2018 [217]	Inference cost: few microseconds per image	unstated	stated; unstated; unstated	Sec. 6 (Discussion)	about ~5 hours (70 epochs); 1x NVIDIA GeForce GTX 780 (desktop PC) (stated; Sec. 6 (Discussion))
essick2020 [212]	Latency: ~5 sec for feature extractors to process O(10^3) channels (5 s)	unstated	stated; unstated; streaming	Sec. IV	O(hours) runtime of training jobs; unstated (stated; Sec. IV (Timeseries Production / online mode))

Model	Headline: latency if stated, else inference cost	Hardware	Grade; scope; mode	Location	Training cost (hardware; grade; location)
boudart2022glitch [220]	training cost only	-	-	-	roughly 2 hours hours (200 epochs); 1x Tesla P100 GPU (16GB) (stated; Sec. III (CNN training), text near Fig. 6)
choudhary2022 [204]	Inference cost: 30 sec per SGP map generated (classification of 3000 SGP maps: 90 sec)	1 core of Intel Xeon Gold 6142 CPU	stated; unstated; batch	Tab. II	20 min for training on 10000 samples; 1 core of Intel Xeon Gold 6142 CPU (stated; Tab. II)
sakai2022 [194]	training cost only	-	-	-	approximately 1.0 h/100 epochs (VAE, mini-batch 128); 2x NVIDIA GeForce RTX 2080 Ti, Intel Xeon CPU E5-2637 v4 (core 8), 125 GB main memory (stated; Results, 'Training process of our architecture')
yan2022gan [182]	Inference cost: 5 minutes to generate 1000 512x512 spectrogram images (ProGAN generator)	1x NVIDIA Tesla V100 (32GB) (stated for all experiments; not restated for generation)	stated; unstated; unstated	Sec. 4.1 (Data augmentation)	3 days per Gravity Spy class (ProGAN, fixed number of epochs, each class trained separately); 1x NVIDIA Tesla V100 (32GB) (stated; Sec. 4.1 (Data augmentation))
fernandes2023 [178]	training cost only	-	-	-	between 1 and 2 minutes training time per model; 1x NVIDIA RTX A4000 (16 GB VRAM) (stated; Sec. IV B (supervised models with transfer learning))
laguarta2023 [199]	Inference cost: 1 h of data in 11 s s to FD-encode 1 h of auxiliary-channel data (numba, O(N^2 log N))	unstated (Fig. 2 benchmarks on Intel Xeon E5-2630 v4 CPU)	stated; end to end; batch	Sec. II A, text after Fig. 2	-
arutkeerthi2024 [207]	Inference cost: 30 ms to classify one signal (glitch or not) from a resized chirp volume (0.005 compute-s per s)	unstated	stated; network; unstated	Sec. VIII (Conclusion)	-
li2024ctsae [195]	training cost only	-	-	-	approximately 4 days to complete 200 epochs days; 8x NVIDIA Tesla V100 (stated; Sec. 4 (implementation details))
licciardi2024 [181]	Inference cost: 0.211 ± 0.003 s per sample (Q-transform, GwPy, CPU; WST 0.197 ± 0.007 s on CPU and 0.063 ± 0.002 s on GPU; preprocessing only, classifier not timed)	CPU (model unstated); GPU (model unstated) for the WST PyTorch timing	stated; end to end; unstated	Sec. IV ('Computational Cost' subsection)	-
vandyke2025 [205]	training cost only	-	-	-	approximately 12 hours per model from initialization (batch size 32, 30 epochs, 32-bit precision); 1x NVIDIA H100 (Kamiak cluster, WSU) (stated; Appendix A.4 (Training))
deshmukh2026 [196]	Inference cost: between 10-15 mins (evaluation including generation of t-SNE plots)	1x DGX A100 GPU (same resource)	stated; end to end; batch	Sec. 3.1	around 24 hrs (LoRA fine-tuning of AST, 4000-8000 glitches per class); 1x DGX A100 GPU (stated; Sec. 3.1)
dooney2026 [224]	Inference cost: under 22 seconds to generate 1000 glitches	CPU (model unstated)	stated; network; batch	Abstract; Sec. IV; Sec. VI (Conclusions)	approximately 12 hours (500 epochs, batch 64; weights at epoch 210 used); 1x NVIDIA A100 GPU (stated; Sec. III B (Training procedure))
fernandes2026 [189]	Inference cost: 0.04 s per glitch (ResNet, the best model; tree models < 0.01 to 0.01 s), averaged over 20 samples	CPU (model unstated)	stated; network; unstated	Tab. V	940 s (ResNet; tree models 0.07 to 4.38 s); 1x NVIDIA GeForce GTX 1050 Ti (used to accelerate ResNet and XGBoost training) (stated; Tab. V; Sec. III)
manoharan2026 [222]	Inference cost: figure only; per-model points not read seconds per sample (total predict time / N, N=10112)	unstated (identical batch size and hardware settings across models)	estimated; end to end; batch	Fig. 5, Sec. C (Performance vs. Inference Time)	figure only; per-model points not read (axis 10^2 to 10^4 s) seconds (wall-clock, single model fit incl. validation and early stopping, excl. hyperparameter search); unstated (estimated; Fig. 4, Sec. B (Performance vs. Training Time))

8.4 What cannot be compared, and why

Gravity Spy accuracies differ in release (O1 20-class, O1 22-class, v1.0/1.1/1.2, O3 23-class), in split (random, stratified, own cleaning) and in averaging (sample-weighted accuracy versus macro-F1), and several are validation rather than test numbers. They are graded conditional at best; the one internally fair comparison against the Gravity Spy model itself is a paper that retrains the previous Gravity Spy model on its own split. Clustering scores (NMI, ARI, silhouette), generator-fidelity scores and auxiliary-channel classifiers are on different axes.

The 42 models below have no shared axis; their headline metric is listed with the reason.

Model	Headline metric (location; how)	Why not comparable	gravyflow
powell2015 [216]	per-morphology cluster fractions on simulated SG/Gaussian/ringdown transients (PCAT/LIB/WDF-ML) (Tables 2-3; stated)	Simulated transients in Gaussian noise; each method analyses a different subset.	*pending*
powell2016 [219]	fraction of detected transients correctly classified: L1 95 / 98 / 95 %, H1 99 / 95 / ~92 % (PCAT / PC-LIB / WDF-ML) (Secs 4-5; stated)	Real ER7 glitches scored against the authors' visual inspection; PCAT and PC-LIB each miss a different subset of the transients.	*pending*
bahaadini2018 [173]	none extracted	Not read (paywalled journal version).	*pending*
bahaadini2018direct [191]	NMI 0.598 / ARI 0.455 (DIRECT) vs autoencoder 0.545 / 0.324 on 5 held-out Gravity Spy 1.0 classes (Table 1; stated)	Clustering of unseen classes; embedding size chosen on the evaluation classes.	*pending*
cavaglia2018 [215]	genetic-programming recall 0.934, precision 0.937 (magnetometer-glitch attribution) (Table 2; stated)	Auxiliary-channel origin attribution for one glitch class.	*pending*
llorensmonteagudo2018 [218]	dictionary-learning accuracy 96.0 % on 3,000 simulated glitches (SNR 1-400) (Sec 5.2; stated)	Three simulated morphologies in Gaussian noise.	*pending*
razzano2018 [217]	accuracy 0.998 on 6 simulated glitch families + noise (Table 4; stated)	Fully synthetic, near-saturated problem.	*pending*

Model	Headline metric (location; how)	Why not comparable	gravyflow
coughlin2019 [192]	data reduction at 50 % retention: ~0.99 (Raven Peck), ~0.99 (Water Jet) with DIRECT (Fig 3 caption; stated)	Similarity search; no classification or detection metric.	*pending*
colgan2020 [213]	auxiliary-channel glitch vs clean: AUC ~0.93 (ER14), ~0.91 (O3) (Fig 6; read-from-figure); accuracy 0.799 (O3) (Sec III; stated)	Balanced glitch / glitch-free samples from auxiliary channels; accuracy depends on the 50:50 prior.	*pending*
essick2020 [212]	iDQ glitch-detection efficiency 10-40 % at clean-time FAP 0.1-1 % (Sec V; stated)	Auxiliary-channel data-quality statistic scored over clean-time samples.	*pending*
ding2021 [200]	none extracted	Anomaly detection with only a loss-based candidate-vs-noise comparison on 77 O3 public alerts; no efficiency or FAR.	*pending*
mesuga2021 [176]	AUC 0.990 (VGG19), averaging unstated, Gravity Spy 22-class Kaggle copy (Fig 7; stated)	AUC only; the accuracy in the Fig 9 title is not the matrix accuracy.	*pending*
sankarapandian2021 [201]	linear-probe accuracy 81.6 % on Gravity Spy latents (Table 1; stated)	Unsupervised encoder with linear probe; split unstated.	*pending*
bahaadini2022 [193]	NMI 0.61 (VAT + discriminative) vs DIRECT 0.53 on 10 held-out Gravity Spy v1.2 classes (Table III; stated)	Clustering of held-out classes.	*pending*
boudart2022glitch [220]	glitch-containing maps identified >95.5 %; chirp maps 90.2 %, background maps 99.7 % (Sec III B; stated)	Cross-correlated H1-L1 maps of minute-long transients: real O3a background and glitches, simulated chirps.	*pending*
choudhary2022 [204]	75 % TPR increment (relative or absolute unstated) over chi-square vetoes at FPR 1e-2 (80-140 Msun, SNR 3-8) (Abstract, Fig 5; stated)	BBH-vs-blip veto ROC on simulated BBH and Gravity Spy blips.	*pending*
colgan2022 [214]	scattered-light vs clean accuracy 97.1 % (test, 25 of 39,147 aux channels) (Sec IV A; stated)	Balanced auxiliary-channel classification on one day of L1.	*pending*
dooney2022 [223]	discriminative score WGAN vs DVGAN (e.g. pulse 0.029 vs 0.009); BBH match 0.9918 vs 0.9917 (Table I; stated)	Generator fidelity metrics.	*pending*
sakai2022 [194]	unsupervised cluster-purity 90.9 % (36 clusters vs 22 labels, Gravity Spy O1) (Discussion; stated)	Custom purity score on training data with C chosen by the same labels.	*pending*
talpini2022 [183]	none extracted	Abstract-level; no numbers.	*pending*
alvarez2023 [202]	GSpyNetTree signal-vs-glitch accuracy ~97 % (LM / HM / EHM, Inception-V3) (Fig 5; stated)	Glitch classes plus injected CBC; balanced classes, no FAR.	*pending*
apostol2023 [221]	DeepWaves ensemble 10-fold accuracy 0.91 on Gravity Spy O1 metadata (22 classes) (Table 4; stated)	Tabular metadata, not images; features include GPS time and ID.	*pending*
biskin2023 [184]	none extracted	Not read.	*pending*
jarov2023 [203]	GSpySVG GW accuracy 98.2 % (LM) / 96.2 % (HM); glitches 197/200 (Table IV; computed from counts)	Signal-vs-glitch on ~500 injections and 200 glitches per classifier.	*pending*
laguarta2023 [199]	177 of 2,688 glitches flagged as anomalies by manual t-SNE inspection (Sec V; stated)	Unsupervised autoencoder evaluated on its training data; anomalies picked by eye from t-SNE, with no ground truth.	*pending*
shah2023 [208]	random-forest accuracy 0.93 (H1) / 0.94 (L1) (Sec 5.2; stated); macro recall ~0.81 / ~0.85 (Fig 4; computed from the median diagonals)	Six-class custom set with synthetic GWs in Gaussian noise beside real glitches.	*pending*
arutkeerthi2024 [207]	glitch vs merger accuracy ~96 % on 308 validation samples (Sec VII; stated)	Small set with augmentation that can leak events across splits.	*pending*
koyama2024 [185]	none extracted	Not read.	*pending*
li2024ctsae [195]	ARI 0.409 / NMI 0.636 (CTSAE) vs DIRECT 0.214 / 0.428 on Gravity Spy O3 test (23 classes) (Table 2; stated)	Unsupervised clustering.	*pending*
magee2024interp [206]	signal-vs-glitch TPR 0.941-0.999 at FPR 0-0.004 (simulated) (Figs 3-4; read-from-figure)	Signal-consistency classifier on simulated glitches.	*pending*
malz2024 [186]	conformal prediction sets: macro-F1 0.456, mean set size 4.26 of 22 at alpha 0.1 (Table II; stated)	Set-valued uncertainty on top of Gravity Spy; not a single-label score.	*pending*
bhandari2025 [187]	none extracted	Not read.	*pending*
chan2025 [210]	localisation recall at IoU 0.3: GW 93.7 %, Blip 94.3 %, Scattering 52.4 % (Table 3; stated)	Object localisation on 1 s Q-transforms.	*pending*
gupta2025ae [211]	none extracted	Abstract-level; custom 1,200-sample set.	*pending*
oshino2025 [198]	8 spectral clusters from 45,345 KAGRA O3GK glitches; Blips 79.2 % (Table 2; stated)	Unsupervised clustering into a custom KAGRA taxonomy; judged only by Davies-Bouldin index, silhouette and visual inspection.	*pending*
singh2025 [188]	none extracted	Not read.	*pending*
deshmukh2026 [196]	none extracted	Silhouette and t-SNE only; no numbers.	*pending*
dooney2026 [224]	Gravity Spy accuracy on GlitchGAN samples 79.3 % vs 89.6 % on real reconstructions (n = 700) (Tables IV-V; stated)	Generator fidelity scored by the Gravity Spy classifier; depends on injected SNR.	*pending*
fernandes2026 [189]	Virgo O3b 7-class ResNet34 accuracy 0.983, F1 0.977 (1,679 test) (Sec IV; stated)	Virgo merged-class taxonomy; F1 is macro-averaged.	*pending*
kim2026kagra [197]	KAGRA O3GK clustering: DEC NMI 0.367, ConvAE+PCA ARI 0.436 (Table 4; stated)	Custom KAGRA taxonomy.	*pending*
manoharan2026 [222]	tabular Gravity Spy O3 metadata: XGBoost weighted F1 ~0.853 (Figs 4-5; read-from-figure)	Tabular metadata including timestamps.	*pending*
xie2026 [190]	none extracted	Abstract-level; no numbers.	*pending*

8.5 Suspected errors in the literature

60 suspected errors or internal inconsistencies in 38 papers, selected from the extraction notes (132 notes for this section in the catalogue). Each is a reading of the paper to be checked, not a verdict.

- [216] WDF-ML ">97 %" for both sets vs 91.3 % for SG in set 2 (Table 3).
- [219] WDF-ML H1 class counts sum to 1829, not the 1865 transients classified.
- [219] WDF-ML L1 overall accuracy 95 % follows from the listed misclassifications only if the 8 whistles in the spike classes count as errors (otherwise ~97 %).
- [21] Text says Air Compressor is perfect for both classifiers; Table 4 gives 80 % for classifier 4.
- [21] Architectures optimised for "best classification accuracy" on an unstated split, with no error bars (~966 test samples).
- [174] No validation set while the best of 100 epochs was chosen, i.e. selection on test.
- [174] Paired Doves training size 21 (text) vs 24 (Fig 3 caption).
- [20] "Average accuracy 97.1 %" is the sample-weighted accuracy; the mean of the printed per-class recalls is ~0.90.
- [20] Fig 7 caption says training-set normalisation while titled test set.
- [191] k = 200 is selected from NMI/ARI on the same unknown classes used in Table 1.
- [215] Table 2 caption swaps FN and FP definitions.
- [175] "Above 98.8 %" holds for two of five models (Table 1).
- [175] Dataset ~8,500 vs 7,932 for the same O1 release in Glanzer 2023.

- [217] Per-class accuracy, precision, recall and F1 are identical in every row of Table 4, which is not possible in general for a multi-class matrix.
- [192] Raven Peck reduction given as 99.9 % (text), 99.0 % (caption) and 0.87-0.97 (Table I).
- [192] Raw pixels match or beat DIRECT for Raven Peck (Fig 3).
- [176] Fig 9 title prints 98.98 % accuracy; the diagonal gives ~89 % (the digits match the AUC).
- [176] Table I counts 7,966 images vs 31,868 in the text (four views each).
- [22] Fast Scattering count 55,279 (Sec 3.2) vs 55,211 (conclusions).
- [193] Table III does not state the label fraction behind its NMIs (DIRECT 0.53, VAT 0.61); they match Fig 6 only near 30 % labels.
- [193] Pair accuracy 49.07 % at both 0.1 % and 1 % labels (chance level).
- [220] Conclusion claims a low false-alarm rate on random chirping signals, yet 9.83 % of chirp maps are labelled glitches (Fig 7).
- [204] The 75 % is a relative gain at one bin and FPR without absolute TPRs.
- [223] WGAN bolded for BBH match on a 1e-4 difference with zero stated bounds.
- [194] Eq 1 as printed sums to a value up to 36, not a fraction.
- [182] "Comparable or exceeding transfer learning" rests on a 0.0005 difference inside 0.003 error bars.
- [182] Minority-class F1 rests on 4-9 test images.
- [202] Identical 97 % for three classifiers with different class sets.
- [202] LM GW accuracy 96 % (text) vs 97 % (caption).
- [221] GPS time and Gravity Spy id are among the features (possible label leakage).
- [209] Abstract says "up to 30 %"; Fig 4 gives up to +44 %.
- [178] Text says < 0.2 pp below fine-tuned ResNet50; Table I gives 0.47 pp.
- [178] Conclusion quotes validation F1 gains while the test set shows the opposite ordering.
- [23] Abstract glitch totals (data up to the end of O3) do not match the O3-only Table 1 sums; likely a difference of scope rather than an error.
- [23] 8,427 + 1,203 = 9,630, not 9,631.
- [203] Table II Paired Doves 279 is a typo for 27 (the column then sums to 9,631).
- [203] Abstract "100 % of confirmed events as true signals" vs 19/21 labelled GW.
- [203] Glitch accuracy quoted as 99 %; counts give 98.5 %.
- [199] Scattered Light 28 % + 72 % already makes 100 %, leaving no room for the overlap case.
- [208] "Mean accuracy" is the sample-weighted accuracy; macro recall is 0.81 / 0.85.
- [208] Modes-only results called "quite similar" while Koi Fish recall falls from 0.93 to 0.64.
- [207] Each event contributes two detectors and a shifted copy, which can straddle the split.
- [195] Table 1 split sums to 41,655 vs 41,745 stated.
- [195] Claims superiority over deployed semi-supervised methods, but VAT at 5 % labels is better (Table 2).
- [181] Text says second-order WST "slightly increases" first-order performance; that holds only for S1+S2 (88.57 % vs 88.11 %), since second order alone gives 68.65 % (Table I).
- [181] "More than one order of magnitude" faster holds only for batched GPU WST (100 samples in 0.063 s); per-sample timings give 1.07-3.35x.
- [206] Fig 3 caption FPR 0.7 % vs the printed 0.3 %.
- [206] Toy-pipeline TPR/FPR at rho 7 differ between Sec III C and the Fig 7 caption.
- [180] The two added classes come from a different image pipeline, a possible shortcut feature.
- [179] Stated split (8,439 / 1,687 / 3,538) disagrees with Table 1 sums (6,537 / 1,637 / 4,448).
- [177] O3 model validation 99.8 % here vs 98.8 % in Soni 2021 / Glanzer 2023.
- [177] Train/validation accuracies set beside earlier test accuracies.
- [210] FP percentages for GW (0.1 %) and Blip (0.2 %) do not follow from the counts (1.2 %, 1.9 %).
- [205] Text says CoBiTS does "perceptively better" than matched filtering with glitches present, but newSNR has the higher scattered-light AUC (0.973 vs 0.965; Fig 4).
- [205] Test BBH waveforms were left in the training pool.
- [224] Epoch chosen partly by the UMAP overlap later reported as fidelity evidence.
- [197] Table 3 silhouette scores rank methods although the space they are computed in is not stated (Fig 7 uses each method's own latent space), so they may not be comparable.
- [222] Discussion calls DANet the best model while XGBoost leads Figs 3-4.
- [222] Fig 3 box-plot medians (0.11-0.74) are inconsistent with Figs 4-6 (0.68-0.85).
- [222] Per-sample inference times of ~0.1 s to ~5 min on 9 features are not credible.

9 Denoising and signal extraction

Denoising work splits into waveform reconstruction (recover the injected signal from strain; scored by overlap or match) and noise regression (subtract noise witnessed by auxiliary channels; scored by range gain).

9.1 Families and lineage

9.1.1 Waveform reconstruction

Recurrent denoising autoencoders [24], [225], [226] and an early WaveNet (dilated-convolution) denoiser [227] gave way to convolutional, U-Net and transformer reconstructors on real noise [228], [229], [230], [231], [232], [233], [234].

> **Our results (to be added)**
> gravyflow reproduction: *pending* for 18 models: [24], [225], [227], [226], [228], [235], [236], [231], [232], [234], [237], [238], [239], [229], [230], [233], [240], [241]. Results will be placed on the shared axes of this section beside the published values.

9.1.2 Noise regression with witness channels

DeepClean subtracts noise witnessed by auxiliary channels [25], [242], [243], [244], with non-linear regression studies alongside [245], [246].

	Type	Num Filters / Neurons / Heads	Filter / Pool / Head Size	Dilation	Activation Function
	Convolutional (first, not stated)	-	-	-	ReLU
×110 Layers (11 dilations x 10)	Gated Dilated Residual Layer	128	3, 1	1-1024	Tanh x Sigmoid
	Addition (sum of skips)	-	-	-	ReLU
	Convolutional (1x1)	64	1	-	ReLU
	Convolutional (1x1)	1	1	-	-

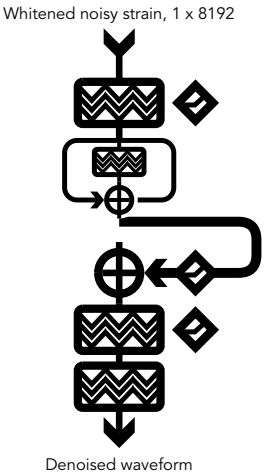


Figure 18: WaveNet denoiser, Model I (Wei & Huerta 2020) [227], collapsed blocks. Architecture from: Fig. 1 (block structure: input conv, ReLU, gated dilated layers with skip connections, summed skips, two 1x1 convolutions) and Sections 2.2.1 and 2.2.2 (dilations, repeats, channel counts). Diagram confidence: **medium**; 6 details the paper leaves unstated are listed in the diagram spec, and are not drawn. Notation: Appendix A.

Type	Num Filters / Neurons / Heads	Filter / Pool / Head Size	Stride	Norm	Activation Function
Convolutional (input)	21	7	1	Batch	Tanh
Down-sampling Convolutions	8-64	7	2	Batch	Tanh
Transposed Convolutions	32-21	7	2	Batch	Tanh
Convolutional (output)	1	7	1	-	-

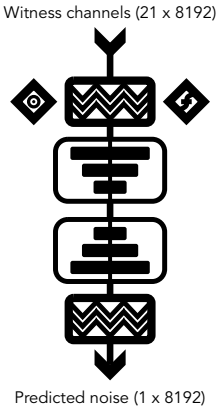


Figure 19: DeepClean (Ormiston et al. 2020) [25], collapsed blocks. Architecture from: Fig. 7 (appendix: layer shapes for the O3 60 Hz dataset, conv, batch norm, tanh per layer) and Sec. III (kernel 7, stride 2, batch norm and tanh on all but the last layer, three down and three up layers). Diagram confidence: **medium**; 5 details the paper leaves unstated are listed in the diagram spec, and are not drawn. Notation: Appendix A.

Our results (to be added)
gravyflow reproduction: *pending* for 7 models: [25], [245], [246], [242], [243], [244], [247]. Results will be placed on the shared axes of this section beside the published values.

9.1.3 Glitch subtraction and gap filling

Removing, bridging or down-weighting transient noise [248], [249], [250], [251].

Our results (to be added)
gravyflow reproduction: *pending* for 4 models: [250], [248], [249], [251]. Results will be placed on the shared axes of this section beside the published values.

9.1.4 Third-generation and space-based extraction

Einstein Telescope and Taiji studies [252], [253], [254].

Our results (to be added)
gravyflow reproduction: *pending* for 3 models: [252], [253], [254]. Results will be placed on the shared axes of this section beside the published values.

9.2 Comparison on the shared axes

Rows are the models graded comparable or conditionally comparable (13 of 32); each axis entry is one row. FAR is converted to per month (30 d) and to Hz where the paper states a rate.

Model	Axis	Published value (location; how obtained)	FAR	Grade: what differs	gravyflow
shen2019 [225]	overlap	EDRDAE 0.994 (quasi-circular) / 0.985 (eccentric) vs DRDAE 0.906 / 0.955 at MF SNR ~6.4, real LOSC noise (Table 2; stated)	-	conditional One SNR, one 4096 s noise stretch; whether Table 2 is a test-set mean is not stated.	*pending*
ormiston2020 [25]	range	O2 H1 BNS range +~20 % (~70 to ~84 Mpc); injection SNR +21.6 % (Fig 4; read-from-figure)	-	conditional DeepClean noise regression on real O2 data; range gain is comparable with later DeepClean papers only.	*pending*
wei2020denoise [227]	overlap	>= 0.97 for SNR >= 12 (Gaussian); GW150914 / GW170104 / GW170814 0.99, GW170608 0.73 (Figs 2-3; stated)	-	conditional Time- and phase-maximised overlap; real-event results are single cases.	*pending*
chatterjee2021 [226]	overlap	>0.95 for >95 % of Gaussian-noise samples (chirp mass 15-50); O1/O2 events 0.967-0.995 (Fig 8; stated)	-	conditional Merger-centred 0.25 s windows, single detector.	*pending*
saleem2023 [242]	range	GstLAL VT ratio DeepClean/raw 1.01-1.026 at FAR 1e-5 to 1e-6 Hz on ~20 d of O3 (Fig 4; read-from-figure)	-	conditional DeepClean 60 Hz subtraction measured through a production search; relative to LIGO's own cleaned frames.	*pending*
	PE	P-P combined KS p 0.0011 (before) vs 0.0008 (after), 78 BBH (Fig 6; stated)	-		*pending*
chatterjee2024harm [231]	overlap	86 % of 1,000 O3b injections (SNR 8-30) with overlap > 0.9; real events 0.74-0.98 vs cWB / LALInference (Secs 3.2-3.3, Table 2; stated)	-	conditional Higher-mode reconstruction in real noise.	*pending*
lin2024 [234]	overlap	BBH mean overlap ~0.75 (2 s) / ~0.88 (0.25 s) at SNR 25; real events with 0.25 s overlap > 0.5: H1 37.8 %, L1 48.8 % (Fig 3, Sec III B; read-from-figure)	-	conditional Noise-weighted overlap on real O2/O3 noise with known signal position.	*pending*
wang2022waveformer [229]	overlap	94.58 % of O3b injections (SNR > 6) with overlap > 0.9; GW150914 > 0.99 (Sec 3.3; stated)	-	conditional Overlap on real O3b noise; the headline depends on the unstated low-mass share.	*pending*
	events	IFAR improved for 75 BBH events (custom time shifts on denoised data) (Sec 3.4; stated)	-		*pending*
dooney2025 [230]	overlap	median mismatch 0.9 % (DeepExtractor, simulated glitches in white noise, test SNR 7.5-100; Table I); 0.3 % vs BayesWave 1.9 % on a 360-glitch subset at SNR 15-100 (Sec IV B, Fig 7; stated)	-	conditional Glitch and signal reconstruction in white noise; the BayesWave comparison uses a different (SNR >= 15) subset.	*pending*
kiendrebeogo2025virgo [243]	range	Virgo O3b BNS range +1.3 Mpc (2.5 %) from 13-band DeepClean (Sec III D; stated)	-	conditional DeepClean on Virgo; comparable with the other DeepClean range gains.	*pending*
lee2025 [233]	overlap	74 % of 14,399 O3a injections with overlap > 0.8 (SNR 4-25) (Sec 3.2; stated)	-	conditional Signal always present, with the merger inside a narrow 0.23 s window; the test SNR is weighted towards low values.	*pending*
reissel2025 [244]	range	Hanford O4 BNS range +1.4 % (+4.3 % volume) (Results, Fig 8; stated)	-	conditional Coherence DeepClean on real O4; comparable with other DeepClean gains.	*pending*
pimpalkar2026 [251]	reach@FAR	IAS-HM VT ratio up to ~1.18 (q = 0.25, m1 ~280 Msun) at IFAR > 1 yr, O3b (Fig 7; read-from-figure; text states up to ~15 %)	0.0821 /month; 3.17e-08 Hz	conditional Wavelet-network re-ranking inside the IAS-HM search; relative VT.	*pending*

9.3 Latency and compute cost

17 of the 32 models in this section state a latency, inference cost or training cost; 10 papers were read and state none, and 5 were not read for runtime. Headline: the model's latency if it states one, else its inference cost; normalised values in brackets (Section 13).

Model	Headline: latency if stated, else inference cost	Hardware	Grade; scope; mode	Location	Training cost (hardware; grade; location)
shen2017 [24]	Inference cost: less than 100 milliseconds per 1 s of input data (0.1 compute-s per s)	unstated (a GPU)	stated; network; unstated	Results, computational performance paragraph (p. 3-4)	-
ormiston2020 [25]	Inference cost: a few seconds seconds for inference on 1024-3600 s of data	1x NVIDIA TITAN X (Pascal) GPU	stated; unstated; batch	Sec. 'Noise subtraction pipeline' (unnumbered), last paragraph before 'Pipeline performance on LIGO data'	about 2-6 minutes minutes for 300-1024 s of training data (8 s segments, 7.75 s overlap), including data pre-processing; 1x NVIDIA TITAN X (Pascal) GPU (stated; Sec. 'Noise subtraction pipeline' (unnumbered), last paragraph before 'Pipeline performance on LIGO data')
torresforne2020 [250]	Inference cost: ~1.5 s to process 1 s of data sampled at 8192 Hz (1.5 compute-s per s)	MacBook Pro, 2.3 GHz Intel Core i5, 4 cores, 16 GB RAM	stated; unstated; unstated	Sec. V (Summary)	-
vajente2020 [245]	training cost only	-	-	-	approximately ten minutes (optimisation of parametrised non-stationary subtraction, 600 s of training data); 1x Nvidia Titan GPU (stated; Sec. IV (applications))
wei2020denoise [227]	training cost only	-	-	-	a few minutes (transfer-learning update when the noise PSD changes); one 'inexpensive' GPU (model unstated) (stated; Sec. 2 (training procedure, PSD update paragraph))
chatterjee2021 [226]	Inference cost: around 0.5 milli-seconds to extract a waveform (0.002 compute-s per s)	1x Tesla K40 GPU	stated; network; unstated	Sec. IV (Discussion and Conclusion)	around 8.5 hours; 1x P100 GPU (stated; Sec. II C 1)
mogushi2021nnetfix [249]	Inference cost: of the order of a few seconds seconds CPU time to reconstruct the data, for gate durations up to hundreds of milliseconds	CPU (model and count unstated)	stated; unstated; batch	Sec. 5 (Conclusion)	-

Model	Headline: latency if stated, else inference cost	Hardware	Grade; scope; mode	Location	Training cost (hardware; grade; location)
bacon2022 [228]	Inference cost: approximately 20 ms to denoise one time series instance (0.02 compute-s per s)	unstated (presumably the V100 GPU)	stated; network; unstated	Sec. 3.5 (Model size and training time)	approximately 480 s for 8000 instances, 50 epochs, including data read; 1x NVIDIA Tesla V100-SXM2-32GB (stated; Sec. 3.5 (Model size and training time))
murali2022 [235]	training cost only	-	-	-	a few minutes minutes to retrain; GPU-based HPC system (model and count unstated) (stated; Sec. Conclusion)
saleem2023 [242]	Latency: approximately 1 second seconds, added by waiting for a future 1 s frame to avoid edge effects (1 s)	unstated	stated; unstated; streaming	Sec. VI B; Fig. 7 caption	-
lin2024 [234]	training cost only	-	-	-	about 120 hours hours wall time; 8x NVIDIA RTX 3080 GPUs (CICA cluster) (stated; Sec. II (training; before Sec. III))
wang2022waveformer [229]	training cost only	-	-	-	approximately 24 hours for 300,000 iterations; 8x NVIDIA V100 32GB (stated; Sec. 3.1 (implementation/ training))
xu2024denoise [254]	Inference cost: within 0.01 seconds (to predict the merger time for 30 days of inspiral data)	unstated	stated; unstated; unstated	Sec. I (Introduction)	10 hours (100 epochs); 3x NVIDIA A100 40GB (stated; Sec. IV (training details))
dooney2025 [230]	Inference cost: 0.14 s per glitch sample on a CPU (0.07 compute-s per s)	CPU (model unstated)	stated; unstated; unstated	Sec. IV A	approximately 8 (24 epochs) hours on simulated data; 1x NVIDIA A100 GPU (80 GB) (stated; Sec. III A (DeepExtractor))
kiendrebeogo2025virgo [243]	Inference cost: few seconds to minutes to perform cleaning on several hours of data seconds to minutes per several hours of data	1x NVIDIA A100 Tensor Core GPU (given as an example)	stated; unstated; batch	Sec. I (Introduction)	a few minutes to train on roughly one hour of data minutes per ~1 h of data; 1x NVIDIA A100 Tensor Core GPU (given as an example) (stated; Sec. I (Introduction))
lee2025 [233]	training cost only	-	-	-	about a day wall time; 8x NVIDIA Tesla V100 GPUs (32 cores), TensorFlow (stated; Sec. 2.3 (training))
reissel2025 [244]	training cost only	-	-	-	between several and up to 15 minutes minutes for witness-channel selection, training and inference together; 1x NVIDIA TITAN X Pascal GPU (IGWN computing grid) (stated; Sec. IV (CDC pipeline description), paragraph before Sec. V)

9.4 What cannot be compared, and why

Overlap is defined differently across papers (maximised over time and phase or not, whitened or not, normalised or not, over a full window or a short window around the peak), and the SNR distribution of the test set is often weighted towards loud signals. Noise-regression papers report range gains on particular days of data, which compare only within DeepClean-type studies.

The 19 models below have no shared axis; their headline metric is listed with the reason.

Model	Headline metric (location; how)	Why not comparable	gravyflow
shen2017 [24]	SMTDAE 0.986 vs PCA 0.814, dictionary 0.691 at MF SNR 12.9 (single example, real LIGO noise; trained on white noise) (Fig 2; stated)	One hand-shown example per SNR; no test-set statistics.	*pending*
torresforne2020 [250]	blip residual SNR estimator 1.2-2.9 after combined dictionaries (16 test blips) (Table I; stated)	Self-defined estimators; no ground-truth overlap.	*pending*
vajente2020 [245]	70 Msun BBH range 729.3 to 754.6 Mpc (+11 % volume), non-neural regression (Sec IV 3; stated)	Main method is not a neural network; the GRU appendix has no numbers.	*pending*
mogushi2021 [248]	mSTFT overlap of network estimate vs extracted glitch: median 0.65 (scattered light), 0.86 (loud glitches) (Secs 3.1-3.2; stated)	Witness-channel glitch subtraction scored in a time-frequency overlap; not a waveform overlap.	*pending*
mogushi2021nnetfix [249]	gap-filling R^2 0.81-0.83; reconstruction efficiency 0.31 to >0.95 (Secs 2-3; stated)	Gated-data reconstruction scored by R^2 and fractional match gain.	*pending*
yu2021noise [246]	~10x angular-noise ASD reduction at 10 Hz (simulated bilinear coupling) (Sec 3; stated)	Simulated noise regression; the authors report no broadband gain on real data.	*pending*
bacon2022 [228]	6.6 % of test signals with overlap < 0.75 (L1 O1 noise) (Sec 3.1; stated)	The overlap of Eq 8 is not normalised by the denoised norm and is unbounded.	*pending*
murali2022 [235]	template overlap 30.5-90 % per detector on 8 O1/O2 BBH events; single-detector FAP 0.01-5 % (Table I; stated)	Real-event overlaps against best-fit templates; FAP from the authors' own 8 s segment background.	*pending*
ghalsasi2023 [236]	G2Net subset validation AUC 0.834 (EfficientNet) vs 0.833 (U-Net N2C + EfficientNet) (Table I; stated)	Denoiser judged by a downstream AUC on G2Net data.	*pending*
chatterjee2024rec [232]	GW190412-like injection, SNR 22: 0.88 (XPHM), 0.71 (SEOBNRv4PHM), 0.87 (NRHybSur) (Sec 3.2; stated)	Single-realisation overlaps; uncertainty intervals without a coverage test.	*pending*
ma2024et [252]	early-inspiral fraction with overlap > 0.9: ~70 % to ~95 % with transformer + regularity model (ET, SNR 20-80) (Sec IV A, Fig 10; stated)	Non-maximised overlap on loud synthetic ET BNS; test SNR up to twice the training maximum.	*pending*
ma2024mix [253]	both of two coincident BBHs separated (overlap > 0.9) in ~83 % of 1000 samples (ET) (Fig 6; read-from-figure)	Source separation in ET noise with a hard overlap criterion.	*pending*
ma2024msnr2 [237]	+1.3 % detection rate over MSNRnet; 0 false triggers in ~5 months of O3b (abstract) (abstract; stated)	Abstract-level; detection rate on its own event scheme.	*pending*
ma2024tsr [238]	false alarms -25 %, detections +5 %, cost -90 % vs MSNRnet (abstract) (abstract; stated)	Abstract-level; relative to its own predecessor.	*pending*
sakai2024 [239]	none extracted	Not read.	*pending*
xu2024denoise [254]	91.98 % of Taiji MBHB test samples with overlap >= 0.99 (SNR 10-50) (Sec V A; stated)	Space-based (Taiji) simulated noise; threshold chosen on the test data.	*pending*
jia2026 [240]	none extracted	Not read; category unverified.	*pending*

Model	Headline metric (location; how)	Why not comparable	gravyflow
`kelleter2026nnnn` [247]	Newtonian-noise residual 13 % (CNN) with WF/CNN residual ratio 25-80 (synthetic ET seismic field) (Sec 5.3; stated)	Seismic-array regression for ET; not strain data. Flagged marginal by its dossier.	*pending*
`kumar2026` [241]	none extracted	Not read.	*pending*

9.5 Suspected errors in the literature

32 suspected errors or internal inconsistencies in 21 papers, selected from the extraction notes (79 notes for this section in the catalogue). Each is a reading of the paper to be checked, not a verdict.

- [24] Overlap is higher at the lower SNR (0.994 at 6.44 vs 0.986 at 12.9), a single-example artefact.
- [24] Baselines differ from those re-quoted in shen2019 for the same SNR.
- [225] Split given as 75/25; 9,861 / 2,459 is 80/20.
- [225] Text claims lower MSE on both classes; eccentric MSE is higher for EDRDAE (0.005 vs 0.004).
- [25] Absolute ranges appear only in the figure.
- [250] 85/15 split of 100 glitches stated, but 16 test glitches listed.
- [227] Abstract says >= 0.99 for all four events; Fig 3 prints 0.73 for GW170608.
- [227] Conclusion gives spin-precessing overlap >= 0.97; Sec 3.3 gives a minimum of 0.93.
- [226] Text says all O1/O2 events reach >= 0.97; Fig 8 shows 0.967 and 0.968.
- [226] Training GPU given as P100 and as K40.
- [248] Excess power removed: 20-70 % (abstract) vs 10-70 % (conclusions).
- [249] Median OLR example in Sec 4 contradicts Table 9.
- [228] The "~1 % of SNR > 8 signals" figure is derived by subtracting two different fractions.
- [242] P-P plots called "excellent agreement" while both combined p-values reject calibration at p < 0.01.
- [242] VT ratio dips below 1 around 1e-9 Hz although an improvement is claimed across the FAR range.
- [242] Batch arithmetic (1993 / 32 = 62.3, not 80 batches) does not close.
- [231] Abstract range 0.85-0.98 vs Table 2 values 0.74, 0.79, 0.84.
- [231] Reconstructed SNR exceeds the reference (fractions 1.10-1.15), unexplained.
- [231] Waveforms simulated to 2048 Hz at a 2048 Hz sampling rate.
- [232] 0.71 described as generalising well.
- [234] Real-event percentages are not integer fractions of the stated totals.
- [234] NSBH/BNS test SNR starts at 8, below the 10-35 training range.
- [253] Peak-time difference direction differs between Figs 4-6 and the Fig 5 caption.
- [229] T^2/delta with T = 30 d gives ~2.1e6 yr, not the 4e5 yr stated.
- [254] "Detection accuracy" is the TPR; the matrices give 99.3-99.8 % accuracy.
- [254] Introduction says merger time within 0.01 s; results give errors of hours.
- [230] "0.3 % vs 1.9 %" and the 0.9 % headline come from different test sets.
- [243] P-P described as well calibrated while the combined KS p-values are 0.0212 and 1e-4.
- [243] A 55-56 Hz band contribution does not match any band in the 13-band list.
- [233] Training SNR given as 4-12 and test as 4-25.
- [247] Abstract "factors of 15-80" vs 10-110 and "up to 40" in the captions.
- [251] Abstract "up to ~15 %" vs the Fig 7 central peak near 18 %.

10 Parameter estimation

Parameter estimation (PE) moved from point regressors on simulated data to amortised posterior estimators validated against stochastic samplers on real events.

10.1 Families and lineage

10.1.1 Neural posterior estimation (DINGO and relatives)

Normalising flows trained on simulations [26], amortised over noise with group-equivariant tricks [27], corrected by importance sampling [28], extended to binary neutron stars [255] and to transformer embeddings [256] related flows appear for lensing, tests of GR, echoes, 3G detectors and spectrogram inputs [257], [258], [259], [260], [261].

> **Our results (to be added)**
> gravyflow reproduction: *pending* for 11 models: [26], [27], [28], [255], [260], [262], [257], [256], [261], [259], [258]. Results will be placed on the shared axes of this section beside the published values.

10.1.2 Variational and other amortised posteriors

Conditional VAEs [263], [264], ratio and likelihood estimation [265], [266], reduced-basis posteriors [267] and a WaveNet with a normalising flow [268].

> **Our results (to be added)**
> gravyflow reproduction: *pending* for 10 models: [267], [268], [265], [263], [269], [270], [264], [271], [272], [266]. Results will be placed on the shared axes of this section beside the published values.

10.1.3 Sky localisation

Rapid localisation networks compared with BAYESTAR [273], [274], [275] or with Bilby [276].

	Type	Num Filters / Neurons / Heads	Norm	Activation Function
	Permute	-	-	-
	Linear	-	-	-
×15 Coupling Transforms	Spline Coupling Transform	512	Batch	ELU
	Permute	-	-	-
	Linear	-	-	-

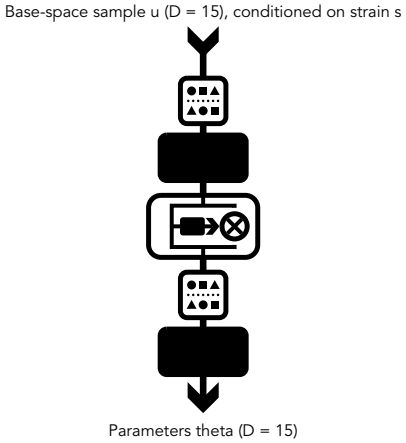


Figure 20: Green & Gair 2020 neural spline flow [26], collapsed blocks. Architecture from: Paper Fig. 1 (overall structure of the neural spline flow with hyperparameters n_flows = 15, n_hidden = 512, n_blocks = 10, K = 9, n_SVD = 100) and its caption (residual blocks of two fully-connected hidden layers, batch normalisation and ELU before each linear transformation, conditioning on the strain). Diagram confidence: **high**; 5 details the paper leaves unstated are listed in the diagram spec, and are not drawn. Notation: Appendix A.

	Type	Num Filters / Neurons / Heads	Norm	Activation Function
	Dense (SVD)	400 per detector	-	Linear
×6 Residual Blocks (1024)	Residual Block	1024	Batch	ELU
×6 Residual Blocks (512)	Residual Block	512	Batch	ELU
×6 Residual Blocks (256)	Residual Block	256	Batch	ELU
×6 Residual Blocks (128)	Residual Block	128	Batch	ELU
×30 Coupling Transforms (5 residual blocks each)	Spline Coupling	-	-	-

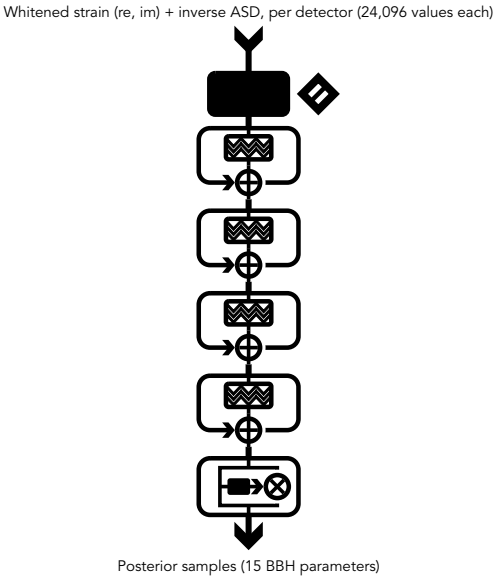


Figure 21: Dax et al. 2021 DINGO (embedding network + spline flow) [27], collapsed blocks. Architecture from: Supplement 'Neural network' section: embedding layer 1 (SVD), residual MLP of six blocks each of 1024, 512, 256 and 128 hidden dimensions, 30 rational-quadratic spline coupling transforms with 5 residual blocks each; Fig. 1 flow chart (embedding network, GNPE, 128-d context, flow). Diagram confidence: **medium**; 4 details the paper leaves unstated are listed in the diagram spec, and are not drawn. Notation: Appendix A.

Type	Num Filters / Neurons / Heads	Filter / Pool / Head Size	Stride	Activation Function
Convolutional ×6	96	64-16	1-4	LeakyReLU
Flatten	-	-	-	-
FC Head (r1, r2, q)	4096, 2048, 1024	-	-	LeakyReLU
Dense	960 / 30 / 30	-	-	-

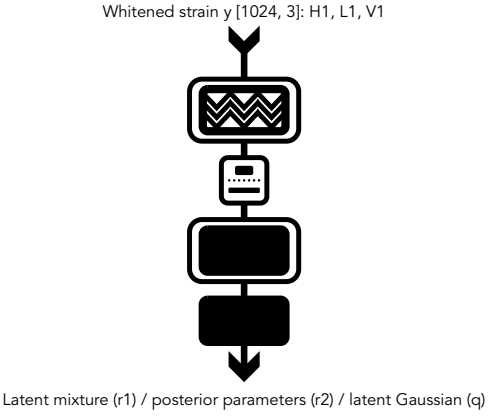


Figure 22: VItamin CVAE (Gabbard et al. 2022) [263], collapsed blocks. Architecture from: Paper Table III (layer sizes of the three networks r_theta1, r_theta2 and q_phi) and the Network design text; Fig. 1 gives the wiring. Diagram confidence: **high**; 4 details the paper leaves unstated are listed in the diagram spec, and are not drawn. Notation: Appendix A.

> **Our results (to be added)**
> gravyflow reproduction: *pending* for 4 models: [273], [276], [274], [275]. Results will be placed on the shared axes of this section beside the published values.

10.1.4 Point estimators

Regressors and classifiers that return parameter values without calibrated posteriors [277], [278], [279], [280], [281], [282], [283], [283], [284] also fits an optional flow on frozen features (combined P-P p 0.13).

> **Our results (to be added)**
> gravyflow reproduction: *pending* for 8 models: [277], [278], [284], [279], [280], [282], [281], [283]. Results will be placed on the shared axes of this section beside the published values.

10.2 Comparison on the shared axes

Rows are the models graded comparable or conditionally comparable (11 of 33); each axis entry is one row. FAR is converted to per month (30 d) and to Hz where the paper states a rate.

Model	Axis	Published value (location; how obtained)	FAR	Grade: what differs	gravyflow
`delaunoy2020` [265]	PE	coverage of 90 % intervals 89.8-93.3 % on 1000 signals (Table 1; stated)	-	conditional Neural ratio estimation in Gaussian noise; slightly conservative.	*pending*
`green2020` [26]	PE	P-P on 100 injections: per-parameter KS p 0.043 (phi_JL) to 0.90; agrees with dynesty on GW150914 (Figs 2-3; stated)	-	conditional Gaussian noise at the GW150914 PSD; 100 injections give low power.	*pending*
`dax2021` [27]	PE	combined P-P p 0.46 (1000 injections); mean JSD vs LALInference 0.0009 nat over 8 GWTC-1 events (Figs 2, 4; stated)	-	comparable The reference NPE result: P-P and JSD against MCMC on real events.	*pending*
`dax2022` [28]	PE	median IS sample efficiency 10.9 % (XPHM) / 4.4 % (SEOBNRv4PHM) on 42 events; GW150914 mean JSD 0.5e-3 nat after IS (Tables I-II; stated)	-	comparable Real GWTC-2/3 events against LALInference and Bilby evidences.	*pending*
`gabbard2022pe` [263]	PE	P-P p-values best 0.918 / worst 0.047 (VItamin) over 250 tests; JS divergence vs Dynesty between CPNest and ptemcee (Fig 4-5; stated)	-	conditional Gaussian design noise, BBH, three detectors; calibration comparable in kind to dax2021.	*pending*
`dax2024` [255]	PE	mean IS sample efficiency 31-48.5 % in four BNS experiments; sky area ~30 % smaller than BAYESTAR (Experiments; stated)	-	conditional Stationary Gaussian noise assumption; GW170817 reproduced.	*pending*
`hu2024` [260]	PE	combined P-P p 0.78 (SNR 20-50), 0.078 (SNR 200-500) on 300 3G injections (Fig 3; stated)	-	conditional Simulated third-generation network; importance sampling still costs CPU hours.	*pending*
`chan2025lens` [257]	PE	P-P KS p-values 0.020 (a1) to 0.952 (q) on 1e3 lensed injections (O5 design) (Fig 2; stated)	-	conditional Lensed-signal NPE; calibration comparable in kind to other NPE work on the same priors.	*pending*
`kofler2025` [256]	PE	median IS sample efficiency 4.2 % on 48 O3 events (baseline 1.4 % on 30 HLV events) (Main text; stated)	-	conditional Dingo-specific sample efficiency; depends on priors and detector count.	*pending*
`lanchares2025` [261]	PE	combined P-P p 0.54 (500 injections); mean JSD vs LVK 8.0e-2 nat (3.6e-2 without the localisation subspace RA, dec, t_c) on 33 events (Figs 4-6, Secs III-IV; stated)	-	conditional Time-frequency NPE; real-event agreement on a prior-restricted subset.	*pending*
`zhang2026` [258]	PE	PIT KS p 0.36-0.91 for nine ppE parameters (1000 injections) (Fig 5; stated)	-	conditional Gaussian design noise plus GW150914; ppE deviation parameters only.	*pending*

10.3 Latency and compute cost

30 of the 33 models in this section state a latency, inference cost or training cost; 2 papers were read and state none, and 1 were not read for runtime. Headline: the model's latency if it states one, else its inference cost; normalised values in brackets (Section 13).

Model	Headline: latency if stated, else inference cost	Hardware	Grade; scope; mode	Location	Training cost (hardware; grade; location)
`chatterjee2019` [273]	Latency: 0.0189 (about 0.018 in abstract) s per signal incl. feature extraction, 8192 sectors (0.0189 s)	14 Intel Xeon CPU cores	stated; end to end; unstated	Abstract; Sec. IX	-
`shen2019pe` [268]	Inference cost: within two (2) milliseconds per event (median values and posterior distributions)	1x NVIDIA V100	stated; network; unstated	Abstract; Sec. 4.2	about 1-2 days; 1x NVIDIA V100 (stated; Sec. 2.3 (Dataset Preparation and Training))
`delaunoy2020` [265]	Inference cost: less than a minute per event	unstated	stated; unstated; unstated	Sec. 5 Results	-
`green2020` [26]	Inference cost: 5,000 posterior samples per second	unstated	stated; network; batch	Sec. on inference for GW150914; Conclusions	≈ 6 days; 1x NVIDIA Quadro P4000 (stated; Results paragraph (main text))
`marulanda2020` [277]	Inference cost: about 6% greater (FCNN incl. spectrogram computation vs time-domain CNN) percent relative execution time	unstated	stated; end to end; unstated	Abstract; Results (execution-time paragraph); Conclusions	-
`singh2020` [278]	training cost only	-	-	-	approximately 40 hours for 30 epochs; Intel Core i7-7700HQ laptop with NVIDIA GTX 1060 6 GB, 16 GB RAM (stated; Sec. 4 (model training description))
`dax2021` [27]	Inference cost: 20 seconds per event (50,000 posterior samples, 30 GNPE iterations)	1x NVIDIA A100 (implied by footnote 3, which contrasts V100's	stated; unstated; batch	Abstract; Results (GWTC-1 analysis); footnote 3	roughly 10 days; 1x NVIDIA A100 (stated; Main text (Network/training paragraph); Supplemental Material 'Training' (450 epochs, batch 4096))

Model	Headline: latency if stated, else inference cost	Hardware	Grade; scope; mode	Location	Training cost (hardware; grade; location)
		roughly a minute with 20 seconds)			
kolmus2021 [276]	Inference cost: roughly 5 minutes of computing time (100,000 importance-sampling steps)	16-core AMD Ryzen 5950X CPU, NVIDIA 3090 RTX GPU, 64 GB RAM	stated; end to end; batch	Fig. 4 caption; Sec. IV C	-
chatterjee2022 [274]	Latency: ~1.2 s total inference (sample generation, KDE fit, density over adaptive mesh) (1.2 s)	1x P100 GPU	stated; end to end; unstated	Sec. 3; Abstract (around 1 sec)	~1.5 hours; 1x P100 GPU (stated; Sec. 3 (training/inference))
dax2022 [28]	Latency: approximately 20 seconds (low-latency Dingo samples, no importance sampling) (20 s)	unstated	stated; network; batch	Main text footnote 3; Discussion ('a matter of seconds'); Intro ('typically under a minute')	7 minutes (GNPE proxy density network, 20 epochs); 1 hour (Bilby-IS flow, 60 epochs) minutes / hour on A100; A100 GPU (count unstated) (stated; Tab. III)
gabbard2022pe [263]	Inference cost: 1 × 10^-1 seconds for 8000 posterior samples (VItamin)	unstated (1x NVIDIA Tesla V100 named in Methods for training; consumer GPUs said to be equally fast)	stated; network; batch	Tab. I	on the order of 7 days; 1x NVIDIA Tesla V100 (16/32 GB) (stated; Results text; Methods (Training procedure))
mcleod2022 [269]	Inference cost: ~1.4 milliseconds per sample prediction (0.0056 compute-s per s)	1x NVIDIA P100 GPU and Intel Gold 6140 CPU	stated; network; unstated	Sec. III (Prediction Results); Sec. IV (Discussion); Sec. V (Conclusion gives ~1 ms)	1 hour (typically); 1x NVIDIA P100 (stated; Sec. II (Network Architecture); Sec. IV (Discussion))
andrescarcasona2023pe [270]	Inference cost: 0.05 s to generate 10,000 posterior samples	1x NVIDIA GeForce RTX 2080 Ti	stated; network; unstated	Sec. 5 (real events, Fig. 4)	-
chatterjee2023skyloc [275]	Latency: about 1.2 s total (5000 samples, KDE, adaptive refinement) (1.2 s)	unstated	stated; end to end; unstated	Sec. 5	less than 1 hour; 1x NVIDIA Tesla P100 (stated; Sec. 5 / Discussion)
ravichandran2023 [284]	Inference cost: 0.82 – 2 ms per batch of 500 images (testing time, both problems)	48 cores of AMD EPYC 7352 CPU	stated; network; batch	Sec. III A (Training and Testing)	8 seconds per batch (detection problem), batch size 500; 48 cores of AMD EPYC 7352 CPU (stated; Sec. III A (Training and Testing))
badanerin2024 [264]	Inference cost: around 4 s per waveform (N=8000 samples)	1x NVIDIA A40	stated; network; unstated	App. A; Tab. I	7 to 8 hours; 1x NVIDIA A40 (stated; Sec. V; App. A)
dax2024 [255]	Latency: ~1 second (inference latency for pre-merger estimates) (1 s)	unstated (H100 used for IS timing)	stated; unstated; streaming	Fig. 2 caption	-
hu2024 [260]	Inference cost: within a second (network sampling) + O(1)-O(10) CPU hours importance sampling second / CPU hours	1x NVIDIA 3080Ti GPU (sampling); CPU for IS	stated; end to end; batch	Sec. 3.1 Parameter estimation	2-3 weeks; 1x NVIDIA A100 GPU (stated; Sec. 2 (training the PE model))
vanstraalen2024 [262]	training cost only	-	-	-	between four and five days to fully train a network; 1x NVIDIA A100-PCIE-40GB (stated; Sec. III (parameter estimation network training))
benedikt2025 [280]	Inference cost: 10.6 ± 0.5 ms per event, batch size 1 (standard forward pass) (0.0106 compute-s per s)	1x NVIDIA A100	stated; network; unstated	Tab. 9; Sec. 5.4	79.64 scratch vs 13.05 pretrained (small, SNR 10); 242.82 vs 169.63 (large, SNR 10); 348.69 vs 87.92 (large, SNR 30) minutes to reach target validation loss; 1x NVIDIA A100 (stated; Tab. 1)
chan2025lens [257]	Inference cost: tens of seconds per event s per event	unstated	stated; unstated; unstated	Sec. IV; Abstract	approximately 10 days (450 epochs); 1x NVIDIA A100 PCIe (stated; Sec. III A; App. A)
iwanaga2025 [282]	Inference cost: approximately six minutes (total estimation time incl. Monte Carlo uncertainty evaluation)	unstated	stated; end to end; batch	Sec. III A and Conclusions	approximately one hour (50 epochs); 1x NVIDIA GeForce RTX 3090 (stated; Sec. II B (Training and validation))
khalouei2025 [281]	Inference cost: on the order of 100 ms per event (average over 100 repeated inferences)	unstated (presumed the training GPU)	stated; network; batch	Sec. 3; Sec. 4	less than one hour; 1x NVIDIA GeForce RTX 2070 (8 GB) (stated; Sec. 2.3 Training; Sec. 4 Conclusions)
kofler2025 [256]	Latency: under five seconds (initial low-latency samples) (5 s)	unstated (64 CPU cores used for subsequent IS)	stated; network; batch	Results; Sec. on inference	approximately 9.5 days (183 epochs); Tab. IV: 9d 9h days; 8x NVIDIA A100-SXM4-80GB (stated; Main text (Results); Tab. IV; Supplemental 'Training')
lanchares2025 [261]	Inference cost: 1.13 ± 0.13 seconds per 10,000 posterior samples	1x NVIDIA A100	stated; network; batch	Results text, Sec. III (GP15 vs Bilby comparison paragraph)	36 hours per stage; 1x NVIDIA A100 (Artemisa cluster, Valencia) (stated; Sec. II (end of Methodology, training paragraph))
papalini2025 [272]	Inference cost: ~1 second to generate a full posterior with 10^4 samples	unstated	stated; network; batch	Sec. 4 Results; Conclusion ('~1 s')	around 3 days (300 epochs); 1x NVIDIA A30 (stated; Sec. 3.4 (Training))
emma2026 [266]	Inference cost: approximately an order of magnitude larger than Whittle-likelihood runs ratio of total wall time	unstated	stated; end to end; batch	Sec. IV (GW analysis wall time)	below one hour for ~ 2000 simulations to several hours for 10^5 simulations hours; unstated (stated; Sec. IV (Configuring RNLE for efficient GW analysis); Fig. 5 black squares)
luo2026 [259]	Inference cost: less than 20 minutes per event (echo search / Bayesian inference)	1x NVIDIA RTX 5090	stated; unstated; batch	Sec. I (Introduction); Sec. III (Results); Sec. IV (Conclusion)	about 9–10 days (full training procedure); 2x NVIDIA RTX 5090 (one per network: initialization and main GNPE) (stated; Sec. III (Results, first paragraph))
sakellariou2026 [283]	Inference cost: approximately 40 ms per strain (DeepModel-Hybrid; 36 ms for the baseline DeepModel) (0.04 compute-s per s)	unstated	stated; network; unstated	Sec. 3 (Results, computational-efficiency paragraph)	-
zhang2026 [258]	Inference cost: approximately 0.4 (less than 0.4) seconds for complete posterior sampling of a single event	1x NVIDIA RTX A6000	stated; network; batch	Sec. on training/ implementation; Sec. C 'Computational	approximately 8 to 13 hours per ppE parameter estimator; 1x NVIDIA RTX A6000 (stated; Sec. on implementation; Sec. C)

Model	Headline: latency if stated, else inference cost	Hardware	Grade; scope; mode	Location	Training cost (hardware; grade; location)
				performance comparison'; Conclusion	

10.4 What cannot be compared, and why

Calibration (P-P or coverage) and divergence from a reference sampler are the only shared axes, and they require the same priors, waveform model and noise. Point regressors report mean errors on their own grids and are not comparable with posterior estimators.

The 22 models below have no shared axis; their headline metric is listed with the reason.

Model	Headline metric (location; how)	Why not comparable	gravyflow
`chatterjee2019` [273]	sky-sector accuracy 62-85 % (SNR 20-35) and 77-91 % (SNR 50-55) (Table V; stated)	Sky localisation as sector classification.	*pending*
`chua2019` [267]	posterior-mean error ~2 % of prior width (quoted as 98 % accuracy); sigma overestimated up to 3x (Sec "Example results"; stated)	Reduced-basis toy without detector strain.	*pending*
`shen2019pe` [268]	GW150914 m1 38.9 vs LIGO 35.6 Msun; 5 events within 90 % intervals (Table 1; stated)	Five-event comparison and a qualitative P-P plot.	*pending*
`marulanda2020` [277]	mass MAPE ~5-7 % at SNR 25, ~80-90 % at SNR 5 (Figs 6, 9; read-from-figure)	Point regression on interleaved training and validation grids.	*pending*
`singh2020` [278]	multi-label accuracy ~0.90 on a sparse grid (Sec 5.1; stated)	Element-wise multi-label accuracy dominated by true negatives.	*pending*
`kolmus2021` [276]	mean absolute sky error ~32 / 14 / 9.5 deg at SNR 10 / 15 / 20 (Fig 2 left; read-from-figure)	Sky localisation seeding importance sampling.	*pending*
`chatterjee2022` [274]	90 % areas with medians close to BAYESTAR (2000 injections per class) (Fig 4; read-from-figure)	Sky areas against BAYESTAR; sky-only P-P with KS p 0.075-0.835 (Fig 7).	*pending*
`mcleod2022` [269]	mean error 10.5 % (Mc), 24.3 % (q) on simulations; 6.2 % / 8.5 % vs LALInference medians on 10 events (Sec III; stated)	Gaussian posteriors for two parameters.	*pending*
`andrescarcasona2023pe` [270]	P-P inside the 3-sigma band for 7 parameters (100 signals); deviations ~0.1 (Fig 3; read-from-figure)	Narrow BBH prior, 100 test signals, three real events compared by eye.	*pending*
`chatterjee2023skyloc` [275]	P-P deviates from the 95 % band at every lead time except 0 s (Fig 5; stated)	Pre-merger BNS sky areas in Gaussian design noise vs BAYESTAR.	*pending*
`ravichandran2023` [284]	eccentric-signal detection accuracy 0.914, TPR 0.862 (Abstract, Fig 8; stated)	Gaussian-noise classifier.	*pending*
`soma2023` [279]	chirp mass / tidal R^2 0.95 / 0.94 (independent); classifier macro precision 0.96 at SNR 20 (Secs 3.2, 4.3.1; stated)	No separate classifier test set described (66,000 train + 9,000 validation per class).	*pending*
`badanerin2024` [264]	~95 % of true lensing parameters inside 95 % intervals at SNR 8-40 (Fig 6; stated)	Two lensing parameters with all others fixed to truth in the Bilby comparison.	*pending*
`jenner2024` [271]	none extracted	Abstract-level.	*pending*
`vanstraalen2024` [262]	best pre-merger 90 % sky area 363 deg^2 (70 Hz cut), 141 deg^2 (100 Hz) (Sec VI; stated)	Pre-merger BNS NPE in Gaussian design noise.	*pending*
`benedikt2025` [280]	MAE m1 5.44 Msun, chi_eff 0.03 at SNR 10 (ET noise) (Table 2; stated)	Point estimates on an ET grid; MC-dropout widths are not calibrated.	*pending*
`iwanaga2025` [282]	simulated GW150914: chirp mass 27.6 (+0.8/-1.0) vs LALInference 27.9 (+2.3/-1.8) Msun; chi_eff -0.09 (+0.07/-0.09) vs -0.09 (+0.19/-0.17) (Table II; stated)	Noise-scatter intervals on one simulated event; no posterior.	*pending*
`khalouei2025` [281]	eccentricity MAE 0.042-0.049 (SNR 10-20), 0.018-0.025 (SNR 20-60) (Table 1; stated)	Point regression in Gaussian O4 noise.	*pending*
`papalini2025` [272]	combined P-P KS p 0.489 (64 injections, three overlapping BBHs in ET) (Fig 6; stated)	Overlapping-signal NPE for ET in simulated Gaussian noise; 64 test injections.	*pending*
`emma2026` [266]	JSD to the Whittle posterior at or below the 10/N = 0.003 threshold for all parameters in the 10D analysis, but up to 0.027 (15D) and 0.060 (11D) for some (Sec V, Table III; stated)	Likelihood-estimation method compared with Whittle, not with other SBI.	*pending*
`luo2026` [259]	combined P-P p 0.154 (20 parameters, 1000 simulations) (Fig 3; stated)	Echo search; Bayes factors from an unstated evidence estimate.	*pending*
`sakellariou2026` [283]	MAE m1 2.28 Msun, dL 40.6 Mpc on a discrete synthetic grid (Table 2; stated)	Grid-split regression with min-max normalised strain.	*pending*

10.5 Suspected errors in the literature

24 suspected errors or internal inconsistencies in 19 papers, selected from the extraction notes (92 notes for this section in the catalogue). Each is a reading of the paper to be checked, not a verdict.

- [273] Abstract "very high accuracy (> 90%)" for 18, 50 and 128 sectors vs 62-91 % exact-sector test accuracy in noise (Table V) and 85 % at 128 sectors without noise (Table IV); only the within-one-sector accuracy (92-98.5 %) exceeds 90 % in every case.
- [268] Training noise starts at the GPS time of GW150914 (leakage).
- [268] The "GW190521" row mixes GW190521_074359 masses with the SNR of GW190521.
- [265] The coverage table is called Table 5 in the text.
- [26] KS "confirms" calibration while phi_JL has p = 0.043 and no multiple-testing correction is applied.
- [278] "Four detections" with seven events listed.
- [27] The O(day) MCMC baseline is cited, not measured.
- [276] Outliers removed from the importance-sampling results without a count.
- [270] Spin "magnitude" prior U(-1, 1) is a signed component.
- [275] "A few milliseconds" inference vs 1.2 s per event including KDE.
- [275] Lead times 14 s (Sec 3) vs 15 s (figures); SNR range 8-40 vs 9-40.
- [284] "FPR 0.138" in the abstract is the false-negative rate; the matrix gives FPR 0.028.
- [264] Table I saving column (79.7 %) does not follow from its time columns (44 %).
- [260] "Seconds" applies to flow sampling only.
- [280] Abstract chi_eff MAE ~1e-3 vs Table 2 0.03.
- [280] MC-dropout sigma is 1.5-2x smaller than the actual error.

- [257] Calibration called good while a1 p = 0.020 and theta_jn p = 0.056 over 17 parameters.
- [281] A large train/validation MSE gap (0.003 vs 0.11) is not discussed.
- [256] The 1.4 % to 4.2 % headline compares a 30-event HLV baseline with the 48-event median that includes two-detector events (the three-detector median is also 4.2 %).
- [261] Abstract says all three-detector GWTC events; only 33 that fit the priors were used.
- [283] Baseline flow CRPS 3.3e14 and p-value 1.3e-75 indicate a diverged baseline.
- [283] The total loss averages MAEs in different units.
- [258] 90 % widths for dchi3-dchi7 (1.71-1.83) match the 1.8 of a uniform prior on the normalised [-1, 1] range, so "consistent" means uninformative.
- [258] "Over 10 hours" per MCMC event holds for two of nine parameters.

11 Space-based detectors

Space-based detector studies use simulated LISA, Taiji or TianQin data, mostly as time-delay-interferometry (TDI) channels, and are outside gravyflow's present detector set.

11.1 Families and lineage

11.1.1 Massive black hole binaries and EMRIs

Detection, inference and extraction of massive black hole binaries and extreme-mass-ratio inspirals in simulated LISA, Taiji and TianQin data [285], [286], [287], [288], [289], [290], [291], [292].

> **Our results (to be added)**
> gravyflow reproduction: *pending* for 9 models: [293], [289], [291], [285], [287], [290], [288], [292], [286]. Results will be placed on the shared axes of this section beside the published values.

11.1.2 Instrumental artefacts and separation

Glitches and overlapping sources in TDI data [294], [295], [296], and quantum-network feasibility studies [297], [298].

> **Our results (to be added)**
> gravyflow reproduction: *pending* for 5 models: [294], [295], [297], [298], [296]. Results will be placed on the shared axes of this section beside the published values.

11.2 Comparison on the shared axes

Rows are the models graded comparable or conditionally comparable (0 of 14); each axis entry is one row. FAR is converted to per month (30 d) and to Hz where the paper states a rate.

11.3 Latency and compute cost

10 of the 14 models in this section state a latency, inference cost or training cost; 4 papers were read and state none, and 0 were not read for runtime. Headline: the model's latency if it states one, else its inference cost; normalised values in brackets (Section 13).

Model	Headline: latency if stated, else inference cost	Hardware	Grade; scope; mode	Location	Training cost (hardware; grade; location)
`caramete2020` [293]	training cost only	-	-	-	about 22 hours (training time of the 2-class LL BiLSTM on raw time-series values; about 4 minutes when trained on spectral features); unstated (stated; Sec. 3.3 (text discussing Fig. 19); Sec. 4 (Conclusions))
`zhang2022emri` [289]	Inference cost: within one second seconds to classify three months' data (7,864,320 s) (1.27e-07 compute-s per s)	1x GPU Tesla V100 PCIe 16 GB (the implementation hardware of footnote 2; not restated for inference)	stated; unstated; batch	Sec. VI (Conclusions)	about 10.5 days to train the final CNN (about 2 hours wall clock per epoch); GPU Tesla V100 PCIe 16 GB (1), CPU Intel Xeon Gold 6140 (72 threads), 256 GB RAM (stated; Sec. IV A (Training phase) and footnote 2)
`liu2023` [291]	Inference cost: SSL and LSTM time cost more than 70% lower than LDASoft (LISA MDC) percent reduction in time cost	unstated	stated; unstated; unstated	Sec. IV, text around Fig. 9	-
`ruan2023` [285]	Inference cost: within ten seconds (a year-long Sangria dataset) s per year of data (3.17e-07 compute-s per s)	unstated	stated; unstated; batch	Sec. VI	10 hours; 1x NVIDIA GeForce RTX 2080 (8GB) (stated; Sec. IV)
`ruan2023pe` [287]	Inference cost: about twenty seconds to draw 50,000 posterior samples (4 parameters) for one MBHB signal (4.07e-05 compute-s per s)	unstated	stated; unstated; unstated	Abstract; Sec. I; Sec. IV; Sec. V	about 5 days (3,200 epochs, batch size 512, 60,000 signals generated per epoch); 1x NVIDIA Tesla A40 GPU (stated; Sec. IV)
`zhao2023` [288]	Inference cost: 1.9 s per batch of 256 samples (474 days of data) (4.64e-08 compute-s per s)	unstated (training on a single NVIDIA V100)	stated; network; batch	Tab. 3; Sec. 2.7	-
`ruan2024` [286]	Inference cost: 0.01 seconds per 4.7-day LISA segment (81,920 samples at 5 s cadence) (2.44e-08 compute-s per s)	NVIDIA Tesla A40 (1; 'all training and testing' on it)	stated; unstated; unstated	Sec. IV (Results, p.~8) and Sec. V	nearly three days (500 epochs) wall time; NVIDIA Tesla A40 (1) (stated; Sec. IV)
`houba2025separation` [295]	Inference cost: within seconds (model inference on a standard laptop)	a standard laptop (model unstated)	stated; unstated; unstated	Sec. V (Conclusion)	approximately 4 hours (25 epochs, 25,000 mixtures of 2-hour segments, batch size 16); 1x MacBook Pro, Apple

Model	Headline: latency if stated, else inference cost	Hardware	Grade; scope; mode	Location	Training cost (hardware; grade; location)
					M2 Max, 32 GB unified memory (stated; Appendix A)
`isfan2026qnn` [298]	training cost only	-	-	-	one hour of QPU time (IQM's free hour) judged enough to run the sQNN training and prediction; IQM Quantum Computers Garnet (estimated; Sec. 3.4)
`xu2026ssl_lisa` [296]	training cost only	-	-	-	approximately 6 hours (100 epochs, batch size 64); 4x NVIDIA A100 40 GB (stated; Sec. IV (Training strategy))

11.4 What cannot be compared, and why

All space-based results use simulated data; where a false-alarm figure is given it is a per-sample fraction, none is a rate-based FAR, and the detectors are not in gravyflow's present scope.

The 14 models below have no shared axis; their headline metric is listed with the reason.

Model	Headline metric (location; how)	Why not comparable	gravyflow
`caramete2020` [293]	4-class mass-ratio accuracy 95.57 % (Sec 3.4; stated)	Simulated LISA-like signals, noiseless except a random-noise subset of class D; cloning before the split leaks samples.	*pending*
`zhang2022emri` [289]	TianQin EMRI TAP ~0.88-0.92 at FAP 1 % (SNR 50-120); ~0.75 claimed for an astrophysical population (Sec V A, Fig 3; stated)	Per-3-month-sample FAP in Gaussian TianQin noise.	*pending*
`liu2023` [291]	AUC 0.85-0.87 across TDI generations (LISA/Taiji MBHB, SNR 30-50) (Fig 4; read-from-figure)	Space-based synthetic Gaussian noise; AUC at an unspecified SNR mix.	*pending*
`ruan2023` [285]	TPR at per-sample FPR 1e-3: 0.997 (z = 1) to 0.627 (z = 16) (Sec V A; stated); 15 of 15 Sangria (LDC-2) mergers, no false alarms (Sec V B; stated)	LISA; FPR resolved to ~3e-4 by 3,000 negatives per redshift.	*pending*
`ruan2023pe` [287]	P-P within 3-sigma bands (1000 signals); >4 orders of magnitude prior-volume reduction for SNR > 100 (Figs 2-3; stated)	Simulated LISA MBHB PE.	*pending*
`yun2023` [290]	Taiji EMRI TPR 94.2 % at 1 % FPR (SNR 50-100) (Abstract, Sec IV; stated)	6,000 samples in total; TPR at 1 % rests on a few tens of noise samples.	*pending*
`zhao2023` [288]	TPR at 1 % FPR: EMRI 98.2-99.7 %, MBHB 99.99-100 % (SNR 30-50) (Sec 2.5; stated); MBHB extraction overlap > 0.99 for all test samples (SNR 50) (Sec 2.4; stated)	LISA long-wavelength synthetic data; per-sample FPR.	*pending*
`houba2024glitch` [294]	LISA anomaly detection 99.99915 % of anomalous segments; glitch localisation accuracy 92.1 % (Secs III-IV; stated)	LISA TDI simulations with an own 25-class taxonomy.	*pending*
`houba2024mbh` [292]	>= 95 % of m1 = 3e5 Msun mergers at 15 Gpc flagged one month before merger (Fig 8; read-from-figure)	LISA pre-merger detection without a merger-free false-positive rate.	*pending*
`ruan2024` [286]	TPR > 0.9 at per-segment FPR 0.01 for SNR > 25; detection 53.5 h before merger (median) (Sec IV A, Fig 3; Sec IV B, Fig 6; stated)	Pre-merger LISA MBHB classifier; per-segment FPR.	*pending*
`houba2025separation` [295]	MBHB median match 0.994 with an overlapping glitch (SNR 50); > 0.95 above SNR ~15 (Figs 8, 10; read-from-figure)	LISA source separation on simulated TDI.	*pending*
`isfan2025quantum` [297]	98.4 % of Sangria blind windows labelled GW (no negative class); 5 of 6 mergers (Fig 10, Fig 14; stated)	Quantum NN on LISA Sangria; "accuracy" without negatives is a recall.	*pending*
`isfan2026qnn` [298]	hardware accuracy 18.7 % (ibm_kyoto) vs 100 % on simulator (Sec 3.1; stated)	Hardware-feasibility study; no detection performance.	*pending*
`xu2026ssl_lisa` [296]	94.8 / 97.3 / 98.4 % of MBHB extractions above 0.9 at SNR 30 / 40 / 50 (Taiji) (Sec V A; stated)	Self-supervised extraction on simulated Taiji data.	*pending*

11.5 Suspected errors in the literature

19 suspected errors or internal inconsistencies in 12 papers, selected from the extraction notes (63 notes for this section in the catalogue). Each is a reading of the paper to be checked, not a verdict.

- [293] Class D alone carries added noise, a confound.
- [293] Per-class percentages in the text are precisions, not the recalls described.
- [289] Astrophysical-population TAP "about 75 %" at FAP 1 % vs ~0.56 read from Fig 3.
- [289] TAP below FAP at high masses in Fig 4(b).
- [291] Text says AUC rises monotonically with epochs; Fig 4 is non-monotonic.
- [291] Time-cost reduction "over 70 %" vs ~59-63 % in Fig 9.
- [291] Test SNR given as 30-50 and as "to 20".
- [285] All 15 Sangria mergers recovered although the lowest-SNR ones sit where TPR is 0.33-0.60 (Fig 5).
- [285] TPR at FPR 1e-3 rests on 3 false positives of 3,000.
- [290] Inset said to zoom TPR 0-0.01; the axes show FPR 0-0.1.
- [288] EMRI TPR 99.7 % at 1 % FPR vs ~0.991 read from the Fig 5a ROC.
- [288] Abstract "at least 95 % similarity" vs 92 % > 0.95 for EMRI.
- [294] 99.99915 % on <= 20,000 anomalous samples is finer than the test-set resolution.
- [294] Glitches labelled as bursts: 0.065 % stated vs 27 / 19,228 = 0.14 % from Fig 9.
- [292] No false-positive rate is reported although 20,000 merger-free samples are described.
- [295] Hallucination area 6.88e-5 (text) vs 6.47e-5 (Fig 11).
- [297] Window length and step differ between Sec 3.2 and Table 2.
- [298] Abstract gives 20 % accuracy and different cloud costs from the body.
- [296] Overlap thresholds differ by class (0.9 vs 0.8), so the class fractions are not on one scale.

12 Other: alerts, generators, lensing, backgrounds

Models that do not fit the categories above: classifiers of public-alert products, generative models of signals and glitches, lensing and stochastic-background studies, a hardware study and a review.

12.1 Families and lineage

12.1.1 Alert-product classifiers

GWSkyNet and its successors classify public-alert sky maps rather than strain [299], [300], [301].

> **Our results (to be added)**
> gravyflow reproduction: *pending* for 3 models: [299], [300], [301]. Results will be placed on the shared axes of this section beside the published values.

12.1.2 Generative models

GANs for bursts, blips, tomtes, BBH signals and Gravity Spy images [302], [303], [304], [305].

> **Our results (to be added)**
> gravyflow reproduction: *pending* for 4 models: [302], [303], [304], [305]. Results will be placed on the shared axes of this section beside the published values.

12.1.3 Lensing, backgrounds and miscellany

Lensing identification [306], [307], stochastic backgrounds [308], [309], equation-of-state inference from CCSN [310], space-based extraction [311], FPGA acceleration [312] and a review [313].

> **Our results (to be added)**
> gravyflow reproduction: *pending* for 8 models: [306], [313], [310], [312], [308], [307], [311], [309]. Results will be placed on the shared axes of this section beside the published values.

12.2 Comparison on the shared axes

Rows are the models graded comparable or conditionally comparable (0 of 15); each axis entry is one row. FAR is converted to per month (30 d) and to Hz where the paper states a rate.

12.3 Latency and compute cost

8 of the 15 models in this section state a latency, inference cost or training cost; 6 papers were read and state none, and 1 were not read for runtime. Headline: the model's latency if it states one, else its inference cost; normalised values in brackets (Section 13).

Model	Headline: latency if stated, else inference cost	Hardware	Grade; scope; mode	Location	Training cost (hardware; grade; location)
`cuoco2020` [313]	Latency: within 8 seconds (iDQ Ordered Veto List glitch identification for GW170817)	unstated	stated; unstated; streaming	Sec. 2.2 (iDQ paragraph, GW170817)	minutes minutes (train and run, a few thousand noise transients); unstated (LIGO-Virgo computing clusters) (stated; Sec. 2.2 (RF and GP glitch-origin algorithms, ref))
`mcginn2021` [302]	Inference cost: fractions of a second s to generate waveforms	unstated	stated; network; unstated	Sec. 1 (Introduction)	O(10) hours (500 epochs); 1x GeForce RTX 2080 Ti (stated; Sec. 3.1 (Architecture details))
`que2021` [312]	Latency: 0.40 us (FPGA design, 16-bit fixed point, batch 1; 39.7 ms on CPU, 32.1 ms on GPU; Tab. IV gives 0.867 us for the four-layer design) (4e-07 s)	1x Xilinx U250 FPGA (300 MHz)	stated; network; streaming	Tab. III	-
`lopez2022gan` [303]	training cost only	-	-	-	≈ 7.75 hours (500 epochs); 1x GPU TITAN V (stated; Sec. III C 2 (CT-GAN training procedure))
`powell2022` [304]	Inference cost: less than a second s to convert GAN-generated glitch images into time series	unstated	stated; unstated; unstated	Sec. 4 (Applications)	-
`chan2024` [301]	Latency: <= 66 seconds, between release of the public alert and completion of GWSkyNet analysis, for 95% of O4 public alerts (66 s)	unstated	stated; end to end; streaming	Sec. 3 (O4 results, text near Table 2)	-
`dooney2024` [305]	Inference cost: approximately 18 s to generate 2,500 BBH signals	CPU (model unstated)	stated; network; batch	Sec. IV C (Fitting-factor results)	-
`xu2024` [311]	Inference cost: < 10^-2 seconds per signal extraction (6.25e-08 compute-s per s)	unstated	stated; unstated; unstated	Sec. V (ideal-case results)	-

12.4 What cannot be compared, and why

These models do not share axes with each other or with the detection categories.

The 15 models below have no shared axis; their headline metric is listed with the reason.

Model	Headline metric (location; how)	Why not comparable	gravyflow
`cabero2020` [299]	GWSkyNet test accuracy 93.5 % (413 candidates); O3 retractions 10 of 22 (Sec 4; stated)	Classifier of alert sky maps, not strain.	*pending*

Model	Headline metric (location; how)	Why not comparable	gravyflow
`kim2020lens` [306]	lensed classification 96.7-98.3 % (optimal SNR 10-50); 62-68 % at SNR 12-13 (Fig 5, Sec 4.3; stated)	Balanced simulated lensing classification.	*pending*
`abbott2021` [300]	GWSkyNet-Multi hierarchical accuracy 93.4 % (427 test events); 36 of 40 O3a alerts (Sec 3-4; stated)	Alert-product classifier.	*pending*
`cuoco2020` [313]	none extracted	Review article; numbers are secondary.	*pending*
`edwards2021` [310]	18-way EoS top-1 accuracy 0.64-0.72, top-5 0.91-0.97 (noise-free) (Table II; stated)	Noise-free CCSN waveform classification.	*pending*
`mcginn2021` [302]	network SNR ~7.3-7.6 at 50 % true-alarm probability for the uniform-trained CNN at FAP 1e-3 (5 burst classes) (Fig 11; read-from-figure)	Generator paper; detection test in Gaussian noise with per-sample FAP.	*pending*
`que2021` [312]	LSTM autoencoder AUC 0.763 (SNR distribution unstated); FPGA latency 0.867 us (Fig 9, Table IV; stated)	Hardware-acceleration study.	*pending*
`utina2021` [308]	none extracted	Abstract-level stochastic-background study.	*pending*
`kim2022lens` [307]	learner AUC > 0.99 (simulated); 1 of 46 GWTC-1/2 events primarily classified lensed, then judged likely unlensed (Fig 5, Sec 3, Table 3; stated)	Lensing search; "ROC" uses a set-level false-alarm probability.	*pending*
`lopez2022gan` [303]	Gravity Spy labels > 70 % of 1000 GAN H1 blips as Blip (Fig 10; read-from-figure)	Generator fidelity.	*pending*
`powell2022` [304]	real-data CNN classifies GAN glitches with 99.0 % mean accuracy (22 classes) (Abstract, Fig 5; stated)	Generator fidelity.	*pending*
`chan2024` [301]	GWSkyNet II detection probability 92.5 % (fine-tuned) vs 68.3 % at FAP 20 % among O4 MDC candidates (Sec 3; stated)	Post-trigger alert classifier; 96 noise transients in the test set.	*pending*
`dooney2024` [305]	downstream CNN AUC 0.802 (cDVGAN simplex) vs 0.770 best baseline (SNR 1-16) (Table I; stated)	Generator judged by a downstream classifier on a private set.	*pending*
`xu2024` [311]	fraction with overlap >= 0.9: 1.0 (ideal) to 0.56 (4x acceleration noise) and 0.31 (long 5e-14 m/s^2 glitch) for Taiji MBHB (Tables II-V; stated)	Space-based extraction robustness study.	*pending*
`einsle2025` [309]	BBH background Omega ~1e-9 at 25 Hz reaches log10 BF 3 with 23.7 d (A+ Gaussian) (Fig 3; stated)	Stochastic-background task.	*pending*

12.5 Suspected errors in the literature

11 suspected errors or internal inconsistencies in 8 papers, selected from the extraction notes (33 notes for this section in the catalogue). Each is a reading of the paper to be checked, not a verdict.

- [299] The 93.5 % headline uses a simulated real class; the real retraction test is 10/22.
- [306] Named VGG-19 but the appendix lists 10 conv layers.
- [300] Fig 7(c) BBH matrix duplicates the glitch panel.
- [300] F1 in Table 1 not reproducible from Fig 7.
- [300] 36/40 vs 6 + 29 = 35 correct.
- [312] Same design latency 0.40 us (Table III) vs 0.867 us (Table IV).
- [303] ~500 Blips in the text vs 736 in Fig 10 at the same SNR.
- [303] Pearson coefficients printed as percentages.
- [301] Baseline beats fine-tuned in both detector-count subsets while losing overall.
- [305] The abstract's maximum 4.2 % AUC gain (0.802 vs 0.770) is over the weakest baseline's best (McGANn); cWGAN reaches 0.788 and the authors' cDVGAN2 0.791 (Table I).
- [309] MSE 0.01 on log spectra is a ~26 % factor, not "about 3 %".

13 Latency and compute cost

Speed is half of the case most papers make for machine learning in gravitational-wave analysis, so every paper in the archive was read for three quantities: **latency**, the time from the arrival of the data to an alert or an output; **inference cost**, the compute time to analyse a stretch of data; and **training cost**. Each number is recorded as the paper gives it, with its statistic (single value, mean, median, percentile), the hardware it was measured on, its scope (**network**: the forward pass alone; **end to end**: with whitening, input and output, and clustering), its mode (**streaming** or **batch**), its location in the paper, and a grade: **stated** by the paper, **derived** from numbers it states, or **estimated**. A second reader checked every record and every “none stated” against the paper: of the 492 records, 230 were confirmed as extracted, 244 were corrected and 18 were added by the check. Where a paper gives the duration of data analysed, the inference cost is also normalised to compute-seconds per second of strain over the detector network (cs/s below), and a latency to seconds.

Why there is no single efficiency ratio. Normalisation removes the units, not the hardware. The records span a two-core laptop CPU, single consumer and data-centre GPUs, an FPGA, and allocations of hundreds of GPUs on a supercomputer; the device count and generation, the batch size, and whether whitening, data loading and trigger clustering are counted each move a number by orders of magnitude. A latency also depends on the buffering and the detection threshold behind it. We therefore tabulate the numbers and rank none of them. A model is set against a matched-filter, burst or sampling pipeline only where the paper itself makes that comparison, and the tables say whether both ran on the same hardware.

13.1 Coverage

Of the 313 models in this review, 172 state at least one of the three quantities (492 records), 104 papers were read and state none, and 37 were not read for runtime because their paper is not in the archive. Inference cost is the quantity most papers give (126 models) and latency the rarest (22 models). 162 records carry a normalised value, but for inference cost only 61 of the 126 models give the duration of data needed to normalise one. 120 records quote a baseline pipeline. A model counts under a grade in Table 4 when at least one of its records of that quantity has that grade.

Table 4. Models stating each quantity, by the grade of their records, and the number of records.

Quantity	Stated	Derived	Estimated	Any grade	Records
Latency	22	1	0	22	39
Inference cost	124	4	2	126	270
Training cost	117	5	3	120	183

Each review section above carries its own table of the models' headline numbers. The rest of this section collects the latencies, the inference costs and the comparisons with baseline pipelines across sections.

13.2 Stated latencies

Twenty-two models state a latency (Table 5). The streaming detection searches report seconds: 1 s end to end for GPU inference served as a stream in a denoising-plus-BBH search, whose denoiser alone adds a 0.75 s aggregation latency [68] (Methods B.3, Fig 11; Discussion), a median of 3.114 s on one A30 GPU for Aframe [9] (Tab III), and 10.2 s from signal arrival to alert for a BNS search, a conservative estimate at 10 false alarms per month that leaves out whitening, resampling and formatting of the input [5] (Abstract; Sec V B). Amortised parameter estimation gives samples in seconds: about 20 s for low-latency DINGO samples before importance sampling [28] (footnote 3) and about 1.2 s for a sky-localisation flow [274] (Sec 3). At the other extreme an LSTM autoencoder runs its network in 0.40 µs on an FPGA, against 39.7 ms for the same model on a CPU and 32.1 ms on a GPU [312] (Tab III); that paper's own Table IV gives 0.867 µs for what appears to be the same design, a disagreement listed among the suspected errors of Section 12. The scope and mode columns matter here: a network-only latency on a dedicated device and an end-to-end latency on a shared cluster measure different things.

13.3 Inference cost

Table 6 gives one row per model: its headline inference-cost record, chosen as a normalised record first, then by grade, then in the paper's order. The last column counts the model's inference-cost records; the zoo site's runtime page lists every record, with its arithmetic and the check's verdict. Values in cs/s are compute-seconds per second of strain over the detector network, on the hardware named in the same row: a multi-GPU allocation that finishes a month of data in minutes and a laptop that keeps up with real time are both summarised by one number, and the hardware column is what tells them apart.

13.4 Comparisons with baseline pipelines

68 papers set their runtime against a baseline (Table 7, 95 distinct comparisons). Few of them ran both on the same hardware. The clearest like-for-like comparison is the early one: a CNN (Deep Filtering) classifying a 1 s input in 6.7 ms on two cores of a laptop CPU, against 1.1 s for time-domain matched filtering with the same template bank on the same CPU [1] (Sec V; Fig 14). Most later comparisons cross hardware. A spatiotemporal-graph ensemble processes a 4096 s segment in 20.95 s on one A100, against 4.5 hours per segment for PyCBC on four CPUs [67] (Results, PyCBC comparison). Aframe's latency budget of 3.114 s, and its median of 3.9 s (90th percentile 4.3 s) in a real-time mock-data-challenge replay that includes GraceDB processing, are set beside 12.3 s (median) and 41.4 s (90th percentile) for matched-filter pipelines in that challenge, on hardware the paper does not state [9] (Tab III; Sec VIII). The BNS search's 10.2 s is set beside 16 s on average for PyCBC Live [5] (Abstract; Sec V B). In parameter estimation, DINGO with importance sampling typically takes under an hour per event with IMRPhenomXPHM and about 10 hours with SEOBNRv4PHM on one A100 and 64 CPU cores, against an estimated several months per event for SEOBNRv4PHM with stochastic samplers; Bilby-dynesty took about a day per run on 10 CPUs at reduced settings (nlive 1000, nact 5) and about a week at the settings of the paper's reference runs [28] (Introduction; Results; Supplemental Material). MLGWSC-1 timed the entries on the challenge's own node, but approximated PyCBC and cWB as 16-core CPU jobs: on the real-noise dataset PyCBC took 163102 s per month of data, against 3514 s for the TPI FSU Jena network and 5516 s for Virgo-AUTh, each on one RTX 2070 Super [8] (Tab IV). Its rows are drawn in Figure 23 instead of Table 7.

13.5 Sensitivity against cost on MLGWSC-1 (historical)

The one place where sensitivity and cost share a contract is MLGWSC-1: the challenge's evaluation code on one month of test data per dataset, at a fixed false-alarm rate. Figure 23 joins the sensitive distance at one false alarm per month of each comparable-grade row to the inference-cost record that times it: 26 points from 3 papers. No comparable MLGWSC-1 entry states a latency, so there is no latency counterpart: the latencies above come from other benchmarks, populations and hardware.

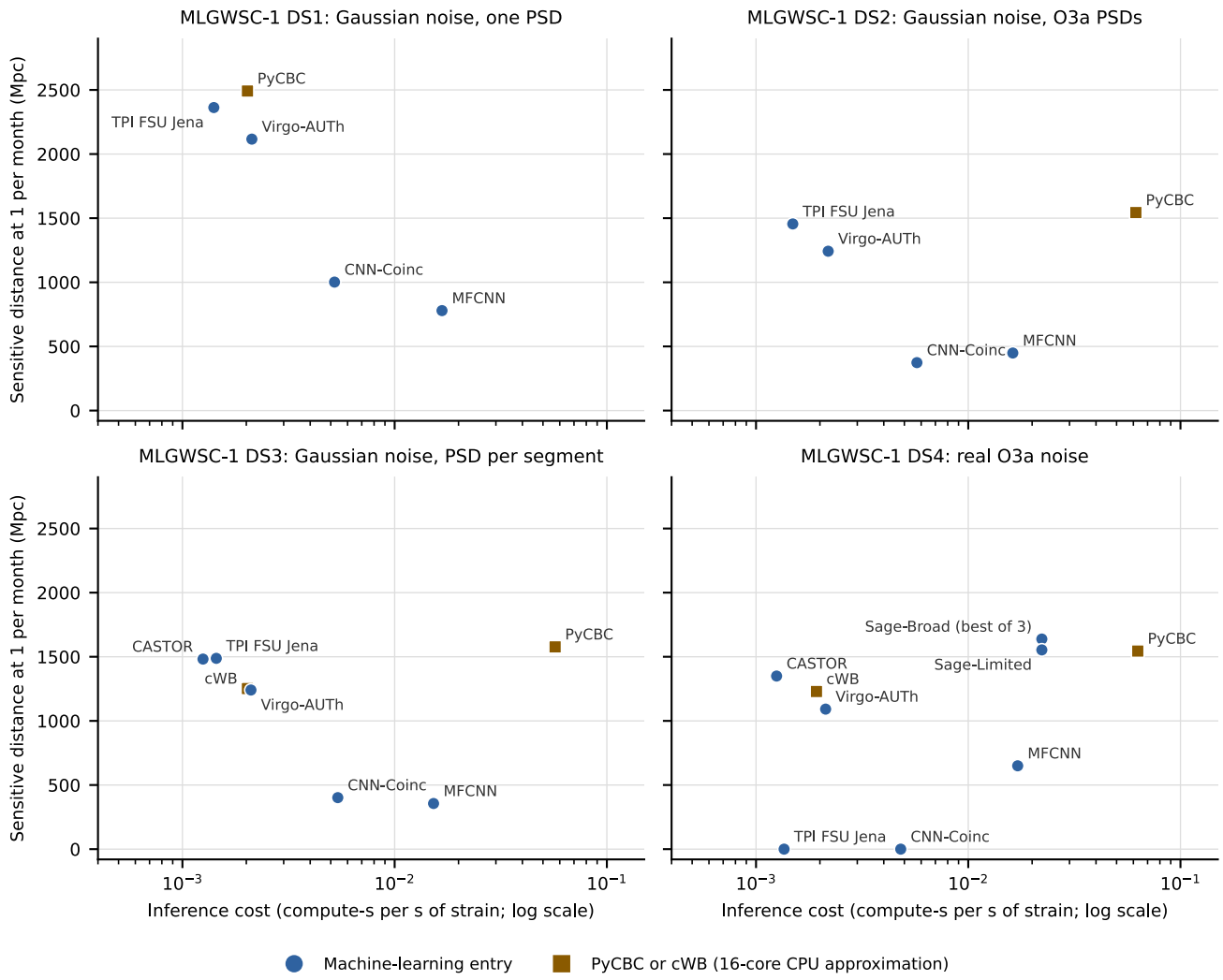


Figure 23: Historical record: sensitive distance at one false alarm per month against inference cost, a panel per MLGWSC-1 dataset, MLGWSC-1 rows only [8], [14], [72]. The challenge's entries were timed on its node, with one RTX 2070 Super per entry except MFCNN (eight), and PyCBC and cWB as 16-core CPU approximations (Tab IV); Sage (Quadro RTX 8000 and 16 CPU cores; Sec III, Sec V) and CASTOR (one A100; Tab I, Sec IV A) were timed on their authors' hardware. Points on different hardware share the axes, not a scale: no ratio between them is meaningful. Two DS4 entries have no detections at this rate and sit at 0 Mpc.

Table 5. Every stated latency: the value as the paper gives it, normalised to seconds where possible, with the hardware, grade, scope and mode, and the baseline the paper quotes. Area is the review section.

Model	Area	Value as stated	Seconds	Statistic	Hardware	Grade; scope; mode	Location	Baseline the paper quotes
`schafer2020` [5]	CBC	10.2 s between signal arrival and alert (conservative estimate at FAR 10 per month, true positives)	10.2 s	mean	unstated	stated; unstated; streaming	Abstract; Sec. V B; Sec. VI	PyCBC Live: 16 (reducible to 10 at double compute) s average latency (same hardware: unstated)
`gunny2021` [68]	CBC	1 s	1 s	single	unstated (V100 16GB GPUs per Methods B.4, count varies with Fig. 9 configuration)	stated; end to end; streaming	Sec. Discussion/conclusion; Methods B.4 and Fig. 9	-
`gunny2021` [68]	CBC	0.75 s	0.75 s	single	unstated	stated; end to end; streaming	Sec. II / Methods B.3, Fig. 11	-
`lin2021` [37]	CBC	about 20 seconds	20 s	single	moderate GPU-equipped personal computer (AMD RX 480 GPU, Intel i5 6500 CPU as stated for training; inference machine not separately specified)	stated; end to end; unstated	Abstract; Sec. III B; Sec. V	-
`marx2024` [9]	CBC	3.114 (+0.006 / -0.001) seconds	3.114 s	median	1x NVIDIA A30 GPU (24 GB)	stated; end to end; streaming	Tab. III; Sec. VIII	matched filtering pipelines (reported in the MDC study): 12.3 median, 41.4 p90 seconds (same hardware: unstated)
`marx2024` [9]	CBC	3.9 seconds	3.9 s	median	unstated	stated; end to end; streaming	Sec. VIII	matched filtering pipelines: 12.3 seconds (median) (same hardware: unstated)
`marx2024` [9]	CBC	4.3 seconds	4.3 s	p90	unstated	stated; end to end; streaming	Sec. VIII	matched filtering pipelines: 41.4 seconds (90th percentile) (same hardware: unstated)
`marx2024` [9]	CBC	less than 0.4 seconds (all computational steps combined: data reading/writing, GPU transfer, whitening, event identification)	0.4 s	single	1x NVIDIA A30 GPU (24 GB)	stated; end to end; streaming	Sec. VIII; Tab. III lower section	-
`mcleod2025` [16]	CBC	~0.5 seconds (inference step)	0.5 s	single	1x NVIDIA A100 GPU	stated; network; streaming	Sec. V E	-
`mcleod2025` [16]	CBC	1 second (unavoidable, from 1 s moving average of predictions)	1 s	single	unstated	stated; network; streaming	Sec. V E; Sec. VI	-
`gupta2026` [11]	CBC	~4.7 s	4.7 s	median	unstated	stated; end to end; streaming	Sec. V B	match-filtering CBC pipelines in production (Chaudhary et al. Table 2, O3 MDC): ~10 s (same hardware: no)
`rai2026` [99]	CBC	< 45 ms total pipeline (classifier + FastSHAP explainer)	-	max	standard laptop, Intel Core i7 class CPU, no GPU	stated; network; unstated	Sec. 4.2; Tab. 3; Tab. A.4	Aframe: ~3.1 s total pipeline latency (same hardware: no)
`baltus2022` [108]	early	around 2 s to load 300 s of data for 3 detectors with GWpy on the LIGO servers	2 s	single	LIGO servers (unstated)	stated; end to end; streaming	Sec. III A	-
`baltus2022` [108]	early	3.15 (GPU) / 3.64 (CPU) s per analysis step (load + PSD + whitening + peak removal + network)	3.15 s	single	GTX 750 GPU or unstated CPU	derived; end to end; streaming	Sec. III A	-
`baltus2022` [108]	early	20 seconds of delay when requiring 5 consecutive triggers with 5 s window steps	20 s	single	unstated	stated; end to end; streaming	Sec. III A (Performance of the network)	-
`skliris2024` [135]	burst	order of 1 second s	1 s	single	unstated	stated; end to end; streaming	Abstract / Sec. I	current LIGO-Virgo low-latency unmodelled burst searches: tens of seconds to minutes (hundreds of dedicated CPUs) latency (same hardware: no)
`cuoco2026` [137]	burst	about three seconds (conditioning reads ahead of what it emits)	3 s	single	unstated	stated; end to end; streaming	Sec. VII (Conclusion, closing paragraph)	-
`bayley2020` [164]	CW	~13 hours to efficiency curve and candidates after end of run, parallelised over 200 CPUs with Viterbi-map network trained and tested	4.68e+04 s	single	200 CPUs (+1 GPU)	stated; end to end; batch	Sec. VI	-
`bayley2020` [164]	CW	~9.5 hours total run time on S6 on 200 CPUs when a network trained on a previous run is reused	3.42e+04 s	single	200 CPUs	stated; end to end; batch	Sec. VI	-
`bayley2020` [164]	CW	~3.5 hours after the end of the run if narrow-banding is done periodically during the run (proposed)	1.26e+04 s	single	unstated	stated; end to end; batch	Sec. VI B	-
`essick2020` [212]	glitch	O(10^-1) sec after receiving features (online timeseries production)	-	single	unstated	stated; unstated; streaming	Sec. IV	-

Model	Area	Value as stated	Seconds	Statistic	Hardware	Grade; scope; mode	Location	Baseline the paper quotes
essick2020 [212]	glitch	~5 sec for feature extractors to process O(10^3) channels	5 s	single	unstated	stated; unstated; streaming	Sec. IV	KleineWelle (earlier iDQ feature extractor, O1-O2): ~32 sec (same hardware: unstated)
essick2020 [212]	glitch	within 8 sec of the candidate being reported to GraceDB (GW170817 noise transient)	8 s	single	unstated	stated; end to end; streaming	Sec. V (Examples, GW170817)	-
essick2020 [212]	glitch	within 7 sec (veto of the 4 February 2017 oLIB candidate)	7 s	single	unstated	stated; end to end; streaming	Sec. V (Examples, Fig. 8)	-
saleem2023 [242]	denoise	~1-2 s seconds, overall latency of cleaned output	-	single	unstated	stated; end to end; streaming	Abstract; Sec. VI B (Fig. 8, footnote 4)	-
saleem2023 [242]	denoise	approximately 1 second seconds, added by waiting for a future 1 s frame to avoid edge effects	1 s	single	unstated	stated; unstated; streaming	Sec. VI B; Fig. 7 caption	-
chatterjee2019 [273]	PE	0.0189 (about 0.018 in abstract) s per signal incl. feature extraction, 8192 sectors	0.0189 s	single	14 Intel Xeon CPU cores	stated; end to end; unstated	Abstract; Sec. IX	BAYESTAR: 10 s - 1 min s (same hardware: unstated)
chatterjee2022 [274]	PE	~1.2 s total inference (sample generation, KDE fit, density over adaptive mesh)	1.2 s	mean	1x P100 GPU	stated; end to end; unstated	Sec. 3; Abstract (around 1 sec)	BAYESTAR: a few seconds s (same hardware: unstated)
dax2022 [28]	PE	approximately 20 seconds (low-latency Dingo samples, no importance sampling)	20 s	single	unstated	stated; network; batch	Main text footnote 3; Discussion ('a matter of seconds'); Intro ('typically under a minute')	-
chatterjee2023skyloc [275]	PE	about 1.2 s total (5000 samples, KDE, adaptive refinement)	1.2 s	mean	unstated	stated; end to end; unstated	Sec. 5	BAYESTAR: about 1 second average in O3 s (same hardware: unstated)
dax2024 [255]	PE	~1 second (inference latency for pre-merger estimates)	1 s	single	unstated (H100 used for IS timing)	stated; unstated; streaming	Fig. 2 caption	Bayestar: localization in seconds (no number) s (same hardware: unstated)
kofler2025 [256]	PE	under five seconds (initial low-latency samples)	5 s	single	unstated (64 CPU cores used for subsequent IS)	stated; network; batch	Results; Sec. on inference	-
cuoco2020 [313]	other	8 seconds	-	single	unstated	stated; unstated; streaming	Sec. 2.2 (iDQ paragraph, GW170817)	-
que2021 [312]	other	39.7 ms	0.0397 s	single	1x Intel E2620 CPU (F32, AVX2)	stated; network; streaming	Tab. III	-
que2021 [312]	other	32.1 ms	0.0321 s	single	1x NVIDIA TITAN X GPU (F32, cuDNN)	stated; network; streaming	Tab. III	-
que2021 [312]	other	0.40 microseconds	4e-07 s	single	1x Xilinx Alveo U250 FPGA (16-bit fixed)	stated; network; streaming	Tab. III	same LSTM autoencoder on Intel E2620 CPU (and TITAN X GPU 32.1 ms): 39.7 ms (same hardware: no)
que2021 [312]	other	0.867 microseconds	8.67e-07 s	single	1x Xilinx Alveo U250 FPGA at 300 MHz	stated; network; streaming	Tab. IV (This work, four layers: 32,8,8,32)	prior FPGA LSTM anomaly detection, ref (Kintex7 K410T, single layer): 4.27 us (same hardware: no)
que2021 [312]	other	0.343 microseconds	3.43e-07 s	single	1x Xilinx Alveo U250 FPGA at 300 MHz	stated; network; streaming	Tab. IV (This work, single layer, 32 hidden)	prior FPGA LSTM physics design, ref (KU115, single layer, 16 hidden units): 1.35 us (same hardware: no)
chan2024 [301]	other	<= 66 seconds, between release of the public alert and completion of GWSkyNet analysis, for 95% of O4 public alerts	66 s	single	unstated	stated; end to end; streaming	Sec. 3 (O4 results, text near Table 2)	-

Table 6. Inference cost, the headline record per model (126 models, 270 records). cs/s: compute-seconds per second of strain, where the paper gives the duration analysed.

Model	Area	Value as stated	cs/s	Hardware	Grade; scope; mode	Location	Records
`george2018plb` [29]	CBC	85 milliseconds per second of data (each CNN)	0.085	1x CPU (model unstated)	stated; network; unstated	Sec. 3 (Results), final paragraph	2
`george2018` [1]	CBC	6.7 milliseconds per input of 1 second duration (simple classifier and predictor, 2 MB each)	0.0067	1x Intel Core i7-6500U CPU (2 cores, per Fig. 14 caption)	stated; network; batch	Sec. V (Discussion); Fig. 14	4
`george2018b` [3]	CBC	85 milliseconds per second of data (each CNN)	0.085	1x CPU core (model unstated)	stated; network; unstated	Sec. III (Results), final paragraph	2
`kim2019` [101]	CBC	tens of milliseconds to score ~45,000 samples (6 features each)	-	unstated	stated; network; batch	Abstract; Sec. II; Sec. V	1
`lin2019` [42]	CBC	approximately 0.027 seconds per sample (4 s single-detector sample at 16384 Hz)	0.0135	1 core of an Intel Xeon E7-4830 2.13 GHz CPU (one-core computation, Sec. 5)	stated; unstated; unstated	Sec. 4 'Efficiency' paragraph; Sec. 5	1
`li2020` [46]	CBC	18.765 milliseconds per sample (0.03 s detection window)	-	12x Intel Xeon E7-4830 CPUs, 2.31 GHz	stated; unstated; unstated	Sec. 4 (end, after Sec. 4.2)	1
`schafer2020` [5]	CBC	206 ms per network pass covering 250 ms of data	0.824	Intel i7-6600U dual-core (4 threads) laptop CPU	stated; network; streaming	Sec. V B	1
`gunny2021` [68]	CBC	2 seconds of strain per second (1000 inferences per second)	2	4x NVIDIA T4	stated; end to end; batch	Sec. HEPCloud scaling, Fig. 10; Methods B.6	2
`huerta2021` [64]	CBC	7 min for the whole of August 2017 (H1 and L1)	0.01004	64x NVIDIA V100 (16 nodes x 4, HAL cluster); ensemble of 4 networks, 16 GPUs per network	stated; end to end; batch	Sec. Results, Fig. 2a, Fig. 6	2
`krastev2021` [31]	CBC	~3 milliseconds to process 10 s of GW data (with both the detection and regression CNNs)	-	GPUs (model and count unstated)	stated; network; unstated	Sec. 3.4	2
`lin2021` [37]	CBC	about 20 seconds per 8 s data chunk (MC sampling of size 4096, batch size 32)	2.5	moderate GPU-equipped PC (the stated training machine has an AMD RX 480 GPU and an Intel i5 6500 CPU; the inference machine is not separately specified)	derived; end to end; batch	Sec. III B 'Performance against Real Events'	1
`schafer2021a` [6]	CBC	about 1 hour to evaluate an entire month of data	0.00139	1x NVIDIA RTX 2070 SUPER	stated; network; batch	Sec. II B	1
`schafer2021b` [7]	CBC	< 12 hours to generate O(1000) months of time-shifted background (sub-network C on stored sub-network A/B outputs)	-	1x NVIDIA RTX 2070 Super	stated; network; batch	Sec. III D (Background Estimation)	1
`wei2021ensemble` [63]	CBC	within 14 hours for 200 hours of data	0.28	4x NVIDIA V100 (one HAL node)	stated; end to end; batch	Sec. IV (Results, summary); Fig. 5 caption	1
`alhassan2022` [93]	CBC	4 minutes and 57 seconds per hour of data min:s per hour of ET mock data	0.0825	Core i7 MacBook Pro (16 GB, 2.6 GHz), CPU, not parallelised	stated; end to end; unstated	Sec. 4 (continuous-data inference) and Sec. 5 (conclusions)	1
`chaturvedi2022` [65]	CBC	within 50 seconds for one month (August 2017) of H1+L1 advanced LIGO data	0.00299	160x NVIDIA A100 (20 ThetaGPU nodes x 8 GPUs)	stated; unstated; batch	Abstract; Sec. 1; Fig. 7 caption; Sec. 3.4	1
`ma2022` [38]	CBC	less than 2 milliseconds ('mini-seconds' in the paper; typical delay time of the ensemble model)	-	1x NVIDIA 1080Ti GPU	stated; network; unstated	Sec. III C 2 (Base learner construction), before subsection 3 'Sub-ensemble model construction'	1
`mlgwsc2022` [8]	CBC	43331 s per month of two-detector data (mean of foreground 42842 and background 43820)	0.1337	8x NVIDIA RTX 2070 Super (all 8 GPUs used), 2x Intel Xeon Silver 4215	stated; end to end; batch	Tab. IV (A: MFCNN, dataset 1); Sec. V C	22
`alhassan2023pymerger` [94]	CBC	1.9 minutes per hour of data (TSDCD, three sub-detectors) min per hour of data	0.0317	Core i7 MacBook Pro (16 GB, 2.6 GHz) CPU	stated; end to end; unstated	Sec. 5; Sec. 6 (PyMerger description); conclusions	2
`andrescarcasona2023` [78]	CBC	of the order of 2,000 CPU-hours for the entire O3 scan CPU-hours	0.5376	Intel Xeon E5-2680 v4 @ 2.40 GHz (core count unstated)	stated; end to end; batch	Sec. IV (Results)	1
`jadhav2023` [81]	CBC	~16 hours run-time for multi-GPU inference on images of 8.8 days of H1 and L1 data	0.303	4x NVIDIA Tesla P100	stated; network; batch	Sec. VI B	2
`nousi2022` [12]	CBC	less than 2 hours to evaluate one month of test data	-	1x NVIDIA A6000 GPU (same as training)	stated; end to end; batch	Sec. I (Introduction)	1
`tian2023` [66]	CBC	165 seconds	0.00535	1024x NVIDIA A100 (Polaris)	stated; network; batch	Sec. 3 Results, 'AI performance'	4
`tian2024gnn` [67]	CBC	3.19 hours	0.00466	128x NVIDIA A100 (Polaris)	stated; network; batch	Methods 'AI inference at scale'	4
`trovato2023` [74]	CBC	about 4 hours per model for the full 4,216,489 s data set	0.00342	NVIDIA Tesla V100S (count unstated)	stated; end to end; batch	Sec. 5.2	4
`badger2024` [87]	CBC	24.6 s to reconstruct all 35 O3b confident events s	-	AMD EPYC 7502 CPU	stated; network; batch	Sec. III / Sec. IV	2
`marx2024` [9]	CBC	about 500 seconds of two-detector data analysed per second per GPU	0.002	NVIDIA 16 GB Tesla V100, per GPU (DGX server with 8x V100, Triton inference server)	stated; end to end; batch	Sec. VIII (page 7)	3
`menendez2024` [79]	CBC	of the order of 2000 CPU hours per search (entire O3 analysis)	0.5376	Intel Xeon E5-2680 v4 2.4 GHz CPU (core count unstated)	stated; end to end; batch	Sec. 5 (Results)	1
`wang2023tf` [91]	CBC	0.1868 seconds for 173 GW signals (approximately 1 ms each)	-	unstated	stated; network; batch	Sec. II (identification time); footnote 7	1
`zelenka2024` [69]	CBC	~40 times ratio of estimated PyCBC runtime to this work's runtime (datasets 2-4)	-	8x Intel Xeon Silver 4215 cores + 8x NVIDIA RTX 2070 (challenge hardware, for the ML runs)	stated; end to end; batch	Sec. IV B	2

Model	Area	Value as stated	cs/s	Hardware	Grade; scope; mode	Location	Records
mcleod2025 [16]	CBC	16,457 inferences per second required for the template bank at 16 Hz with the paper's clustering	0.4773	1x NVIDIA A100 GPU	derived; network; streaming	Sec. V E	2
nagarajan2025 [14]	CBC	≈ 16 hours to generate triggers on 1 month of testing data	0.02222	1x NVIDIA Quadro RTX 8000 GPU plus 16 CPU cores	stated; end to end; batch	Sec. III (preamble, before III A); Sec. V 'Evaluation speed'	1
tiki2025 [80]	CBC	under 5.0 hours for 22.7 days of noise livetime (plus 13 events)	0.00918	1x NVIDIA A100 (Delta)	stated; end to end; batch	Sec. 2.3 'Benchmark on real O3b data'; footnote 1	2
yamamoto2025 [70]	CBC	2802 s runtime, foreground, MLGWSC-1 dataset 1	-	Intel Xeon Platinum 8468 (2.1 GHz, 48 core) + 1x NVIDIA H100 (GPU count implied single, stated as 'a GPU NVIDIA H100')	stated; end to end; batch	Tab. III	3
chatterjee2026 [72]	CBC	≈ 0.9 hrs for one month of continuous MLGWSC-1 data	0.00125	1x NVIDIA A100 80 GB	stated; unstated; batch	Tab. I; Sec. IV A	2
gupta2026 [11]	CBC	~450 s of two-detector strain analysed per second per GPU	0.00222	2x NVIDIA A30 (24 GB), Triton IaaS via hermes	stated; end to end; batch	Sec. V A	3
mobilia2026 [86]	CBC	~3 ms per image (TT-SNR map)	-	off-the-shelf CPUs, no particular optimisation (model unstated)	stated; network; unstated	Conclusions (Sec. IX)	1
rai2026 [99]	CBC	34.11 ms per 0.25 s window (100-run average)	-	standard laptop, Intel Core i7 class CPU, no GPU	stated; network; unstated	Sec. 4.2; Tab. 3; Tab. A.4	1
weerasinghe2026 [62]	CBC	3.04 ms per event (100 ms post-merger window, ensemble of 5 CNNs)	-	unstated (GPU with host-device transfers implied)	stated; unstated; unstated	Abstract (3.0 ms); Tab. 1; Sec. 5.3.1	2
baltus2021 [106]	early	~0.005 s to analyse one 50 s frame	0.0001	1x Nvidia GeForce RTX 2070 SUPER	stated; network; unstated	Sec. IV A (Performance of the CNNs); footnote 4	1
baltus2021b [107]	early	0.005 s to analyze a 50 second frame	0.0001	1x Nvidia GeForce RTX 2070 SUPER	stated; network; unstated	Sec. IV (Results and discussion)	1
wei2021 [114]	early	8 ms per prediction (INT8 quantized ResNet-50, batch of 16 images)	0.008	Intel Xeon E5-2620 CPU (AVX2), count unstated	stated; network; batch	Sec. 4	2
yu2021early [112]	early	~30 ms to classify one 256 s time series (CNN-class)	0.000117	1x NVIDIA Tesla P100-PCIE-16GB	stated; network; unstated	Fig. 12, Sec. VI (Discussion)	2
baltus2022 [108]	early	0.01 s to predict one 300 s frame (up to 3 detectors)	0.002	1x GeForce GTX 750 GPU	stated; network; streaming	Sec. III A (Performance of the network)	3
martins2024 [109]	early	27.53 ± 3.69 ms per sample	9.18e-05	1x AMD EPYC 7551P 32-core (all 32 cores)	stated; network; unstated	Tab. 3, Sec. 5.2	14
papigkiotis2026 [111]	early	~0.1 s per 40 s observational time window (TSFEL feature extraction)	0.025	CPU (model and count unstated)	stated; unstated; streaming	Sec. V C 1	2
lopez2021b [116]	burst	3 ms per 2 s long image	0.0015	1x Nvidia Quadro P5000 (named in the same sentence as training)	stated; network; batch	Sec. VIII (Conclusion)	2
meijer2023cosmicstrings [145]	burst	10 milliseconds per example (three data streams of 8 seconds)	0.00125	Intel E5-2670 CPU and 1x NVIDIA Tesla V100 16GB GPU (the machine the CNNs were run on, Sec. IV B)	stated; network; unstated	Sec. V A (Numerical Results); Sec. VI (Conclusions)	1
modafferi2023 [144]	burst	c_CNN ≈ 4 × 10^-4 s per sample, averaged over a batch of 10^4 via Model.predict_on_batch	-	1x Tesla V100-PCIE-32GB	stated; network; batch	Sec. III E	2
attadio2024 [143]	burst	20 minutes (denoise O(5000) maps)	-	1x NVIDIA L40S 48 GB	stated; network; batch	Sec. IV F (Computational load); App. C	3
morales2024 [122]	burst	about 5 s for testing (test set of the phenomenological-waveform analysis)	-	1x NVIDIA T4 (Google Colaboratory)	stated; network; batch	Sec. 3.1.3	1
raikman2024 [136]	burst	4,500 GPU hours	5.13e-05	GPU (model and count unstated)	stated; unstated; batch	Sec. III C (Computational limitations)	1
skliris2024 [135]	burst	51 ms per 1 s of whitened data	0.051	3.5 GHz Xeon E3-1240v5 quad-core CPU, 32 GB RAM	stated; network; unstated	Sec. V B (Inference and Training Times)	3
cuoco2026 [137]	burst	about 10 ms per second of analyzed data, two detectors together	0.01	1 core of 3.0 GHz AMD EPYC 7313	stated; end to end; streaming	Sec. VII (Conclusion, closing paragraph)	2
ratner2026 [138]	burst	5 hours for three billion windows	-	1x T4 GPU	stated; network; batch	Sec. VI (Impact of trigger)	1
cain2025 [148]	anomaly	164.4 ± 1.0 ms per 32-second segment (total processing)	0.00514	1x AMD Ryzen 9 9950X CPU, 64 GB RAM, no GPU	stated; end to end; unstated	Sec. 4.4 (Computational Efficiency)	3
dreissigacker2019 [155]	CW	58.8 s (full all-sky DNN search, T = 1e5 s)	0.000588	unstated	stated; network; batch	Tab. VII	5
miller2019 [171]	CW	about 2 minutes (CNNs analysing full data set and generating triggers) min per 1 week of data, Hz	0.000198	unstated	stated; unstated; batch	Sec. VI (Search for a remnant of GW170817)	4
bayley2020 [164]	CW	~9 hours, narrow-banding SFTs (single CPU)	0.000799	1 CPU	stated; end to end; batch	Tab. II	6
dreissigacker2020 [156]	CW	roughly two times faster than the matched-filtering search (training, executing and following up, T = 1e5 s) ratio of total computing cost	-	unstated	stated; end to end; batch	Sec. VI (Discussion)	1
yamamoto2021cw [169]	CW	6.4e6 x (FAP_EP/1e-2) s total NN time (Eq. 5.4)	0.3815	1x GeForce GTX 1080Ti	stated; network; batch	Sec. V A, Eq. 5.4	5
yamamoto2022cw [170]	CW	1.02e8 s total CNN time (Eq. 5.6; abstract: O(1e8) s)	6.08	1x GeForce GTX 1080Ti	stated; network; batch	Sec. V, Eq. 5.6	4
yamamoto2023 [172]	CW	7.66 x 10^-2 s for 500 test data (residual network; shallower CNN 2.54 x 10^-2 s, deeper CNN 7.96 x 10^-2 s; the non-Gaussian-statistic grid-search baseline takes 1.13 x 10^4 s on a CPU)	-	GPU Quadro GV100 (16.6 TFLOPS single precision)	stated; network; batch	Tab. IV; Sec. IV C	4
joshi2024large [159]	CW	~27 s to evaluate ~50 000 input windows across 20-1000 Hz (batch size 256) for a ten-day, two-detector (H1, L1) search	3.13e-05	unstated for inference (training on NVIDIA A100-SXM4 40 GB)	stated; network; batch	Sec. IV C	1

Model	Area	Value as stated	cs/s	Hardware	Grade; scope; mode	Location	Records
tenorio2025g2net_cw [163]	CW	5 days (JunKoda), cK ~ 50 s per band days on a GPU for the full dataset	-	1x GPU (model unstated)	stated; unstated; unstated	Sec. 5 (Comparison to CW searches), after Eq. 8	6
george2017tl [174]	glitch	a few milliseconds ms per input spectrogram	-	GPU (model unstated)	stated; network; unstated	Sec. 4 (Discussion)	1
llorensmonteagudo2018 [218]	glitch	about 36 hours hours to reconstruct 3000 glitches, each in one 16384-sample patch, by all three learned dictionaries	43.2	1x AMD Phenom II x4 processor, 12 GB RAM	stated; network; batch	Sec. 6 (Conclusions)	1
razzano2018 [217]	glitch	few microseconds per image	-	unstated	stated; unstated; unstated	Sec. 6 (Discussion)	1
essick2020 [212]	glitch	O(sec) runtime of evaluation and calibration jobs	-	unstated	stated; network; streaming	Sec. IV (online mode)	1
choudhary2022 [204]	glitch	30 sec per SGP map generated	-	1 core of Intel Xeon Gold 6142 CPU	stated; unstated; batch	Tab. II	3
yan2022gan [182]	glitch	5 minutes to generate 1000 512x512 spectrogram images (ProGAN generator)	-	1x NVIDIA Tesla V100 (32GB) (stated for all experiments; not restated for generation)	stated; unstated; unstated	Sec. 4.1 (Data augmentation)	1
laguarta2023 [199]	glitch	1 h of data in 11 s s to FD-encode 1 h of auxiliary-channel data (numba, O(N^2 log N))	-	unstated (Fig. 2 benchmarks on Intel Xeon E5-2630 v4 CPU)	stated; end to end; batch	Sec. II B (Fractal dimension), text after Fig. 2	1
arutkeerthi2024 [207]	glitch	30 ms to classify one signal (glitch or not) from a resized chirp volume	0.005	unstated	stated; network; unstated	Sec. VIII (Conclusion)	2
licciardi2024 [181]	glitch	0.211 ± 0.003 s per sample (Q-transform, GwPy)	-	CPU (model unstated)	stated; end to end; unstated	Sec. IV ('Computational Cost' subsection)	4
deshmukh2026 [196]	glitch	between 10-15 mins (evaluation including generation of t-SNE plots)	-	1x DGX A100 GPU (same resource)	stated; end to end; batch	Sec. 3.1	1
dooney2026 [224]	glitch	under 22 seconds to generate 1000 glitches	-	CPU (model unstated)	stated; network; batch	Abstract; Sec. IV; Sec. VI (Conclusions)	2
fernandes2026 [189]	glitch	0.04 s per glitch (ResNet, CPU, averaged over 20 samples; tree-based baselines < 0.01 s)	-	CPU (model unstated)	stated; network; unstated	Tab. V	5
manoharan2026 [222]	glitch	figure only; per-model points not read seconds per sample (total predict time / N, N=10112)	-	unstated (identical batch size and hardware settings across models)	estimated; end to end; batch	Fig. 5, Sec. C (Performance vs. Inference Time)	1
shen2017 [24]	denoise	less than 100 milliseconds per 1 s of input data	0.1	unstated (a GPU)	stated; network; unstated	Results, computational performance paragraph (p. 3-4)	1
ormiston2020 [25]	denoise	a few seconds seconds for inference on 1024-3600 s of data	-	1x NVIDIA TITAN X (Pascal) GPU	stated; unstated; batch	Sec. 'Noise subtraction pipeline' (unnumbered), last paragraph before 'Pipeline performance on LIGO data'	1
torresforne2020 [250]	denoise	~1.5 s to process 1 s of data sampled at 8192 Hz	1.5	MacBook Pro, 2.3 GHz Intel Core i5, 4 cores, 16 GB RAM	stated; unstated; unstated	Sec. V (Summary)	1
chatterjee2021 [226]	denoise	around 0.5 milli-seconds to extract a waveform	0.002	1x Tesla K40 GPU	stated; network; unstated	Sec. IV (Discussion and Conclusion)	1
mogushi2021nnetfix [249]	denoise	of the order of a few seconds seconds CPU time to reconstruct the data, for gate durations up to hundreds of milliseconds	-	CPU (model and count unstated)	stated; unstated; batch	Sec. 5 (Conclusion)	1
bacon2022 [228]	denoise	approximately 20 ms to denoise one time series instance	0.02	unstated (presumably the V100 GPU)	stated; network; unstated	Sec. 3.5 (Model size and training time)	1
xu2024denoise [254]	denoise	within 0.01 seconds (to predict the merger time for 30 days of inspiral data)	-	unstated	stated; unstated; unstated	Sec. I (Introduction)	1
dooney2025 [230]	denoise	0.14 s per glitch sample on a CPU	0.07	CPU (model unstated)	stated; unstated; unstated	Sec. IV A	2
kiendrebeogo2025virgo [243]	denoise	few seconds to minutes to perform cleaning on several hours of data seconds to minutes per several hours of data	-	1x NVIDIA A100 Tensor Core GPU (given as an example)	stated; unstated; batch	Sec. I (Introduction)	1
chatterjee2019 [273]	PE	0.0003 s per test sample (classification)	-	14 Intel Xeon CPU cores	stated; network; unstated	Sec. IX	1
shen2019pe [268]	PE	within two (2) milliseconds per event (median values and posterior distributions)	-	1x NVIDIA V100	stated; network; unstated	Abstract; Sec. 4.2	1
delaunoy2020 [265]	PE	less than a minute per event	-	unstated	stated; unstated; unstated	Sec. 5 Results	1
green2020 [26]	PE	5,000 posterior samples per second	-	unstated	stated; network; batch	Sec. on inference for GW150914; Conclusions	1
marulanda2020 [277]	PE	about 6% greater (FCNN incl. spectrogram computation vs time-domain CNN) percent relative execution time	-	unstated	stated; end to end; unstated	Abstract; Results (execution-time paragraph); Conclusions	1
dax2021 [27]	PE	20 seconds per event (50,000 posterior samples, 30 GNPE iterations)	-	1x NVIDIA A100 (implied by footnote 3, which contrasts V100's roughly a minute with 20 seconds)	stated; unstated; batch	Abstract; Results (GWTC-1 analysis); footnote 3	2
kolmus2021 [276]	PE	roughly 5 minutes of computing time (100,000 importance-sampling steps)	-	16-core AMD Ryzen 5950X CPU, NVIDIA 3090 RTX GPU, 64 GB RAM	stated; end to end; batch	Fig. 4 caption; Sec. IV C	2
chatterjee2022 [274]	PE	a few ms to generate 5000 samples	-	1x P100 GPU	stated; network; unstated	Sec. 3	1
dax2022 [28]	PE	less than 1 hour per event, total (IMRPhenomXPHM)	-	1x NVIDIA A100 GPU + 64 CPU cores	stated; end to end; batch	Main text, Results (Dingo-IS timing) and Discussion	5

Model	Area	Value as stated	cs/s	Hardware	Grade; scope; mode	Location	Records
gabbard2022pe [263]	PE	1 × 10^-1 seconds for 8000 posterior samples (VItamin)	-	unstated (1x NVIDIA Tesla V100 named in Methods for training; consumer GPUs said to be equally fast)	stated; network; batch	Tab. I	1
mcleod2022 [269]	PE	~1.4 milliseconds per sample prediction	0.0056	1x NVIDIA P100 GPU and Intel Gold 6140 CPU	stated; network; unstated	Sec. III (Prediction Results); Sec. IV (Discussion); Sec. V (Conclusion gives ~1 ms)	1
andrescarcasona2023pe [270]	PE	0.05 s to generate 10,000 posterior samples	-	1x NVIDIA GeForce RTX 2080 Ti	stated; network; unstated	Sec. 5 (real events, Fig. 4)	1
chatterjee2023skyloc [275]	PE	a few ms (sample generation)	-	NVIDIA P100 12GB PCIe	stated; network; unstated	Sec. 1	1
ravichandran2023 [284]	PE	0.82 – 2 ms per batch of 500 images (testing time, both problems)	-	48 cores of AMD EPYC 7352 CPU	stated; network; batch	Sec. III A (Training and Testing)	1
badanerin2024 [264]	PE	around 4 s per waveform (N=8000 samples)	-	1x NVIDIA A40	stated; network; unstated	App. A; Tab. I	2
dax2024 [255]	PE	roughly one second for 1e4 effective samples (importance sampled)	-	1x NVIDIA H100 GPU	stated; end to end; batch	Main text, Experiments; Abstract ('one second')	6
hu2024 [260]	PE	within a second (network sampling) + O(1)-O(10) CPU hours importance sampling second / CPU hours	-	1x NVIDIA 3080Ti GPU (sampling); CPU for IS	stated; end to end; batch	Sec. 3.1 Parameter estimation	2
benedikt2025 [280]	PE	10.6 ± 0.5 ms per event, batch size 1 (standard forward pass)	0.0106	1x NVIDIA A100	stated; network; unstated	Tab. 9; Sec. 5.4	2
chan2025lens [257]	PE	tens of seconds per event s per event	-	unstated	stated; unstated; unstated	Sec. IV; Abstract	2
iwanaga2025 [282]	PE	approximately six minutes (total estimation time incl. Monte Carlo uncertainty evaluation)	-	unstated	stated; end to end; batch	Sec. III A and Conclusions	1
khalouei2025 [281]	PE	on the order of 100 ms per event (average over 100 repeated inferences)	-	unstated (presumed the training GPU)	stated; network; batch	Sec. 3; Sec. 4	1
kofler2025 [256]	PE	five to ten minutes minutes for 10^5 importance-sampled posterior samples (phase reconstruction + IS)	-	64 CPU cores	stated; end to end; batch	Main text (Results); Supplemental Material (inference)	1
lanchares2025 [261]	PE	1.13 ± 0.13 seconds per 10,000 posterior samples	-	1x NVIDIA A100	stated; network; batch	Results text, Sec. III (GP15 vs Bilby comparison paragraph)	1
papalini2025 [272]	PE	~1 second to generate a full posterior with 10^4 samples	-	unstated	stated; network; batch	Sec. 4 Results; Conclusion ('~1 s')	2
emma2026 [266]	PE	approximately an order of magnitude larger than Whittle-likelihood runs ratio of total wall time	-	unstated	stated; end to end; batch	Sec. IV (GW analysis wall time)	2
luo2026 [259]	PE	less than 20 minutes per event (echo search / Bayesian inference)	-	1x NVIDIA RTX 5090	stated; unstated; batch	Sec. I (Introduction); Sec. III (Results); Sec. IV (Conclusion)	1
sakellariou2026 [283]	PE	approximately 36 ms per strain (DeepModel)	0.036	unstated	stated; network; unstated	Sec. 3 (Results, computational-efficiency paragraph)	2
zhang2026 [258]	PE	approximately 0.4 (less than 0.4) seconds for complete posterior sampling of a single event	-	1x NVIDIA RTX A6000	stated; network; batch	Sec. on training/implementation; Sec. C 'Computational performance comparison'; Conclusion	2
zhang2022emri [289]	space	within one second seconds to classify three months' data (7,864,320 s)	1.27e-07	1x GPU Tesla V100 PCIe 16 GB (the implementation hardware of footnote 2; not restated for inference)	stated; unstated; batch	Sec. VI (Conclusions)	1
liu2023 [291]	space	SSL and LSTM time cost more than 70% lower than LDASoft (LISA MDC) percent reduction in time cost	-	unstated	stated; unstated; unstated	Sec. IV, text around Fig. 9	1
ruan2023 [285]	space	within ten seconds (a year-long Sangria dataset) s per year of data	3.17e-07	unstated	stated; unstated; batch	Sec. VI	2
ruan2023pe [287]	space	about twenty seconds to draw 50,000 posterior samples (4 parameters) for one MBHB signal	4.07e-05	unstated	stated; unstated; unstated	Abstract; Sec. I; Sec. IV; Sec. V	1
zhao2023 [288]	space	1.9 s per batch of 256 samples (474 days of data)	4.64e-08	unstated (training on a single NVIDIA V100)	stated; network; batch	Tab. 3; Sec. 2.7	3
ruan2024 [286]	space	0.01 seconds per 4.7-day LISA segment (81,920 samples at 5 s cadence)	2.44e-08	NVIDIA Tesla A40 (1; 'all training and testing' on it)	stated; unstated; unstated	Sec. IV (Results, p.~8) and Sec. V	1
cuoco2020 [313]	other	less than a few milliseconds ms per posterior	-	unstated	stated; network; unstated	Sec. 5.1 (CVAE, ref)	4
mcginn2021 [302]	other	fractions of a second s to generate waveforms	-	unstated	stated; network; unstated	Sec. 1 (Introduction)	1
powell2022 [304]	other	less than a second s to convert GAN-generated glitch images into time series	-	unstated	stated; unstated; unstated	Sec. 4 (Applications)	1
dooney2024 [305]	other	approximately 18 s to generate 2,500 BBH signals	-	CPU (model unstated)	stated; network; batch	Sec. IV C (Fitting-factor results)	2
xu2024 [311]	other	< 10^-2 seconds per signal extraction	6.25e-08	unstated	stated; unstated; unstated	Sec. V (ideal-case results)	1

Table 7. Comparisons with baseline pipelines, one row per distinct baseline in a paper (MLGWSC-1's own timings are in Figure 23). "Same hardware" is the paper's statement as read; "unstated" means the paper does not say.

Model	Quantity	The model, as stated	Hardware	Baseline pipeline	Baseline, as stated	Same hardware	Location
george2018plb [29]	Inference cost	85 milliseconds per second of data (each CNN)	1x CPU (model unstated)	matched filtering used for comparison	over 2 s to analyse 1 s inputs on our CPU	yes	Sec. 3 (Results), final paragraph
george2018 [1]	Inference cost	6.7 milliseconds per input of 1 second duration (simple classifier and predictor, 2 MB each)	1x Intel Core i7-6500U CPU (2 cores, per Fig. 14 caption)	time-domain matched filtering, same template bank (estimated by the authors)	1.1 s per 1 s input on the same CPU (2 cores)	yes	Sec. V (Discussion); Fig. 14
george2018b [3]	Inference cost	85 milliseconds per second of data (each CNN)	1x CPU core (model unstated)	time-domain matched filtering used for comparison	over 2 s to analyse 1 s inputs on our CPU	yes	Sec. III (Results), final paragraph
rebei2018 [33]	Training cost	thirty minutes (training stage, TensorFlow + Horovod distributed training)	NVIDIA Tesla V100 GPUs on Blue Waters and the NCSA Innovative Systems Lab (count unstated)	same training without Horovod scaling	ten hours	unstated	Sec. VI (list of algorithm improvements)
kim2019 [101]	Inference cost	tens of milliseconds to score ~45,000 samples (6 features each)	unstated	GstLAL ranking (comparison is sensitivity only)	not stated	unstated	Abstract; Sec. II; Sec. V
lin2019 [42]	Inference cost	approximately 0.027 seconds per sample (4 s single-detector sample at 16384 Hz)	1 core of an Intel Xeon E7-4830 2.13 GHz CPU (one-core computation, Sec. 5)	matched filtering	about 261 seconds per sample	yes	Sec. 4 'Efficiency' paragraph; Sec. 5
li2020 [46]	Inference cost	18.765 milliseconds per sample (0.03 s detection window)	12x Intel Xeon E7-4830 CPUs, 2.31 GHz	matched filtering	1899.350 ms per sample	yes	Sec. 4 (end, after Sec. 4.2)
schafer2020 [5]	Latency	10.2 s between signal arrival and alert (conservative estimate at FAR 10 per month, true positives)	unstated	PyCBC Live	16 (reducible to 10 at double compute) s average latency	unstated	Abstract; Sec. V B; Sec. VI
schafer2020 [5]	Inference cost	206 ms per network pass covering 250 ms of data	Intel i7-6600U dual-core (4 threads) laptop CPU	PyCBC Live (same BNS bank, 1960 templates per detector)	3920 templates at 6300 templates per core in real time = 0.622 cores core-seconds per second (derived from stated numbers)	unstated	Sec. V B
krastev2021 [31]	Inference cost	~3 milliseconds to process 10 s of GW data (with both the detection and regression CNNs)	GPUs (model and count unstated)	conventional matched filtering and Bayesian inference	not stated (qualitative: computationally prohibitive in real time)	unstated	Sec. 3.4
chaturvedi2022 [65]	Inference cost	within 50 seconds for one month (August 2017) of H1+L1 advanced LIGO data	160x NVIDIA A100 (20 ThetaGPU nodes x 8 GPUs)	same ensemble in TensorFlow (no TensorRT)	3X slower than the TensorRT ensemble (about 150 s by arithmetic 3 x 50 s) s for the month	yes	Abstract; Sec. 1; Fig. 7 caption; Sec. 3.4
alhassan2023pymerger [94]	Inference cost	1.9 minutes per hour of data (TSDCD, three sub-detectors) min per hour of data	Core i7 MacBook Pro (16 GB, 2.6 GHz) CPU	Chaturvedi et al. 2022 NVIDIA TensorRT AI ensemble (cited, not run here)	50 50 s for one month of advanced LIGO data	no	Sec. 5; Sec. 7 (PyMerger description); conclusions
andrescarcasona2023 [78]	Inference cost	of the order of 2,000 CPU-hours for the entire O3 scan CPU-hours	Intel Xeon E5-2680 v4 @ 2.40 GHz (core count unstated)	known matched-filtering pipelines	stated as a major improvement in CPU-time no number given	unstated	Sec. IV (Results)
tian2024gnn [67]	Inference cost	20.95 seconds per 4096 s segment	1x NVIDIA A100	PyCBC (filtering, coincidence, ranking, small injection set)	4.5 hours per 4096 s segment on 4 CPUs, 8GB RAM (derived 15.82 cpu-s per s: 4.5 x 3600 x 4 / 4096)	no	Results (PyCBC comparison, Table 3 discussion)
badger2024 [87]	Inference cost	24.6 s to reconstruct all 35 O3b confident events s	AMD EPYC 7502 CPU	traditional matched-filtering pipelines	paper says potentially faster, but states an explicit speed study is needed no number	unstated	Sec. III / Sec. IV
marx2024 [9]	Latency	3.114 (+0.006 / -0.001) seconds	1x NVIDIA A30 GPU (24 GB)	matched filtering pipelines (reported in the MDC study)	12.3 median, 41.4 p90 seconds	unstated	Tab. III; Sec. VIII
marx2024 [9]	Latency	3.9 seconds	unstated	matched filtering pipelines	12.3 seconds (median)	unstated	Sec. VIII
marx2024 [9]	Latency	4.3 seconds	unstated	matched filtering pipelines	41.4 seconds (90th percentile)	unstated	Sec. VIII
zelenka2024 [69]	Inference cost	~40 times ratio of estimated PyCBC runtime to this work's runtime (datasets 2-4)	8x Intel Xeon Silver 4215 cores + 8x NVIDIA RTX 2070 (challenge hardware, for the ML runs)	PyCBC	~40x this work's runtime ratio	no	Sec. IV B
yamamoto2025 [70]	Inference cost	2819 s runtime, average of foreground and background, MLGWSC-1 dataset 1	Intel Xeon Platinum 8468 (2.1 GHz, 48 core) + 1x NVIDIA H100	PyCBC (MLGWSC-1 run, approximated)	5249 s average (foreground 5406, background 5092; approximated)	no	Tab. III
chatterjee2026 [72]	Training cost	≈ 1.5 hrs per run	1x NVIDIA A100 80 GB	GW-Whisper (learned model, not a classical pipeline)	≈ 8 hrs per run	yes	Tab. I; Sec. III B 2
chatterjee2026 [72]	Inference cost	≈ 0.9 hrs for one month of continuous MLGWSC-1 data	1x NVIDIA A100 80 GB	GW-Whisper (learned model, value from a cited reference)	≈ 18 hrs for one month of MLGWSC-1 data	unstated	Tab. I; Sec. IV A
gupta2026 [11]	Latency	~4.7 s	unstated	match-filtering CBC pipelines in production (Chaudhary et al. Table 2, O3 MDC)	~10 s	no	Sec. V B

Model	Quantity	The model, as stated	Hardware	Baseline pipeline	Baseline, as stated	Same hardware	Location
rai2026 [99]	Inference cost	34.11 ms per 0.25 s window (100-run average)	standard laptop, Intel Core i7 class CPU, no GPU	Aframe (NN inference)	< 10 ms (GPU)	no	Sec. 4.2; Tab. 3; Tab. A.4
rai2026 [99]	Latency	< 45 ms total pipeline (classifier + FastSHAP explainer)	standard laptop, Intel Core i7 class CPU, no GPU	Aframe	~3.1 s total pipeline latency	no	Sec. 4.2; Tab. 3; Tab. A.4
weerasinghe2026 [62]	Inference cost	3.04 ms per event (100 ms post-merger window, ensemble of 5 CNNs)	unstated (GPU with host-device transfers implied)	simplified Lorentzian-template matched filter (reduced-order, 100 basis vectors), CPU-only	0.65 ms	no	Abstract (3.0 ms); Tab. 1; Sec. 5.3.1
baltus2021 [106]	Inference cost	~0.005 s to analyse one 50 s frame	1x Nvidia GeForce RTX 2070 SUPER	matched filter, single optimal template	~0.05 s on a 50 s frame (lower bound; whole bank not applied)	no	Sec. IV A (Performance of the CNNs); footnote 4
baltus2021b [107]	Inference cost	0.005 s to analyze a 50 second frame	1x Nvidia GeForce RTX 2070 SUPER	matched filtering, one template	0.05 s on a 50 second frame (lower bound; whole bank needed)	no	Sec. IV (Results and discussion)
yu2021early [112]	Inference cost	~30 ms to classify one 256 s time series (CNN-class)	1x NVIDIA Tesla P100-PCIE-16GB	GstLAL	~6 s typical latency	unstated	Fig. 12, Sec. VI (Discussion)
modafferi2023 [144]	Inference cost	c_CNN ≈ 4 × 10^-4 s per sample, averaged over a batch of 10^4 via Model.predict_on_batch	1x Tesla V100-PCIE-32GB	2Fmax or B_tS/G (rectangular windows: LALSuite CPU on Intel Xeon Gold 6130 2.10GHz, PyFstat GPU on the V100); B_tS/G exponential windows	≈1e-2 s (rectangular, CPU or GPU); >15 s CPU and ≈3e-2 s V100 (exponential) s per sample	yes for the GPU figures (same V100); CPU figures on a Xeon	Sec. III E
modafferi2023 [144]	Inference cost	cCNN × N ≈ 1.3 hours, first-stage CNN over N ≈ 1.15 × 10^7 templates (exponential-window test set)	1x Tesla V100-PCIE-32GB	B_tS/G single-stage search (exponential windows)	≈95 hours (GPU); ≳4 × 10^4 hours (CPU, Intel Xeon Gold 6130 2.10GHz) hours	yes for GPU (V100); CPU on a Xeon	Sec. IV B (Two-stage filtering)
skliris2024 [135]	Inference cost	51 ms per 1 s of whitened data	3.5 GHz Xeon E3-1240v5 quad-core CPU, 32 GB RAM	standard LIGO-Virgo low-latency unmodelled burst searches	several hundred dedicated CPUs	no	Sec. V B (Inference and Training Times)
skliris2024 [135]	Inference cost	three or four A100-SXM4 GPUs to run full analysis incl. FAR estimation (~10^3 time shifts) in real time	A100-SXM4	standard algorithms	several hundred dedicated CPUs	no	Sec. V B (Inference and Training Times)
skliris2024 [135]	Latency	order of 1 second s	unstated	current LIGO-Virgo low-latency unmodelled burst searches	tens of seconds to minutes (hundreds of dedicated CPUs) latency	no	Abstract / Sec. I
dreissigacker2019 [155]	Inference cost	58.8 s (full all-sky DNN search, T = 1e5 s)	unstated	WEAVE fully coherent matched filter	6.7e6 (one search of T=1e5 s; ~78 days) s on a single CPU core	no	Tab. VII; Tab. II (WEAVE cost)
dreissigacker2019 [155]	Inference cost	196 s (full all-sky DNN search, T = 1e6 s)	unstated	WEAVE fully coherent matched filter	3.9e9 (~45000 days) s on a single CPU core	no	Tab. VII; Tab. II (WEAVE cost)
miller2019 [171]	Inference cost	a microsecond per time/frequency map (CNN) us per map	unstated	FH / GFH (Frequency Hough / generalised FH) analysis of a map	~1 minute min per map	unstated	Sec. V (Search design)
bayley2020 [164]	Inference cost	~386 hours total computing time, raw SFTs to results, no trained networks	1 CPU and 1 GPU	Hough searches (fastest) and Einstein@Home (slowest), expected cost for the first 4 months of O1 (S6 MDC paper)	0.9e6 (Hough); 100e6-170e6 (Einstein@Home) core-hours	unstated	Sec. VI B
bayley2020 [164]	Inference cost	~100-200 core-hours (equivalent cost of the SOAP + CNN search for the first 4 months of O1)	unstated	Hough searches (fastest) and Einstein@Home (slowest), first 4 months of O1 (S6 MDC paper)	0.9e6 (Hough); 100e6-170e6 (Einstein@Home) core-hours	unstated	Sec. VI B
dreissigacker2020 [156]	Inference cost	roughly two times faster than the matched-filtering search (training, executing and following up, T = 1e5 s) ratio of total computing cost	unstated	WEAVE matched-filter search	1x (reference) relative	unstated	Sec. VI (Discussion)
yamamoto2021cw [169]	Inference cost	9.0e22 x (rNN/1e-3 rad)^2 x (FAP_EP/1e-2) FLOP (coherent matched-filter follow-up, Eq. 5.12)	unstated	full coherent all-sky matched filter search (monochromatic source)	2.2e8 s on a 1 PFlops machine (1.7e12 FLOP per grid point per FFT; Tobs 1e7 s, fs 1024 Hz)	no	Sec. V A, Eq. 5.12; Sec. III (matched-filter cost)
yamamoto2022cw [170]	Inference cost	1.07e19 FLOP (preprocess, Eq. 5.4); 1.4e5 core-hours	Intel E5-2670 assumed, 20.8 GFlops per core	Table VII: FrequencyHough 9e6; SkyHough 2.5e6; time-domain F-statistic 2.4e7; SOAP 1-2e2	9e6 / 2.5e6 / 2.4e7 / 1-2e2 core-hours	yes (same assumed E5-2670 floating-point-only estimate)	Sec. V, Eqs. 5.4 and 5.7, Table VII
yamamoto2023 [172]	Inference cost	2.54 x 10^-2 s for 500 test data (shallower CNN)	GPU Quadro GV100 (16.6 TFLOPS single precision)	non-Gaussian statistic grid search	1.13 x 10^4 s	no	Tab. IV

Model	Quantity	The model, as stated	Hardware	Baseline pipeline	Baseline, as stated	Same hardware	Location
tenorio2025g2net_cw [163]	Inference cost	5 days (JunKoda, a non-neural track-statistic solution), cK ~ 50 s per band days on a GPU for the full dataset	1x GPU (model unstated)	standard semicoherent F-statistic CW search (GCT/Weave): ct*nt per band	1e4 - 1e7 s per band (ct ~ 10-100 ns per template, nt ~ 1e12-1e15)	no	Sec. 5 (Comparison to CW searches), after Eq. 8
tenorio2025g2net_cw [163]	Inference cost	no competition-model cost here: about 100 million CPU hours or 1 million GPU hours is the budget of the GCT/Weave reference searches CPU hours / GPU hours	CPU or GPU, unstated	GCT / Weave deep all-sky CW searches	1e8 CPU hours or 1e6 GPU hours hours	unstated	Sec. 3 (Dataset description)
essick2020 [212]	Latency	~5 sec for feature extractors to process O(10^3) channels	unstated	KleineWelle (earlier iDQ feature extractor, O1-O2)	~32 sec	unstated	Sec. IV
laguarta2023 [199]	Inference cost	1 h of data in 11 s s to FD-encode 1 h of auxiliary-channel data (numba, O(N^2 log N))	unstated (Fig. 2 benchmarks on Intel Xeon E5-2630 v4 CPU)	earlier O(N^3) FD implementation of a cited reference (Algorithm 1)	1 h of data in 1 h 1 h of data in 1 h	unstated	Sec. II B, text after Fig. 2
licciardi2024 [181]	Inference cost	0.197 ± 0.007 s per sample (WST, Kymatio NumPy)	CPU (model unstated)	Q-transform (GwPy)	0.211 ± 0.003 s per sample	yes	Sec. IV ('Computational Cost' subsection)
licciardi2024 [181]	Inference cost	0.063 ± 0.002 s per sample (WST, Kymatio PyTorch)	GPU (model unstated, 'not state-of-the-art')	Q-transform (GwPy, CPU)	0.211 ± 0.003 s per sample	no	Sec. IV ('Computational Cost' subsection)
licciardi2024 [181]	Inference cost	258,761 (Q-transform) vs 14,374 (WST) elementary operations and function calls (cProfile)	unstated	Q-transform	258,761 operations	unstated	Sec. IV ('Computational Cost' subsection)
shen2017 [24]	Inference cost	less than 100 milliseconds per 1 s of input data	unstated (a GPU)	PCA denoising	two minutes per 1 s of input data min per s	unstated	Results, computational performance paragraph (p. 3-4)
dooney2025 [230]	Inference cost	0.14 s per glitch sample on a CPU	CPU (model unstated)	BayesWave	approximately one hour per glitch hour per glitch	unstated	Sec. IV A
chatterjee2019 [273]	Latency	0.0189 (about 0.018 in abstract) s per signal incl. feature extraction, 8192 sectors	14 Intel Xeon CPU cores	BAYESTAR	10 s - 1 min s	unstated	Abstract; Sec. IX
shen2019pe [268]	Inference cost	within two (2) milliseconds per event (median values and posterior distributions)	1x NVIDIA V100	PyCBC Inference	not stated (qualitative comparison of posteriors only) n/a	unstated	Abstract; Sec. 4.2
delaunoy2020 [265]	Inference cost	less than a minute per event	unstated	MCMC	more than a day per event	unstated	Sec. 5 Results
green2020 [26]	Inference cost	5,000 posterior samples per second	unstated	bilby / dynesty	days (BBH), weeks (BNS) per event	unstated	Sec. on inference for GW150914; Conclusions
marulanda2020 [277]	Inference cost	about 6% greater (FCNN incl. spectrogram computation vs time-domain CNN) percent relative execution time	unstated	time-domain CNN (TCNN), the paper's own comparison network	1 (reference) relative	unstated	Abstract; Results (execution-time paragraph); Conclusions
dax2021 [27]	Inference cost	20 seconds per event (50,000 posterior samples, 30 GNPE iterations)	1x NVIDIA A100 (implied by footnote 3, which contrasts V100's roughly a minute with 20 seconds)	LALInference MCMC	O(day) per event; inference times reduced by factors of 10^3-10^4 per event	unstated	Abstract; Results (GWTC-1 analysis); footnote 3
kolmus2021 [276]	Inference cost	roughly 5 minutes of computing time (100,000 importance-sampling steps)	16-core AMD Ryzen 5950X CPU, NVIDIA 3090 RTX GPU, 64 GB RAM	Bilby (at convergence)	between 2.5 and 7 hours per run (Fig. 4: at least 3 hours) hours	yes	Fig. 4 caption; Sec. III C; Sec. IV C
kolmus2021 [276]	Inference cost	a few minutes minutes	AMD Ryzen 5950X CPU, NVIDIA 3090 RTX GPU	BAYESTAR	within a minute per event	unstated	Abstract; Sec. I (BAYESTAR)
chatterjee2022 [274]	Latency	~1.2 s total inference (sample generation, KDE fit, density over adaptive mesh)	1x P100 GPU	BAYESTAR	a few seconds s	unstated	Sec. 3; Abstract (around 1 sec)
dax2022 [28]	Inference cost	less than 1 hour per event, total (IMRPhenomXPHM)	1x NVIDIA A100 GPU + 64 CPU cores	LALInference/Bilby-dynesty (stochastic samplers)	Bilby-dynesty about 1 day per run on 10 CPUs (nlive=1000, nact=5); about a week for nlive=4000, nact=50 per run	no	Main text, Results (Dingo-IS timing) and Discussion; Supplemental Material (stochastic samplers)
dax2022 [28]	Inference cost	approximately 10 hours per event total (SEOBNRv4PHM)	1x NVIDIA A100 GPU + 64 CPU cores	stochastic samplers, SEOBNRv4PHM	several months per event per event	unstated	Main text, Results and Introduction
gabbard2022pe [263]	Inference cost	1 × 10^-1 seconds for 8000 posterior samples (VItamin)	unstated (1x NVIDIA Tesla V100 named in Methods for training; consumer GPUs said to be equally fast)	Dynesty / emcee / ptemcee / CPNest	median 45607 / 19821 / 41151.0 / 83807 (min-max: 21564-261268 / 16712-39930 / 2392-501632 / 10309-437008) seconds	unstated	Tab. I
mcleod2022 [269]	Inference cost	~1.4 milliseconds per sample prediction	1x NVIDIA P100 GPU and Intel Gold 6140 CPU	LALInference	hours or days (full PE); 'several orders of magnitude faster' hours-days	no	Sec. III (Prediction Results); Sec. IV (Discussion); Sec. V (Conclusion gives ~1 ms)

Model	Quantity	The model, as stated	Hardware	Baseline pipeline	Baseline, as stated	Same hardware	Location
andrescarcasona2023pe [270]	Inference cost	0.05 s to generate 10,000 posterior samples	1x NVIDIA GeForce RTX 2080 Ti	Bayesian inference (LVK published PE)	several hours to days (MCMC); 1-10 hours for neural+importance sampling (Dax et al.) hours/days	no	Sec. 5 (real events, Fig. 4)
chatterjee2023skyloc [275]	Latency	about 1.2 s total (5000 samples, KDE, adaptive refinement)	unstated	BAYESTAR	about 1 second average in O3 s	unstated	Sec. 5
chatterjee2023skyloc [275]	Inference cost	a few ms (sample generation)	NVIDIA P100 12GB PCIe	BAYESTAR	about 1 s s	unstated	Sec. 1
badanerin2024 [264]	Inference cost	around 4 s per waveform (N=8000 samples)	1x NVIDIA A40	Bilby (standalone)	54045 (mean); 27197 min; 145063 max s	no	App. A; Tab. I
badanerin2024 [264]	Inference cost	28159 s Bilby run with CVAE-informed priors (47.9% saving)	ICCUB Nyx cluster, dual AMD EPYC 7763 (64 cores each)	Bilby standalone	54045 s	yes	Tab. I
dax2024 [255]	Latency	~1 second (inference latency for pre-merger estimates)	unstated (H100 used for IS timing)	Bayestar	localization in seconds (no number) s	unstated	Fig. 2 caption
dax2024 [255]	Inference cost	roughly one second for 1e4 effective samples (importance sampled)	1x NVIDIA H100 GPU	ROQ-based inference	6 minutes localization; 2 hours full inference per event	unstated	Main text, Experiments; Abstract ('one second')
hu2024 [260]	Inference cost	within a second (network sampling) + O(1)-O(10) CPU hours importance sampling second / CPU hours	1x NVIDIA 3080Ti GPU (sampling); CPU for IS	Bilby (nested sampling)	3500 CPU hours	no	Sec. 3.1 Parameter estimation
hu2024 [260]	Inference cost	less than one second (EOS inference)	unstated (single GPU)	traditional stochastic sampling for EOS	tens to hundreds of CPU hours CPU hours	no	Sec. 3.2
benedikt2025 [280]	Inference cost	119.4 ± 0.3 (132.0 ± 0.3 with MC Dropout) ms per batch of 250	1x NVIDIA A100	Shen et al. 2022 (BNN)	2 ms per event ms	unstated	Tab. 9; Sec. 5.4
chan2025lens [257]	Inference cost	tens of seconds per event s per event	unstated	conventional samplers (bilby)	weeks weeks	unstated	Sec. IV; Abstract
chan2025lens [257]	Inference cost	~40 s per event (representative lensed injection, network SNR 16.18, with importance sampling, efficiency 2.3%)	unstated	bilby	weeks (Abstract) weeks	unstated	Sec. III C
iwanaga2025 [282]	Inference cost	approximately six minutes (total estimation time incl. Monte Carlo uncertainty evaluation)	unstated	LALInference/Bayesian methods	several days per event (BBH) per event	unstated	Sec. III A and Conclusions
kofler2025 [256]	Training cost	approximately 9.5 days (183 epochs); Tab. IV: 9d 9h days	8x NVIDIA A100-SXM4-80GB	Dingo baseline NPE	3d 1h (273 epochs) training time	yes	Main text (Results); Tab. IV; Supplemental 'Training'
kofler2025 [256]	Inference cost	five to ten minutes minutes for 10^5 importance-sampled posterior samples (phase reconstruction + IS)	64 CPU cores	Dingo with GNPE	roughly 1 h for 10^5 importance-sampled posterior samples on one NVIDIA A100 GPU and 64 CPUs hour	unstated	Main text (Results); Supplemental Material (inference)
lanchares2025 [261]	Inference cost	1.13 ± 0.13 seconds per 10,000 posterior samples	1x NVIDIA A100	parallel Bilby (reference samples)	hours to days to converge, with 640 CPUs hours-days	no	Results text, Sec. III (GP15 vs Bilby comparison paragraph)
emma2026 [266]	Inference cost	approximately an order of magnitude larger than Whittle-likelihood runs ratio of total wall time	unstated	Whittle likelihood	1x relative wall time	unstated	Sec. IV (GW analysis wall time)
emma2026 [266]	Inference cost	~ 3–5 ratio of per-likelihood-evaluation time, RNLE over Whittle likelihood	unstated	Whittle likelihood (Bilby)	1x relative per-evaluation time	unstated	Sec. IV; Fig. 7 black squares
luo2026 [259]	Inference cost	less than 20 minutes per event (echo search / Bayesian inference)	1x NVIDIA RTX 5090	traditional Bayesian inference	several hours hours	unstated	Sec. I (Introduction); Sec. III (Results); Sec. IV (Conclusion)
sakellariou2026 [283]	Inference cost	approximately 36 ms per strain (DeepModel)	unstated	traditional Bayesian inference	hours or even days (Sec. 1); 'orders of magnitude' slower hours-days	no	Sec. 3 (Results, computational-efficiency paragraph)
zhang2026 [258]	Inference cost	approximately 0.4 (less than 0.4) seconds for complete posterior sampling of a single event	1x NVIDIA RTX A6000	MCMC (Bilby with Dynesty nested sampling)	over 10 hours per event; 20,943 to 44,818 s across parameters (Table II) seconds / hours	unstated	Sec. on training/ implementation; Sec. C 'Computational performance comparison'; Conclusion
zhang2026 [258]	Inference cost	0.21 to 0.39 (per ppE parameter: 0.31, 0.39, 0.34, 0.35, 0.38, 0.38, 0.21, 0.35, 0.36) seconds (NPE run time per parameter estimator)	1x NVIDIA RTX A6000	MCMC (Dynesty)	20943 to 44818 (27449, 26831, 33846, 28501, 22799, 36205, 44818, 20943, 29021) seconds run time	unstated	Tab. II
liu2023 [291]	Inference cost	SSL and LSTM time cost more than 70% lower than LDASoft (LISA MDC) percent reduction in time cost	unstated	LDASoft (globalfit-based)	reference time cost (relative)	unstated	Sec. IV, text around Fig. 9
ruan2023 [285]	Inference cost	several seconds to analyse a year of LISA Sangria data s per year of data	unstated	global-fit MBHB analysis (multi-core)	several hours for a year of data hours	no	Abstract; Sec. V; Sec. VI (global-fit cost)

Model	Quantity	The model, as stated	Hardware	Baseline pipeline	Baseline, as stated	Same hardware	Location
`ruan2023` [285]	Inference cost	within ten seconds (a year-long Sangria dataset) s per year of data	unstated	global fit for a MBHB (multi-core processor)	several hours hours	no	Sec. VI
`zhao2023` [288]	Inference cost	1.9 s per batch of 256 samples (474 days of data)	unstated (training on a single NVIDIA V100)	traditional MLDC/LDC searches (grid search, MCMC)	a few hours; 2-36 core-hours (Tab. 2) hours / core-hours	no	Tab. 3; Sec. 2.7
`zhao2023` [288]	Inference cost	< 10^-2 s to perform extraction and detection	unstated	traditional approaches	several hours hours	no	Sec. 1
`que2021` [312]	Latency	0.40 microseconds	1x Xilinx Alveo U250 FPGA (16-bit fixed)	same LSTM autoencoder on Intel E2620 CPU (and TITAN X GPU 32.1 ms)	39.7 ms	no	Tab. III
`que2021` [312]	Latency	0.867 microseconds	1x Xilinx Alveo U250 FPGA at 300 MHz	prior FPGA LSTM anomaly detection, ref (Kintex7 K410T, single layer)	4.27 us	no	Tab. IV (This work, four layers: 32,8,8,32)
`que2021` [312]	Latency	0.343 microseconds	1x Xilinx Alveo U250 FPGA at 300 MHz	prior FPGA LSTM physics design, ref (KU115, single layer, 16 hidden units)	1.35 us	no	Tab. IV (This work, single layer, 32 hidden)

14 Code and reproducibility

106 of the 313 models have a public repository recorded as official (third-party reimplementations and empty repositories are not counted; for some models the only repository listed is an upstream framework, a data release, trigger-generator documentation or a related model's code, marked in Table 9 where identified), and 29 of those release trained weights (2 more release partial weights). Licences were read from the repositories; a missing licence means none was found, which does not grant reuse. The per-section counts are in Table 8 and the repositories in Table 9.

Table 8. Code and weights by section.

Section	Models	With code	With weights
Compact-binary detection	92	30	10
Early warning	9	2	0
Burst and unmodelled transients	33	9	3
Anomaly and multi-signal detection	11	5	1
Continuous and long-duration waves	18	4	0
Glitch classification and data quality	56	19	7
Denoising and signal extraction	32	10	1
Parameter estimation	33	18	4
Space-based detectors	14	3	0
Other: alerts, generators, lensing, backgrounds	15	6	3

Table 9. Official repositories (first listed per model).

Model	Repository	Licence	Last commit	Weights
`abbott2021` [300]	github.com/GWML/GWSkyNet	MIT	2021	yes
`alhassan2023pymerger` [94]	github.com/wathela/PyMerger	NOASSERTION (LICENSE file present, SPDX not recognised)	2024	yes
`andrews2022` [82]	github.com/mbandrews/gw-analysis	none found	2021	no
`apostol2023` [221]	github.com/DS4AI-UPB/ML4GW	none found	2023	no
`arutkeerthi2024` [207]	github.com/Arut123/Glitch-veto-GW-using-chirp	none found	2025	yes
`aveiro2022` [90]	doi.org/10.5281/zenodo.6222936 (upstream YOLOv5 v6.1 release, not the authors' code)	Zenodo record: "Other (Open)"	2022	no
`badger2024` [87]	github.com/Rahul-Srinivasan/GW_DictionaryLearning	none found	2025	no
`bahaadini2018` [173]	github.com/Gravity-Spy/GravitySpy	GPL-3.0	2024	yes
`bayley2020` [164]	github.com/jcbayley/soapcw	MIT	2026	no
`beheshtipour2020` [165]	github.com/matterport/Mask_RCNN (upstream Mask R-CNN implementation the paper cites, not the authors' code)	NOASSERTION	2024	no
`beheshtipour2020b` [166]	github.com/matterport/Mask_RCNN (upstream Mask R-CNN implementation, not the authors' code)	NOASSERTION	2024	no
`benedikt2025` [280]	github.com/GraviBERT/GraviBERT (+1)	MIT	2026	yes
`beveridge2025o3` [15]	github.com/damonbeveridge/new_BBH_O3_candidate-UWA (data release: parameter-estimation configuration files and posterior samples)	none found	2025	no
`cabero2020` [299]	github.com/GWML/GWSkyNet	MIT	2021	yes
`cain2025` [148]	github.com/jericho-cain/cwt-lstm-ae-grav-wav	MIT	2026	no
`cavaglia2018` [215]	github.com/kstaats/karoo_gp	other (non-standard licence listed by GitHub)	2022	n/a
`chan2024` [301]	github.com/nayyer-raza/GWSkyNet-Multi (code and weights of the GWSkyNet-Multi classifier, not this model; the paper links no repository)	MIT	2025	yes
`chan2025lens` [257]	github.com/dingo-lensing/dingo-lensing	MIT	2026	no
`chatterjee2022` [274]	github.com/chayanchatterjee/GW-SkyLocator	GPL-3.0	2024	yes
`chatterjee2023skyloc` [275]	github.com/chayanchatterjee/GW-SkyLocator	GPL-3.0	2024	no
`chatterjee2024` [71]	github.com/ondrzel/ml-gw-search/tree/main	Apache-2.0	2023	no
`chatterjee2024harm` [231]	github.com/chayanchatterjee/AWaRe	GPL-3.0	2025	unverified
`chatterjee2024rec` [232]	github.com/chayanchatterjee/AWaRe	GPL-3.0	2025	unverified
`chatterjee2026` [72]	github.com/chayanchatterjee/castor	Apache-2.0	2026	no
`cheung2025unet` [160]	github.com/damondmc/cwunet	none found	2025	no
`chua2019` [267]	github.com/vallis/truebayes	MIT	2019	yes
`cirfeta2026` [153]	github.com/lucacirfeta/dante-gravi-signal-ml	GPL-3.0	2026	unverified
`cuoco2026` [137]	github.com/elenacuoco/p4TSA (+1)	GPL-3.0	2026	no
`dax2021` [27]	github.com/dingo-gw/dingo	MIT	2026	no
`dax2022` [28]	github.com/dingo-gw/dingo	MIT	2026	no
`dax2024` [255]	github.com/dingo-gw/dingo	MIT	2026	unverified
`ding2021` [200]	github.com/JulianZDing/UniMAP	none found	2021	no
`dodia2021specgrav` [95]	github.com/HrithikaDodia/SpecGrav	none found	2021	no
`dooney2024` [305]	git.ligo.org/tom.dooney/cdvgan_paper	none found	-	no
`dooney2025` [230]	git.ligo.org/tom.dooney/deepextractor.git	MIT	-	no
`dooney2026` [224]	github.com/tomdooney95/glitchgan (+2)	none found	2026	yes
`emma2026` [266]	github.com/MattiaEmma2022/sbilby	MIT	2026	no
`essick2020` [212]	git.ligo.org/reed.essick/iDQ	none found	-	no
`fernandes2026` [189]	github.com/tttiago/glitchgram	none found	2026	no

Model	Repository	Licence	Last commit	Weights
gabbard2022pe [263]	github.com/hagabbar/VItamin	MIT	2022	no
george2018glitch [175]	github.com/Gravity-Spy/GravitySpy (Gravity Spy project code, not this paper's; the paper links no repository)	GPL-3.0	2024	unverified
glanzer2023 [23]	zenodo.org/records/5649212 (data release of classifications; +1)	unknown (Zenodo)	2021	n/a
green2020 [26]	github.com/stephengreen/lfi-gw (+2)	MIT	2021	unverified
gunny2021 [68]	github.com/fastmachinelearning/gw-iaas (+1)	MIT	2022	no
gupta2026 [11]	github.com/ML4GW/aframe	MIT	2026	no
hu2024 [260]	github.com/MarinerQ/river	none found	2025	no
huerta2021 [64]	doi.org/10.26311/ZWFW-3N68	unknown	2021	unverified
iess2020 [118]	wdf.readthedocs.io/ (WDF trigger-generator documentation, not the CNN code)	unknown	-	n/a
iess2023 [119]	wdf.readthedocs.io/ (WDF trigger-generator documentation, not the CNN/LSTM code)	unknown	-	n/a
inguglia2026 [17]	github.com/ginguglia/MADGRAV	MIT	2026	yes
insley2026flows [104]	github.com/gwastro/pycbc (the host pipeline; the paper gives no code link)	GPL-3.0	2026	no
jadhav2020 [102]	github.com/shreejit-92/MLStat-1 (data release)	none stated	2020	no
kelleter2026nnnn [247]	github.com/lc316353/Newtonian-Noise-Simulation (+1)	none stated (no LICENSE file)	2026	no
kiendrebeogo2025virgo [243]	github.com/weizmannk/Virgo-DeepClean	AGPL-3.0	2026	no
kofler2025 [256]	github.com/dingo-gw/dingo-T1 (+2)	MIT	2026	yes
kolmus2021 [276]	github.com/akolmus/swiftsky	none found	2021	no
koloniari2024 [13]	github.com/vivinousi/gw-detection-deep-learning (code of the earlier AresGW version, ref. 74)	Apache-2.0	2025	yes
lanchares2025 [261]	github.com/Daniel-Lanchares/dtempest	none found	2026	no
li2024ctsae [195]	github.com/Zod-L/CTSAE	Apache-2.0	2024	no
liu2023 [291]	github.com/dingo-gw/dingo (+1) (the DINGO framework the paper modified; no repository for this paper)	MIT	2026	no
luo2026 [259]	github.com/dingo-gw/dingo (the dingo framework the paper builds on; no repository for this paper)	MIT	2026	no
ma2024mix [253]	github.com/z2062358453/gravitational-wave-separation (not linked from the paper)	none found	2024	yes
malz2024 [186]	doi.org/10.5281/zenodo.14066270	CC BY 4.0	2024	no
manoharan2026 [222]	github.com/rudhresh1997/glitch-tabular-prd	MIT	2026	no
martins2024 [109]	github.com/anaismartins/deploying-custom-nn-in-fpga	MIT	2025	no
marx2024 [9]	github.com/ML4GW/aframe (+2)	MIT	2026	no
marx2025aframeO3 [10]	github.com/ML4GW/aframe	MIT	2026	no
mcginn2021 [302]	github.com/jmcginn/McGANn	none found	2021	no
mcisaac2022 [103]	github.com/connor-mcisaac/chisqnet (+1)	GPL-3.0	2022	yes
mcleod2025 [16]	github.com/alistair-mcleod/infernus (+1)	MIT	2026	no
mlgwsc2022 [8]	github.com/gwastro/ml-mock-data-challenge-1	Apache-2.0	2022	yes
morales2024 [122]	github.com/ManuelDMorales/resnet50-sngw-hff (+2)	GPL-3.0 (per paper)	unchecked	unverified
moreno2022 [140]	github.com/eric-moreno/Anomaly-Detection-Autoencoder (+1)	none (no licence file; all rights reserved by default)	2022	yes
nagarajan2025 [14]	github.com/nnarenraju/sage	GPL-3.0	2026	no
nousi2022 [12]	github.com/vivinousi/gw-detection-deep-learning (+1)	Apache-2.0	2025	yes
ormiston2020 [25]	github.com/rormiston/DeepClean (+3)	GPL-3.0	2018	no
papalini2025 [272]	github.com/luciapapalini/kenn-gw-transformer	MIT	-	unverified
pracchia2025 [128]	git.ligo.org/maxime.fays/gwpyxel	none shown	2026	no
que2021 [312]	github.com/walkieq/RNN_HLS	Apache-2.0	2024	unverified
raikman2023 [146]	github.com/ML4GW/gwak	MIT	2026	partial
raikman2024 [136]	github.com/ML4GW/gwak	MIT	2026	partial
reissel2025 [244]	github.com/ML4GW/DeepClean (+1)	none stated	2023	unverified
saleem2023 [242]	github.com/ML4GW/DeepClean (+2)	none declared (no LICENSE file; GitHub licence field null)	2022	no
schafer2020 [5]	github.com/gwastro/bns-machine-learning-search	none declared (no LICENSE file; GitHub licence field null)	2020	yes
schafer2021a [6]	github.com/gwastro/ml-training-strategies	GPL-3.0	2022	no
schafer2021b [7]	github.com/gwastro/ml-training-strategies (release of the single-detector predecessor; not cited by this paper)	GPL-3.0	2022	no
singh2020 [278]	github.com/SSingh087/DL-estimating-parameters-of-GWs	MIT	2022	no
skliris2024 [135]	github.com/VasSkliris/mly-methods-paper (+3)	none listed	2024	yes
soni2021 [22]	github.com/Gravity-Spy/GravitySpy	GPL-3.0	2024	yes
srivastava2024 [180]	kaggle.com/code/divyansh050/vitb32-ligo/notebook	none found	-	no
tian2023 [66]	github.com/mtian8/Higher_Order_GW_Spatiotemporal_GNN	none found	2024	unverified
tian2024gnn [67]	github.com/mtian8/gw_spatiotemporal_gnn	MIT	2024	unverified
tiki2025 [80]	github.com/victoria-tiki/AttenGW (+1)	MIT	2026	yes
wang2020 [83]	github.com/WuShichao/mfcnn_catalog (+1) (trigger list)	none listed	2020	no

Model	Repository	Licence	Last commit	Weights
`wang2022waveformer` [229]	github.com/AI-HPC-Research-Team/LIGO_noise_suppression	Apache-2.0	2024	no
`weerasinghe2026` [62]	github.com/sathnidu9/postmerger-gw-eos-inference	MIT	2025	no
`wu2024` [179]	github.com/Gravity-Spy/gravityspy-ligo-pipeline	GPL-3.0	2025	yes
`xu2026ssl_lisa` [296]	github.com/TriangleDataCenter/Triangle-Simulator (+1) (data simulator used, not the network's code)	GPL-3.0	2026	no
`yamamoto2025` [70]	github.com/tsyamamoto21/gw-hybridmfcnn-stellarmassbbh	none declared	2025	yes
`yan2022gan` [182]	github.com/tkarras/progressive_growing_of_gans (upstream ProGAN the paper used, not the authors' code)	NOASSERTION (NVIDIA research licence)	2022	no
`yan2025deepgrav` [139]	github.com/yan123yan/DeepGrav	MIT	2026	yes
`yu2021early` [112]	github.com/hangyu45 (+1) (author's account; the paper gives no code)	none for relevant repos (account has GPL-3.0 repos for other papers)	2026	no
`zelenka2024` [69]	github.com/ondrzel/ml-gw-search	Apache-2.0	2023	yes
`zevin2017` [20]	github.com/Gravity-Spy/GravitySpy (+3)	GPL-3.0	2024	yes
`zevin2024` [177]	github.com/Gravity-Spy/GravitySpy (+2)	GPL-3.0	2024	yes
`zhao2023` [288]	github.com/AI-HPC-Research-Team/Space_signal_detection_all4	MIT	2023	no

15 Reproduction programme in gravyflow

The original catalogue set an itinerary of sixteen steps, ordered so that shared library features are built once and then reused. The added models are placed on the same steps. Effort follows the original scale: S is model code and configuration on the existing data path (days), M one new library feature or substantial data work (one to two weeks), L several new features, an external pipeline or dataset, or a new physics domain; for added models it is estimated from the one-off efforts of their adaptation tags. Every model's reproduction is pending.

Table 10. Itinerary.

Step	Title	Shared adaptations	Orig.	Added	Why (first catalogue)
0	Retrieve the unread papers	x-retrieve	11	30	Nine entries have no architecture at all; their adaptations cannot be stated until the text is read. Cheap for the owner (institutional access), and it unblocks later steps.
1	Close the shipped families	n-real, n-sim-glitch, wf-subst	4	1	George-Huerta (both nets) and Gabbard are shipped; MLy is shipped without its incoherent and single-detector WNB noise classes. IncoherentGenerator now exists, so MLy's types 2/3 and George-Huerta's real-noise follow-up are the cheapest completions of what is already there.
2	1D CNN ports on the current data path	wf-subst, t-multiclass, det-hlv, in-window, t-ensemble	11	7	Whitened-strain 1D CNNs that need only model code, a declared approximant substitution and at most a multi-class head or longer window. The most cited early literature lives here.
3	Search-grade evaluation: sensitive distance and continuous search	e-far, e-sensitivity, n-real	1	3	The post-2021 literature reports sensitive distance vs FAR on continuous data with time slides; the validator has FAR and time slides but not sensitive distance or a streamed search. Building this once makes every later CBC entry comparable to MLGWSC-1 and to PyCBC.
4	MLGWSC-1 benchmark and its entries	n-mlgwsc, wf-xphm, n-psd-aug, t-custom-layer	6	1	One benchmark covers AresGW, CNN-Coinc, TPI FSU Jena, MFCNN and is the common yardstick of CASTOR, GW-Whisper, Sage and Yamamoto 2025. DS1 is reachable now (non-spinning, no higher modes, so its XPHM reduces to the IMRPhenomXAS gravyflow offers); DS2-DS4 need XPHM, PSD variation and real O3a noise.
5	Per-timestep outputs (WaveNet family)	t-sequence, n-real, t-ensemble	3	3	Gebhard 2019 and the Huerta/Wei WaveNet ensembles emit a score per time step on real noise; one dense-target feature serves all of them and the U-Net segmentation models of step 9.
6	Matched-filter front ends	in-mf-series	6	2	MatchedFilterLayer and TemplateBank exist; exposing per-template SNR series and SNR maps unlocks MFCNN, Beveridge, McLeod (BNS), Yamamoto 2025, Mobilia and Yan 2021.
7	Burst, anomaly and autoencoder models	t-ae, wf-pheno-burst, in-pearson, t-recurrent	6	5	Extends MLy: noise-only autoencoders (Moreno), per-class autoencoders (GWAK), generalised-burst training (McGinn), unsupervised coincident anomaly detection (Ratner) and synthetic transient classification (Nigam).
8	Denoising and waveform extraction	t-ae, e-overlap, t-recurrent	8	7	Strain-in, clean-waveform-out models trained against WHITENED_INJECTIONS and scored by overlap.
9	Time-frequency images and pretrained backbones	in-qtransform, in-stft, in-wavelet, in-pretrained, t-sequence	6	5	Q-transform/CQT/CWT images and ImageNet backbones serve the ResNet-50 O2/O3 searches, the spectrogram early-warning nets and, in step 10, all of Gravity Spy.
10	Glitch classification (Gravity Spy family)	n-glitch, in-qtransform, t-multiclass, e-multiclass, n-external	17	16	Twenty-odd glitch papers share one dataset (Gravity Spy) and one input (multi-duration Omega scans). Either a Zenodo image loader or a Q-transform over Gravity Spy times.
11	Early warning and long BNS windows	in-window, in-multirate, e-early, wf-subst	6	3	BNS inspirals need 10-300 s windows, multi-rate inputs, partial-inspiral SNR labels and time-before-merger metrics.
12	Core-collapse supernovae and catalogue waveforms	wf-catalogue, wf-pheno-burst, t-multiclass	7	5	Supernova detectors need numerical waveforms from files (or Astone's phenomenological model), distance scaling and 3-class heads.
13	Exotic training: Bayesian, GAN, VAE, transformers, foundation models	t-generative, t-recurrent, in-pretrained	7	7	Each needs a custom train step or heavy pretrained weights; worth doing after the data and evaluation foundations exist.
14	Auxiliary-channel and pipeline-coupled models	in-aux, in-pipeline-triggers, t-nonneural	18	18	DeepClean, iDQ, EMU and cWB-coupled classifiers need non-public channels or an external pipeline; they are reference entries until gravyflow reads auxiliary data.

Step	Title	Shared adaptations	Orig.	Added	Why (first catalogue)
15	Out of core scope: space-based, CW, long transients, skymaps, PE	det-new, wf-cw, e-pe	21	62	Recorded for completeness; each needs a new physics domain.

Table 11. Shared adaptations: state in gravyflow (first catalogue) and how many of the 313 models need each.

Tag	Feature	State in gravyflow	Effort once	Models
in-window	Window length	Have: any onsource length, fixed CBC buffer B(W)=2^ceil(log2 2W). Gap: windows of 10-300 s are unmeasured for memory and generation cost.	S	25
in-multirate	Multi-rate input	Have: one sample rate per dataset, one resampler. Gap: a window cut into segments at different rates (Schafer 2020, Nagarajan 2025).	M	2
in-stft	STFT spectrogram	Have: gf.spectrogram / SPECTROGRAM_ONSOURCE (|STFT|^2 of whitened data, Hann frames). Gap: per-paper window, overlap, log scaling, band crop, image resize.	S	20
in-qtransform	Q-transform / Omega-scan images	Gap: no Q-transform in gravyflow; Gravity Spy-style multi-duration (0.5/1/2/4 s) images, CQT, colour maps and resizing to backbone inputs.	M	39
in-wavelet	Wavelet transforms	Gap: wavelet packet, CWT (Morlet), scattering transform, cWB-like WDF time-frequency maps.	M	14
in-freqdomain	Frequency-domain input	Gap: real/imaginary whitened rfft as model input (DINGO, CW nets); PSD-as-input channels.	S	14
in-mf-series	Matched-filter front end	Have: MatchedFilter, MatchedFilterLayer (returns network-SNR logits), TemplateBank. Gap: per-template SNR time series and template-by-time SNR maps as model inputs, top-k template selection, learnable template layer (MFCNN).	M	12
in-pearson	Cross-detector correlation	Have: ROLLING_PEARSON_ONSOURCE (circular shift; lag convention not pinned). Gap: frequency-domain correlation features.	S	1
in-aux	Auxiliary witness channels	Gap: gravyflow reads strain only; no auxiliary-channel acquisition (needs LocalFrames on a LIGO cluster, non-public channels).	L	15
in-pipeline-triggers	External pipeline triggers	Gap: cWB, Omicron, PyCBC or BAYESTAR products as inputs (cWB summary statistics, reconstructed waveforms, skymaps).	L	29
in-pretrained	Pretrained backbones	Gap: ImageNet / audio pretrained weights (keras.applications under the JAX backend, Whisper, YOLO) and their input normalisation.	M	27
det-hlv	H1/L1/V1 network	Have: H1, L1, V1 for synthetic and real noise. Gap: transient (glitch/event) mode is H1/L1 only; zero-filled missing detectors.	S	16
det-new	New detectors	Gap: KAGRA, Einstein Telescope, LISA/TianQin/Taiji responses and PSDs (TDI for space).	L	32
wf-subst	Approximant substitution	Have: ripple IMRPhenomD, XAS, Pv2, TaylorF2, D_NRTidalv2. Gap: SEOBNR/EOB, SpinTaylorT4, PhenomPNRT and surrogates are absent; substitute and declare the departure (as the George-Huerta example does) or bridge to LALSimulation.	S	31
wf-xphm	IMRPhenomXPHM / higher modes	Gap: XPHM deliberately not offered (ripple loses LAL parity at equal masses with spin); MLGWSC-1 and O3 searches such as AresGW and GW-Whisper use it.	M	17
wf-ecc	Eccentric / beyond-GR waveforms	Gap: CBCGenerator refuses eccentricity; no ppE/beyond-GR models.	L	5
wf-catalogue	Waveform-from-file injection	Gap: injecting numerical waveforms from a catalogue (CCSN simulations, NR, Richers 2017) with sky projection and SNR/HRSS scaling.	M	10
wf-pheno-burst	Extra burst families	Have: sine-Gaussian, chirplet, ringdown, periodic, WNB, IncoherentGenerator. Gap: Gaussian pulse, cosmic-string cusp, phenomenological CCSN (Astone), scattered-light / whistle shapes, minute-long chirps, magnetar spin-down.	M	15
wf-cw	Continuous waves	Gap: CW signals with Doppler and spin-down over days (LALSuite/PyFstat).	L	17
n-real	Real GWOSC noise	Have: REAL and PSEUDO_REAL noise O1-O3, flags and vetoes, split by GPS. Gap: real noise is read sequentially (not index-addressed); detectors are always two footprints apart (no coincident mode for training on coincident noise).	S	71
n-glitch	Gravity Spy glitches	Have: TransientDataObtainer with Gravity Spy classes (GlitchType) H1/L1 O1-O3. Gap: glitch-labelled examples mixed as a noise class at set ratios; multi-class labels through ComposedDataset (outside the batch contract).	S	47
n-sim-glitch	Simulated glitches	Gap: single-detector sine-Gaussian/Gaussian/ringdown/scattered-light glitches labelled noise (IncoherentGenerator or a per-detector injection labelled noise).	S	14
n-psd-aug	PSD augmentation	Gap: recolouring with sampled PSDs (DINGO, Sage, MLGWSC-1 DS2/DS3); PSEUDO_REAL is the nearest relative.	M	4
n-mlgwsc	MLGWSC-1 datasets	Gap: the challenge's four datasets (DS1-DS4) and its injection/evaluation contract; all four use XPHM, and DS3-DS4 add precession and higher modes.	M	7
n-external	External datasets	Gap: a loader for a published dataset (Gravity Spy Zenodo images, Codabench GWAK, ET MDC, LDC, Richers catalogue).	M	12
t-multiclass	Multi-class training	Have: ComposedDataset pools and POOL_LABEL. Gap: >2-class heads through the recipe and the batch contract.	S	47
t-regression	Regression targets	Have: mass regression (George-Huerta predictor), WaveformParameters labels. Gap: SNR, chirp-mass, time-to-merger and sky targets in the recipe.	S	15
t-sequence	Per-timestep / dense targets	Gap: per-sample or per-pixel labels (WaveNet-style per-timestep classification, U-Net masks, bounding boxes).	M	21
t-ae	Autoencoder / reconstruction	Have: WHITENED_INJECTIONS as a clean target. Gap: noise-only reconstruction training and an anomaly-score validator path.	M	31
t-curriculum	Curricula and schedules	Have: gf.Curriculum (SNR or any Distribution, by epoch). Gap: step/trigger-based curricula, curricula on frequency cutoff or mass, LR schedules beyond Keras.	S	9
t-ensemble	Ensembles	Gap: training and scoring N independently seeded members, voting and coincidence rules.	S	12
t-custom-layer	Custom layers	Gap: layers Keras 3 lacks or the examples do not use: DAIN, GroupNorm (present in Keras), periodic conv, SE/shrinkage, quadratic layer, inception/residual blocks.	S	12
t-recurrent	Recurrent / attention nets	Have: Keras 3 LSTM/GRU/MultiHeadAttention run on JAX. Gap: none in the library; only example code and memory measurement for long sequences.	S	28
t-generative	GAN / VAE / flow / Bayesian training	Gap: custom train steps (WGAN-GP, VAE ELBO, flipout Bayesian layers, normalising flows) under Keras 3 JAX.	L	35
t-unsupervised	Unsupervised / similarity	Gap: clustering, contrastive pairs, IIC, UMAP/KDE outlier scoring.	M	13
t-nonneural	Non-neural models	Gap: trees, GMMs, genetic programming, logistic regression: outside Keras; would need a scikit-learn/XGBoost adapter to the validator.	M	26
e-far	FAR and efficiency	Have: Validator: exact FAR in Hz with Garwood intervals, efficiency by SNR bin, ROC/AUC, time slides on multi-detector real noise (num_time_slides, cluster_seconds). Gap: streamed continuous-data search with clustering over long stretches.	S	22
e-sensitivity	Sensitive distance / volume	Gap: sensitive distance or volume vs FAR over a population (MLGWSC-1 metric), hrss50 for bursts.	M	23
e-multiclass	Multi-class evaluation	Gap: confusion matrices, per-class accuracy/F1, NMI/ARI for clustering.	S	13
e-overlap	Waveform overlap	Gap: match/overlap of reconstructed vs injected waveform (the matched-filter code has the pieces).	S	12
e-events	Real events	Have: score_real_events (confident GWTC events, test split). Gap: catalogue masses on event records; O4 events.	S	17
e-early	Early-warning metrics	Gap: detection time before merger, partial-inspiral SNR labels.	M	10

Tag	Feature	State in gravyflow	Effort once	Models
e-pe	PE diagnostics	Gap: P-P plots, JSD against samplers, sky-area metrics.	L	23
x-retrieve	Full text needed	The paper was not read (paywall or unextractable PDF); the owner or an institutional copy is needed before any adaptation can be stated.	-	42

Table 12. Per-model index: section, comparability, adaptation effort, itinerary step and reproduction status.

Model	Year	Section	Origin	Grade	Effort	Step	gravyflow
abbott2021 [300]	2021	other	added	not comparable	L	14	*pending*
alfaidi2024 [110]	2024	early	added	conditional	M	11	*pending*
alhassan2022 [93]	2022	cbc	original	not comparable	M	15	*pending*
alhassan2023pymerger [94]	2023	cbc	added	not comparable	L	15	*pending*
alvares2021 [89]	2021	cbc	added	not comparable	M	9	*pending*
alvarez2022cwbml [19]	2022	burst	added	comparable	L	14	*pending*
alvarez2023 [202]	2023	glitch	original	not comparable	M	10	*pending*
andrescarcasona2023 [78]	2023	cbc	original	conditional	-	0	*pending*
andrescarcasona2023pe [270]	2023	pe	added	not comparable	L	15	*pending*
andrews2022 [82]	2022	cbc	original	conditional	S	2	*pending*
antelis2021 [131]	2021	burst	added	not comparable	L	14	*pending*
apostol2023 [221]	2023	glitch	added	not comparable	M	10	*pending*
arutkeerthi2024 [207]	2024	glitch	added	not comparable	S	10	*pending*
ashton2025 [105]	2025	cbc	added	not comparable	L	14	*pending*
astone2018 [115]	2018	burst	original	conditional	M	12	*pending*
attadio2024 [143]	2024	burst	original	not comparable	L	15	*pending*
aveiro2022 [90]	2022	cbc	original	not comparable	L	9	*pending*
bacon2022 [228]	2022	denoise	original	not comparable	S	8	*pending*
badanerin2024 [264]	2024	pe	added	not comparable	L	15	*pending*
badger2024 [87]	2024	cbc	added	conditional	M	8	*pending*
bahaadini2017 [21]	2017	glitch	original	conditional	S	10	*pending*
bahaadini2018 [173]	2018	glitch	original	not comparable	S	10	*pending*
bahaadini2018direct [191]	2018	glitch	original	not comparable	M	10	*pending*
bahaadini2022 [193]	2022	glitch	added	not comparable	M	10	*pending*
baltus2021 [106]	2021	early	original	not comparable	M	11	*pending*
baltus2021b [107]	2021	early	original	not comparable	-	0	*pending*
baltus2022 [108]	2022	early	original	conditional	M	11	*pending*
bayley2020 [164]	2020	cw	original	conditional	L	15	*pending*
beheshtipour2020 [165]	2020	cw	added	not comparable	L	15	*pending*
beheshtipour2020b [166]	2020	cw	added	not comparable	L	15	*pending*
benedikt2025 [280]	2025	pe	added	not comparable	L	15	*pending*
beveridge2023 [77]	2023	cbc	original	conditional	M	6	*pending*
beveridge2025o3 [15]	2025	cbc	added	comparable	M	6	*pending*
bhandari2025 [187]	2025	glitch	added	not comparable	-	0	*pending*
bini2023 [209]	2023	glitch	original	conditional	M	14	*pending*
biskin2023 [184]	2023	glitch	added	not comparable	-	0	*pending*
boudart2022 [142]	2022	burst	original	not comparable	L	15	*pending*
boudart2022glitch [220]	2022	glitch	original	not comparable	L	15	*pending*
bresten2019 [40]	2019	cbc	original	conditional	M	13	*pending*
cabero2020 [299]	2020	other	original	not comparable	L	15	*pending*
cain2025 [148]	2025	anomaly	added	not comparable	M	7	*pending*
caramete2020 [293]	2020	space	added	not comparable	L	15	*pending*
casallaslagos2026 [121]	2026	burst	added	not comparable	L	12	*pending*
cavaglia2018 [215]	2018	glitch	added	not comparable	L	14	*pending*
cavaglia2020 [130]	2020	burst	original	not comparable	L	14	*pending*
chan2020 [117]	2020	burst	original	conditional	M	12	*pending*
chan2024 [301]	2024	other	original	not comparable	L	15	*pending*
chan2025 [210]	2025	glitch	added	not comparable	M	10	*pending*
chan2025lens [257]	2025	pe	added	conditional	L	15	*pending*
chatterjee2019 [273]	2019	pe	added	not comparable	S	15	*pending*
chatterjee2021 [226]	2021	denoise	original	conditional	M	8	*pending*
chatterjee2022 [274]	2022	pe	added	not comparable	L	15	*pending*
chatterjee2023skyloc [275]	2023	pe	original	not comparable	L	15	*pending*
chatterjee2024 [71]	2024	anomaly	original	comparable	L	13	*pending*
chatterjee2024harm [231]	2024	denoise	added	conditional	M	8	*pending*
chatterjee2024rec [232]	2024	denoise	added	not comparable	M	8	*pending*
chatterjee2026 [72]	2026	cbc	original	comparable	M	4	*pending*
chaturvedi2022 [65]	2022	cbc	added	not comparable	M	5	*pending*
chauhan2020 [45]	2021	cbc	added	not comparable	S	2	*pending*
cheung2025unet [160]	2025	cw	added	conditional	L	15	*pending*

Model	Year	Section	Origin	Grade	Effort	Step	gravyflow
choudhary2022 [204]	2022	glitch	added	not comparable	S	10	*pending*
chua2019 [267]	2019	pe	added	not comparable	L	15	*pending*
cirfeta2026 [153]	2026	anomaly	added	not comparable	L	14	*pending*
colgan2020 [213]	2020	glitch	original	not comparable	L	14	*pending*
colgan2022 [214]	2022	glitch	added	not comparable	L	14	*pending*
coughlin2019 [192]	2019	glitch	original	not comparable	M	10	*pending*
cuoco2020 [313]	2021	other	original	not comparable	-	15	*pending*
cuoco2026 [137]	2026	burst	original	conditional	L	14	*pending*
dax2021 [27]	2021	pe	original	comparable	L	15	*pending*
dax2022 [28]	2022	pe	added	comparable	L	15	*pending*
dax2024 [255]	2024	pe	added	conditional	L	15	*pending*
deighan2020 [35]	2020	cbc	added	not comparable	S	2	*pending*
delaunoy2020 [265]	2020	pe	added	conditional	L	15	*pending*
deshmukh2026 [196]	2026	glitch	added	not comparable	M	10	*pending*
dicesare2026fh [168]	2026	cw	added	not comparable	L	15	*pending*
ding2021 [200]	2021	glitch	added	not comparable	M	7	*pending*
dodia2021residual [48]	2021	cbc	added	not comparable	S	2	*pending*
dodia2021specgrav [95]	2021	cbc	added	not comparable	S	9	*pending*
dominguez2023 [161]	2023	cw	added	not comparable	L	15	*pending*
dooney2022 [223]	2022	glitch	added	not comparable	L	13	*pending*
dooney2024 [305]	2024	other	original	not comparable	L	13	*pending*
dooney2025 [230]	2025	denoise	original	conditional	M	8	*pending*
dooney2026 [224]	2026	glitch	added	not comparable	L	13	*pending*
dreissigacker2019 [155]	2019	cw	original	comparable	L	15	*pending*
dreissigacker2020 [156]	2020	cw	original	comparable	L	15	*pending*
duraisamy2023 [162]	2023	cw	added	not comparable	-	0	*pending*
edwards2021 [310]	2021	other	original	not comparable	M	12	*pending*
einsle2025 [309]	2025	other	added	not comparable	M	15	*pending*
emma2026 [266]	2026	pe	added	not comparable	L	15	*pending*
essick2020 [212]	2020	glitch	original	not comparable	L	14	*pending*
fan2019 [41]	2019	cbc	original	not comparable	S	2	*pending*
fan2021_2d [96]	2021	cbc	original	not comparable	-	0	*pending*
fayad2024 [147]	2024	anomaly	original	not comparable	M	13	*pending*
fernandes2023 [178]	2023	glitch	original	conditional	S	10	*pending*
fernandes2025 [134]	2025	burst	added	conditional	L	14	*pending*
fernandes2026 [189]	2026	glitch	added	not comparable	M	10	*pending*
g2net2022_acm [97]	2021	cbc	added	not comparable	-	0	*pending*
gabbard2018 [2]	2018	cbc	original	conditional	-	1	*pending*
gabbard2022pe [263]	2022	pe	original	conditional	L	15	*pending*
garg2024 [56]	2023	cbc	added	not comparable	-	0	*pending*
gebhard2019 [4]	2019	cbc	original	conditional	M	5	*pending*
george2017tl [174]	2017	glitch	added	conditional	L	10	*pending*
george2018 [1]	2018	cbc	original	not comparable	-	1	*pending*
george2018b [3]	2018	cbc	original	not comparable	S	1	*pending*
george2018glitch [175]	2017	glitch	original	conditional	S	10	*pending*
george2018plb [29]	2017	cbc	added	not comparable	S	1	*pending*
ghalsasi2023 [236]	2023	denoise	added	not comparable	L	9	*pending*
glanzer2023 [23]	2023	glitch	original	conditional	S	10	*pending*
green2020 [26]	2020	pe	original	conditional	L	15	*pending*
gunny2021 [68]	2021	cbc	original	not comparable	-	14	*pending*
guo2025 [149]	2025	anomaly	added	not comparable	M	7	*pending*
gupta2025ae [211]	2025	glitch	added	not comparable	-	0	*pending*
gupta2026 [11]	2026	cbc	added	comparable	M	3	*pending*
gupta2026q [61]	2026	cbc	added	not comparable	L	13	*pending*
houba2024glitch [294]	2024	space	added	not comparable	L	15	*pending*
houba2024mbh [292]	2024	space	added	not comparable	L	15	*pending*
houba2025separation [295]	2025	space	added	not comparable	L	15	*pending*
hu2024 [260]	2024	pe	added	conditional	L	15	*pending*
huerta2021 [64]	2021	cbc	original	not comparable	M	5	*pending*
iess2020 [118]	2020	burst	original	not comparable	M	12	*pending*
iess2023 [119]	2023	burst	original	not comparable	M	12	*pending*
imran2024 [150]	2024	anomaly	added	not comparable	-	0	*pending*
inguglia2025 [152]	2025	anomaly	original	not comparable	L	15	*pending*
inguglia2026 [17]	2026	anomaly	added	comparable	L	7	*pending*
insley2026flows [104]	2026	cbc	added	conditional	L	14	*pending*
isfan2025quantum [297]	2025	space	added	not comparable	L	15	*pending*

Model	Year	Section	Origin	Grade	Effort	Step	gravyflow
isfan2026qnn [298]	2026	space	added	not comparable	L	15	*pending*
iwanaga2025 [282]	2025	pe	added	not comparable	M	15	*pending*
jadhav2020 [102]	2020	cbc	added	conditional	L	14	*pending*
jadhav2023 [81]	2023	cbc	original	conditional	L	13	*pending*
jarov2023 [203]	2023	glitch	original	not comparable	S	10	*pending*
jenner2024 [271]	2024	pe	added	not comparable	-	0	*pending*
jia2026 [240]	2026	denoise	added	not comparable	-	0	*pending*
jiang2022 [51]	2022	cbc	original	not comparable	-	0	*pending*
jiang2022real [52]	2022	cbc	added	not comparable	-	0	*pending*
joshi2023 [157]	2023	cw	added	comparable	L	15	*pending*
joshi2024large [159]	2024	cw	added	comparable	L	15	*pending*
joshi2025vit [158]	2025	cw	added	comparable	L	15	*pending*
kelleter2026nnnn [247]	2026	denoise	added	not comparable	L	15	*pending*
khalouei2025 [281]	2025	pe	added	not comparable	L	15	*pending*
kiendrebeogo2025virgo [243]	2025	denoise	added	conditional	L	14	*pending*
kim2015grb [100]	2015	cbc	added	not comparable	L	14	*pending*
kim2019 [101]	2019	cbc	added	not comparable	L	14	*pending*
kim2020lens [306]	2020	other	added	not comparable	M	15	*pending*
kim2022lens [307]	2022	other	added	not comparable	M	15	*pending*
kim2026kagra [197]	2026	glitch	added	not comparable	L	15	*pending*
kofler2025 [256]	2025	pe	original	conditional	L	15	*pending*
kolmus2021 [276]	2021	pe	added	not comparable	L	15	*pending*
koloniari2024 [13]	2024	cbc	original	conditional	L	4	*pending*
koyama2024 [185]	2024	glitch	added	not comparable	-	0	*pending*
krastev2020 [30]	2020	cbc	original	not comparable	S	2	*pending*
krastev2021 [31]	2021	cbc	original	not comparable	M	11	*pending*
kumar2023 [57]	2023	cbc	added	not comparable	-	0	*pending*
kumar2026 [241]	2026	denoise	added	not comparable	-	0	*pending*
laguarta2023 [199]	2023	glitch	original	not comparable	L	14	*pending*
lanchares2025 [261]	2025	pe	added	conditional	L	15	*pending*
lee2025 [233]	2025	denoise	added	conditional	M	8	*pending*
li2020 [46]	2020	cbc	original	not comparable	M	9	*pending*
li2024ctsae [195]	2024	glitch	added	not comparable	L	10	*pending*
licciardi2024 [181]	2024	glitch	original	conditional	M	14	*pending*
lin2019 [42]	2019	cbc	original	not comparable	M	9	*pending*
lin2021 [37]	2021	cbc	original	not comparable	L	13	*pending*
lin2024 [234]	2024	denoise	added	conditional	M	8	*pending*
liu2023 [291]	2023	space	added	not comparable	L	15	*pending*
llorensmonteagudo2018 [218]	2018	glitch	added	not comparable	M	10	*pending*
lopez2021a [126]	2021	burst	original	not comparable	-	0	*pending*
lopez2021b [116]	2021	burst	original	conditional	L	12	*pending*
lopez2022gan [303]	2022	other	added	not comparable	L	13	*pending*
lopez2022gmm [133]	2022	burst	original	conditional	L	14	*pending*
luo2019 [43]	2019	cbc	original	not comparable	-	0	*pending*
luo2026 [259]	2026	pe	added	not comparable	L	15	*pending*
ma2022 [38]	2022	cbc	original	not comparable	S	2	*pending*
ma2024et [252]	2024	denoise	added	not comparable	L	15	*pending*
ma2024mix [253]	2024	denoise	added	not comparable	L	15	*pending*
ma2024msnr2 [237]	2024	denoise	added	not comparable	-	0	*pending*
ma2024tempfree [85]	2023	cbc	added	not comparable	L	8	*pending*
ma2024tsr [238]	2024	denoise	added	not comparable	-	0	*pending*
ma2025et [88]	2025	cbc	added	not comparable	-	0	*pending*
ma2025overlap [154]	2025	anomaly	added	not comparable	-	0	*pending*
magee2024interp [206]	2024	glitch	added	not comparable	M	6	*pending*
malz2024 [186]	2024	glitch	added	not comparable	M	10	*pending*
manoharan2026 [222]	2026	glitch	added	not comparable	M	10	*pending*
marianer2021 [141]	2021	burst	original	not comparable	L	10	*pending*
martins2024 [109]	2024	early	added	not comparable	M	11	*pending*
marulanda2020 [277]	2020	pe	added	not comparable	S	15	*pending*
marx2024 [9]	2024	cbc	original	comparable	M	4	*pending*
marx2025aframeO3 [10]	2025	cbc	added	comparable	M	3	*pending*
mcginn2021 [302]	2021	other	original	not comparable	M	7	*pending*
mcisaac2022 [103]	2022	cbc	added	conditional	L	14	*pending*
mcleod2022 [269]	2022	pe	added	not comparable	L	15	*pending*
mcleod2025 [16]	2025	cbc	original	comparable	L	6	*pending*
meijer2023cosmicstrings [145]	2023	burst	added	not comparable	L	12	*pending*

Model	Year	Section	Origin	Grade	Effort	Step	gravyflow
meijer2024 [36]	2024	cbc	added	not comparable	M	2	*pending*
menendez2022sub [53]	2022	cbc	added	not comparable	-	0	*pending*
menendez2024 [79]	2024	cbc	original	conditional	M	9	*pending*
menendezvazquez2021 [73]	2021	cbc	original	conditional	M	9	*pending*
mesuga2021 [176]	2021	glitch	added	not comparable	L	10	*pending*
miller2019 [171]	2019	cw	added	not comparable	L	15	*pending*
mishra2021 [132]	2021	burst	added	conditional	L	14	*pending*
mlgwsc2022 [8]	2022	cbc	original	comparable	L	4	*pending*
mobilia2026 [86]	2026	cbc	original	not comparable	M	6	*pending*
modafferi2023 [144]	2023	burst	original	not comparable	L	15	*pending*
mogushi2021 [248]	2021	denoise	original	not comparable	L	14	*pending*
mogushi2021nnetfix [249]	2021	denoise	original	not comparable	S	8	*pending*
morales2024 [122]	2024	burst	added	not comparable	L	12	*pending*
morawski2020cw [167]	2019	cw	added	not comparable	L	15	*pending*
morawski2021 [151]	2021	anomaly	added	not comparable	-	0	*pending*
moreno2022 [140]	2022	burst	original	conditional	M	7	*pending*
murali2022 [235]	2022	denoise	added	not comparable	-	0	*pending*
nagarajan2025 [14]	2025	cbc	original	comparable	L	4	*pending*
nigam2020 [125]	2020	burst	original	not comparable	S	7	*pending*
nousi2022 [12]	2023	cbc	original	comparable	M	4	*pending*
nunes2024 [127]	2024	burst	added	not comparable	-	0	*pending*
ormiston2020 [25]	2020	denoise	original	conditional	L	14	*pending*
oshino2025 [198]	2025	glitch	added	not comparable	-	0	*pending*
papalini2025 [272]	2025	pe	added	not comparable	-	0	*pending*
papigkiotis2026 [111]	2026	early	added	not comparable	L	11	*pending*
pimpalkar2026 [251]	2026	denoise	added	conditional	L	14	*pending*
powell2015 [216]	2015	glitch	original	not comparable	M	14	*pending*
powell2016 [219]	2016	glitch	original	not comparable	L	14	*pending*
powell2022 [304]	2022	other	added	not comparable	L	13	*pending*
pracchia2025 [128]	2025	burst	original	conditional	M	12	*pending*
qiu2022 [39]	2022	cbc	original	not comparable	S	2	*pending*
que2021 [312]	2021	other	added	not comparable	M	13	*pending*
rai2026 [99]	2026	cbc	added	not comparable	S	9	*pending*
raikman2023 [146]	2023	anomaly	original	conditional	M	7	*pending*
raikman2024 [136]	2024	burst	original	comparable	L	7	*pending*
ratner2026 [138]	2026	burst	original	conditional	M	7	*pending*
ravichandran2023 [284]	2023	pe	added	not comparable	L	15	*pending*
razzano2018 [217]	2018	glitch	original	not comparable	S	10	*pending*
rebei2018 [33]	2018	cbc	added	not comparable	L	2	*pending*
reissel2025 [244]	2025	denoise	added	conditional	L	14	*pending*
ruan2023 [285]	2023	space	original	not comparable	L	15	*pending*
ruan2023pe [287]	2023	space	added	not comparable	L	15	*pending*
ruan2024 [286]	2024	space	added	not comparable	L	15	*pending*
sakai2022 [194]	2022	glitch	original	not comparable	M	13	*pending*
sakai2024 [239]	2024	denoise	added	not comparable	-	0	*pending*
sakellariou2026 [283]	2026	pe	added	not comparable	S	15	*pending*
saleem2023 [242]	2023	denoise	original	conditional	L	14	*pending*
sankarapandian2021 [201]	2021	glitch	added	not comparable	L	13	*pending*
santos2020 [47]	2020	cbc	added	not comparable	S	2	*pending*
sasaoka2023_cbc [98]	2023	cbc	original	not comparable	-	0	*pending*
sasaoka2023_ccsn [120]	2023	burst	original	not comparable	-	0	*pending*
sasaoka2023attr [92]	2023	cbc	added	not comparable	M	9	*pending*
schafer2020 [5]	2020	cbc	original	comparable	L	11	*pending*
schafer2021a [6]	2021	cbc	original	conditional	S	2	*pending*
schafer2021b [7]	2021	cbc	original	conditional	M	3	*pending*
schmitt2019 [44]	2019	cbc	original	not comparable	-	0	*pending*
shah2023 [208]	2023	glitch	added	not comparable	L	10	*pending*
sharma2023 [58]	2023	cbc	added	not comparable	-	0	*pending*
sharma2024 [59]	2024	cbc	added	not comparable	-	0	*pending*
shen2017 [24]	2017	denoise	original	not comparable	S	8	*pending*
shen2019 [225]	2019	denoise	original	conditional	S	8	*pending*
shen2019pe [268]	2019	pe	added	not comparable	L	15	*pending*
silver2025 [76]	2025	cbc	original	not comparable	M	2	*pending*
singh2020 [278]	2020	pe	added	not comparable	S	15	*pending*
singh2025 [188]	2025	glitch	added	not comparable	-	0	*pending*
skliris2024 [135]	2024	burst	original	comparable	S	1	*pending*

Model	Year	Section	Origin	Grade	Effort	Step	gravyflow
soma2023 [279]	2023	pe	added	not comparable	S	15	*pending*
soni2021 [22]	2021	glitch	original	conditional	S	10	*pending*
srivastava2024 [180]	2024	glitch	original	conditional	S	10	*pending*
sun2026 [123]	2026	burst	added	conditional	L	12	*pending*
szczepanczyk2022 [18]	2022	burst	original	comparable	L	14	*pending*
talpini2022 [183]	2022	glitch	added	not comparable	-	0	*pending*
tenorio2025g2net_cw [163]	2025	cw	added	conditional	L	15	*pending*
tian2023 [66]	2023	cbc	added	conditional	M	5	*pending*
tian2024gnn [67]	2023	cbc	added	conditional	M	5	*pending*
tiki2025 [80]	2025	cbc	added	conditional	M	3	*pending*
torresforne2020 [250]	2020	denoise	added	not comparable	M	8	*pending*
trovato2023 [74]	2023	cbc	original	conditional	M	10	*pending*
ussipov2024 [60]	2024	cbc	added	not comparable	-	0	*pending*
utina2021 [308]	2021	other	added	not comparable	S	15	*pending*
vajente2020 [245]	2020	denoise	original	not comparable	L	14	*pending*
vandyke2025 [205]	2025	glitch	added	conditional	S	10	*pending*
vanstraalen2024 [262]	2024	pe	added	not comparable	L	15	*pending*
verma2021 [49]	2021	cbc	original	not comparable	S	2	*pending*
verma2022 [32]	2022	cbc	original	not comparable	S	2	*pending*
vinciguerra2017 [129]	2017	burst	added	not comparable	L	14	*pending*
wang2020 [83]	2020	cbc	original	not comparable	M	6	*pending*
wang2022waveformer [229]	2024	denoise	original	conditional	L	8	*pending*
wang2023tf [91]	2024	cbc	original	conditional	M	9	*pending*
wang2025exotic [75]	2025	cbc	original	conditional	M	2	*pending*
wang2026ccsn_scl [124]	2026	burst	added	not comparable	L	12	*pending*
weerasinghe2026 [62]	2026	cbc	added	not comparable	M	15	*pending*
wei2020denoise [227]	2020	denoise	original	conditional	S	8	*pending*
wei2021 [114]	2021	early	original	not comparable	L	11	*pending*
wei2021ensemble [63]	2021	cbc	original	conditional	M	5	*pending*
wei2021forecast [113]	2021	early	original	not comparable	M	11	*pending*
wu2024 [179]	2024	glitch	original	conditional	S	10	*pending*
xia2021 [34]	2021	cbc	original	conditional	S	2	*pending*
xie2026 [190]	2026	glitch	added	not comparable	-	0	*pending*
xu2024 [311]	2024	other	original	not comparable	L	15	*pending*
xu2024denoise [254]	2024	denoise	added	not comparable	L	15	*pending*
xu2026ssl_lisa [296]	2026	space	added	not comparable	L	15	*pending*
yamamoto2021cw [169]	2021	cw	added	conditional	L	15	*pending*
yamamoto2022cw [170]	2022	cw	added	conditional	L	15	*pending*
yamamoto2023 [172]	2023	cw	original	not comparable	S	15	*pending*
yamamoto2025 [70]	2025	cbc	original	comparable	M	6	*pending*
yan2021 [84]	2021	cbc	original	not comparable	M	6	*pending*
yan2022 [54]	2022	cbc	original	not comparable	-	0	*pending*
yan2022gan [182]	2022	glitch	added	conditional	L	10	*pending*
yan2025deepgrav [139]	2025	burst	added	not comparable	M	7	*pending*
yu2021early [112]	2021	early	original	conditional	L	14	*pending*
yu2021noise [246]	2021	denoise	original	not comparable	L	14	*pending*
yun2023 [290]	2023	space	added	not comparable	L	15	*pending*
zelenka2024 [69]	2024	cbc	added	comparable	M	4	*pending*
zevin2017 [20]	2017	glitch	original	conditional	M	10	*pending*
zevin2024 [177]	2024	glitch	original	conditional	S	10	*pending*
zhan2021 [50]	2021	cbc	added	not comparable	S	2	*pending*
zhang2022 [55]	2022	cbc	original	not comparable	-	0	*pending*
zhang2022emri [289]	2022	space	added	not comparable	L	15	*pending*
zhang2026 [258]	2026	pe	added	conditional	L	15	*pending*
zhao2023 [288]	2023	space	original	not comparable	L	15	*pending*

16 Open questions

Which benchmark contract should gravyflow's validator reproduce first? MLGWSC-1 is the only cross-paper detection contract with published code; the LVK injection campaigns are the production-grade alternative. Reproducing both would let every reproduced detector be placed on the comparable rows of Section Section 3.

Per-window false-alarm probabilities. Most detection papers report them. A rate needs the window stride, the clustering rule and the background duration, which many papers omit; our reproductions can supply the rate for the paper's own network, but the paper's numbers stay as reported.

Gravity Spy release and split. Glitch accuracies differ by release and split. A fixed gravyflow split of one release (with macro-F1 as well as accuracy) would make the reproduced classifiers comparable among themselves.

Entries to resolve. 37 entries were read at abstract level or not at all (none has an arXiv version); most need institutional access. The sweep left open whether george2018plb should merge with george2018b, and it found that ma2024tempfree and the MSNRnet-1 citation share a journal article number (Phys. Rev. D 109, 043009); they are the same article (the published version, retitled "Using deep learning to predict matched signal-to-noise ratio of gravitational waves"), so neither citation is wrong. Two marginal entries are kept and flagged: kelleter2026nnnn (seismic-array Newtonian-noise regression) and jia2026 (data type unverified).

Suspected errors. The highlighted inconsistencies are the synthesiser's reading of the extraction notes. Before any is cited outside this review it should be checked against the paper, and where it matters, raised with the authors.

Out-of-scope domains. Space-based detectors, continuous waves and full parameter estimation need new detector responses, waveform families or diagnostics in gravyflow. Whether they enter the reproduction programme is a scope decision.

References

[1] D. George and E. A. Huerta, 'Deep neural networks to enable real-time multimessenger astrophysics (Deep Filtering)', *Phys. Rev. D 97, 044039 (2018)*, 2018, doi: 10.1103/PhysRevD.97.044039.

[2] H. Gabbard, M. Williams, F. Hayes, and C. Messenger, 'Matching matched filtering with deep networks for gravitational-wave astronomy', *Phys. Rev. Lett. 120, 141103 (2018)*, 2018, doi: 10.1103/PhysRevLett.120.141103.

[3] D. George and E. A. Huerta, 'Deep Learning for real-time gravitational wave detection and parameter estimation: Results with Advanced LIGO data', *Phys. Lett. B 778, 64-70 (2018)*, 2018, doi: 10.1016/j.physletb.2017.12.053.

[4] T. D. Gebhard, N. Kilbertus, I. Harry, and B. Scholkopf, 'Convolutional neural networks: a magic bullet for gravitational-wave detection?', *Phys. Rev. D 100, 063015 (2019)*, 2019, doi: 10.1103/PhysRevD.100.063015.

[5] M. B. Schafer, F. Ohme, and A. H. Nitz, 'Detection of gravitational-wave signals from binary neutron star mergers using machine learning', *Phys. Rev. D 102, 063015 (2020)*, 2020, [Online]. Available: https://arxiv.org/abs/2006.01509

[6] M. B. Schafer, O. Zelenka, A. H. Nitz, F. Ohme, and B. Bruegmann, 'Training Strategies for Deep Learning Gravitational-Wave Searches', 2021. [Online]. Available: https://arxiv.org/abs/2106.03741

[7] M. B. Schafer and A. H. Nitz, 'From One to Many: A Deep Learning Coincident Gravitational-Wave Search', 2021. [Online]. Available: https://arxiv.org/abs/2108.10715

[8] M. B. Schafer, O. Zelenka, and A. H. Nitz, 'MLGWSC-1: The first Machine Learning Gravitational-Wave Search Mock Data Challenge', 2022. [Online]. Available: https://arxiv.org/abs/2209.11146

[9] E. Marx, W. Benoit, and A. Gunny, 'A machine-learning pipeline for real-time detection of gravitational waves from compact binary coalescences (Aframe)', 2024. [Online]. Available: https://arxiv.org/abs/2403.18661

[10] E. Marx, W. Benoit, T. Blodgett, D. Chatterjee, E. de Bruin, and S. Henderson, 'A machine learning-enabled search for binary black hole mergers in LIGO-Virgo-KAGRAs third observing run', *arXiv preprint*, 2025, [Online]. Available: https://arxiv.org/abs/2505.21261

[11] B. Gupta *et al.*, 'AI-enabled gravitational-waves searches for binary neutron stars at optimal sensitivity', *arXiv preprint*, 2026, [Online]. Available: https://arxiv.org/abs/2607.01372

[12] P. Nousi, A. E. Koloniari, N. Passalis, P. Iosif, N. Stergioulas, and A. Tefas, 'Deep Residual Networks for Gravitational Wave Detection (AresGW)', *Phys. Rev. D 108, 042002 (2023)*, 2023, [Online]. Available: https://arxiv.org/abs/2211.01520

[13] A. E. Koloniari, E. C. Koursoumpa, and P. Nousi, 'New Gravitational Wave Discoveries Enabled by Machine Learning (AresGW upgrade)', *Mach. Learn.: Sci. Technol. 6(1), 015054 (2025)*, 2024, [Online]. Available: https://arxiv.org/abs/2407.07820

[14] N. Nagarajan, 'Identifying and Mitigating Machine Learning Biases for the Gravitational Wave Detection Problem (Sage)', 2025. [Online]. Available: https://arxiv.org/abs/2501.13846

[15] D. Beveridge, A. McLeod, L. Wen, W. Guo, and A. Wicenec, 'Searching for binary black hole mergers with deep learning in Advanced LIGO's third observing run', *arXiv preprint v2 (Jul 2026); DOI 10.1103/3yph-35r4 given on the arXiv record (journal not stated in the text)*, 2025, doi: 10.1103/3yph-35r4.

[16] A. McLeod, 'Binary Neutron Star Merger Search Pipeline Powered by Deep Learning', *Phys. Rev. D 111, 024035 (2025)*, 2025, [Online]. Available: https://arxiv.org/abs/2409.06266

[17] G. Inguglia, H. Haigh, U. Dupletsa, and A. Longo, 'MADGRAV: a multilevel anomaly-detection pipeline for gravitational-wave searches applied to LIGO data', *arXiv preprint*, 2026, [Online]. Available: https://arxiv.org/abs/2609.39583

[18] M. J. Szczepanczyk, 'Search for gravitational-wave bursts in the third Advanced LIGO-Virgo run with coherent WaveBurst enhanced by Machine Learning', 2022. [Online]. Available: https://arxiv.org/abs/2210.01754

[19] T. Mishra, B. O'Brien, M. Szczepanczyk, G. Vedovato, S. Bhaumik, and V. Gayathri, 'Search for binary black hole mergers in the third observing run of Advanced LIGO-Virgo using coherent WaveBurst enhanced with machine learning', *Phys. Rev. D 105, 083018 (2022); arXiv:2201.01495*, 2022, doi: 10.1103/PhysRevD.105.083018.

[20] M. Zevin, S. Coughlin, and S. Bahaadini, 'Gravity Spy: Integrating Advanced LIGO Detector Characterization, Machine Learning, and Citizen Science', *Class. Quantum Grav. 34, 064003 (2017)*, 2017, [Online]. Available: https://arxiv.org/abs/1611.04596

[21] S. Bahaadini, 'Deep Multi-view Models for Glitch Classification', *ICASSP 2017*, 2017, [Online]. Available: https://arxiv.org/abs/1705.00034

[22] S. Soni, 'Discovering features in gravitational-wave data through detector characterization, citizen science and machine learning', *Class. Quantum Grav. 38, 195016 (2021)*, 2021, [Online]. Available: https://arxiv.org/abs/2103.12104

[23] J. Glanzer, 'Data quality up to the third observing run of Advanced LIGO: Gravity Spy glitch classifications', *Class. Quantum Grav. 40, 065004 (2023)*, 2023, [Online]. Available: https://arxiv.org/abs/2208.12849

[24] H. Shen, D. George, E. A. Huerta, and Z. Zhao, 'Denoising Gravitational Waves using Deep Learning with Recurrent Denoising Autoencoders', *ICASSP 2019*, 2017, [Online]. Available: https://arxiv.org/abs/1711.09919

[25] R. Ormiston, T. Nguyen, M. Coughlin, R. X. Adhikari, and E. Katsavounidis, 'Noise Reduction in Gravitational-wave Data via Deep Learning (DeepClean)', *Phys. Rev. Research 2, 033066 (2020)*, 2020, [Online]. Available: https://arxiv.org/abs/2005.06534

[26] S. R. Green and J. Gair, 'Complete parameter inference for GW150914 using deep learning', 2020. [Online]. Available: https://arxiv.org/abs/2008.03312

[27] M. Dax, 'Real-time gravitational-wave science with neural posterior estimation (DINGO)', *Phys. Rev. Lett. 127, 241103 (2021)*, 2021, [Online]. Available: https://arxiv.org/abs/2106.12594

[28] M. Dax, S. R. Green, J. Gair, M. Pürrer, J. Wildberger, and J. H. Macke, 'Neural Importance Sampling for Rapid and Reliable Gravitational-Wave Inference', *Phys. Rev. Lett. 130, 171403 (2023)*, 2022, doi: 10.1103/PhysRevLett.130.171403.

[29] D. George and E. A. Huerta, 'Deep Learning for Real-time Gravitational Wave Detection and Parameter Estimation with LIGO Data', *NIPS 2017 Workshop on Deep Learning for Physical Sciences (DLPS), arXiv 1711.07966v2*, 2017, [Online]. Available: https://arxiv.org/abs/1711.07966

[30] P. G. Krastev, 'Real-time detection of gravitational waves from binary neutron stars using artificial neural networks', *Phys. Lett. B 803, 135330 (2020)*, 2020, doi: 10.1016/j.physletb.2020.135330.

[31] P. G. Krastev, K. Gill, V. A. Villar, and E. Berger, 'Detection and Parameter Estimation of Gravitational Waves from Binary Neutron-Star Mergers in Real LIGO Data using Deep Learning', *Phys. Lett. B 815, 136161 (2021)*, 2021, [Online]. Available: https://arxiv.org/abs/2012.13101

[32] C. Verma, 'Detection of Gravitational Wave Signals from Precessing Binary Black Hole Systems using Convolutional Neural Network', 2022. [Online]. Available: https://arxiv.org/abs/2206.12673

[33] A. Rebei, E. A. Huerta, S. Wang, S. Habib, R. Haas, and D. Johnson, 'Fusing numerical relativity and deep learning to detect higher-order multipole waveforms from eccentric binary black hole mergers', *Phys. Rev. D 100, 044025 (2019)*, 2018, doi: 10.1103/PhysRevD.100.044025.

[34] H. Xia, 'Improved deep learning techniques in gravitational-wave data analysis', *Phys. Rev. D 103, 024040 (2021)*, 2021, [Online]. Available: https://arxiv.org/abs/2011.04418

[35] D. S. Deighan, S. E. Field, C. D. Capano, and G. Khanna, 'Genetic-algorithm-optimized neural networks for gravitational wave classification', *Neural Computing and Applications volume 33, pages 13859 - 13883 (2021)*, 2020, doi: 10.1007/s00521-021-06024-4.

[36] Q. Meijer, M. van der Sluys, and S. Caudill, 'Robustness of Deep Learning Models to Precession in Gravitational-Wave Searches for Intermediate-Mass Black Hole Binaries', *arXiv preprint*, 2024, [Online]. Available: https://arxiv.org/abs/2410.07855

[37] Y.-C. Lin and J.-H. P. Wu, 'Detection of gravitational waves using Bayesian neural networks', *Phys. Rev. D 103, 063034 (2021)*, 2021, doi: 10.1103/PhysRevD.103.063034.

[38] C. Ma, W. Wang, H. Wang, and Z. Cao, 'Ensemble of deep convolutional neural networks for real-time gravitational wave signal recognition', *Phys. Rev. D 105, 083013 (2022)*, 2022, doi: 10.1103/PhysRevD.105.083013.

[39] R. Qiu, 'Deep Learning Detection and Classification of Gravitational Waves from Neutron Star-Black Hole Mergers', 2022. [Online]. Available: https://arxiv.org/abs/2210.15888

[40] C. Bresten, 'Detection of gravitational waves using topological data analysis and convolutional neural network: An improved approach', 2019. [Online]. Available: https://arxiv.org/abs/1910.08245

[41] X. Fan, J. Li, X. Li, Y. Zhong, and J. Cao, 'Applying deep neural networks to the detection and space parameter estimation of compact binary coalescence with a network of gravitational wave detectors', *Sci. China Phys. Mech. Astron. 62, 969512 (2019)*, 2019, doi: 10.1007/s11433-018-9321-7.

[42] B.-J. Lin, X.-R. Li, and W.-L. Yu, 'Binary neutron stars gravitational wave detection based on wavelet packet analysis and convolutional neural networks', *Frontiers of Physics 15(2): 24602 (2020)*, 2019, doi: 10.1007/s11467-019-0935-y.

[43] H.-M. Luo, W. Lin, Z.-C. Chen, and Q.-G. Huang, 'Extraction of gravitational wave signals with optimized convolutional neural network', *Frontiers of Physics 15(1)*, 2019, doi: 10.1007/s11467-019-0936-x.

[44] A. Schmitt, K. Fu, S. Fan, and Y. Luo, 'Investigating Deep Neural Networks for Gravitational Wave Detection in Advanced LIGO Data', *Proc. 2nd ICCSSE (ACM)*, 2019, doi: 10.1145/3339363.3339377.

[45] Y. Chauhan, 'Deep Learning Techniques to make Gravitational Wave Detections from Weak Time-Series Data', *International Astronomy and Astrophysics Research Journal, Vol. 3, no. 2, Apr. 2021, pp. 1-9*, 2020, [Online]. Available: https://arxiv.org/abs/2007.05889

[46] X.-R. Li *et al.*, 'Some optimizations on detecting gravitational wave using convolutional neural network', *Front. Phys. 15(5), 54501 (2020)*, 2020, doi: 10.1007/s11467-020-0966-4.

[47] G. R. Santos, M. P. Figueiredo, A. de P. Santos, P. Protopapas, and T. A. E. Ferreira, 'Gravitational Wave Detection and Information Extraction via Neural Networks', *arXiv preprint*, 2020, [Online]. Available: https://arxiv.org/abs/2003.09995

[48] H. Dodia, 'Detecting residues of cosmic events using residual neural network', *arXiv preprint*, 2021, [Online]. Available: https://arxiv.org/abs/2101.00195

[49] C. Verma, 'Employing Deep Learning for Detection of Gravitational Waves from Compact Binary Coalescences', 2021. [Online]. Available: https://arxiv.org/abs/2110.01883

[50] C. Zhan, M. Jia, C. Ma, Z. Lu, and W. Lin, 'The response of the Convolutional Neural Network to the transient noise in Gravitational wave detection', *arXiv preprint*, 2021, [Online]. Available: https://arxiv.org/abs/2103.03557

[51] L. Jiang and Y. Luo, 'Convolutional Transformer for Fast and Accurate Gravitational Wave Detection', *ICPR 2022*, 2022, doi: 10.1109/ICPR56361.2022.9956104.

[52] M.-Q. Jiang, N. Yang, and J. Li, 'Identify real gravitational wave events in the LIGO-Virgo catalog GWTC-1 and GWTC-2 with convolutional neural network', *Frontiers of Physics 17, 54501 (2022)*, 2022, doi: 10.1007/s11467-021-1150-1.

[53] M. Vazquez and M. Perez, 'Search for subsolar mass black holes in LIGO/Virgo using O3 data and the implementation of Machine Learning algorithms', *Proceedings*, 2022.

[54] R.-Q. Yan *et al.*, 'Gravitational Wave Detection Based on Squeeze-and-excitation Shrinkage Networks and Multiple Detector Coherent SNR', *Res. Astron. Astrophys. 22(11)*, 2022, doi: 10.1088/1674-4527/ac846c.

[55] Y. Zhang, H. Xu, M. Liu, C. Liu, Y. Zhao, and J. Zhu, 'Deep learning model based on a bidirectional gated recurrent unit for the detection of gravitational wave signals', *Phys. Rev. D 106, 122002 (2022)*, 2022, doi: 10.1103/PhysRevD.106.122002.

[56] S. Garg *et al.*, 'Comparison of training methods for Convolutional Neural Network model for Gravitational-Wave detection from Neutron Star-Black Hole Binaries', *38th International Cosmic Ray Conference (ICRC2023), PoS 444 (MM&GW); pre-published 18 Aug 2023, published 27 Sep 2024*, 2023, doi: 10.22323/1.444.1536.

[57] L. Kumar, S. K. Sahay, and H. G. Kusneniwar, 'An investigation of two-step cascaded CNN for the detection of gravitational wave signal from two different astrophysical sources', *Procedia Computer Science 222 (2023) 676-684 (INNS DLIA workshop at IJCNN 2023 per dblp)*, 2023, doi: 10.1016/j.procs.2023.08.205.

[58] Sharma and Chandra, 'Searching for massive black hole binaries with a transfer learning algorithm', *Phys. Rev. D 108, 124061 (2023)*, 2023, doi: 10.1103/PhysRevD.108.124061.

[59] Sharma and Kaur, 'Detection of Gravitational Waves Using Machine Learning', *IEEE INCET 2024*, 2024, doi: 10.1109/INCET61516.2024.10593087.

[60] Ussipov and Zhanabaev, 'Classification of Gravitational Waves from Black Hole-Neutron Star Mergers with Machine Learning', *J. Astron. Space Sci. 41, 149 (2024)*, 2024, doi: 10.5140/JASS.2024.41.3.149.

[61] A. Gupta and N. Jindal, 'A hybrid quantum-classical deep learning approach for gravitational wave detection', *Quantum Machine Intelligence (2026), article 10.1007/s42484-026-00393-6*, 2026, doi: 10.1007/s42484-026-00393-6.

[62] R. Weerasinghe, 'Real-Time Multi-Mode Post-Merger Gravitational Wave Detection using Convolutional Neural Networks: Methodology Development for Third-Generation Detectors', *arXiv preprint*, 2026, [Online]. Available: https://arxiv.org/abs/2601.00985

[63] W. Wei, A. Khan, and E. A. Huerta, 'Deep Learning Ensemble for Real-time Gravitational Wave Detection of Spinning Binary Black Hole Mergers', *Phys. Lett. B 812, 136029 (2021)*, 2021, [Online]. Available: https://arxiv.org/abs/2010.15845

[64] E. A. Huerta, A. Khan, and X. Huang, 'Accelerated, scalable and reproducible AI-driven gravitational wave detection', *Nature Astronomy 5, 1062-1068 (2021)*, 2021, doi: 10.1038/s41550-021-01405-0.

[65] P. Chaturvedi, A. Khan, M. Tian, E. A. Huerta, and H. Zheng, 'Inference-optimized AI and high performance computing for gravitational wave detection at scale', *Front. Artif. Intell. 5:828672 (2022)*, 2022, doi: 10.3389/frai.2022.828672.

[66] M. Tian, E. A. Huerta, and H. Zheng, 'AI ensemble for signal detection of higher order gravitational wave modes of quasi-circular, spinning, non-precessing binary black hole mergers', *arXiv preprint*, 2023, [Online]. Available: https://arxiv.org/abs/2310.00052

[67] M. Tian, E. A. Huerta, H. Zheng, and P. Kumar, 'Physics-inspired spatiotemporal-graph AI ensemble for the detection of higher order wave mode signals of spinning binary black hole mergers', *Mach. Learn.: Sci. Technol. 5 (2024) 025056*, 2023, doi: 10.1088/2632-2153/ad4c37.

[68] A. Gunny, 'Hardware-accelerated Inference for Real-Time Gravitational-Wave Astronomy', 2021. [Online]. Available: https://arxiv.org/abs/2108.12430

[69] O. Zelenka, B. Bruegmann, and F. Ohme, 'Convolutional Neural Networks for signal detection in real LIGO data', *Phys. Rev. D 110, 024024 (2024)*, 2024, doi: 10.1103/PhysRevD.110.024024.

[70] T. S. Yamamoto, 'Hybrid algorithm combining matched filtering and convolutional neural networks for searching gravitational waves from binary black hole mergers', 2025. [Online]. Available: https://arxiv.org/abs/2512.12399

[71] C. Chatterjee, 'Pre-trained Audio Transformer as a Foundational AI Tool for Gravitational Waves (GW-Whisper)', 2024. [Online]. Available: https://arxiv.org/abs/2412.20789

[72] C. Chatterjee, A. Petulante, H. Fu, Y. Hu, R. Lau, and K. Jani, 'A Fast and Scalable Transformer Pipeline for Binary Black Hole Detection (CASTOR)', 2026. [Online]. Available: https://arxiv.org/abs/2609.00339

[73] A. Menendez-Vazquez, M. Kolstein, M. Martinez, and L. M. Mir, 'Searches for compact binary coalescence events using neural networks in the LIGO/Virgo second observation period', *Phys. Rev. D 103, 062004 (2021)*, 2021, doi: 10.1103/PhysRevD.103.062004.

[74] A. Trovato, 'Neural network time-series classifiers for gravitational-wave searches in single-detector periods', *Class. Quantum Grav. 41, 125003 (2024)*, 2023, [Online]. Available: https://arxiv.org/abs/2307.09268

[75] Y.-X. Wang, 'Search for exotic gravitational wave signals beyond general relativity using deep learning', *Phys. Rev. D 112, 024030 (2025)*, 2025, [Online]. Available: https://arxiv.org/abs/2410.20129

[76] E. Silver, 'A Search for Binary Black Hole Mergers in LIGO O1-O3 Data with Convolutional Neural Networks', 2025. [Online]. Available: https://arxiv.org/abs/2512.17204

[77] D. Beveridge, L. Wen, and A. Wicenec, 'Novel deep learning approach to detecting binary black hole mergers (earlier: Detection of BBH mergers from the SNR time series using deep learning)', *Phys. Rev. D 111, 024005 (2025) (from search results)*, 2023, [Online]. Available: https://arxiv.org/abs/2308.08429

[78] M. Andres-Carcasona, A. Menendez-Vazquez, M. Martinez, and L. M. Mir, 'Searches for mass-asymmetric compact binary coalescence events using neural networks in the LIGO/Virgo third observation period', *Phys. Rev. D 107, 082003 (2023)*, 2023, doi: 10.1103/PhysRevD.107.082003.

[79] A. Menendez-Vazquez, 'Searches for Compact Binary Coalescence Events Using Neural Networks in LIGO/Virgo Third Observation Period', *Class. Quantum Grav. 41, 135018 (2024)*, 2024, [Online]. Available: https://arxiv.org/abs/2401.12912

[80] V. Tiki and E. Huerta, 'A Practical Study of Lightweight Neural Gravitational-Wave Detection in Real LIGO Noise: Models, Ablations, and Open Software', *arXiv preprint*, 2025, [Online]. Available: https://arxiv.org/abs/2512.12513

[81] S. Jadhav, M. Shrivastava, and S. Mitra, 'Towards a robust and reliable deep learning approach for detection of compact binary mergers in gravitational wave data', *Mach. Learn.: Sci. Technol. 4, 045028 (2023)*, 2023, [Online]. Available: https://arxiv.org/abs/2306.11797

[82] M. Andrews, M. Paulini, L. Sellers, A. Bobrick, G. Martire, and H. Vestal, 'DeepSNR: A deep learning foundation for offline gravitational wave detection', 2022. [Online]. Available: https://arxiv.org/abs/2207.04749

[83] H. Wang, S. Wu, Z. Cao, X. Liu, and J.-Y. Zhu, 'Gravitational-wave signal recognition of LIGO data by deep learning (MFCNN)', *Phys. Rev. D 101, 104003 (2020)*, 2020, doi: 10.1103/PhysRevD.101.104003.

[84] J. Yan *et al.*, 'Generalized Approach to Matched Filtering using Neural Networks', 2021. [Online]. Available: https://arxiv.org/abs/2104.03961

[85] C. Ma, S. Wang, W. Wang, and Z. Cao, 'Matched filtering for gravitational wave detection without template bank driven by deep learning template prediction model bank', *Phys. Rev. D 109, 043009 (2024); arXiv v1 5 Sep 2023 (the authors later call the framework MSNRnet)*, 2023, doi: 10.1103/PhysRevD.109.043009.

[86] L. Mobilia, 'Deep Learning Search for Gravitational Waves from Compact Binary Coalescence (TT-SNR map + CNN)', 2026. [Online]. Available: https://arxiv.org/abs/2603.09386

[87] C. Badger *et al.*, 'Rapid detection of gravitational waves from binary black hole mergers using sparse dictionary learning', *arXiv preprint (v3, 21 Mar 2025); no journal reference on the arXiv record*, 2024, [Online]. Available: https://arxiv.org/abs/2405.17721

[88] Ma and Liu, 'Denoising and detection for binary black hole gravitational waves in the context of the Einstein Telescope', *Sci. China Phys. Mech. Astron. 2025*, 2025, doi: 10.1007/s11433-025-2673-5.

[89] J. D. Álvares, J. A. Font, F. F. Freitas, O. G. Freitas, A. P. Morais, and S. Nunes, 'Exploring gravitational-wave detection and parameter inference using Deep Learning methods', *Class. Quantum Grav. 38 (2021) 155010 (arXiv:2011.10425)*, 2021, doi: 10.1088/1361-6382/ac0455.

[90] J. Aveiro *et al.*, 'Identification of binary neutron star mergers in gravitational-wave data using object-detection machine learning models (YOLO)', *Phys. Rev. D 106, 084059 (2022)*, 2022, doi: 10.1103/PhysRevD.106.084059.

[91] Y.-X. Wang, 'Rapid identification of time-frequency domain gravitational wave signals from binary black holes using deep learning', *Chin. Phys. C 48, 125107 (2024)*, 2024, [Online]. Available: https://arxiv.org/abs/2305.19003

[92] S. Sasaoka, N. Koyama, D. Dominguez, Y. Sakai, K. Somiya, and Y. Omae, 'Comparative study of 1D and 2D convolutional neural network models with attribution analysis for gravitational wave detection from compact binary coalescences', *Phys. Rev. D 109, 043011 (2024)*, 2023, doi: 10.1103/PhysRevD.109.043011.

[93] W. Alhassan, 'Detection of Einstein Telescope gravitational wave signals from binary black holes using deep learning', 2022. [Online]. Available: https://arxiv.org/abs/2211.13789

[94] W. Alhassan, T. Bulik, and M. Suchenek, 'PyMerger: Detecting Binary Black Hole merger from Einstein Telescope Using Deep Learning', *arXiv preprint (astro-ph.IM, Oct 2023); later versions list the software paper Alhassan et al. 2024; no journal reference in this PDF*, 2023, [Online]. Available: https://arxiv.org/abs/2310.10409

[95] H. Dodia, H. Tandel, and L. D'Mello, 'SpecGrav -- Detection of Gravitational Waves using Deep Learning', *arXiv preprint*, 2021, [Online]. Available: https://arxiv.org/abs/2107.03607

[96] S. Fan, Y. Wang, Y. Luo, A. Schmitt, and S. Yu, 'Improving Gravitational Wave Detection with 2D Convolutional Neural Networks', *ICPR 2020, pp. 7103-7110*, 2021, doi: 10.1109/ICPR48806.2020.9412180.

[97] D. Zhang, 'Detecting Gravitational Waves using Constant-Q Transform and Convolutional Neural Networks', *2021 4th International Conference on Computational Intelligence and Intelligent Systems (CIIS 2021), Tokyo; ACM, published 20 Nov 2021 (ACM DL record 2022)*, 2021, doi: 10.1145/3507623.3507630.

[98] S. Sasaoka *et al.*, 'Deep Learning for Detecting Gravitational Waves from Compact Binary Coalescences and Its Visualization by Grad-CAM', *PoS(ICRC2023) 1498*, 2023, doi: 10.22323/1.444.1498.

[99] R. Rai, R. Verma, and Somya, 'CASPER: Interpretable ResNet based Classifier with FastShap Explainer for Gravitational Wave Detection', *arXiv preprint*, 2026, [Online]. Available: https://arxiv.org/abs/2606.17214

[100] K. Kim *et al.*, 'Application of Artificial Neural Network to Search for Gravitational-Wave Signals Associated with Short Gamma-Ray Bursts', *Class. Quantum Grav. 32 (2015) 245002*, 2015, doi: 10.1088/0264-9381/32/24/245002.

[101] K. Kim, T. G. F. Li, R. K. L. Lo, S. Sachdev, and R. S. H. Yuen, 'Ranking Candidate Signals with Machine Learning in Low-Latency Search for Gravitational-Waves from Compact Binary Mergers', *Phys. Rev. D 101, 083006 (2020)*, 2019, doi: 10.1103/PhysRevD.101.083006.

[102] S. Jadhav, N. Mukund, B. Gadre, S. Mitra, and S. Abraham, 'Improving significance of binary black hole mergers in Advanced LIGO data using deep learning : Confirmation of GW151216', *Phys. Rev. D 104, 064051 (2021)*, 2020, doi: 10.1103/PhysRevD.104.064051.

[103] C. McIsaac and I. Harry, 'Using machine learning to auto-tune chi-squared tests for gravitational wave searches', *Phys. Rev. D 105, 104056 (2022)*, 2022, doi: 10.1103/PhysRevD.105.104056.

[104] S. Insley, M. J. Williams, R. Dhurkunde, and I. Harry, 'Normalizing flows for density estimation in multi-detector gravitational-wave searches', *arXiv preprint*, 2026, [Online]. Available: https://arxiv.org/abs/2604.26581

[105] G. Ashton, A.-K. Malz, and N. Colombo, 'Enhancing gravitational-wave detection: a machine learning pipeline combination approach with robust uncertainty quantification', *Phys. Rev. D (DOI 10.1103/yfb3-fgf2); arXiv:2504.17587 (PDF is a later revision citing GWTC-4.0)*, 2025, doi: 10.1103/yfb3-fgf2.

[106] G. Baltus, J. Janquart, M. Lopez, A. Reza, S. Caudill, and J.-R. Cudell, 'Convolutional neural networks for the detection of the early inspiral of a gravitational-wave signal', *Phys. Rev. D 103, 102003 (2021)*, 2021, doi: 10.1103/PhysRevD.103.102003.

[107] G. Baltus, J.-R. Cudell, J. Janquart, M. Lopez, S. Caudill, and A. Reza, 'Detecting the early inspiral of a gravitational-wave signal with convolutional neural networks (CBMI 2021)', *CBMI 2021*, 2021, doi: 10.1109/CBMI50038.2021.9461919.

[108] G. Baltus, J. Janquart, M. Lopez, H. Narola, and J.-R. Cudell, 'Convolutional neural network for gravitational-wave early alert: Going down in frequency', *Phys. Rev. D 106, 042002 (2022)*, 2022, doi: 10.1103/PhysRevD.106.042002.

[109] A. Martins, M. Lopez, Q. Meijer, G. Baltus, M. van der Sluys, and C. Van Den Broeck, 'Improving early detection of gravitational waves from binary neutron stars using CNNs and FPGAs', *arXiv preprint*, 2024, [Online]. Available: https://arxiv.org/abs/2409.05068

[110] R. Alfaidi and C. Messenger, 'Long Short-Term Memory for Early Warning Detection of Gravitational Waves', *arXiv preprint (gr-qc, 7 Feb 2024); no journal reference found in the PDF*, 2024, [Online]. Available: https://arxiv.org/abs/2402.04589

[111] G. Papigkiotis, G. Vardakas, and N. Stergioulas, 'GW-FALCON: A Novel Feature-Driven Deep Learning Approach for Early Warning Alerts of BNS and NSBH Inspirals in Next-Generation GW Observatories', *arXiv preprint*, 2026, [Online]. Available: https://arxiv.org/abs/2602.15073

[112] H. Yu and R. X. Adhikari, 'Early warning of coalescing neutron-star and neutron-star-black-hole binaries from nonstationary noise background using neural networks', *Phys. Rev. D 104, 062004 (2021)*, 2021, [Online]. Available: https://arxiv.org/abs/2104.09438

[113] W. Wei and E. A. Huerta, 'Deep learning for gravitational wave forecasting of neutron star mergers', *Phys. Lett. B 816, 136185 (2021)*, 2021, [Online]. Available: https://arxiv.org/abs/2010.09751

[114] W. Wei *et al.*, 'Deep Learning with Quantized Neural Networks for Gravitational-wave Forecasting of Eccentric Compact Binary Coalescence', *Astrophys. J. 919(2), 82 (2021)*, 2021, doi: 10.3847/1538-4357/ac1121.

[115] P. Astone *et al.*, 'A New Method to Observe Gravitational Waves emitted by Core Collapse Supernovae', *Phys. Rev. D 98, 122002 (2018)*, 2018, [Online]. Available: https://arxiv.org/abs/1812.05363

[116] M. Lopez, I. Di Palma, M. Drago, P. Cerda-Duran, and F. Ricci, 'Deep learning for core-collapse supernova detection (arXiv: Deep learning for multimessenger core-collapse supernova detection)', *Phys. Rev. D 103, 063011 (2021)*, 2021, doi: 10.1103/PhysRevD.103.063011.

[117] M. L. Chan, I. S. Heng, and C. Messenger, 'Detection and classification of supernova gravitational wave signals: A deep learning approach', *Phys. Rev. D 102, 043022 (2020)*, 2020, doi: 10.1103/PhysRevD.102.043022.

[118] A. Iess, E. Cuoco, F. Morawski, and J. Powell, 'Core-Collapse Supernova Gravitational-Wave Search and Deep Learning Classification', *Mach. Learn.: Sci. Technol. 1(2), 025014 (2020)*, 2020, doi: 10.1088/2632-2153/ab7d31.

[119] A. Iess, E. Cuoco, F. Morawski, C. Nicolaou, and O. Lahav, 'LSTM and CNN application for core-collapse supernova search in gravitational wave real data', *Astron. Astrophys. 669 (2023)*, 2023, doi: 10.1051/0004-6361/202142525.

[120] S. Sasaoka *et al.*, 'Deep Learning Application for Detecting Gravitational Waves from Core-Collapse Supernovae (ICRC2023)', *PoS(ICRC2023) 1499*, 2023, doi: 10.22323/1.444.1499.

[121] A. Casallas-Lagos, M. J. Szczepańczyk, M. Zanolin, A. Mezzacappa, J. M. Antelis, and D. Murphy, 'Classifying the nuclear equation of state in LVK interferometric noise through core-collapse supernova gravitational-wave signatures using convolutional neural networks', *arXiv preprint*, 2026, [Online]. Available: https://arxiv.org/abs/2607.21924

[122] M. D. Morales, J. M. Antelis, and C. Moreno, 'Residual neural networks to classify the high frequency emission in core-collapse supernova gravitational waves', *Gen. Relativ. Gravit. 57, 162 (2025)*, 2024, doi: 10.1007/s10714-025-03498-8.

[123] T.-Y. Sun, Y. Niu, C.-Y. Jiang, S.-J. Jin, Y. Yuan, and X. Zhang, 'Contrastive self-supervised convolutional autoencoder for core-collapse supernova gravitational-wave detection', *arXiv preprint*, 2026, [Online]. Available: https://arxiv.org/abs/2605.21310

[124] A.-B. Wang, Y. Yuan, H. Cai, and X.-L. Fan, 'Classifying Core-Collapse Supernova Gravitational Waves using Supervised Contrastive Learning', *arXiv preprint*, 2026, [Online]. Available: https://arxiv.org/abs/2601.01376

[125] R. Nigam, A. Mishra, and P. Reddy, 'Transient Classification in low SNR Gravitational Wave data using Deep Learning', 2020. [Online]. Available: https://arxiv.org/abs/2009.12168

[126] M. Lopez, M. Drago, I. Di Palma, F. Ricci, and P. Cerda-Duran, 'Deep learning algorithms for gravitational waves core-collapse supernova detection (CBMI 2021)', *CBMI 2021, pp. 1-6*, 2021, doi: 10.1109/CBMI50038.2021.9461885.

[127] S. Nunes, G. Escrig, O. G. Freitas, J. A. Font, T. Fernandes, and A. Onofre, 'Deep-Learning Classification and Parameter Inference of Rotational Core-Collapse Supernovae', *arXiv preprint*, 2024, doi: 10.1103/PhysRevD.110.064037.

[128] M. Pracchia, 'Short-Duration Gravitational Wave Burst Detection using Convolutional Neural Network', 2025. [Online]. Available: https://arxiv.org/abs/2509.13241

[129] S. Vinciguerra, M. Drago, G. A. Prodi, S. Klimenko, C. Lazzaro, and V. Necula, 'Enhancing the significance of gravitational wave bursts through signal classification', *Class. Quantum Grav. 34 (2017) 094003 (18pp)*, 2017, doi: 10.1088/1361-6382/aa6654.

[130] M. Cavaglia, 'Improving the background of gravitational-wave searches for core collapse supernovae: A machine learning approach', *Mach. Learn.: Sci. Technol. 1, 015005 (2020)*, 2020, [Online]. Available: https://arxiv.org/abs/2002.04591

[131] J. M. Antelis, M. Cavaglia, T. Hansen, M. D. Morales, C. Moreno, and S. Mukherjee, 'Using supervised learning algorithms as a follow-up method in the search of gravitational waves from core-collapse supernovae', *Phys. Rev. D 105, 084054 (2022); arXiv:2111.07219*, 2021, doi: 10.1103/PhysRevD.105.084054.

[132] T. Mishra, B. O'Brien, V. Gayathri, M. Szczepanczyk, S. Bhaumik, and I. Bartos, 'Optimization of model independent gravitational wave search using machine learning', *Phys. Rev. D 104, 023014 (2021)*, 2021, doi: 10.1103/PhysRevD.104.023014.

[133] D. Lopez, 'Utilizing Gaussian mixture models in all-sky searches for short-duration gravitational wave bursts', *Phys. Rev. D 105, 063024 (2022)*, 2022, [Online]. Available: https://arxiv.org/abs/2112.06608

[134] J. Fernandes, A. Pai, and K. Chandra, 'Improving the detection significance of gravitational wave transient searches with CNN models', *arXiv preprint 2505.08332v1*, 2025, [Online]. Available: https://arxiv.org/abs/2505.08332

[135] V. Skliris, M. R. K. Norman, and P. J. Sutton, 'Real-Time Detection of Unmodelled Gravitational-Wave Transients Using Convolutional Neural Networks (MLy)', *Phys. Rev. D 110, 104034 (2024)*, 2024, doi: 10.1103/PhysRevD.110.104034.

[136] R. Raikman, 'A Neural Network-Based Search for Unmodeled Transients in LIGO-Virgo-KAGRA's Third Observing Run', 2024. [Online]. Available: https://arxiv.org/abs/2412.19883

[137] E. Cuoco, 'The Wavelet Detection Filter: a real time unmodeled pipeline for gravitational wave transients, ranking coincidences with a graph neural network', 2026. [Online]. Available: https://arxiv.org/abs/2609.12797

[138] D. Ratner, 'Template-free search for gravitational wave events using coincident anomaly detection', 2026. [Online]. Available: https://arxiv.org/abs/2601.11842

[139] J. Yan, A. P. Leung, D. C. Y. Hui, and S. Kim, 'DeepGrav: Anomalous Gravitational-Wave Detection Through Deep Latent Features', *arXiv preprint*, 2025, [Online]. Available: https://arxiv.org/abs/2503.03799

[140] E. A. Moreno, B. Borzyszkowski, M. Pierini, J.-R. Vlimant, and M. Spiropulu, 'Source-agnostic gravitational-wave detection with recurrent autoencoders', *Mach. Learn.: Sci. Technol. 3, 025005 (2022)*, 2022, doi: 10.1088/2632-2153/ac5435.

[141] T. Marianer, D. Poznanski, and J. X. Prochaska, 'A semisupervised machine learning search for never-seen gravitational-wave sources', *Mon. Not. R. Astron. Soc. 500(4), 5408 (2021)*, 2021, doi: 10.1093/mnras/staa3550.

[142] V. Boudart, 'A machine learning algorithm for minute-long Burst searches (ALBUS)', *Phys. Rev. D 105, 083007 (2022) (per later citation)*, 2022, [Online]. Available: https://arxiv.org/abs/2201.08727

[143] F. Attadio, 'Neural network method to search for long transient gravitational waves', 2024. [Online]. Available: https://arxiv.org/abs/2407.02391

[144] L. M. Modafferi, 'Convolutional neural network search for long-duration transient gravitational waves from glitching pulsars', *Phys. Rev. D 108, 023005 (2023)*, 2023, [Online]. Available: https://arxiv.org/abs/2303.16720

[145] Q. Meijer, M. Lopez, D. Tsuna, and S. Caudill, 'Gravitational-Wave Searches for Cosmic String Cusps in Einstein Telescope Data using Deep Learning', *arXiv preprint*, 2023, doi: 10.1103/PhysRevD.109.022006.

[146] R. Raikman, E. A. Moreno, and E. Govorkova, 'GWAK: Gravitational-Wave Anomalous Knowledge with Recurrent Autoencoders', 2023. [Online]. Available: https://arxiv.org/abs/2309.11537

[147] A. Fayad, 'Unsupervised Learning Approach to Anomaly Detection in Gravitational Wave Data (catalogue title unverified; this is the arXiv title)', 2024. [Online]. Available: https://arxiv.org/abs/2411.19450

[148] J. Cain, 'Template-Free Gravitational Wave Detection with CWT-LSTM Autoencoders: A Case Study of Run-Dependent Calibration Effects in LIGO Data', *Class. Quantum Grav. 43, 035019 (2026)*, 2025, doi: 10.1088/1361-6382/ae415e.

[149] Y.-Y. Guo, S. D. Mohanty, X. Qunying, and Y.-X. Liu, 'Using normal to find abnormal: AI-based anomaly detection in gravitational wave data', *arXiv preprint*, 2025, [Online]. Available: https://arxiv.org/abs/2508.19772

[150] A. R. Imran, 'Utilizing Convolutional Autoencoder for Anomaly Detections in LIGO Spectrogram Data', *Journal of Physics: Conference Series 2734, 012048 (IOP, March 2024)*, 2024, doi: 10.1088/1742-6596/2734/1/012048.

[151] F. Morawski, M. Bejger, E. Cuoco, and L. Petre, 'Anomaly Detection in Gravitational Waves data using Convolutional AutoEncoders', *arXiv preprint*, 2021, doi: 10.1088/2632-2153/abf3d0.

[152] G. Inguglia, 'Towards an anomaly detection pipeline for gravitational waves at the Einstein Telescope', *Phys. Lett. B 874 (2026) 140272*, 2025, [Online]. Available: https://arxiv.org/abs/2511.13154

[153] L. Cirfeta, 'DANTE: A Reference-Guided Unsupervised Pipeline for Extended-Transient Anomaly Characterization in LIGO O4a', *arXiv preprint*, 2026, [Online]. Available: https://arxiv.org/abs/2606.25702

[154] Ma and Zhou, 'Combine deep learning and Bayesian analysis to separate overlapping gravitational wave signals', *Sci. China Phys. Mech. Astron. 2025*, 2025, doi: 10.1007/s11433-024-2594-5.

[155] C. Dreissigacker, R. Sharma, C. Messenger, R. Zhao, and R. Prix, 'Deep-Learning continuous gravitational waves', *Phys. Rev. D 100, 044009 (2019)*, 2019, [Online]. Available: https://arxiv.org/abs/1904.13291

[156] C. Dreissigacker and R. Prix, 'Deep-Learning Continuous Gravitational Waves: Multiple detectors and realistic noise', *Phys. Rev. D 102, 022005 (2020)*, 2020, [Online]. Available: https://arxiv.org/abs/2005.04140

[157] P. M. Joshi and R. Prix, 'Novel neural-network architecture for continuous gravitational waves', *Phys. Rev. D 108, 063021 (2023)*, 2023, doi: 10.1103/PhysRevD.108.063021.

[158] P. M. Joshi and R. Prix, 'Transformer Networks for Continuous Gravitational-wave Searches', *Phys. Rev. D 112, 123052 (2025)*, 2025, doi: 10.1103/xgz9-qdkg.

[159] P. M. Joshi and R. Prix, 'Large-kernel Convolutional Neural Networks for Wide Parameter-Space Searches of Continuous Gravitational Waves', *Phys. Rev. D 110, 124071 (2024)*, 2024, doi: 10.1103/PhysRevD.110.124071.

[160] D. H. T. Cheung, 'Attention U-Net for all-sky continuous gravitational wave searches', *arXiv preprint*, 2025, [Online]. Available: https://arxiv.org/abs/2509.19838

[161] D. S. Dominguez *et al.*, 'Convolutional neural network for continuous gravitational waves detection', *38th International Cosmic Ray Conference (ICRC2023), PoS 444 (MM&GW), pre-published Aug 2023, published 27 Sep 2024*, 2023, doi: 10.22323/1.444.1519.

[162] P. Duraisamy, Y. Natarajan, V. Niranjani, and K. Parvathy, 'Optimized Detection of Continuous Gravitational-Wave Signals using Convolutional Neural Network', *2023 3rd International Conference on Artificial Intelligence and Signal Processing (AISP), IEEE*, 2023, doi: 10.1109/AISP57993.2023.10134809.

[163] R. Tenorio, M. J. Williams, J. Bayley, C. Messenger, M. Demkin, and W. Reade, 'Learning to detect continuous gravitational waves: an open data-analysis competition', *2025 Mach. Learn.: Sci. Technol. 6 040702*, 2025, doi: 10.1088/2632-2153/ae2dbc.

[164] J. Bayley, C. Messenger, and G. Woan, 'A robust machine learning algorithm to search for continuous gravitational waves', *Phys. Rev. D 102, 083024 (2020)*, 2020, [Online]. Available: https://arxiv.org/abs/2007.08207

[165] B. Beheshtipour and M. A. Papa, 'Deep learning for clustering of continuous gravitational wave candidates', *Phys. Rev. D 101, 064009 (2020)*, 2020, doi: 10.1103/PhysRevD.101.064009.

[166] B. Beheshtipour and M. A. Papa, 'Deep learning for clustering of continuous gravitational wave candidates II: identification of low-SNR candidates', *Phys. Rev. D 103, 064027 (2021)*, 2020, doi: 10.1103/PhysRevD.103.064027.

[167] F. Morawski, M. Bejger, and P. Cieciel\{a}g, 'Convolutional neural network classifier for the output of the time-domain F-statistic all-sky search for continuous gravitational waves', *arXiv preprint*, 2019, doi: 10.1088/2632-2153/ab86c7.

[168] M. Di Cesare, P. Astone, R. De Rosa, D. Keitel, C. Palomba, and M. Serra, 'Alternative neural-network follow-up for all-sky FrequencyHough in continuous gravitational-wave searches', *arXiv preprint*, 2026, [Online]. Available: https://arxiv.org/abs/2609.06211

[169] T. S. Yamamoto and T. Tanaka, 'Use of Excess Power Method and Convolutional Neural Network in All-Sky Search for Continuous Gravitational Waves', *Phys. Rev. D 103, 084049 (2021)*, 2020, doi: 10.1103/PhysRevD.103.084049.

[170] T. S. Yamamoto, A. L. Miller, M. Sieniawska, and T. Tanaka, 'Assessing the impact of non-Gaussian noise on convolutional neural networks that search for continuous gravitational waves', *Phys. Rev. D 106, 024025 (2022)*, 2022, doi: 10.1103/PhysRevD.106.024025.

[171] A. L. Miller, P. Astone, S. D'Antonio, S. Frasca, G. Intini, and I. La Rosa, 'How effective is machine learning to detect long transient gravitational waves from neutron stars in a real search?', *Phys. Rev. D (2019) 100, 062005*, 2019, doi: 10.1103/PhysRevD.100.062005.

[172] T. S. Yamamoto, 'Deep learning for intermittent gravitational wave signals', *Phys. Rev. D 107, 044032 (2023)*, 2023, [Online]. Available: https://arxiv.org/abs/2208.13156

[173] S. Bahaadini *et al.*, 'Machine learning for Gravity Spy: Glitch classification and dataset', *Information Sciences 444, 172-186 (2018)*, 2018, doi: 10.1016/j.ins.2018.02.068.

[174] D. George, H. Shen, and E. A. Huerta, 'Deep Transfer Learning: A new deep learning glitch classification method for advanced LIGO', *arXiv 1706.07446v1 (22 Jun 2017); published as Phys. Rev. D 97, 101501(R) (2018)*, 2017, doi: 10.1103/PhysRevD.97.101501.

[175] D. George, H. Shen, and E. A. Huerta, 'Glitch Classification and Clustering for LIGO with Deep Transfer Learning', *Phys. Rev. D 97, 101501 (2018)*, 2018, [Online]. Available: https://arxiv.org/abs/1711.07468

[176] R. Mesuga and B. J. Bayanay, 'A Deep Transfer Learning Approach on Identifying Glitch Wave-form in Gravitational Wave Data', *arXiv preprint*, 2021, [Online]. Available: https://arxiv.org/abs/2107.01863

[177] M. Zevin, 'Gravity Spy: Lessons Learned and a Path Forward', *Eur. Phys. J. Plus 139, 100 (2024)*, 2024, [Online]. Available: https://arxiv.org/abs/2308.15530

[178] T. S. Fernandes, 'Convolutional Neural Networks for the classification of glitches in gravitational-wave data streams', 2023. [Online]. Available: https://arxiv.org/abs/2303.13917

[179] Y. Wu, 'Advancing Glitch Classification in Gravity Spy: Multi-view Fusion with Attention-based Machine Learning for Advanced LIGO's Fourth Observing Run', *Class. Quantum Grav. 42, 165015 (2025)*, 2024, [Online]. Available: https://arxiv.org/abs/2401.12913

[180] D. Srivastava and Niedzielski, 'Vision Transformer for Transient Noise Classification', *Acta Astronomica 74(3), 231-238 (2024)*, 2024, [Online]. Available: https://arxiv.org/abs/2510.06273

[181] A. Licciardi, 'Wavelet Scattering Transform for Gravitational Waves Analysis. An Application to Glitch Characterization', 2024. [Online]. Available: https://arxiv.org/abs/2411.19122

[182] J. Yan, A. P. Leung, and C. Y. Hui, 'On Improving the Performance of Glitch Classification for Gravitational Wave Detection by using Generative Adversarial Networks', *Mon. Not. R. Astron. Soc. 515, 4606-4621 (2022)*, 2022, doi: 10.1093/mnras/stac1996.

[183] Talpini and Razzano, 'Deep convolutional neural network to characterize transient noise in gravitational-wave detectors', *Nuovo Cimento C 2022*, 2022, doi: 10.1393/ncc/i2022-22154-0.

[184] O. T. Biskin, I. Kirbas, and A. Celik, 'A fast and time-efficient glitch classification method: A deep learning-based visual feature extractor for machine learning algorithms', *Astronomy and Computing 42, 100683 (2023; online Dec 2022)*, 2023, doi: 10.1016/j.ascom.2022.100683.

[185] N. Koyama *et al.*, 'Enhancing the rationale of convolutional neural networks for glitch classification in gravitational wave detectors: a visual explanation', *Mach. Learn.: Sci. Technol. 5 (2024) 035028*, 2024, doi: 10.1088/2632-2153/ad6391.

[186] A.-K. Malz, G. Ashton, and N. Colombo, 'Classification uncertainty for transient gravitational-wave noise artefacts with optimised conformal prediction', *Phys. Rev. D 111 (2025) 084078*, 2024, doi: 10.1103/PhysRevD.111.084078.

[187] M. Bhandari, A. Thapa, L. Regmi, S. Sharma, P. Paudel, and S. P. Panday, 'Interpreting decisions of lightweight CNN for glitch classification in time frequency images: Enhancing black-hole merger detection in O1 and O2 runs', *Physics of the Dark Universe 48, 101914 (May 2025)*, 2025, doi: 10.1016/j.dark.2025.101914.

[188] Singh and Sharma, 'Gravitational Wave and Glitch Spectrogram Classification using Convolutional Neural Networks', *IEEE ICSCDS 2025*, 2025, doi: 10.1109/ICSCDS65426.2025.11167769.

[189] T. Fernandes, F. Di Renzo, A. Onofre, A. Torres-Forne, and J. A. Font, 'Automatic classification pipeline for glitches in the Virgo detector', *arXiv preprint 2604.13687v2 (20 Aug 2026); published as doi 10.1103/ys5f-2yqk (journal not stated in the arXiv text)*, 2026, doi: 10.1103/ys5f-2yqk.

[190] Q.-Y. Xie, R.-C. Kang, J.-T. Li, and S.-D. Zhao, 'Application of fusion-view and optimization convolutional neural network in glitch classification', *Physica Scripta 101(11), 116004 (2026)*, 2026, doi: 10.1088/1402-4896/ae4e62.

[191] S. Bahaadini, 'DIRECT: Deep Discriminative Embedding for Clustering of LIGO Data', 2018. [Online]. Available: https://arxiv.org/abs/1805.02296

[192] S. B. Coughlin, 'Classifying the unknown: discovering novel gravitational-wave detector glitches using similarity learning', *Phys. Rev. D 99, 082002 (2019)*, 2019, [Online]. Available: https://arxiv.org/abs/1903.04058

[193] S. Bahaadini, Y. Wu, S. Coughlin, M. Zevin, and A. K. Katsaggelos, 'Discriminative Dimensionality Reduction using Deep Neural Networks for Clustering of LIGO Data', *arXiv preprint*, 2022, [Online]. Available: https://arxiv.org/abs/2205.13672

[194] Y. Sakai, 'Unsupervised Learning Architecture for Classifying the Transient Noise of Interferometric Gravitational-wave Detectors', *Scientific Reports 12, 9935 (2022)*, 2022, [Online]. Available: https://arxiv.org/abs/2111.10053

[195] Y. Li, Y. Wu, and A. K. Katsaggelos, 'Cross-Temporal Spectrogram Autoencoder (CTSAE): Unsupervised Dimensionality Reduction for Clustering Gravitational Wave Glitches', *CVPR Workshops 2024*, 2024, [Online]. Available: https://arxiv.org/abs/2404.15552

[196] S. Deshmukh, C. Chatterjee, A. Petulante, T. A. Ferreira, and K. Jani, 'The Sound of Noise: Leveraging the Inductive Bias of Pre-trained Audio Transformers for Glitch Identification in LIGO', *arXiv preprint*, 2026, [Online]. Available: https://arxiv.org/abs/2601.20034

[197] Y. Kim, K. Jung, and K. Kwak, 'Unsupervised Deep Learning Method for Clustering KAGRA O3GK Transient Noise Data', *MNRAS preprint (2026); arXiv:2608.24690*, 2026, [Online]. Available: https://arxiv.org/abs/2608.24690

[198] S. Oshino, Y. Sakai, M. Meyer-Conde, T. Uchiyama, Y. Itoh, and Y. Shikano, 'Glitch noise classification in KAGRA O3GK observing data using unsupervised machine learning', *arXiv preprint*, 2025, [Online]. Available: https://arxiv.org/abs/2510.14291

[199] Laguarta, 'Detection of anomalies amongst LIGO's glitch populations with autoencoders', *Class. Quantum Grav.*, 2023, [Online]. Available: https://arxiv.org/abs/2310.03453

[200] J. Ding, R. Ng, and J. McIver, 'UniMAP: Model-free detection of unclassified noise transients in LIGO-Virgo data using the Temporal Outlier Factor', *arXiv preprint*, 2021, doi: 10.1088/1361-6382/ac7278.

[201] S. Sankarapandian and B. Kulis, 'β-Annealed Variational Autoencoder for glitches', *arXiv preprint*, 2021, [Online]. Available: https://arxiv.org/abs/2107.10667

[202] S. Alvarez-Lopez, 'GSpyNetTree: A signal-vs-glitch classifier for gravitational-wave event candidates', 2023. [Online]. Available: https://arxiv.org/abs/2304.09977

[203] S. Jarov, 'A new method to distinguish gravitational-wave signals from detector noise transients with Gravity Spy', 2023. [Online]. Available: https://arxiv.org/abs/2307.15867

[204] S. Choudhary, A. More, S. Suyamprakasam, and S. Bose, 'SiGMa-Net: Deep learning network to distinguish binary black hole signals from short-duration noise transients', *arXiv preprint*, 2022, doi: 10.1103/PhysRevD.107.024030.

[205] M. VanDyke, K. Wu, and S. Bose, 'CoBiTS: Single-detector discrimination of binary black hole signals from glitches using deep learning', *arXiv preprint*, 2025, [Online]. Available: https://arxiv.org/abs/2512.17975

[206] R. Magee, R. Sharma, A. Agrawal, and R. Udall, 'Mitigating the impact of noise transients in gravitational-wave searches using reduced basis timeseries and convolutional neural networks', *arXiv preprint*, 2024, [Online]. Available: https://arxiv.org/abs/2410.15513

[207] N. Arutkeerthi, X. Li, C. Cui, and S. Valluri, 'Machine Learning based Glitch Veto for inspiral binary merger signals using Linear Chirp Transform', *arXiv preprint (2410.20269; downloaded PDF is v2, 27 Jul 2026, which cites 2025-2026 material; v1 was Oct 2024); no journal reference found*, 2024, [Online]. Available: https://arxiv.org/abs/2410.20269

[208] N. Shah, A. M. Knee, J. McIver, and D. Stenning, 'Waves in a Forest: A Random Forest Classifier to Distinguish between Gravitational Waves and Detector Glitches', *arXiv preprint*, 2023, [Online]. Available: https://arxiv.org/abs/2306.13787

[209] S. Bini, G. Vedovato, M. Drago, F. Salemi, and G. A. Prodi, 'An autoencoder neural network integrated into gravitational-wave burst searches to improve the rejection of noise transients', *Class. Quantum Grav. 40, 135008 (2023)*, 2023, doi: 10.1088/1361-6382/acd981.

[210] M. L. Chan, J. McIver, Y. Lecoeuche, D. Raghunathan, S. Álvarez-López, and J. Ding, 'GSpyNetTreeS: a machine learning solution for glitch localization in time and frequency', *arXiv preprint*, 2025, [Online]. Available: https://arxiv.org/abs/2511.16861

[211] R. Gupta and C. Binu, 'Transformer-Based Auto-Encoder for Multi-Class Gravitational Wave Signal Classification', *2025 IEEE MIT Undergraduate Research Technology Conference (URTC), IEEE Xplore 11533073*, 2025, doi: 10.1109/URTC68753.2025.11533073.

[212] R. Essick, P. Godwin, C. Hanna, L. Blackburn, and E. Katsavounidis, 'iDQ: Statistical inference of non-gaussian noise with auxiliary degrees of freedom in gravitational-wave detectors', *Mach. Learn.: Sci. Technol. 2, 015004 (2020)*, 2020, doi: 10.1088/2632-2153/abab5f.

[213] R. E. Colgan, 'Efficient Gravitational-wave Glitch Identification from Environmental Data Through Machine Learning (EMU)', *Phys. Rev. D 101, 102003 (2020)*, 2020, [Online]. Available: https://arxiv.org/abs/1911.11831

[214] R. E. Colgan, Z. Márka, J. Yan, I. Bartos, J. N. Wright, and S. Márka, 'Detecting and Diagnosing Terrestrial Gravitational-Wave Mimics Through Feature Learning', *arXiv preprint*, 2022, doi: 10.1103/PhysRevD.107.062006.

[215] M. Cavaglia, K. Staats, and T. Gill, 'Finding the origin of noise transients in LIGO data with machine learning', *Commun. Comput. Phys., 25 (2019), pp. 963-987*, 2018, doi: 10.4208/cicp.OA-2018-0092.

[216] J. Powell, 'Classification methods for noise transients in advanced gravitational-wave detectors', 2015. [Online]. Available: https://arxiv.org/abs/1505.01299

[217] M. Razzano and E. Cuoco, 'Image-based deep learning for classification of noise transients in gravitational wave detectors', *Class. Quantum Grav. 35, 095016 (2018)*, 2018, [Online]. Available: https://arxiv.org/abs/1803.09933

[218] M. Llorens-Monteagudo, A. Torres-Forne, J. A. Font, and A. Marquina, 'Classification of gravitational-wave glitches via dictionary learning', *Class. Quantum Grav. (DOI 10.1088/1361-6382/ab0657); arXiv v1 9 Nov 2018 shows no journal reference*, 2018, doi: 10.1088/1361-6382/ab0657.

[219] J. Powell, 'Classification methods for noise transients in advanced gravitational-wave detectors II: performance tests on Advanced LIGO data', 2016. [Online]. Available: https://arxiv.org/abs/1609.06262

[220] V. Boudart, 'A convolutional neural network to distinguish glitches from minute-long gravitational wave transients', 2022. [Online]. Available: https://arxiv.org/abs/2210.04588

[221] E.-S. Apostol and C.-O. Truică, 'Efficient Machine Learning Ensemble Methods for Detecting Gravitational Wave Glitches in LIGO Time Series', *arXiv preprint (2311.02106, Nov 2023); no journal reference in the PDF*, 2023, [Online]. Available: https://arxiv.org/abs/2311.02106

[222] R. Manoharan and G. Cleaver, 'Evaluating Deep Learning Models for Multiclass Classification of LIGO Gravitational-Wave Glitches', *arXiv preprint*, 2026, [Online]. Available: https://arxiv.org/abs/2604.08796

[223] T. Dooney, S. Bromuri, and L. Curier, 'DVGAN: Stabilize Wasserstein GAN training for time-domain Gravitational Wave physics', *2022 IEEE International Conference on Big Data (Big Data), pp. 5468-5477, Osaka*, 2022, doi: 10.1109/BigData55660.2022.10021080.

[224] T. Dooney *et al.*, 'Time-Domain Synthesis of Gravitational-Wave Detector Glitches using Class-Conditional Derivative Generative Adversarial Networks', *arXiv 2606.27227v3 (14 Sep 2026); the code repository README states Phys. Rev. D 114, 082005 (2026), doi 10.1103/d5yf-3jb4*, 2026, doi: 10.1103/d5yf-3jb4.

[225] H. Shen, 'Denoising Gravitational Waves with Enhanced Deep Recurrent Denoising Auto-Encoders', *ICASSP 2019*, 2019, [Online]. Available: https://arxiv.org/abs/1903.03105

[226] C. Chatterjee, L. Wen, F. Diakogiannis, and K. Vinsen, 'Extraction of binary black hole gravitational wave signals from detector data using deep learning', *Phys. Rev. D 104, 064046 (2021)*, 2021, doi: 10.1103/PhysRevD.104.064046.

[227] W. Wei and E. A. Huerta, 'Gravitational Wave Denoising of Binary Black Hole Mergers with Deep Learning', *Phys. Lett. B 800, 135081 (2020)*, 2020, [Online]. Available: https://arxiv.org/abs/1901.00869

[228] P. Bacon, A. Trovato, and M. Bejger, 'Denoising gravitational-wave signals from binary black holes with dilated convolutional autoencoder', 2022. [Online]. Available: https://arxiv.org/abs/2205.13513

[229] H. Wang, Y. Zhou, Z. Cao, Z.-K. Guo, and Z. Ren, 'WaveFormer: transformer-based denoising method for gravitational-wave data', 2024. [Online]. Available: https://arxiv.org/abs/2212.14283

[230] T. Dooney, 'DeepExtractor: Time-domain reconstruction of signals and glitches in gravitational wave data with deep learning', *Phys. Rev. D 112, 044022 (2025)*, 2025, [Online]. Available: https://arxiv.org/abs/2501.18423

[231] C. Chatterjee and K. Jani, 'Reconstruction of binary black hole harmonics in LIGO using deep learning', *ApJ 969 25 (2024)*, 2024, doi: 10.3847/1538-4357/ad4602.

[232] C. Chatterjee and K. Jani, 'Navigating Unknowns: Deep Learning Robustness for Gravitational Wave Signal Reconstruction', *2024 ApJ 973 112*, 2024, doi: 10.3847/1538-4357/ad6984.

[233] Y.-D. Lee and H.-J. Yo, 'Denoising gravitational wave with deep learning in the time-frequency domain', *Chin. J. Phys. 98 (2025) 1069-1083 (arXiv:2511.20731v1, 25 Nov 2025)*, 2025, doi: 10.1016/j.cjph.2025.11.013.

[234] Y.-C. Lin and A. K. H. Kong, 'Extract non-Gaussian Features in Gravitational Wave Observation Data Using Self-Supervised Learning', *Phys. Rev. D 111, 063520 (2025)*, 2024, doi: 10.1103/PhysRevD.111.063520.

[235] C. Murali and D. Lumley, 'Detecting and Denoising Gravitational Wave Signals from Binary Black Holes using Deep Learning', *arXiv preprint*, 2022, [Online]. Available: https://arxiv.org/abs/2210.01718

[236] A. Ghalsasi, 'Amplifying the Chirp: Using Deep Learning (U-Nets) to filter signal from noise in LIGO data', *arXiv preprint*, 2023, [Online]. Available: https://arxiv.org/abs/2311.17198

[237] C. Ma, S. Li, Z. Cao, and M. Jia, 'Using deep learning to denoise and detect gravitational waves (MSNRnet-2)', *Phys. Rev. D 110, 063010 (2024), 16 pp (INSPIRE record 2825628)*, 2024, doi: 10.1103/PhysRevD.110.063010.

[238] C. Ma, C. Ma, Z. Cao, and M. Jia, 'Gravitational wave search by time-scale-recursive denoising and matched filtering', *Sci. China Phys. Mech. Astron. 67(12), 129511 (2024), 17 pp (INSPIRE record 2853051)*, 2024, doi: 10.1007/s11433-024-2469-4.

[239] Sakai and Odonchimed, 'Automated design of digital filters using convolutional neural networks for extracting ringdown gravitational waves', *Mach. Learn.: Sci. Technol. 2024*, 2024, doi: 10.1088/2632-2153/ad8b94.

[240] D. Jia *et al.*, 'Dendritic learning-based gravitational waves prediction model', *Information Sciences 740, 123218 (June 2026)*, 2026, doi: 10.1016/j.ins.2026.123218.

[241] L. Kumar and S. K. Sahay, 'Temporal context and noise conditioning in neural gravitational-wave inpainting: An architecture ablation study for sub-second strain gaps', *Astronomy and Computing 57 (October 2026)*, 2026, doi: 10.1016/j.ascom.2026.101174.

[242] M. Saleem, 'Demonstration of Machine Learning-assisted real-time noise regression in gravitational wave detectors', 2023. [Online]. Available: https://arxiv.org/abs/2306.11366

[243] R. W. Kiendrebeogo *et al.*, 'Application of Non-Linear Noise Regression in the Virgo Detector', *Phys. Rev. D 112, 083053 (2025); arXiv:2410.06220*, 2025, doi: 10.1103/hwnc-r7yy.

[244] C. Reissel, S. Soni, M. Saleem, M. Coughlin, P. Harris, and E. Katsavounidis, 'Coherence DeepClean: Toward autonomous denoising of gravitational-wave detector data', *arXiv preprint*, 2025, [Online]. Available: https://arxiv.org/abs/2501.04883

[245] G. Vajente, 'Machine-learning non-stationary noise out of gravitational wave detectors', *Phys. Rev. D 101, 042003 (2020)*, 2020, [Online]. Available: https://arxiv.org/abs/1911.09083

[246] H. Yu and R. X. Adhikari, 'Nonlinear noise regression in gravitational-wave detectors with convolutional neural networks', 2021. [Online]. Available: https://arxiv.org/abs/2111.03295

[247] J. Kelleter, P. Schillings, J. Kuckert, D. Bertram, M. Bachlechner, and A. Stahl, 'NNNN: Neural Networks for Newtonian Noise Mitigation at the Einstein Telescope', *arXiv preprint*, 2026, [Online]. Available: https://arxiv.org/abs/2606.19907

[248] K. Mogushi, 'Reduction of transient noise artifacts in gravitational-wave data using deep learning', 2021. [Online]. Available: https://arxiv.org/abs/2105.10522

[249] K. Mogushi, 'NNETFIX: An artificial neural network-based denoising engine for gravitational-wave signals', 2021. [Online]. Available: https://arxiv.org/abs/2101.04712

[250] A. Torres-Forné, E. Cuoco, J. A. Font, and A. Marquina, 'Application of dictionary learning to denoise LIGO's blip noise transients', *Phys. Rev. D 102, 023011 (2020)*, 2020, doi: 10.1103/PhysRevD.102.023011.

[251] A. Pimpalkar, D. Wadekar, M. H.-Y. Cheung, and E. Berti, 'Sample-efficient non-Gaussian noise reduction in gravitational wave data via learnable wavelets', *arXiv preprint*, 2026, [Online]. Available: https://arxiv.org/abs/2601.14326

[252] C. Ma, X. Yu, Z. Cao, and M. Jia, 'Extraction of binary neutron star gravitational wave waveforms from Einstein Telescope using deep learning', *arXiv preprint 2406.15813v1 (22 Jun 2024); no journal reference on arXiv*, 2024, [Online]. Available: https://arxiv.org/abs/2406.15813

[253] C. Ma, W. Zhou, and Z. Cao, 'Gravitational Wave Mixture Separation for Future Gravitational Wave Observatories Utilizing Deep Learning', *arXiv preprint 2407.13239v1 (18 Jul 2024); no journal reference on arXiv*, 2024, [Online]. Available: https://arxiv.org/abs/2407.13239

[254] Y. Xu, H. Wang, M. Du, B. Liang, and P. Xu, 'Gravitational Wave Signal Denoising and Merger Time Prediction By Deep Neural Network', *Phys. Rev. D 111, 063037 (2025)*, 2024, doi: 10.1103/PhysRevD.111.063037.

[255] M. Dax, S. R. Green, J. Gair, N. Gupte, M. Pürrer, and V. Raymond, 'Real-time gravitational-wave inference for binary neutron stars using machine learning', *Nature 639, 49-53 (2025)*, 2024, doi: 10.1038/s41586-025-08593-z.

[256] A. Kofler, 'Flexible Gravitational-Wave Parameter Estimation with Transformers (Dingo-T1)', *Phys. Rev. Lett. 136, 251401 (2026)*, 2025, [Online]. Available: https://arxiv.org/abs/2512.02968

[257] J. C. L. Chan, L. M. Zertuche, J. M. Ezquiaga, R. K. L. Lo, L. Vujeva, and J. Bowman, 'Identification and characterization of distorted gravitational waves by lensing using deep learning', *arXiv preprint*, 2025, doi: 10.1103/8cz1-kl6n.

[258] Y.-X. Zhang *et al.*, 'Factorized neural posterior estimation for rapid and reliable inference of parameterized post-Einsteinian deviation parameters in gravitational waves', *Commun. Theor. Phys. 78 (2026) 115401; arXiv:2602.01319 (PDF read is v2, 30 Sep 2026)*, 2026, doi: 10.1088/1572-9494/ae8f1e.

[259] J.-W. Luo and Y.-S. Piao, 'Searching for gravitational wave echo with group equivariant neural posterior estimation', *arXiv preprint 2609.18337v2 (28 Sep 2026); no journal reference on arXiv*, 2026, [Online]. Available: https://arxiv.org/abs/2609.18337

[260] Q. Hu *et al.*, 'Decoding Long-duration Gravitational Waves from Binary Neutron Stars with Machine Learning: Parameter Estimation and Equations of State', *Astrophys. J. Lett. 987, L17 (2025); arXiv 2412.03454 (v2 dated 28 Jun 2025)*, 2024, doi: 10.3847/2041-8213/ade42f.

[261] D. Lanchares *et al.*, 'Assessment of normalizing flows for parameter estimation on time-frequency representations of gravitational-wave data', *Phys. Rev. D 114 (2026) 043019; arXiv:2505.08089*, 2025, doi: 10.1103/pz2n-b4m5.

[262] W. van Straalen, A. Kolmus, J. Janquart, and C. Van Den Broeck, 'Pre-Merger Detection and Characterization of Inspiraling Binary Neutron Stars Derived from Neural Posterior Estimation', *arXiv preprint*, 2024, [Online]. Available: https://arxiv.org/abs/2407.10263

[263] H. Gabbard, 'Bayesian parameter estimation using conditional variational autoencoders for gravitational-wave astronomy (VItamin)', *Nature Physics 18, 112-117 (2022)*, 2022, [Online]. Available: https://arxiv.org/abs/1909.06296

[264] R. Bada-Nerin, O. Bulashenko, O. G. Freitas, and J. A. Font, 'Parameter estimation of microlensed gravitational waves with Conditional Variational Autoencoders', *Phys. Rev. D 111, 084067 (2025); arXiv:2412.00566 (v2 14 Apr 2025)*, 2024, doi: 10.1103/PhysRevD.111.084067.

[265] A. Delaunoy, A. Wehenkel, T. Hinderer, S. Nissanke, C. Weniger, and A. R. Williamson, 'Lightning-Fast Gravitational Wave Parameter Inference through Neural Amortization', *arXiv preprint*, 2020, [Online]. Available: https://arxiv.org/abs/2010.12931

[266] M. Emma and G. Ashton, 'RNLE: Residual neural likelihood estimation and its application to gravitational-wave astronomy', *arXiv preprint 2601.13857v1*, 2026, [Online]. Available: https://arxiv.org/abs/2601.13857

[267] A. J. K. Chua and M. Vallisneri, 'Learning Bayesian posteriors with neural networks for gravitational-wave inference', *Phys. Rev. Lett. 124, 041102 (2020)*, 2019, doi: 10.1103/PhysRevLett.124.041102.

[268] H. Shen, E. A. Huerta, E. O'Shea, P. Kumar, and Z. Zhao, 'Statistically-informed deep learning for gravitational wave parameter estimation', *Machine Learning: Science and Technology, Volume 3, Number 1, Year 2022*, 2019, doi: 10.1088/2632-2153/ac3843.

[269] A. McLeod, D. Jacobs, C. Chatterjee, L. Wen, and F. Panther, 'Rapid Mass Parameter Estimation of Binary Black Hole Coalescences Using Deep Learning', *arXiv preprint*, 2022, [Online]. Available: https://arxiv.org/abs/2201.11126

[270] M. Andrés-Carcasona, M. Martinez, and L. M. Mir, 'Fast Bayesian gravitational wave parameter estimation using convolutional neural networks', *Mon. Not. R. Astron. Soc. (MNRAS; arXiv:2309.04303v2, Nov 2023, MNRAS style file)*, 2023, [Online]. Available: https://arxiv.org/abs/2309.04303

[271] K. L. Jenner, 'Convolutional Neural Network for Analysing Gravitational Wave Signals', *Thesis, University of Adelaide (2024)*, 2024.

[272] L. Papalini, F. De Santi, M. Razzano, I. S. Heng, and E. Cuoco, 'Can Transformers help us perform parameter estimation of overlapping signals in gravitational wave detectors?', *arXiv preprint*, 2025, [Online]. Available: https://arxiv.org/abs/2505.02773

[273] C. Chatterjee, L. Wen, K. Vinsen, M. Kovalam, and A. Datta, 'Using Deep Learning to Localize Gravitational Wave Sources', *Phys. Rev. D 100, 103025 (2019)*, 2019, doi: 10.1103/PhysRevD.100.103025.

[274] C. Chatterjee, M. Kovalam, L. Wen, D. Beveridge, F. Diakogiannis, and K. Vinsen, 'Rapid localization of gravitational wave sources from compact binary coalescences using deep learning', *The Astrophysical Journal, Volume 959, Number 1 (2023)*, 2022, doi: 10.3847/1538-4357/ad08b7.

[275] C. Chatterjee and L. Wen, 'Pre-merger sky localization of gravitational waves from binary neutron star mergers using deep learning (GW-SkyLocator)', *Astrophys. J. 959, No. 2 (2023)*, 2023, [Online]. Available: https://arxiv.org/abs/2301.03558

[276] A. Kolmus, G. Baltus, J. Janquart, T. van Laarhoven, S. Caudill, and T. Heskes, 'Swift sky localization of gravitational waves using deep learning seeded importance sampling', *Phys. Rev. D 106, 023032 (2022); arXiv:2111.00833*, 2021, doi: 10.1103/PhysRevD.106.023032.

[277] J. P. Marulanda, C. Santa, and A. E. Romano, 'Deep learning merger masses estimation from gravitational waves signals in the frequency domain', *arXiv preprint*, 2020, doi: 10.1016/j.physletb.2020.135790.

[278] S. Singh, A. Singh, A. Prajapati, and K. N. Pathak, 'Deep Learning for estimating parameters of Gravitational Waves', *arXiv preprint*, 2020, doi: 10.1093/mnras/stab2417.

[279] S. Soma, H. Stöcker, and K. Zhou, 'Mass and tidal parameter extraction from gravitational waves of binary neutron stars mergers using deep learning', *JCAP 01(2024)009*, 2023, doi: 10.1088/1475-7516/2024/01/009.

[280] M. Benedikt and I. D. Saltas, 'GraviBERT: Transformer-based inference for gravitational-wave time series', *arXiv preprint*, 2025, [Online]. Available: https://arxiv.org/abs/2512.21390

[281] E. Khalouei, C. G. Sabiu, H. M. Lee, and A. Gopakumar, 'External Attention Transformer: A Robust AI Model for Identifying Initial Eccentricity Signatures in Binary Black Hole Events in Simulated Advanced LIGO Data', *Journal of Cosmology and Astroparticle Physics 2025(10), 028; arXiv v2 read*, 2025, doi: 10.1088/1475-7516/2025/10/028.

[282] H. Iwanaga, M. Matsuyama, and Y. Itoh, 'Enhancing the reliability of machine learning for gravitational wave parameter estimation with attention-based models', *Physical Review D 112, 083020 (2025); arXiv v2 read*, 2025, doi: 10.1103/blfk-7k9f.

[283] P. N. Sakellariou, S. V. Georgakopoulos, S. Tasoulis, and V. P. Plagianakos, 'Binary Black Hole Parameter Estimation with Hybrid CNN-Transformer Neural Networks', *Astronomy and Computing, vol. 54, 101027 (2026)*, 2026, doi: 10.1016/j.ascom.2025.101027.

[284] A. Ravichandran, A. Vijaykumar, S. J. Kapadia, and P. Kumar, 'Rapid Identification and Classification of Eccentric Gravitational Wave Inspirals with Machine Learning', *arXiv preprint*, 2023, [Online]. Available: https://arxiv.org/abs/2302.00666

[285] W.-H. Ruan, H. Wang, C. Liu, and Z.-K. Guo, 'Rapid search for massive black hole binary coalescences using deep learning', *Phys. Lett. B 841, 137904 (2023)*, 2023, doi: 10.1016/j.physletb.2023.137904.

[286] W.-H. Ruan and Z.-K. Guo, 'Premerger detection of massive black hole binaries using deep learning', *Phys. Rev. D 109, 123031 (2024)*, 2024, doi: 10.1103/PhysRevD.109.123031.

[287] W.-H. Ruan, H. Wang, C. Liu, and Z.-K. Guo, 'Parameter inference for coalescing massive black hole binaries using deep learning', *Universe 2023, 9(9), 407*, 2023, doi: 10.3390/universe9090407.

[288] T. Zhao, R. Lyu, H. Wang, Z. Cao, and Z. Ren, 'Space-based gravitational wave signal detection and extraction with deep neural network', *Commun. Phys. 6, 212 (2023)*, 2023, doi: 10.1038/s42005-023-01334-6.

[289] X.-T. Zhang, C. Messenger, N. Korsakova, M. L. Chan, Y.-M. Hu, and J.-dong Zhang, 'Detecting Gravitational-waves from Extreme Mass Ratio Inspirals using Convolutional Neural Networks', *Phys. Rev. D 105, 123027 (2022); arXiv:2202.07158 (the PDF read is arXiv v1, 15 Feb 2022, so the published text may differ)*, 2022, doi: 10.1103/PhysRevD.105.123027.

[290] Q. Yun, W.-B. Han, Y.-Y. Guo, H. Wang, and M. Du, 'Detecting extreme-mass-ratio inspirals for space-borne detectors with deep learning', *arXiv preprint*, 2023, [Online]. Available: https://arxiv.org/abs/2309.06694

[291] H.-Y. Liu and Y.-T. Wang, 'Self-supervised learning for gravitational wave signal identification', *arXiv preprint 2302.00295v2 (7 Sep 2023); no journal reference on arXiv*, 2023, [Online]. Available: https://arxiv.org/abs/2302.00295

[292] N. Houba, S. H. Strub, L. Ferraioli, and D. Giardini, 'Detection and Prediction of Future Massive Black Hole Mergers with Machine Learning and Truncated Waveforms', *arXiv preprint*, 2024, [Online]. Available: https://arxiv.org/abs/2405.11340

[293] A. Caramete *et al.*, 'Characterization of Gravitational Waves Signals Using Neural Networks', *arXiv preprint*, 2020, [Online]. Available: https://arxiv.org/abs/2009.06109

[294] N. Houba, L. Ferraioli, and D. Giardini, 'Detection and Mitigation of Glitches in LISA Data: A Machine Learning Approach', *Phys. Rev. D 109, 083027 (2024); arXiv preprint 2401.00846*, 2024, doi: 10.1103/PhysRevD.109.083027.

[295] N. Houba, 'Deep source separation of overlapping gravitational-wave signals and non-stationary noise artifacts', *arXiv preprint*, 2025, [Online]. Available: https://arxiv.org/abs/2503.10398

[296] Y. Xu, M. Du, B. Liang, Z. Xiao, T. Zhao, and P. Xu, 'Self-supervised reconstruction of transients in data from space-borne gravitational-wave detectors', *arXiv preprint*, 2026, [Online]. Available: https://arxiv.org/abs/2609.05890

[297] M.-C. Isfan, L.-I. Caramete, A. Caramete, D. Tonoiu, and A. Nicolin-Żaczek, 'Quantum Computing Tools for Fast Detection of Gravitational Waves in the Context of LISA Space Mission', *Classical and Quantum Gravity 42 (2025); arXiv v3 read*, 2025, doi: 10.1088/1361-6382/ae1787.

[298] M.-C. Isfan, L.-I. Caramete, and A. Caramete, 'Evaluating state-of-the-art cloud quantum computers for quantum neural networks in gravitational waves data analysis', *arXiv preprint*, 2026, [Online]. Available: https://arxiv.org/abs/2601.14036

[299] M. Cabero, 'GWSkyNet: a real-time classifier for public gravitational-wave candidates', *Astrophys. J. Lett. 904, L9 (2020)*, 2020, [Online]. Available: https://arxiv.org/abs/2010.11829

[300] T. C. Abbott, E. Buffaz, N. Vieira, M. Cabero, D. Haggard, and A. Mahabal, 'GWSkyNet-Multi: A Machine Learning Multi-Class Classifier for LIGO-Virgo Public Alerts', *Astrophys. J. 927 (2022) (DOI 10.3847/1538-4357/ac5019); arXiv:2111.04015*, 2021, doi: 10.3847/1538-4357/ac5019.

[301] M. L. Chan, 'GWSkyNet II: a refined machine learning pipeline for real-time classification of public gravitational wave alerts', 2024. [Online]. Available: https://arxiv.org/abs/2408.06491

[302] J. McGinn, C. Messenger, M. J. Williams, and I. S. Heng, 'Generalised gravitational wave burst generation with generative adversarial networks (McGANn)', *Class. Quantum Grav. 38, 155005 (2021)*, 2021, doi: 10.1088/1361-6382/ac09cc.

[303] M. Lopez, V. Boudart, K. Buijsman, A. Reza, and S. Caudill, 'Simulating Transient Noise Bursts in LIGO with Generative Adversarial Networks', *Phys. Rev. D 106, 023027 (2022); arXiv v1 shows no journal reference*, 2022, doi: 10.1103/PhysRevD.106.023027.

[304] J. Powell, L. Sun, K. Gereb, P. D. Lasky, and M. Dollmann, 'Generating transient noise artifacts in gravitational-wave detector data with generative adversarial networks', *arXiv preprint*, 2022, doi: 10.1088/1361-6382/acb038.

[305] T. Dooney, 'cDVGAN: One Flexible Model for Multi-class Gravitational Wave Signal and Glitch Generation', *Phys. Rev. D 110, 022004 (2024)*, 2024, [Online]. Available: https://arxiv.org/abs/2401.16356

[306] K. Kim, J. Lee, R. S. H. Yuen, O. A. Hannuksela, and T. G. F. Li, 'Identification of Lensed Gravitational Waves with Deep Learning', *ApJ 915 (2021) 119*, 2020, doi: 10.3847/1538-4357/ac0143.

[307] K. Kim, J. Lee, O. A. Hannuksela, and T. G. F. Li, 'Deep Learning-based Search for Microlensing Signature from Binary Black Hole Events in GWTC-1 and -2', *ApJ 938 (2022) 157*, 2022, doi: 10.3847/1538-4357/ac92f3.

[308] Utina and Marangio, 'Deep learning searches for gravitational wave stochastic backgrounds', *IEEE CBMI 2021*, 2021, doi: 10.1109/CBMI50038.2021.9461904.

[309] H. Einsle, M.-A. Bizouard, T. Regimbau, and M. Sakellariadou, 'Gravitational-wave background detection using machine learning', *arXiv preprint 2506.14764v1*, 2025, [Online]. Available: https://arxiv.org/abs/2506.14764

[310] M. C. Edwards, 'Classifying the Equation of State from Rotating Core Collapse Gravitational Waves with Deep Learning', *Phys. Rev. D 103, 024025 (2021)*, 2021, [Online]. Available: https://arxiv.org/abs/2009.07367

[311] Y. Xu, M. Du, P. Xu, B. Liang, and H. Wang, 'Gravitational Wave Signal Extraction Against Non-Stationary Instrumental Noises with Deep Neural Network', *Phys. Lett. B 858, 139016 (2024)*, 2024, [Online]. Available: https://arxiv.org/abs/2402.13091

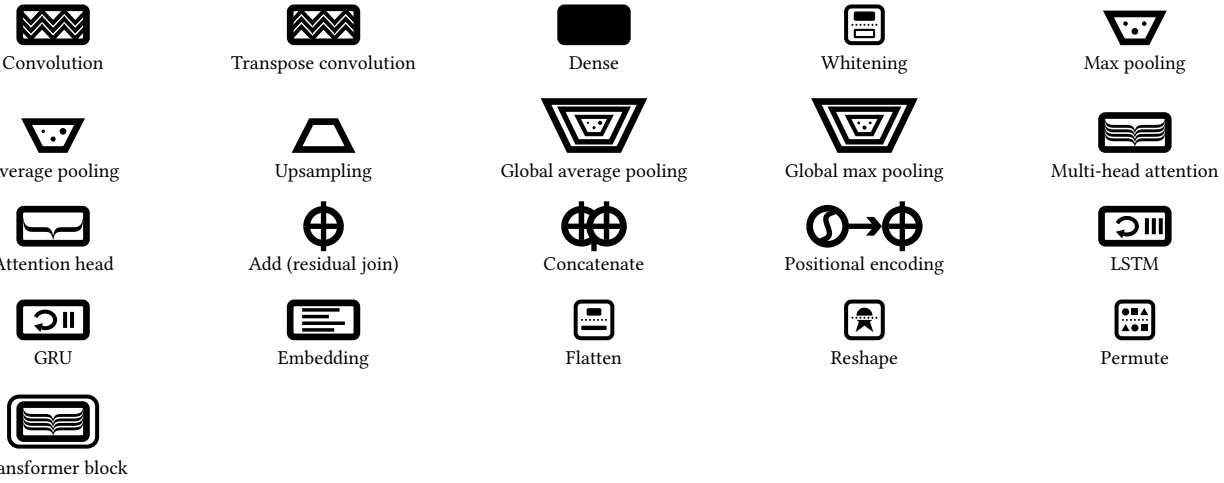
[312] Z. Que, E. Wang, U. Marikar, E. Moreno, J. Ngadiuba, and H. Javed, 'Accelerating Recurrent Neural Networks for Gravitational Wave Experiments', *arXiv preprint*, 2021, doi: 10.1109/ASAP52443.2021.00025.

[313] E. Cuoco, J. Powell, and M. Cavaglia, 'Enhancing Gravitational-Wave Science with Machine Learning (review)', *Mach. Learn.: Sci. Technol. 2, 011002 (2021)*, 2021, [Online]. Available: https://arxiv.org/abs/2005.03745

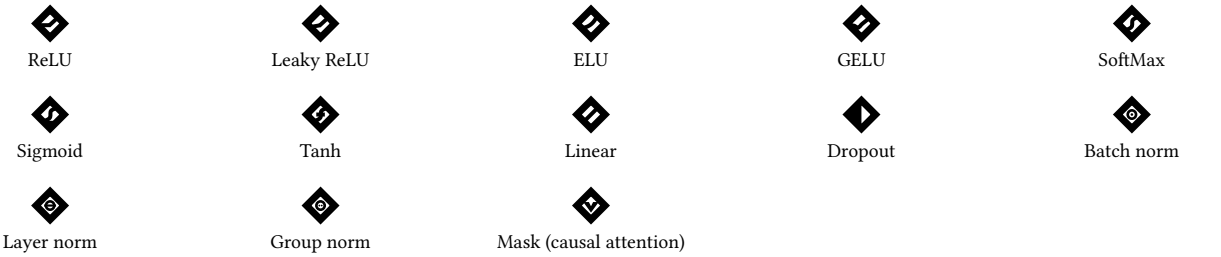
A Diagram notation

The network figures are drawn by the thesis-style generator from each model's diagram spec in the zoo, with blocks collapsed to one icon. A network runs top to bottom along its spine; side diamonds give the activation, regulariser or normalisation applied after a layer; the table beside the spine gives each row's hyperparameters, and a dash-dot bracket marked ×N is a repeated group. The symbols below are the generator's own, shown at half the generator's reference scale. A figure's confidence says how fully the paper specifies the network: **high** when a layer table or labelled figure gives it, **medium** when it is assembled from text and figures with some details unstated (listed in the spec, not drawn).

A.1 Layers and tensor operations

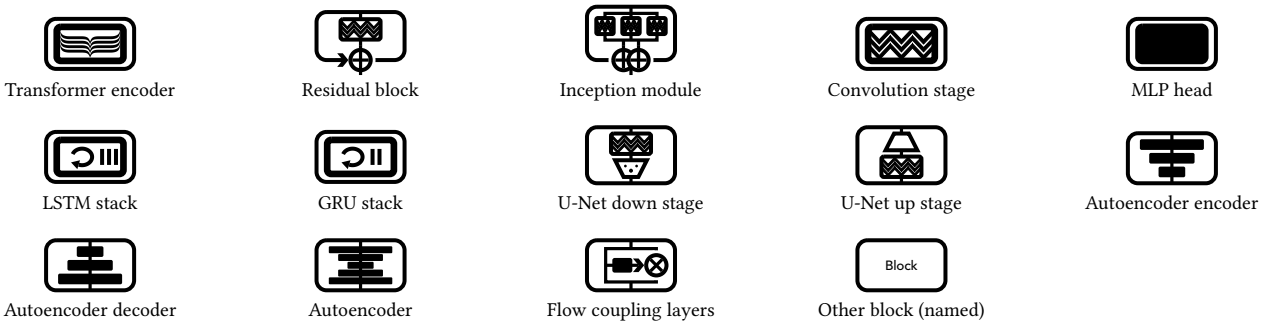


A.2 Side diamonds



A.3 Collapsed blocks

Each block of the figures is drawn closed, as one icon; its table row names the block and its repeat count.



A.4 Spine ends

